

Euclidean Three-Space over the Surreal Numbers

Coordinate geometry, spherical trigonometry,
the rotation group $\mathrm{SO}(3, \mathbf{No})$, and surreal compact groups

22–23 September 2026

Abstract

We develop Euclidean geometry in $\mathbf{No}^3$ in three parts. Every construction of Parts I and II is carried out in a set-sized real closed subfield of $\mathbf{No}$, or in a set-sized real-closed Hahn field when analytic evaluation is needed. Distances, areas, volumes and sphere radii may be infinite or infinitesimal. Angles between directions always have canonical finite surreal representatives, so spatial trigonometry needs no oscillatory value at an infinite argument.

Part I is coordinate geometry. It covers affine incidence, projections, skew lines, sphere intersections, quadrics and rigid motions. On spheres it proves the sine and cosine laws, polar duality, and sharp existence and congruence criteria, including an exceptional continuous SSA family. It also gives spherical centers, Ceva, a branch-correct solid-angle formula, and a finitely additive area for geodesic polygons. Parallel transport has quaternionic holonomy equal to the spherical excess, and a relative planar-limit area estimate holds for arbitrarily thin infinitesimal triangles.

Part II studies the class group $G = \mathrm{SO}(3, \mathbf{No})$. Rotation-matrix entries are bounded, and standard part gives a split extension $G = K \rtimes \mathrm{SO}(3, \mathbb{R})$. The infinitesimal kernel K is uniquely divisible, and each of its elements is a single commutator of elements of K . Every finite subgroup is conjugate to its standard part by an infinitesimal rotation. We treat strong exponential and logarithm maps and the valuation-graded cross-product algebra. For the global rotation exponential we give its fibers and differential; its critical radii are $2\pi\mathbf{Oz}$. Finally, three topologies are separated.

Part III refines Part II over a real closed field with its natural valuation. Single commutators of two valuation-cut subgroups fill exactly the cut subgroup of the summed cut, and the normal closure of an infinitesimal rotation is the full subgroup of its depth. Hence the proper normal subgroups of $\mathrm{SO}(3, F)$ are classified by valuation cuts. Principal subgroups have an explicit perfect residual, which is trivial exactly at order units. Over the full class $\mathbf{No}$, every class homomorphism from $\mathrm{SO}(3, \mathbf{No})$, from $\mathrm{SO}(n, \mathbf{No})$ with $n \geq 3$, or from $\mathrm{SU}(n, \mathbf{No}[i])$ with $n \geq 2$ to a set-sized group factors through standard part. For the circle this fails.

Part V treats every connected real linear algebraic group G with compact connected real points. Exactly the derived infinitesimal subgroup of $G(\mathbf{No})$ is invisible to set-sized groups, so a universal set-sized quotient exists exactly when G is semisimple, and it is then standard part.

The rotation algebra shared by Parts I and II is printed once in Part I: axes, Rodrigues, quaternions, and the Cayley chart. The quaternion double cover is cited from the collection’s surquaternion report.

Status and provenance. An AI-assisted research report built from four manuscripts: sources 04 and 09 of 22 September 2026, merged at 50cb709; source 10 of the same day, added as Part III at 7b5f934; and source 11 of 23 September 2026, added as Part V at 66d7e55; see [section 1.1](#). It has not been refereed, and no statement in it has a Lean formalization. Classical identities, the Bays–Peterzil perfectness precedent and imported analytic foundations are credited, not claimed. No priority claim or resolution of a named open problem is made. The accompanying symbolic checks test selected finite identities only.

Contents

1	Scope, sources, and conventions	7
1.1	Four manuscripts, one report	7
1.2	Relation to the collection	10
1.3	Three levels of assumptions	12
1.4	Notation, and the renamings made by the merge	12
2	Scalar foundations and canonical finite trigonometry	13
2.1	Set-sized workspaces inside the surreal numbers	13
2.2	What an infinite Taylor sum means	15
2.3	Sine, cosine, and inverse tangent	16
I	Coordinate geometry of No^3	17
3	Vectors, distances, orientation, and volume	17
3.1	Areas and tetrahedra	18
3.2	Finite-dimensional minimization does not need completeness	19
4	Affine analytic geometry	19
4.1	Lines, planes, and incidence	19
4.2	Spheres, circles, and discriminants	20
4.3	Quadrics and principal axes	21
4.4	Affine points are not projective ideal points	21
5	Rigid motions, rotations, quaternions, and screws	21
5.1	Axis–angle representation	22
5.2	Quaternion representation	24
5.3	The rational Cayley chart	25
5.4	Screw motions	26
6	Coordinates and ordinary triangle trigonometry in space	26
6.1	Cylindrical and spherical coordinates	26
6.2	Flat triangles retain their usual formulas	26
7	The surreal sphere and its intrinsic angular distance	27
7.1	Minor arcs without a global sine function	27
7.2	Gnomonic and stereographic charts	28
8	Spherical triangles and the Gram-matrix calculus	29
8.1	Polar triangles and the cosine law for angles	30
8.2	Half-angle, right-triangle, and haversine formulas	30

9	Existence, reconstruction, and ambiguous data	31
9.1	SAS, AAA, and ASA	31
9.2	The spherical SSA ambiguity has an exceptional family	32
9.3	Latitude, longitude, and bearings	33
10	Spherical centers, medians, and Ceva	33
10.1	Circumcenter and incenter	33
10.2	Medians and angle bisectors	34
11	Spherical excess, solid angles, and polygon area	34
11.1	L'Huilier's formula	35
11.2	Area as a precise finite-polygon functional	35
11.3	Solid angle of three arbitrary rays	36
12	Differential geometry and parallel transport	37
12.1	Connection, geodesics, and curvature	37
12.2	Explicit parallel transport	37
12.3	A quaternion proof of spherical holonomy	38
12.4	Curvature correction in normal coordinates	38
13	Infinitesimal scales and geometric conditioning	39
13.1	A valuation dictionary	39
13.2	A shape-independent local area estimate	40
13.3	Stable vector data and unstable derived data	41
14	Worked configurations at several surreal scales	41
14.1	Infinite lengths with infinitesimal directions	41
14.2	Nearly parallel lines meet their closest points infinitely far away	42
14.3	Tangency splits at a square-root scale	42
14.4	An almost planar tetrahedron with an infinite circumradius	42
14.5	A right spherical triangle and the flat limit of an infinite sphere	43
14.6	A tiny determinant with almost hemispherical area	43
15	What the development does not silently assume	44
15.1	The intrinsic order topology is not ordinary real topology	44
15.2	Ordinary finite partitions do not automatically define an integral	44
15.3	Finite dimension does not remove size or support conditions	44
II	The rotation group $SO(3, \mathbf{No})$	45
16	Frames, Euler coordinates, and quaternion matrices	45

16.1	Products of reflections and transitivity	45
16.2	Euler coordinates and their degeneracy	46
16.3	Quaternion coordinates	46
16.4	Complex matrices, projective coordinates, and a caution	46
16.5	A quaternion extraction algorithm without a small divisor	47
17	Conjugacy, centralizers, and maximal axis groups	47
18	The standard-part extension	48
19	A perfect, uniquely divisible infinitesimal group	49
20	Finite subgroups have no new surreal forms	50
20.1	Classification and what it does not claim	51
21	The Lie algebra and angular velocity	52
22	Strong exponential, logarithm, and BCH	53
22.1	A support theorem, not a convergence assumption	53
23	Valuation layers and noncommutativity at every scale	55
23.1	Why finite jets are nilpotent but the kernel is perfect	56
23.2	An imported structural perspective	56
24	Finite angles and canonical infinite phases	57
24.1	Finite-angle trigonometry from strong sums	57
24.2	The Ehrlich–Kaplan global extension does exist	57
25	The global rotation exponential and its differential	58
25.1	A radial map on the whole Lie algebra	58
25.2	Fine differentiability	59
26	Three topologies, and what the spin cover means	60
26.1	The full fine topology	60
26.2	The intrinsic topology of a set-sized workspace	61
26.3	The standard-part quotient topology is discrete	62
26.4	Semialgebraic paths, compactness, and the spin fundamental group	62
27	Worked examples at surreal scales	63
27.1	A tiny angle with a macroscopic displacement	63
27.2	A near-half-turn with an infinite Cayley coordinate	63
27.3	Infinitesimally different axes	64
27.4	Noncommutativity first appears at a finer scale	64

27.5	Infinite generators and local sensitivity	64
27.6	A nonreal copy of the octahedral group	64
III	Valuation cuts and the universal set-sized quotient	64
28	Setting, inheritance, and notation for Part III	65
28.1	Scalar hypotheses and the natural valuation	65
28.2	Depth and the infinitesimal kernel	66
28.3	What Part III inherits and what it adds	67
28.4	Notation, and the renamings made for Part III	68
29	Exact commutators at prescribed valuation depths	69
29.1	Arbitrary upper segments and valuation ideals	70
30	Normal generation and the classification of normal subgroups	71
30.1	The full normal-subgroup classification	72
30.2	An exact quotient chart	73
31	Perfect cut subgroups, the lower central series, and the perfect residual	73
31.1	Lower central and derived series	74
32	Recovering the value group from the normal-subgroup lattice	75
33	The full surreal class and the universal set-sized quotient	76
33.1	Foundational conventions	76
33.2	Proper classes inside every positive value interval	77
33.3	The cardinal obstruction in a fixed field	78
33.4	A perfect residual beyond every principal surreal threshold	78
34	Surcomplex special unitary groups and higher-dimensional rotations	78
34.1	The spin bridge for $SU(2)$	79
34.2	Finite near-identity elimination	79
35	Examples, sharp boundaries, and the circle	80
35.1	A rank-one workspace	80
35.2	A rank-two workspace with two principal behaviours	81
35.3	No largest Archimedean scale	81
35.4	The abelian boundary: the theorem fails for the circle	81
IV	Computation, limits, and provenance	82
36	Exact computation and formalization routes	82

36.1 A coordinate-to-invariants algorithm (source 04)	82
36.2 An implementation hierarchy for rotations (source 09)	83
36.3 A layered Lean development for the geometry (source 04)	84
36.4 A Lean development plan for the rotation group (source 09)	84
36.5 A formalization plan for Part III (source 10)	85
36.6 Constructive recipes from Part III (source 10)	85
36.7 The companion checks	86
37 Non-claims and limits, by source	87
37.1 Source 04 (Part I)	87
37.2 Source 09 (Part II)	88
37.3 Source 10 (Part III)	90
37.4 The merge	91
37.5 Source 11 (Part V)	92
38 Conclusion and directions	93
38.1 Questions left open by source 10	95
V Surreal compact groups: small representations and normal structure	96
39 Setting, the main theorem, and notation for Part V	96
39.1 Hypotheses and the main theorem	96
39.2 What Part V inherits and what it adds	97
39.3 Notation, and the renamings made for Part V	98
39.4 Scalars, size and standard part	99
40 Finite Nash charts and transfer at infinitesimal scales	101
40.1 Roots as a finite rescaling device	102
41 Exact normal generation at one valuation scale	102
42 Normal subgroups and valuation segments	104
42.1 Several simple factors and the absence of diagonal normals	105
43 Exact mixed commutator subgroups	106
43.1 The associated graded Lie algebra	107
44 Derived series and perfect residuals	107
45 Why every small representation kills semisimple infinitesimals	109
46 Central tori and the exact obstruction	110

46.1 Splitting the infinitesimal kernel	110
46.2 Logarithmic coordinates on the central infinitesimals	110
46.3 The class quotient and its universal property for observations	111
47 Examples over surreal and surcomplex scalars	112
47.1 Classical and exceptional semisimple groups	112
47.2 An explicit formula for the unitary obstruction	113
47.3 Products and finite observations	113
48 The split contrast: special linear groups have no small images	113
49 Omnific lattices inside compact matrix groups	114
50 Consistency, contrasts, and what remains	116
50.1 Consistency with Parts II and III	116
50.2 Relation to the omnific-groups report	116
50.3 What the results settle, and what remains	117
50.4 Proof audit and foundations checklist	117
A Formula and convention reference for Part II	118
B Source, dependency, and verification ledger	119
B.1 Pins and files	119
B.2 Source 04: inspected sources and dependencies	119
B.3 Source 09: inspected sources and dependency ledger	120
B.4 Source 10: inspected sources and dependency ledger	121
B.5 Source 11: inspected sources and dependency ledger	121

1 Scope, sources, and conventions

An infinite coordinate is not a point at infinity, and an infinitesimal distance is not zero. Those two distinctions are enough to change many geometric interpretations, but they do not destroy Euclidean coordinate algebra. The appropriate starting point is an ordered field equipped with positive square roots, rather than the completeness of the real line.

The central principle of Part I is

Coordinates and lengths can have arbitrary scale; geometric angles can always be finite.

For example, the vectors $(\omega, 1, 0)$ and $(\omega, 0, 0)$ make the nonzero angle $\text{atan}(\omega^{-1})$. The large coordinate does not require evaluation of $\sin \omega$. Likewise, a great-circle arc on a sphere of radius $R = \omega$ can have an infinite length Ra while its central angle a lies in $[0, \pi]$. The dimensionless quantity entering trigonometric functions is a , not Ra .

Part II asks what the group of rotations of this space is. Let $G(F) = \text{SO}(3, F) = \{A \in M_3(F) : A^T A = I, \det A = 1\}$ for a real closed field F ; the group of interest is $G(\mathbf{No})$, read with the size conventions of [section 2](#). There are three complementary answers:

$$G(\mathbf{No}) \cong \text{Sp}(1, \mathbf{No}) / \{1, -1\}, \quad (1.1)$$

$$G(\mathbf{No}) \cong K \rtimes \text{SO}(3, \mathbb{R}), \quad (1.2)$$

$$K \cong (\mathfrak{m}^3, \star), \quad t \star u = \frac{t + u + t \times u}{1 - t \cdot u}. \quad (1.3)$$

Here $\text{Sp}(1, \mathbf{No})$ is the unit group of the Hamilton quaternion algebra, $\mathfrak{m}$ is the class of infinitesimal surreals, and K consists of the rotation matrices infinitesimally close to the identity. The denominator in [equation \(1.3\)](#) is a unit infinitesimally close to one, so the formula is valid on all of $\mathfrak{m}^3$. It is a noncommutative group law, not vector addition. The first isomorphism is the algebraic double cover of the collection's surquaternion report; the second and third are [theorem 18.1](#) and [proposition 18.2](#).

The geometry of a rotation still has three scalar degrees of freedom. What changes is the range of those scalars and the meaning of analysis. Infinitesimal changes of axis and angle give genuinely different rotations even when their standard parts agree. A small rotation can move an infinitely distant point by a finite or infinite distance. Ordinary real paths, compactness arguments, and power-series convergence cannot simply be imported into the full fine topology.

Part III asks which normal subgroups the rotation group has, and what an arbitrary set-sized group can see of it. Over a real closed field $F \supseteq \mathbb{R}$ with its natural valuation, single commutators and normal closures in $G(F)$ are computed exactly in terms of valuation cuts. The resulting classification of normal subgroups and of perfect residuals is a statement about convex subgroups of the value group. For the full class $\mathbf{No}$, a further size argument shows that standard part is the universal set-sized quotient of $\text{SO}(3, \mathbf{No})$.

Part V asks the same question for every connected real linear algebraic group with compact connected real points. Only the derived subgroup of the infinitesimal kernel is invisible to set-sized groups. The infinitesimals of the central torus remain visible, one normal-form coefficient at a time, so standard part is the universal set-sized quotient exactly for semisimple groups ([theorem 39.1](#)). Part V is printed after Part IV so that the numbering of Parts I–IV is unchanged.

1.1 Four manuscripts, one report

This report is built from four manuscripts, written independently against different commits of the repository [VladimirReshetnikov/Surreal](#). Sources 04, 09 and 10 are dated 22 September 2026,

and source 11 is dated 23 September 2026. Sources 04 and 09 were merged into Parts I and II. Source 10 arrived later, as manuscript 03 of a later batch, and was added as Part III. Source 11 arrived as manuscript 06 of batch 29 and was added as Part V. Sources 04 and 09 carry their numbers in the batch in which they arrived; the additions are numbered 10 and 11 here, after the highest existing file prefix, so that numbers do not collide.

Source	Manuscript	Pin	Contribution here
04 (base)	<i>Analytic Geometry and Trigonometry in Surreal Three-Space: finite angles, arbitrary scales, spherical duality, solid angles, rotations, and curvature</i> (34 pages)	d22a5b3	Sections 2 to 15, that is, section 2 and all of Part I; sections 36.1 and 36.3; the first proof of theorem 2.2. Files prefixed 04-three-space-.
09	<i>Rotations of Surreal Three-Space: Algebra, Spin, Infinitesimal Structure, and Topology</i> (31 pages, 19 sections, 26 numbered statements)	dcf8666	All of Part II (sections 16 to 27); in Part I the localization proposition and three-part decomposition of section 2, the hat identities of section 3, and the algebraic-angle, spin-lift, bounds and Cayley-chart statements of section 5; sections 36.2 and 36.4. Files prefixed 09-so3-.
10	<i>Valuation Cuts and the Universal Set-Sized Quotient of Surreal Rotation Groups: exact commutators, normal subgroups, perfect residuals, and surcomplex special unitary groups</i> (24 pages)	465a54b	All of Part III (sections 28 to 35); sections 36.5 and 36.6, the source 10 paragraph of section 36.7, section 37.3, the last part of section 38 with section 38.1, and section B.4. Files prefixed 10-rotation-quotients-.
11	<i>Small Representations and Normal Structure of Surreal Compact Groups: a semisimplicity criterion, valuation-cut classification, and contrasts with split groups and omnific lattices</i> (24 pages)	9693b28	All of Part V (sections 39 to 50) except the framing and the comparisons written for the addition (sections 50.1 and 50.2); the source 11 paragraph of section 36.7, section 37.5 and section B.5. Files prefixed 11-compact-groups-.

Why one report. Sources 04 and 09 overlap in the finite algebra of rotations and are otherwise complementary. Source 04 is a geometry of points, lines, spheres and spherical triangles, for which rotations are one tool; source 09 is a structure theory of the rotation group. Source 04 is the base: it is the wider development and fixes the scalar conventions. Source 09 forms Part II, after the geometry.

Why source 10 is an addition. Source 09’s conclusion names “normal-subgroup classification compatible with convex valuations” as a direction of further work (section 38). Source 10 supplies it for subgroups normal in the full group, and it sharpens two statements of Part II: the inclusion of filtration levels in theorem 23.1 becomes an equality of single-commutator images (remark 23.2), and the sentence that the kernel is “perfect rather than residually nilpotent” receives its exact perfect residual (remark 23.3). It builds directly on theorems 18.1, 19.3 and 23.1 and their proofs. Its main object is $\mathrm{SO}(3, \mathbf{No})$, so it belongs to Part II’s report rather than to the surquaternion report; the circle, where its main theorem fails, is cross-referenced to the trigonometry report.

Why source 11 is an addition. Source 10 asks for “a uniform reductive classification” that separates semisimple from toral factors of compact real algebraic groups (section 38.1). Source 11 proves one for every connected real linear algebraic group with compact connected real points

(theorem 39.1). It also classifies the normal subgroups of the higher-rank groups that Part III leaves unclassified (theorems 42.3 and 42.4), and it reproves theorems 33.2, 34.1 and 34.3 by a uniform second route. Its groups and its scalar field are those of Part III, so it belongs here rather than in the omnific-groups report, whose kernels are constant-term kernels over $\mathbf{Oz}$. Part V is printed after Part IV, not between Parts III and IV, so that no existing section, theorem or equation number changes.

Printed once. The following results are proved in both sources 04 and 09 and appear once here. We say which source's statement and proof are printed.

- (a) The finite phase theorem, theorem 2.2: source 04's statement and proof, with source 09's logarithmic proof kept as a marked second route.
- (b) The classification of distance-preserving maps, theorem 5.1. We print source 04's affine statement, add source 09's orientation clause and its remark on field automorphisms, and give one proof; the two proofs were the same polarization argument.
- (c) The axis theorem and Rodrigues formula, theorem 5.3. The statement is source 09's algebraic-angle form, which contains source 04's finite-angle form equation (5.2). Both proofs were the same determinant argument.
- (d) Finite axis-angle representatives, theorem 5.4. This combines a paragraph of source 04 with the second half of source 09's theorem on finite angles, whose proof is printed.
- (e) The Hamilton product and the quaternion rotation formula, equations (5.6) and (5.7). The two sources give these identically.
- (f) The two-to-one spin map $\mathrm{Sp}(1, F) \rightarrow \mathrm{SO}(3, F)$ with kernel $\{\pm 1\}$, theorem 5.5. Both sources reprove it by the argument that the surquaternion report already gives as the first route of `squat:thm:rotations`. We cite that theorem and do not repeat the proof; only source 09's explicit lift formula is printed.
- (g) The rational Cayley chart, theorem 5.6. Source 04's vector formula equation (5.15) and source 09's matrix formula are the same map, because $\mathbf{t} \times (\mathbf{t} \times \mathbf{x}) = \mathbf{t}(\mathbf{t} \cdot \mathbf{x}) - \|\mathbf{t}\|^2 \mathbf{x}$. The theorem is source 09's; source 04's remark on infinite chart coordinates is kept.
- (h) Scalar foundations. The finite and infinitesimal elements, the standard part, the Hahn valuation, strong summability and the commutative positive-support lemma appear once, in section 2. The noncommutative all-lengths version needed for BCH is lemma 22.1.
- (i) The symmetric spectral theorem over a real closed field. It is proved in section 4.3 (source 04); source 09's proof of lemma 20.1 used the same eigenvector argument and now refers to it.

Source 10 recalls several results of Part II. They are cited, not repeated: the standard-part splitting and kernel (theorem 18.1), the spin map and its surjectivity (theorem 5.5), the canonical near-one lift and the Cayley chart on K with its product law (theorem 5.6 and proposition 18.2), the $\mathrm{SU}(2)$ matrix model (equation (16.4)), and the perfectness of K (theorem 19.3), which is also the case $\Delta = \{0\}$ of theorem 31.2. Two statements appear once with a second proof. The elementwise commutator bound lemma 29.2 is equation (23.3); Part II proves it by BCH in a Hahn workspace, and source 10's finite matrix proof is kept as a marked second route. The scalar identity displayed in the proof of theorem 19.3 is the orthogonal-axis case of lemma 29.1, whose general form and proof are source 10's. The simplicity of $\mathrm{SO}(3, \mathbb{R})$ (lemma 30.5) is classical and is printed with source 10's proof because the classification uses it.

Source 11 recalls results of this report and of the collection. They are cited, or printed once with every proving source named. The size lemma [lemma 33.1](#) is cited; source 11 proves it with the same map. The infinitesimal circle logarithm [lemma 46.2](#) is the infinitesimal factor of the trigonometry report’s circle splitting, which Part III recalls as [equation \(35.1\)](#). The omnific statements of [section 49](#) are credited to `odg:cor:orthogonal`, `ogl:lem:bounded` and `ogl:alg:thm:real`; only the unitary count is new there. For $\mathrm{SO}(3)$, $\mathrm{SO}(n)$ and $\mathrm{SU}(n)$, source 11 reproves [theorems 30.3](#), [30.6](#), [33.2](#), [34.1](#) and [34.3](#) as special cases of its all-type theorems, by Nash charts instead of quaternions and Givens elimination. Those statements are not reprinted; both routes are kept, and each Part V theorem names the Part III case it contains. The commutator bound [lemma 43.1](#) is the identity of source 10’s proof of [lemma 29.2](#), stated for every compact group.

Where the merge had to choose. Symbols were renamed only where the two sources, or a source and the rest of the collection, used one symbol for two things. [Section 1.4](#) lists each renaming. The largest is that source 04’s scalar workspace K is written F here, because K is source 09’s infinitesimal kernel. The word *fine* also changes meaning: source 04 used it for the order topology of one set-sized workspace, source 09 for the full surreal topology. Here *fine* always has the second meaning, in agreement with the surquaternion and foundations reports. Source 04’s usage is written *intrinsic*. Labels from source 04 carry the prefix `e3:`, and labels from source 09 carry `e3:so:`, and labels from source 10 carry `e3:cut:`, and labels from source 11 carry `e3:cpt:`. Source 10’s symbols were renamed where they collided with Parts I and II or with `docs/NOTATION.md`; [section 28.4](#) lists each renaming, and [section 39.3](#) does the same for source 11. Source 11 states its classification of normal subgroups for “every proper class subgroup”; the intended meaning is a proper subgroup, set or class, and [theorem 42.3](#) is worded that way. Text written for the merge and the additions is confined to this section, the framing of [section 5](#), of Part III and of Part V, [section 37](#), the ledgers, short cross-reference remarks and paragraphs (among them [remarks 23.2](#) and [23.3](#) and the one-line comparisons after Part V statements), [sections 50.1](#) and [50.2](#), and the status notes in [section 38](#). It adds no mathematical claim ([section 37.4](#)).

1.2 Relation to the collection

Trigonometry. The trigonometry report [17], in `docs/surcomplex/trigonometry`, develops canonical finite phases, arbitrary-scale plane geometry and the spherical sine and cosine laws. [Theorem 2.2](#) is the Hahn-workspace form of its `trigonometry:thm:polar`. The choice of representatives is its `trigonometry:cor:representatives`, and the planar half-angle chart is its `trigonometry:thm:cayley`. The spherical laws of [theorem 8.2](#) rederive its `trigonometry:thm:spherical` from Gram data, adding polar duality, reconstruction and area. The Ehrlich–Kaplan phase of [section 24](#) is its `trigonometry:def:globaltrig` and `trigonometry:thm:globalexp`. A treatment of $\mathrm{SO}(2, \mathbf{No})$ is being added to that report at the same time as this merge. This report cites only labels present at its pin 50cb709. That treatment is now the trigonometry report’s Section 18 (`trigonometry:rot:sec:main`; numbering at 7b5f934). Part III, added at 7b5f934, cites its canonical circle splitting (`trigonometry:thm:polar`, Theorem 4.2 there, with `trigonometry:eq:circlesplit`) and its layer isomorphism `trigonometry:rot:thm:graded` for the circle counterexample [proposition 35.1](#); both labels were present at source 10’s pin 465a54b. The ledger maps `trigonometry:thm:polar` to Lean declarations on the actual surcomplex field; the circle character of [proposition 35.1](#) has no such mapping.

Part V, added at 66d7e55, uses the infinitesimal factor of the trigonometry report’s circle splitting, the map ℓ_m of `trigonometry:rot:eq:ellm`, as [lemma 46.2](#); its coefficient characters

induce `trigonometry:rot:thm:graded` on layers. Source 11 did not cite the trigonometry report; the credit is added here.

Surquaternions. The report *Surquaternions* (`docs/surquaternions/surquaternions`) [19] already contains the following, cited by label with its theorem numbers at pin 50cb709:

- the rotation double cover, and the isomorphism of $\mathrm{SO}(3, F)$ with $\mathrm{Aut}_F(\mathbb{H}_F)$ (`squat:sub:spin`; `squat:thm:rotations`, Theorem 4.4);
- the quaternion Cayley and stereographic chart (`squat:prop:charts`, Proposition 4.5);
- the standard-part splitting of the unit group (`squat:sub:unitreduction`, §4.3);
- infinitesimal exponential and logarithm, and the filtered group of infinitesimal rotations (`squat:thm:principal-log`, `squat:thm:rotation-log`, `squat:thm:graded`; Theorems 8.1, 8.5, 8.6);
- the all-lengths positive-support lemma (`squat:lem:neumann`, Lemma 7.1) and set discreteness of the fine topology (`squat:thm:discrete`, Theorem 7.3);
- the differential of the four-dimensional radial exponential (`squat:thm:exp-derivative`, Theorem 11.2).

Part II works instead with 3×3 rotation matrices and matrix Cayley coordinates. The two descriptions differ by the factor two of the spin differential (`proposition 21.1`). In particular the rank-one critical spheres of the rotation exponential in `theorem 25.2` are not the rank-two critical spheres of `squat:thm:exp-derivative`. At its pin dcf86662b574, source 09 records these surquaternion results as starting points, not as its own discoveries. At its pin d22a5b35d5b3, source 04 inspected only the trigonometry report and one Lean file. It does not mention the surquaternion report, although that report already contained the double cover and the quaternion Cayley chart at that commit. The citations in `section 5` correct this omission.

Foundations. The foundations report (`docs/foundations-and-computation/foundations`) [20] proves set discreteness of the fine topology (`found:thm:discrete`) and separates intrinsic from full-class subspace topologies (`found:prop:twotopologies`). `Theorem 26.1` specializes the first to rotation coordinates.

Omnific groups and omnific Diophantine geometry. The report `docs/surreal/omnific-groups-and-lattices` [21] studies matrix groups over the omnific integers. Its `ogl:alg:thm:real` characterizes the real algebraic groups with no new omnific points and contains the compact collapse `proposition 49.2`. Its `ogl:el:prop:finiteside` gives a universal set-sized quotient for SL_n of the finite surreals, $n \geq 3$, and its `ogl:el:rem:e3` compares that with Part III. `Section 50.2` records that neither `ogl:el:prop:finiteside` nor `theorem 39.1` implies the other. The report `docs/surreal/omnific-diophantine-geometry` [22] proves that $\mathrm{SO}_n(\mathbf{Oz})$ is the finite group of signed permutation matrices of determinant one (`odg:cor:orthogonal`), which is `theorem 49.3(i)`. Source 11 did not cite either report; both were in the tree at its pin.

Lean. Surreal/Algebra/Geometry.lean develops two-dimensional dot products, determinants, Cauchy–Schwarz, the Ptolemy inequality and the area form of Heron’s identity over ordered fields with nonnegative square roots, for labels of the trigonometry report [18]. It explicitly distinguishes these field-valued lengths from the real-valued distances used by conventional metric-space interfaces. The repository contains no three-dimensional Euclidean or rotation-group Lean development, and no label of this report is mapped to a Lean declaration. Both statements still hold at 7b5f934 and at 66d7e55, and they apply to Parts III and V as well; the Lean modules on rotations added in between concern the planar results of the trigonometry report.

Vector and tensor fields. The report docs/surreal/vector-and-tensor-fields, placed in the same commit as this one, develops finite tensor algebra, metrics and Levi-Civita connections over set-sized real closed surreal fields. The explicit sphere connection, curvature and transport of [section 12](#) are computed directly here and are not derived from that framework.

1.3 Three levels of assumptions

Level	Scalar assumptions	What the assumptions support
Algebraic	Ordered field with positive square roots; real closed when roots of general polynomials are used	Incidence, norms, cross products, rigid motions, Gram matrices, algebraic sine–cosine data, the finite algebra of $SO(3, F)$
Standard part	A real closed field $F \supseteq \mathbb{R}$	The split extension, the infinitesimal kernel, finite subgroups
Trigonometric	A real-closed Hahn workspace with its canonical restricted analytic structure	Finite angle representatives, inverse trigonometric functions, triangle identities, bounded analytic formulas, strong exponential and logarithm
Differential	Explicit restricted-analytic charts and differentiation in an external parameter, or the fine Fréchet derivative	Metric tensors, sphere curvature, finite-angle geodesics, parallel transport, the differential of the rotation exponential

The trigonometric and differential levels do not imply an unrestricted integration theory. Our spherical polygon area is explicitly constructed and proved finitely additive. The distinction matters even for apparently harmless real arguments; [section 15](#) gives concrete counterexamples to naive transfers of calculus.

Source 09 states its hypotheses in the same spirit. The proofs in [sections 5, 16](#) and [17](#) are finite algebra over a real closed field. Standard-part results require $F \supseteq \mathbb{R}$. Strong sums are constructed either in the full surreal normal form or in a named divisible Hahn workspace. Global infinite angles use the specific Ehrlich–Kaplan normalization. Semialgebraic homotopy statements are made over set-sized real closed fields. These are different hypotheses; none is silently substituted for another.

Source 10 adds two more. Its valuation statements ([sections 28 to 32](#)) use a set-sized real closed field $F \supseteq \mathbb{R}$ with its *natural* valuation; the finite-integer ratio property [equation \(28.1\)](#) is essential, and the lattice reconstruction of [section 32](#) needs F to be a set. Its universal quotient theorems ([sections 33 to 34](#)) concern the full class **No** in an NBG-style set/class foundation, with class homomorphisms into a group that is a set in the same foundation. Only the circle counterexample of [section 35.4](#) uses infinitesimal Hahn exp/log.

Source 11 (Part V) works over the full classes **No** and **No**[i] in NBG with choice, with a connected real linear algebraic group whose real points are compact and connected. Its local arguments transfer finite first-order statements about real Nash maps ([section 40](#)). Only its central-torus logarithm uses infinitesimal Hahn exp/log.

1.4 Notation, and the renamings made by the merge

F denotes a real closed scalar field. In Part I it is a set-sized workspace inside **No**, chosen in [section 2](#); in Part II it is any real closed field unless $F \supseteq \mathbb{R}$ is stated. Bold letters denote vectors in F^3 in Part I; Part II, following source 09, writes vectors in plain type. Points can be subtracted

to give vectors, and an origin is chosen whenever coordinates are used. The physical radius of a sphere is $R > 0$. For a spherical triangle with unit direction vectors $\mathbf{u}, \mathbf{v}, \mathbf{w}$, the opposite *angular* side lengths are a, b, c , and the interior angles are α, β, γ . Physical side lengths are Ra, Rb, Rc . We consistently use

$$x = \mathbf{v} \cdot \mathbf{w} = \cos a, \quad y = \mathbf{w} \cdot \mathbf{u} = \cos b, \quad z = \mathbf{u} \cdot \mathbf{v} = \cos c. \quad (1.4)$$

The determinant $\tau = \det[\mathbf{u} \ \mathbf{v} \ \mathbf{w}]$ is oriented; $D = \tau^2$ is the Gram determinant; $d = \sqrt{D} = |\tau|$ is nonnegative. These symbols will not also denote area or a side length. In [section 16](#) the letters a, b, c, d are instead the coordinates of a quaternion $a + bi + cj + dk$.

The merge fixed one meaning per symbol as follows.

Here	Source notation	Reason
$F, F^3, \text{SO}(3, F)$	source 04: $K, K^3, \text{SO}(3, K)$	K is the infinitesimal kernel of Part II
$\mathbb{T}(F)$	source 04: $U(K)$, the unit circle	U denotes elements of K ; $\mathbb{T}$ as in the trigonometry report
$\text{Rot}(n; c, s)$	source 09: $R(n; c, s)$; source 04: $Q_{n, \theta}$	R is a sphere radius
A for rotation matrices in section 5	source 04: Q	Q is also a point
Φ for an isometry	source 04: F	F is the field
$A_0 = \text{st}(A)$, B, B'	source 09: R_0, R, S	R is a radius; S_u a reflection
real rotations		
$\mathcal{E}, \mathcal{J}(w)$	source 09: $E, J(w)$ for the global rotation exponential and its differential	E is the spherical excess, J a Jacobian
dual number κ , $\kappa^2 = 0$	source 09: τ	τ is an oriented determinant
Gn, St	source 04: G, H for the gnomonic and stereographic charts	$G(F) = \text{SO}(3, F)$; H_n is a half-turn
$\sin, \cos, \text{cis}$ on finite angles	source 09: $\sin_0, \cos_0, \text{cis}_0$	one finite-angle convention; $\text{Sin}, \text{Cos}, \text{cis}_{\text{EK}}$ remain the global ones
$\hat{x}$	source 04: $[x]_{\times}$	one name for the cross-product matrix
$f = O(g)$	source 04: $F = O(G)$	letters in use
<i>intrinsic</i> topology of a workspace	source 04: <i>fine</i>	<i>fine</i> means the full surreal topology

The local matrix $E = P - I$ in source 09's positive-root lemma is written Y . Two further symbols keep a double use from their sources. The letter t is the Hahn monomial variable in t^γ ; the vector $\mathbf{t}$, or $t \in F^3$ in Part II, is a Cayley coordinate. $\text{Exp}_P, \text{Log}_P$ with a point subscript are the exponential map of a sphere and its inverse, while $\text{Exp}_m, \text{Log}_m$ are the strong matrix exponential and logarithm.

Part III keeps these conventions. Its own symbols, and the renamings of source 10 that avoid the letters above, are listed in [section 28.4](#). Part V does the same; its renamings are listed in [section 39.3](#).

2 Scalar foundations and canonical finite trigonometry

2.1 Set-sized workspaces inside the surreal numbers

The surreal numbers form a real-closed ordered *class* field, not a set. We use their Conway normal forms but perform each construction in a set-sized field. Let $\Gamma \subset \mathbf{No}$ be a set-sized divisible ordered

additive subgroup and put

$$F = \mathbb{R}((t^\Gamma)), \quad \sum_{\gamma} a_{\gamma} t^{\gamma} \mapsto \sum_{\gamma} a_{\gamma} \omega^{-\gamma} \in \mathbf{No}. \quad (2.1)$$

Supports in the left-hand expression are well ordered. This identifies F with a real-closed subfield of $\mathbf{No}$. Its canonical restricted-analytic structure is an elementary extension of $\mathbb{R}_{\text{an}}$; these structures are compatible under enlargement of Γ [1, §2].

Every finite configuration of surreal coordinates belongs to such a workspace: collect all exponents occurring in its finitely many normal forms and take their rational span. More generally the same argument works for any set of inputs. Neither the support set nor this span is a proper class. All operations used below stay in a workspace or in a specified larger one. Consequently the theorems apply to arbitrary individual surreal coordinates without treating $\mathbf{No}$ as a set or forming a proper-class power set.

Source 09 makes the same point for the rotation group. There is no set containing all surreal numbers, so $\mathbf{No}^3$, $M_3(\mathbf{No})$ and $G(\mathbf{No})$ are proper classes. One may use a class theory such as NBG, or a universe-relative formulation in which the surreal carrier is large relative to the universe containing its individual sign sequences. A matrix still has only nine entries, and every formula for its multiplication, determinant, inverse, or action is finite. Statements about all elements of this class group are schemes of ordinary finite algebra. Statements about a finite subgroup, a sequence, or a set-indexed family retain their ordinary size restriction. We do not form an unrestricted set of all subsets of $\mathbf{No}$, or invoke ordinary compactness for such a nonexistent set-sized space. For finite algebra alone a smaller field suffices.

Proposition 2.1 (Localization of finite algebra). *Every set-sized family of surreal coordinates lies in a set-sized real closed subfield $F \subseteq \mathbf{No}$. One can require $\mathbb{R} \subseteq F$. Thus every set-sized configuration of vectors and matrices can be studied inside an honest real closed field without changing its finite algebraic identities.*

Proof. Adjoin the coordinates to $\mathbb{R}$, obtaining a set-sized ordered subfield. Take its real closure inside the real closed field $\mathbf{No}$: equivalently, take the elements of $\mathbf{No}$ algebraic over that subfield. They form a set and a real closed field. The inclusion preserves the arithmetic and order. It does not assert that the subfield is closed under arbitrary strong sums or inherits the same topology as its intrinsic order topology. $\square$

The Hahn workspace [equation \(2.1\)](#) is larger than this subfield; its purpose is to contain strong sums and analytic evaluations. An arbitrary birthday cutoff is not a substitute for either construction. For example, the finite-birthday surreals are dyadic rationals and are not closed under division. A workspace must have the closure properties actually needed.

We write

$$\mathcal{O} = \{a \in F : |a| < n \text{ for some } n \in \mathbb{N}\}, \quad \mathfrak{m} = \{a \in F : |a| < 1/n \text{ for every } n \geq 1\}.$$

These are respectively the finite elements and the infinitesimals. Every $a \in \mathcal{O}$ has a unique decomposition

$$a = \text{st}(a) + \varepsilon, \quad \text{st}(a) \in \mathbb{R}, \quad \varepsilon \in \mathfrak{m}.$$

The standard-part map is an ordered ring homomorphism on $\mathcal{O}$, with kernel $\mathfrak{m}$. In a Hahn workspace it is the coefficient of t^0 for finite series. The same definitions apply to any ordered field $F \supseteq \mathbb{R}$, with subscripts $\mathcal{O}_F, \mathfrak{m}_F$ where the field matters; $\mathcal{O}$ and $\mathfrak{m}$ without subscript refer to the workspace,

or to all of **No**. In that generality the real cut determined by a bounded a has a real supremum, whose difference from a is infinitesimal. Uniqueness follows because no nonzero real is infinitesimal (source 09).

A surreal has a set-supported normal form $x = \sum_{a \in S} r_a \omega^a$ with $r_a \in \mathbb{R} \setminus \{0\}$ and S reverse well ordered. Splitting the exponents into positive, zero, and negative parts gives

$$x = x^\uparrow + r + \varepsilon. \quad (2.2)$$

The first part is purely infinite, the second is real, and the third is infinitesimal. Purely infinite does not mean positive: a purely infinite number may have either sign. Normal-form and analytic foundations are treated in [1, 3]; only set-supported sums are used.

For $a \neq 0$, let $v(a)$ be the least exponent in its Hahn support, and set $v(0) = +\infty$. In normal-form terms, $v(x) = -\max \text{supp}_\omega(x)$, the negative of the leading exponent, so $v(\omega^{-\gamma}) = \gamma$; the two sources use this one convention, which is also the collection's. For a vector or matrix, v is the minimum of the valuations of its entries. Thus $v(t^\gamma) = \gamma$, positive valuation means infinitesimal magnitude, and $v(ab) = v(a) + v(b)$. If $a, b \neq 0$, the notation $a \sim b$ means $a/b = 1 + \eta$ with $\eta \in \mathfrak{m}$. Later, $f = O(g)$ means $|f| \leq C|g|$ for a positive *real* constant C in the stated range; this is not an assertion about convergence of a sequence of ordinary partial sums.

2.2 What an infinite Taylor sum means

A family of Hahn series is *strongly summable* when the union of its supports is well ordered and every exponent occurs in only finitely many members. The sum is then defined coefficientwise. The positive-support lemma of Hahn-series algebra says that if S is a well-ordered subset of $\Gamma_{>0}$, its additive monoid is well ordered and every exponent has only finitely many decompositions as an unordered finite sum from S . This is the summability mechanism used below; one must not replace it by a claim that $n\delta$ is cofinal in Γ for every $\delta > 0$. The latter claim is false in higher-rank ordered groups. See [4, 2] for the relevant support theory. The all-lengths version for ordered finite words, needed for noncommutative matrix series, is [lemma 22.1](#), which is `squat:lem:neumann` of the surquaternion report.

If f is real analytic near a real number r and $h \in \mathfrak{m}$, define

$$f^F(r + h) = \sum_{n=0}^{\infty} \frac{f^{(n)}(r)}{n!} h^n. \quad (2.3)$$

The positive-support lemma proves strong summability. It also justifies multiplication, substitution, and differentiation of these expressions when used in the corresponding analytic germs. This is consistent with the restricted-analytic structure in [equation \(2.1\)](#).

For a nonzero ordinary real argument, the usual Taylor series about zero need not be a strongly summable Hahn family: infinitely many nonzero real terms have exponent zero. One first evaluates the real analytic function at its real center, and only then evaluates its infinitesimal Taylor expansion. This convention is essential, not a stylistic choice.

2.3 Sine, cosine, and inverse tangent

For $a = r + h \in \mathcal{O}$, define $\sin a$ and $\cos a$ by [equation \(2.3\)](#). In particular,

$$\sin h = h - \frac{h^3}{3!} + \frac{h^5}{5!} - \cdots, \quad (2.4)$$

$$\cos h = 1 - \frac{h^2}{2!} + \frac{h^4}{4!} - \cdots. \quad (2.5)$$

The formal addition identities, together with the ordinary real identities, give

$$\sin(a + b) = \sin a \cos b + \cos a \sin b, \quad \cos(a + b) = \cos a \cos b - \sin a \sin b, \quad (2.6)$$

and $\cos^2 a + \sin^2 a = 1$. In particular $\sin h = h(1 + \mathcal{O}(h^2))$. The real sign and monotonicity properties on any fixed real bounded interval hold with surreal inputs as well. They follow either from the local expansions and endpoint checks, or from restricted-analytic elementarity.

Define $\operatorname{atan} a$ by real-centered Taylor evaluation for finite a , and for infinite a by

$$\operatorname{atan} a = \operatorname{sgn}(a) \frac{\pi}{2} - \operatorname{atan}(1/a). \quad (2.7)$$

It is the unique angle in $(-\pi/2, \pi/2)$ whose tangent is a . Thus inverse tangent accepts arbitrary surreal inputs and always returns a finite surreal angle. Tangent itself is used only where its cosine denominator is nonzero.

Theorem 2.2 (Every direction has a finite phase). *Let $\mathbb{T}(F) = \{(c, s) \in F^2 : c^2 + s^2 = 1\}$, with multiplication $(c, s)(c', s') = (cc' - ss', sc' + cs')$. The map*

$$\theta \longmapsto (\cos \theta, \sin \theta)$$

induces an isomorphism $\mathcal{O}/(2\pi\mathbb{Z}) \simeq \mathbb{T}(F)$. Consequently every unit-circle direction has a unique representative in $[0, 2\pi)$, and also a unique representative in $(-\pi, \pi]$.

Proof. Both coordinates of a unit-circle point are finite. Its standard part is an ordinary real unit-circle point. Rotate this standard part to $(1, 0)$ using an ordinary real phase. The resulting point (C, S) has $C = 1 + \mathfrak{m}$, $S \in \mathfrak{m}$, and $C > 0$. Put $h = \operatorname{atan}(S/C) \in \mathfrak{m}$. Since the unit point with positive first coordinate and slope S/C is unique, $(C, S) = (\cos h, \sin h)$. This proves surjectivity. Addition follows from [equation \(2.6\)](#).

For the kernel, the real standard part of a phase must be $2\pi k$ for an ordinary integer k . Subtract it. An infinitesimal h with $\sin h = 0$ must vanish because $\sin h = h(1 + \mathcal{O}(h^2))$ and the second factor is a unit. Thus the kernel is exactly $2\pi\mathbb{Z}$, not an unspecified group of infinite periods. Finally a finite element can be reduced into either stated interval by an ordinary integer multiple of 2π . $\square$

Second proof (source 09). Write $\operatorname{cis} \theta = \cos \theta + i \sin \theta$ in $F(i)$. For $\theta = r + \varepsilon$ with $r \in \mathbb{R}$ and $\varepsilon \in \mathfrak{m}$, real-centered evaluation gives $\operatorname{cis} \theta = e^{ir} \sum_{k \geq 0} (i\varepsilon)^k / k!$; this is [equation \(24.1\)](#). For a unit surcomplex $z = c + is$, both coordinates are finite. Its standard part z_0 is an ordinary unit complex number. Choose a real angle r with $e^{ir} = z_0$. Then $z/z_0 = 1 + \delta$ for infinitesimal δ , so its strong logarithm is defined. Conjugation and inversion give

$$\overline{\log(z/z_0)} = \log(\overline{z/z_0}) = \log((z/z_0)^{-1}) = -\log(z/z_0).$$

It is therefore $i\varepsilon$ with real infinitesimal ε , and $z = \operatorname{cis}(r + \varepsilon)$. This proves surjectivity. If $\operatorname{cis}(r + \varepsilon) = 1$, standard part gives $r \in 2\pi\mathbb{Z}$; then logarithmic injectivity gives $\varepsilon = 0$. $\square$

The second proof is the argument of `trigonometry:thm:polar`, which proves the same isomorphism over all of **No** [17]. The choice of representatives, including the endpoint correction for infinitesimal excursions, is `trigonometry:cor:representatives`.

For $(X, Y) \neq (0, 0)$, define $\text{atan2}(Y, X) \in (-\pi, \pi]$ as the phase of $(X, Y)/\sqrt{X^2 + Y^2}$. With $r = \sqrt{X^2 + Y^2}$, an explicit formula is

$$\text{atan2}(Y, X) = 2 \text{atan} \frac{Y}{r + X} \quad \text{when } r + X > 0, \quad (2.8)$$

with value π on the negative X -axis. This definition works when either coordinate is infinite or infinitesimal.

For $-1 \leq x \leq 1$, let

$$\text{acos } x = \text{atan2}(\sqrt{1 - x^2}, x) \in [0, \pi], \quad \text{asin } x = \frac{\pi}{2} - \text{acos } x. \quad (2.9)$$

Here the phase at $(1, 0)$ is zero. These inverse functions include their endpoints. For $-1 < x \leq 1$, another exact formula is $\text{acos } x = 2 \text{atan} \sqrt{(1 - x)/(1 + x)}$.

Remark 2.3 (Infinite input is a separate question). The present construction does not assert that no distinguished global sine or cosine exists. Ehrlich and Kaplan construct such functions using the initial integer part of **No** [3, §11]; their construction supplies additional structure and agrees with the finite theory used here. What is unnecessary for geometry is choosing a phase for every infinite input. The naive zero-centered sine series at ω is not a strongly summable Hahn series, regardless of whether a separate global definition is chosen. Part II uses that global phase explicitly, with its period class, in [section 24](#).

Part I

Coordinate geometry of **No**³

Part I is source 04. Rotations enter as one tool among several, and the shared rotation algebra of both sources is in [section 5](#).

3 Vectors, distances, orientation, and volume

For $\mathbf{x} = (x_1, x_2, x_3)$ and $\mathbf{y} = (y_1, y_2, y_3)$ set

$$\mathbf{x} \cdot \mathbf{y} = \sum_{j=1}^3 x_j y_j, \quad \|\mathbf{x}\| = \sqrt{\mathbf{x} \cdot \mathbf{x}}, \quad d(P, Q) = \|Q - P\|.$$

The cross product is given by the usual determinant formula. We use the orientation for which $\mathbf{e}_1 \times \mathbf{e}_2 = \mathbf{e}_3$.

Theorem 3.1 (Euclidean vector identities over the scalar field). *For all vectors in F^3 ,*

$$\|\mathbf{x} \times \mathbf{y}\|^2 = \|\mathbf{x}\|^2 \|\mathbf{y}\|^2 - (\mathbf{x} \cdot \mathbf{y})^2, \quad (3.1)$$

$$\mathbf{x} \times (\mathbf{y} \times \mathbf{z}) = \mathbf{y}(\mathbf{x} \cdot \mathbf{z}) - \mathbf{z}(\mathbf{x} \cdot \mathbf{y}), \quad (3.2)$$

$$(\mathbf{x} \times \mathbf{y}) \cdot (\mathbf{z} \times \mathbf{w}) = (\mathbf{x} \cdot \mathbf{z})(\mathbf{y} \cdot \mathbf{w}) - (\mathbf{x} \cdot \mathbf{w})(\mathbf{y} \cdot \mathbf{z}). \quad (3.3)$$

Moreover $|\mathbf{x} \cdot \mathbf{y}| \leq \|\mathbf{x}\| \|\mathbf{y}\|$ and $\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|$. For nonzero $\mathbf{x}, \mathbf{y}$, equality in the latter holds exactly when $\mathbf{y} = \lambda \mathbf{x}$ for some $\lambda > 0$.

Proof. The three displayed identities follow by expansion in coordinates over any commutative ring. Positivity of the left side of [equation \(3.1\)](#) and nonnegative square roots give Cauchy–Schwarz. Squaring the proposed triangle inequality reduces it to Cauchy–Schwarz. Equality requires $\mathbf{x} \cdot \mathbf{y} = \|\mathbf{x}\| \|\mathbf{y}\|$, hence $\mathbf{x} \times \mathbf{y} = 0$. The two vectors are then proportional, and the positive dot product makes the proportionality factor positive. The converse is direct. Source 09 proves Cauchy–Schwarz by a second finite route: for $\mathbf{y} \neq 0$, minimizing the quadratic $\|\mathbf{x} - t\mathbf{y}\|^2$ algebraically at $t = (\mathbf{x} \cdot \mathbf{y})/(\mathbf{y} \cdot \mathbf{y})$ gives the inequality. $\square$

Thus d is a positive-definite, symmetric, triangle-inequality distance with values in $F_{\geq 0}$. It is a *field-valued* distance, not an ordinary real-valued metric. The coordinate formulas do not implicitly make F^3 a Banach space over $\mathbb{R}$.

The cross product is represented by the matrix

$$\hat{x} = \begin{pmatrix} 0 & -x_3 & x_2 \\ x_3 & 0 & -x_1 \\ -x_2 & x_1 & 0 \end{pmatrix}, \quad \hat{x} \mathbf{y} = \mathbf{x} \times \mathbf{y}. \quad (3.4)$$

Source 09 records the following polynomial identities, valid over every field of characteristic different from two:

$$\hat{x}^2 = \mathbf{x} \mathbf{x}^T - \|\mathbf{x}\|^2 I, \quad \hat{x}^3 = -\|\mathbf{x}\|^2 \hat{x}, \quad (3.5)$$

$$[\hat{x}, \hat{y}] = \widehat{\mathbf{x} \times \mathbf{y}}. \quad (3.6)$$

Applied to a vector, the first identity is the case of equal first two arguments of the vector triple product [equation \(3.2\)](#). No limiting process is involved.

For nonzero vectors define their unoriented angle by

$$\text{ang}(\mathbf{x}, \mathbf{y}) = \text{atan2}(\|\mathbf{x} \times \mathbf{y}\|, \mathbf{x} \cdot \mathbf{y}) \in [0, \pi]. \quad (3.7)$$

Lagrange’s identity shows that its cosine is the normalized dot product and its sine is the normalized cross-product norm. If the vectors lie in a plane with specified unit normal $\mathbf{n}$, their oriented angle is instead $\text{atan2}(\mathbf{n} \cdot (\mathbf{x} \times \mathbf{y}), \mathbf{x} \cdot \mathbf{y})$.

3.1 Areas and tetrahedra

The area of a flat triangle PQR and the oriented volume of a tetrahedron $PQRS$ are

$$\text{Area}(PQR) = \frac{1}{2} \|(\mathbf{Q} - \mathbf{P}) \times (\mathbf{R} - \mathbf{P})\|, \quad V_{\text{or}}(PQRS) = \frac{1}{6} \det[\mathbf{Q} - \mathbf{P} \ \mathbf{R} - \mathbf{P} \ \mathbf{S} - \mathbf{P}]. \quad (3.8)$$

The unoriented volume is $|V_{\text{or}}|$. These are algebraic definitions: finite dissection and linear change-of-variables identities follow from multilinearity. They preserve orientation and multiplicity information at infinitesimal scales.

If M is the matrix of the three edge vectors from P , then

$$\det(M^T M) = \det(M)^2 = 36 \text{Vol}(PQRS)^2.$$

In terms of the six pairwise distances d_{ij} , this becomes

$$288 \operatorname{Vol}(PQRS)^2 = \det \begin{pmatrix} 0 & 1 & 1 & 1 & 1 \\ 1 & 0 & d_{01}^2 & d_{02}^2 & d_{03}^2 \\ 1 & d_{01}^2 & 0 & d_{12}^2 & d_{13}^2 \\ 1 & d_{02}^2 & d_{12}^2 & 0 & d_{23}^2 \\ 1 & d_{03}^2 & d_{13}^2 & d_{23}^2 & 0 \end{pmatrix}. \quad (3.9)$$

Indeed $2M_i \cdot M_j = d_{0i}^2 + d_{0j}^2 - d_{ij}^2$; subtracting the corresponding rows and columns in the displayed determinant reduces it to eight times $\det(M^\top M)$.

3.2 Finite-dimensional minimization does not need completeness

Let $V \subset F^3$ have an orthonormal basis $\mathbf{e}_1, \dots, \mathbf{e}_k$. Gram–Schmidt constructs one whenever a finite independent spanning family is given; it uses only finitely many nonzero divisions and positive square roots, so it constructs orthonormal frames in F^3 exactly as over $\mathbb{R}$, with no completeness assumption. The orthogonal projection

$$\operatorname{proj}_V(\mathbf{x}) = \sum_{j=1}^k (\mathbf{x} \cdot \mathbf{e}_j) \mathbf{e}_j$$

satisfies, for $\mathbf{y} \in V$,

$$\|\mathbf{x} - \mathbf{y}\|^2 = \|\mathbf{x} - \operatorname{proj}_V \mathbf{x}\|^2 + \|\operatorname{proj}_V \mathbf{x} - \mathbf{y}\|^2.$$

It is therefore the unique nearest point. This proves a useful minimization theorem by a finite identity, without a compactness argument or an infimum construction. The same method gives least squares for full-column-rank matrices: the minimizer of $\|A\mathbf{x} - \mathbf{b}\|^2$ is $(A^\top A)^{-1} A^\top \mathbf{b}$.

4 Affine analytic geometry

4.1 Lines, planes, and incidence

A line is $L = P + F\mathbf{u}$ with $\mathbf{u} \neq 0$. A plane is $\Pi = \{X : \mathbf{n} \cdot X = h\}$ with $\mathbf{n} \neq 0$. A line meets this plane uniquely when $\mathbf{n} \cdot \mathbf{u} \neq 0$, at parameter

$$t = \frac{h - \mathbf{n} \cdot P}{\mathbf{n} \cdot \mathbf{u}}. \quad (4.1)$$

A zero denominator means that the line lies in the plane if the numerator also vanishes, and is disjoint from it otherwise. An infinitesimal nonzero denominator is not a reason to declare parallelism: it can produce an infinitely distant intersection.

Two nonparallel planes have line direction $\mathbf{n}_1 \times \mathbf{n}_2$. Three planes meet uniquely exactly when their normal determinant is nonzero. All remaining cases are ordinary finite-rank alternatives. For two lines $P + F\mathbf{u}$ and $Q + F\mathbf{v}$ with $\mathbf{u} \times \mathbf{v} \neq 0$, they intersect if and only if

$$(Q - P) \cdot (\mathbf{u} \times \mathbf{v}) = 0.$$

Otherwise they are skew.

The perpendicular foot and distance from P to Π are

$$P_{\Pi} = P - \frac{\mathbf{n} \cdot P - h}{\|\mathbf{n}\|^2} \mathbf{n}, \quad d(P, \Pi) = \frac{|\mathbf{n} \cdot P - h|}{\|\mathbf{n}\|}. \quad (4.2)$$

For a line $Q + F\mathbf{u}$, they are

$$P_L = Q + \frac{(P - Q) \cdot \mathbf{u}}{\|\mathbf{u}\|^2} \mathbf{u}, \quad d(P, L) = \frac{\|(P - Q) \times \mathbf{u}\|}{\|\mathbf{u}\|}.$$

Reflection in Π sends P to $2P_{\Pi} - P$. These formulas are valid at every surreal scale.

Theorem 4.1 (Distance and closest points of nonparallel lines). *Let $\mathbf{r} = Q - P$ and put*

$$A = \mathbf{u}^2, \quad B = \mathbf{u} \cdot \mathbf{v}, \quad C = \mathbf{v}^2, \quad D_L = AC - B^2 > 0,$$

where $\mathbf{u}^2$ means $\mathbf{u} \cdot \mathbf{u}$. The unique closest parameters on $P + t\mathbf{u}$ and $Q + s\mathbf{v}$ are

$$t = \frac{C(\mathbf{r} \cdot \mathbf{u}) - B(\mathbf{r} \cdot \mathbf{v})}{D_L}, \quad s = \frac{B(\mathbf{r} \cdot \mathbf{u}) - A(\mathbf{r} \cdot \mathbf{v})}{D_L}. \quad (4.3)$$

Their distance is

$$d(L_1, L_2) = \frac{|\mathbf{r} \cdot (\mathbf{u} \times \mathbf{v})|}{\|\mathbf{u} \times \mathbf{v}\|}. \quad (4.4)$$

Proof. The difference $\mathbf{r} + s\mathbf{v} - t\mathbf{u}$ must be perpendicular to both direction vectors. Its two dot products give the linear system $At - Bs = \mathbf{r} \cdot \mathbf{u}$, $Bt - Cs = \mathbf{r} \cdot \mathbf{v}$, whose solution is [equation \(4.3\)](#). The remaining difference is parallel to $\mathbf{u} \times \mathbf{v}$, giving [equation \(4.4\)](#). Every other pair of points adds a vector in $\text{span}(\mathbf{u}, \mathbf{v})$ to this perpendicular difference. Pythagoras proves minimality and uniqueness of the two parameters. $\square$

For parallel lines use the point-to-line distance; the closest pair is then not unique. Angles between unoriented lines or planes are conventionally chosen acute, by taking an absolute value in the normalized dot product. Oriented normal vectors instead retain the supplementary-angle distinction. The acute line–plane angle θ for line direction $\mathbf{u}$ and plane normal $\mathbf{n}$ satisfies $\sin \theta = |\mathbf{u} \cdot \mathbf{n}| / (\|\mathbf{u}\| \|\mathbf{n}\|)$. The line–normal angle is its complement.

4.2 Spheres, circles, and discriminants

The sphere with center C and radius $R > 0$ is $\{X : \|X - C\|^2 = R^2\}$. For the line $P + t\mathbf{u}$, let t_0 be the parameter of its perpendicular foot from C , and let $h = d(C, L)$. Completing the square gives

$$\|\mathbf{u}\|^2 (t - t_0)^2 = R^2 - h^2. \quad (4.5)$$

There are two, one, or no intersections according as $R^2 - h^2$ is positive, zero, or negative. The two parameters, when distinct, are $t_0 \pm \sqrt{R^2 - h^2} / \|\mathbf{u}\|$. A plane at distance h from the center cuts a circle of radius $\sqrt{R^2 - h^2}$ when $h < R$, a point when $h = R$, and nothing when $h > R$.

For two spheres with distinct centers C_1, C_2 , let $d = \|C_2 - C_1\| > 0$ and $\mathbf{e} = (C_2 - C_1)/d$. Their intersection lies in the radical plane perpendicular to $\mathbf{e}$, at signed distance

$$q = \frac{R_1^2 - R_2^2 + d^2}{2d}$$

from C_1 . Its candidate circle has center $C_1 + q\mathbf{e}$ and squared radius $R_1^2 - q^2$. The sign of this expression gives the complete classification. Coincident centers must be handled separately. These formulas never compare an infinitesimal with zero by rounding it away.

Four affinely independent points determine a unique sphere. Translate the first point to zero and write the other three as columns of M . Its center $\mathbf{c}$ solves

$$M^\top \mathbf{c} = \frac{1}{2}(\|M_1\|^2, \|M_2\|^2, \|M_3\|^2)^\top. \quad (4.6)$$

The radius is $\|\mathbf{c}\|$. Small nonzero $\det M$ can make this center infinite even when all four input points are finite.

4.3 Quadrics and principal axes

A quadric has an equation

$$\mathbf{x}^\top A \mathbf{x} + 2\mathbf{b}^\top \mathbf{x} + c = 0, \quad A = A^\top.$$

Over a real-closed field, a symmetric matrix has an orthonormal eigenbasis. Here is a proof that does not use compactness of a real unit sphere. A root λ of its characteristic polynomial lies in $F[i]$, and there is a nonzero vector z with $Az = \lambda z$. Since $\bar{z}^\top Az = \lambda \bar{z}^\top z$ is real and $\bar{z}^\top z = \sum |z_j|^2 > 0$, $\lambda \in F$. Its eigenspace has a nonzero real vector. Normalize it, restrict to its invariant perpendicular complement, and continue by induction. The argument works verbatim for symmetric $d \times d$ matrices over any real closed field; Part II uses it in [lemma 20.1](#).

After an orthogonal change of coordinates the quadric becomes $\sum \lambda_j y_j^2 + 2 \sum b'_j y_j + c = 0$. For $\lambda_j \neq 0$, completing the square removes the corresponding linear term. If all λ_j are nonzero this gives the familiar central forms, classified by coefficient signs and the remaining constant. Ellipsoids, hyperboloids, cones, empty loci, and point loci are distinguished exactly as over $\mathbb{R}$, with arbitrary surreal semiaxes. If a null direction retains a linear term, solving for that coordinate gives a paraboloid-type form. If all linear terms in the nullspace vanish, those directions are cylindrical. Thus the classification is algebraic, while the scales of the axes carry additional information.

4.4 Affine points are not projective ideal points

Homogeneous coordinates embed F^3 into $\mathbb{P}^3(F)$ by $\mathbf{x} \mapsto [x_1 : x_2 : x_3 : 1]$. An affine point such as $(\omega, 0, 0)$ is not $[1 : 0 : 0 : 0]$: the last homogeneous coordinate distinguishes them exactly. Projective completion is a separate incidence construction. It does not identify a very large finite-coordinate point with an ideal direction.

5 Rigid motions, rotations, quaternions, and screws

This section prints once the rotation algebra that both sources develop ([section 1.1](#)). Except for [theorem 5.4](#), everything here is finite algebra over a real closed field F . Source 04 stated it over its Hahn workspace, which is such a field; source 09 stated it over an arbitrary real closed field. We write $G(F) = \text{SO}(3, F)$ and use the cross-product matrix $\hat{x}$ of [equation \(3.4\)](#).

Theorem 5.1 (Classification of field-distance isometries). *Every map $\Phi : F^3 \rightarrow F^3$ preserving $d(P, Q) = \|P - Q\|$ has the form*

$$\Phi(\mathbf{x}) = A\mathbf{x} + \mathbf{b}, \quad A^\top A = I.$$

In particular, it is automatically onto. No continuity or affine assumption is needed. If Φ fixes the origin, it is F -linear with orthogonal matrix A , and it preserves the orientation of the standard basis exactly when $\det A = 1$. Conversely, every such matrix preserves the norm and fixes the origin.

Proof. Subtract $\Phi(0)$ and put $\mathbf{q}_j = \Phi(\mathbf{e}_j) - \Phi(0)$. Distances between $0, \mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$ and polarization,

$$(\Phi(\mathbf{x}) - \Phi(0)) \cdot (\Phi(\mathbf{y}) - \Phi(0)) = \frac{1}{2}(\|\mathbf{x}\|^2 + \|\mathbf{y}\|^2 - \|\mathbf{x} - \mathbf{y}\|^2) = \mathbf{x} \cdot \mathbf{y},$$

show $\mathbf{q}_i \cdot \mathbf{q}_j = \delta_{ij}$. For arbitrary $\mathbf{x}$, the same identity with $\mathbf{y} = \mathbf{e}_j$ gives $(\Phi(\mathbf{x}) - \Phi(0)) \cdot \mathbf{q}_j = x_j$. The $\mathbf{q}_j$ form an orthonormal basis, so $\Phi(\mathbf{x}) - \Phi(0) = \sum x_j \mathbf{q}_j$. Its matrix is orthogonal and therefore invertible. Its determinant is 1 or -1 , and preserving the orientation of the given basis is precisely the first alternative. $\square$

The theorem excludes a potential ambiguity. A field automorphism applied to coordinates may carry squared distances to their images under that automorphism. That is a semilinear transformation of a different geometry, not an isometry preserving every surreal distance as an actual value. The rotations of F^3 are therefore exactly the elements of $G(F)$.

Proposition 5.2 (Bounds and equivariance). *For $A \in G(F)$, every entry satisfies $|A_{ij}| \leq 1$, and*

$$(Ax) \times (Ay) = A(x \times y), \quad A\hat{x}A^{-1} = \widehat{Ax}.$$

Thus no rotation matrix has an infinite entry, even when its vectors have infinite coordinates.

Proof. Each column has sum of squares one. The cross-product identity follows by pairing both sides with Az : both scalar triple products are multiplied by $\det A = 1$. The hat identity follows immediately. $\square$

5.1 Axis-angle representation

Theorem 5.3 (Axis theorem and Rodrigues formula). *Every $A \in G(F)$ fixes a unit vector n . There exist $c, s \in F$ with $c^2 + s^2 = 1$ such that*

$$A = \text{Rot}(n; c, s) = cI + (1 - c)nn^\top + s\hat{n}. \quad (5.1)$$

If $A \neq I$, its fixed space is exactly Fn . Replacing (n, s) by $(-n, -s)$ leaves the rotation unchanged. Conversely $\text{Rot}(n; c, s) \in G(F)$ for every unit n and every (c, s) on the unit circle. When $(c, s) = (\cos \theta, \sin \theta)$ for a finite angle θ , which is always possible over a Hahn workspace by [theorem 5.4](#), the formula reads

$$\text{Rot}(n; \cos \theta, \sin \theta) \mathbf{x} = (\cos \theta) \mathbf{x} + (\sin \theta)(n \times \mathbf{x}) + (1 - \cos \theta)(n \cdot \mathbf{x})n. \quad (5.2)$$

Proof. Since $A^\top = A^{-1}$ and $\det A = 1$,

$$\det(A - I) = \det(A^\top - I) = \det(A^{-1}(I - A)) = \det(I - A) = -\det(A - I).$$

Characteristic zero implies $\det(A - I) = 0$. Normalize a nonzero fixed vector to obtain n . Its perpendicular plane is invariant under A . In an oriented orthonormal basis $(u, n \times u, n)$, the restriction to that plane is an orthogonal 2×2 matrix of determinant one, necessarily $\begin{pmatrix} c & -s \\ s & c \end{pmatrix}$. This gives [equation \(5.1\)](#). Conversely, on $n^\perp$ the operator $Jx = n \times x$ satisfies $J^2 = -I$, so $cI + sJ$ is an

orientation-preserving orthogonal map of that plane; on the axis the formula is the identity. Hence $\text{Rot}(n; c, s) \in G(F)$.

The determinant of the planar matrix minus the identity is $(c - 1)^2 + s^2 = 2(1 - c)$. It vanishes only when $c = 1, s = 0$, which gives the identity rotation. Otherwise the perpendicular plane contains no fixed vector. (Source 04's variant: if two independent vectors were fixed, their cross product would also be fixed, forcing $A = I$.) Expanding $nn^\top$ and $\hat{n}$ on a vector gives [equation \(5.2\)](#). $\square$

The pair (c, s) is an *algebraic angle*. It needs neither a sine function on all of F nor a real number measuring arc length. It is often the safest primary representation of a rotation over an arbitrary real closed field. In the basis used in the proof,

$$\text{tr } A = 1 + 2c, \quad -1 \leq \text{tr } A \leq 3, \quad \chi_A(X) = (X - 1)(X^2 - 2cX + 1). \quad (5.3)$$

Over $F(i)$, the eigenvalues are $1, c + is, c - is$. A half-turn is exactly a rotation with $c = -1, s = 0$, so

$$H_n = 2nn^\top - I, \quad H_n^2 = I, \quad \text{tr } H_n = -1. \quad (5.4)$$

Its axis is an unoriented line. These are all nonidentity involutions.

For a non-half-turn, the skew part is $A - A^\top = 2s\hat{n}$. At a half-turn it vanishes, although the rotation is not the identity. Algorithms that infer the axis only from the skew part must handle this case separately.

Theorem 5.4 (Every rotation has a finite axis-angle description). *Let F be a real-closed Hahn workspace as in [section 2](#); by localization the statement applies to $G(\mathbf{No})$. Every $A \in G(F)$ has a representation*

$$A = \text{Rot}(n; \cos \theta, \sin \theta), \quad 0 \leq \theta \leq \pi.$$

For $A \neq I$, the number θ is unique. The axis is oriented uniquely when $0 < \theta < \pi$, by the requirement $\sin \theta > 0$, and is unoriented at $\theta = \pi$. Thus every rotation, even one involving infinitely separated spatial points, has a finite axis-angle description.

Proof. By [theorem 5.3](#) and the finite phase theorem [theorem 2.2](#), $A = \text{Rot}(n; \cos \theta, \sin \theta)$ for some finite θ . Reduce θ by an ordinary integral multiple of 2π to $(-\pi, \pi]$. To handle an infinitesimal excursion past an endpoint, shift once more by 2π ; the representative must lie in the actual surreal interval, not merely have standard part there. Reverse the axis if the angle is negative, obtaining $[0, \pi]$. On $(0, \pi)$ the finite sine is positive: away from the endpoints its standard part is positive, and at an infinitesimal endpoint displacement the leading Taylor term decides the sign. Equality of two circle values is exactly equality modulo $2\pi\mathbb{Z}$. With the axis-reversal convention, this gives uniqueness of θ and the stated axis ambiguity. $\square$

For unit vectors $\mathbf{u} \neq -\mathbf{v}$, put $c = \mathbf{u} \cdot \mathbf{v}$ and $\mathbf{k} = \mathbf{u} \times \mathbf{v}$. The shortest-angle rotation taking $\mathbf{u}$ to $\mathbf{v}$ is

$$A(\mathbf{u}, \mathbf{v}) = I + \hat{\mathbf{k}} + \frac{\hat{\mathbf{k}}^2}{1 + c}. \quad (5.5)$$

The formula also gives the identity when $\mathbf{u} = \mathbf{v}$. Using $\mathbf{k} \times \mathbf{u} = \mathbf{v} - c\mathbf{u}$ and $\mathbf{k} \times (\mathbf{k} \times \mathbf{u}) = -(1 - c^2)\mathbf{u}$ verifies the mapping. At an antipodal pair the formula is undefined for a genuine reason: there are many half-turns taking one vector to the other.

5.2 Quaternion representation

Write a quaternion as $(a, \mathbf{b}) \in \mathbb{H}_F = F \oplus F^3$ with multiplication

$$(a, \mathbf{b})(c, \mathbf{d}) = (ac - \mathbf{b} \cdot \mathbf{d}, a\mathbf{d} + c\mathbf{b} + \mathbf{b} \times \mathbf{d}). \quad (5.6)$$

The basis $(1, i, j, k)$ satisfies $i^2 = j^2 = k^2 = ijk = -1$. This is an associative algebra; associativity follows either from Hamilton's relations and the faithful matrix representation [equation \(16.4\)](#), or from the triple-product identity. Quaternion conjugation and norm square are

$$\overline{(a, \mathbf{b})} = (a, -\mathbf{b}), \quad N(a, \mathbf{b}) = a^2 + \|\mathbf{b}\|^2.$$

Conjugation reverses products, $q\bar{q} = N(q)$, and $N(pq) = N(p)N(q)$. Since a nonzero sum of squares is positive, $q^{-1} = \bar{q}/N(q)$ for every nonzero q . The center is exactly F . These are ordered-field facts, not real-analytic facts [\[8, 19\]](#).

Put $\text{Sp}(1, F) = \{q \in \mathbb{H}_F : N(q) = 1\}$. Unit quaternions act on pure vectors by conjugation, $p(q)(\mathbf{x}) = q(0, \mathbf{x})q^{-1}$. Direct expansion gives

$$p(a, \mathbf{b}) \mathbf{x} = (a^2 - \|\mathbf{b}\|^2)\mathbf{x} + 2(\mathbf{b} \cdot \mathbf{x})\mathbf{b} + 2a(\mathbf{b} \times \mathbf{x}), \quad (5.7)$$

which is `squat:eq:rodrigues` of the surquaternion report. It preserves the imaginary subspace and the norm. Its determinant is +1 without an appeal to connectedness. For $\mathbf{b} \neq 0$, put $n = \mathbf{b}/\|\mathbf{b}\|$ and compare with [equation \(5.1\)](#) using $c = a^2 - \|\mathbf{b}\|^2$ and $s = 2a\|\mathbf{b}\|$. If $\mathbf{b} = 0$, the map is the identity.

Theorem 5.5 (The spin exact sequence; `squat:thm:rotations`). *Conjugation gives a surjective group homomorphism with precisely two elements in every fiber:*

$$1 \longrightarrow \{1, -1\} \longrightarrow \text{Sp}(1, F) \xrightarrow{p} G(F) \longrightarrow 1. \quad (5.8)$$

In particular, $\text{Spin}(3, F) \cong \text{Sp}(1, F)$.

This is Theorem 4.4 (`squat:thm:rotations`) of the surquaternion report [\[19\]](#), which also identifies $G(F)$ with the automorphism group of $\mathbb{H}_F$. Both sources reprove it by the argument of that theorem's first surjectivity route: a quaternion commuting with every pure vector is central, and a fixed axis gives a lift. The proof is not repeated here. Source 09's explicit lift of $\text{Rot}(n; c, s)$ is used below. If $c \neq -1$,

$$a = \sqrt{(1+c)/2} > 0, \quad u = \frac{s}{2a} n \quad (5.9)$$

give $a^2 + \|u\|^2 = 1$, $a^2 - \|u\|^2 = c$ and $2au = sn$, so (a, u) is a lift. If $c = -1$, the unit imaginary quaternion $(0, n)$ lifts H_n . For a finite angle, source 04's lift is $(\cos(\theta/2), \sin(\theta/2)\mathbf{n})$, which gives [equation \(5.2\)](#). The standard Clifford-algebra spin group of the positive definite ternary form has this Hamilton realization; [equation \(5.8\)](#) supplies an entirely explicit model without Clifford terminology.

The phrase “double cover” here denotes this two-to-one algebraic group homomorphism, not an unproved topological covering assertion in the fine topology. The category in which it is a two-sheeted covering with the familiar fundamental groups is identified in [theorem 26.2](#).

5.3 The rational Cayley chart

Theorem 5.6 (The matrix Cayley chart). *For $t \in F^3$, define*

$$\text{Cay}(t) = (I + \hat{t})(I - \hat{t})^{-1} = \frac{(1 - \|t\|^2)I + 2t t^\top + 2\hat{t}}{1 + \|t\|^2}. \quad (5.10)$$

This is a bijection from F^3 onto the rotations which are not half-turns. Its inverse is

$$\hat{t} = (A - I)(A + I)^{-1}, \quad (5.11)$$

$$t = \frac{(A_{32} - A_{23}, A_{13} - A_{31}, A_{21} - A_{12})^\top}{1 + \text{tr } A}. \quad (5.12)$$

Moreover, whenever $1 - t \cdot u \neq 0$,

$$\text{Cay}(t) \text{Cay}(u) = \text{Cay}\left(\frac{t + u + t \times u}{1 - t \cdot u}\right). \quad (5.13)$$

If $1 - t \cdot u = 0$, the product is a half-turn.

Proof. The determinant of $I - \hat{t}$ is $1 + \|t\|^2 > 0$. Using [equation \(3.5\)](#) gives the displayed rational expression. Equivalently it is the rotation of the unit quaternion

$$q(t) = \frac{(1, t)}{\sqrt{1 + \|t\|^2}}.$$

Its scalar coordinate is nonzero, so the rotation is not a half-turn. Conversely, the lift [equation \(5.9\)](#) of a non-half-turn has nonzero scalar part, and dividing its imaginary part by that scalar gives t . Solving the matrix equation gives [equation \(5.11\)](#); taking skew coordinates gives [equation \(5.12\)](#). In particular,

$$\text{tr } \text{Cay}(t) = \frac{3 - \|t\|^2}{1 + \|t\|^2}, \quad 1 + \text{tr } \text{Cay}(t) = \frac{4}{1 + \|t\|^2}, \quad \det(\text{Cay}(t) + I) = \frac{8}{1 + \|t\|^2}. \quad (5.14)$$

Finally, before normalization, $(1, t)(1, u) = (1 - t \cdot u, t + u + t \times u)$. A nonzero scalar part gives the quotient in [equation \(5.13\)](#). If that scalar vanishes, the product is a nonzero imaginary quaternion and its rotation is a half-turn. $\square$

In vector form, and with $\mathbf{t} = \tan(\theta/2)\mathbf{n}$ for a rotation of finite angle θ away from half-turns, source 04 writes the same chart as

$$\text{Cay}(\mathbf{t}) \mathbf{x} = \mathbf{x} + \frac{2}{1 + \|\mathbf{t}\|^2}(\mathbf{t} \times \mathbf{x} + \mathbf{t} \times (\mathbf{t} \times \mathbf{x})). \quad (5.15)$$

This rational chart can have infinite coordinates near a half-turn, although every entry of the rotation matrix remains between -1 and 1 ([proposition 5.2](#)). It is the three-dimensional analogue of the planar half-angle chart `trigonometry:thm:cayley`.

This is a chart on $G(F)$, not the quaternion stereographic chart $(1 - x)^{-1}(1 + x)$ on $\text{Sp}(1, F)$ of `squat:prop:charts`. The two are related, but their parameters have different half-angle factors. Confusing them changes the composition law and the leading infinitesimal rotation by a factor of two.

The half-turn locus is naturally $\mathbb{P}^2(F)$, by $[n] \mapsto 2nn^\top - I$ after normalization. The chart and its translates show that the group has algebraic dimension three. After extending to $F(i)$, its standard algebraic form is PGL_2 ; thus the algebraic group is connected and absolutely simple of adjoint type. These are algebraic assertions. They do not imply that its F -point group is abstractly simple or that its fine topology is connected. For $F \supsetneq \mathbb{R}$, and for **No**, [sections 18](#) and [26](#) show that neither holds.

5.4 Screw motions

Let $\Phi(\mathbf{x}) = A\mathbf{x} + \mathbf{b}$ with $A \in \text{SO}(3, F)$ nonidentity. Take its unit axis $\mathbf{n}$, put $h = \mathbf{b} \cdot \mathbf{n}$, and $\mathbf{b}_\perp = \mathbf{b} - h\mathbf{n}$. On $\mathbf{n}^\perp$, $I - A$ is invertible, so there is a unique $\mathbf{a} \perp \mathbf{n}$ satisfying $(I - A)\mathbf{a} = \mathbf{b}_\perp$. Therefore

$$\Phi(\mathbf{a} + t\mathbf{n}) = \mathbf{a} + (t + h)\mathbf{n}.$$

The motion is rotation about the line $\mathbf{a} + F\mathbf{n}$, followed by translation h along it. With $0 < \theta \leq \pi$,

$$\mathbf{a} = \frac{1}{2} (\mathbf{b}_\perp + \cot(\theta/2) \mathbf{n} \times \mathbf{b}_\perp). \quad (5.16)$$

This follows by inverting $(1 - \cos \theta)I - \sin \theta J$ on the perpendicular plane, with $J\mathbf{x} = \mathbf{n} \times \mathbf{x}$. An infinitesimal rotation with appreciable transverse translation can have an infinitely distant screw axis. The identity rotation is treated separately as a pure translation.

6 Coordinates and ordinary triangle trigonometry in space

6.1 Cylindrical and spherical coordinates

For $\mathbf{x} = (x, y, z)$ put $r = \sqrt{x^2 + y^2}$. When $r > 0$, cylindrical coordinates are

$$x = r \cos \phi, \quad y = r \sin \phi, \quad z = z, \quad \phi = \text{atan2}(y, x) \pmod{2\pi}.$$

At $r = 0$, azimuth is undetermined. There is no restriction on the scale of r or z .

For $\rho = \|\mathbf{x}\| > 0$, define colatitude $\theta = \arccos(z/\rho) \in [0, \pi]$. Away from the polar axis choose the same azimuth ϕ . Then

$$(x, y, z) = (\rho \sin \theta \cos \phi, \rho \sin \theta \sin \phi, \rho \cos \theta). \quad (6.1)$$

Every nonzero vector has this representation. At the poles the azimuth is free; at the origin both angles are free. The radial coordinate may be infinite, but both angular coordinates are finite. Choosing an azimuth branch introduces a coordinate seam, not a singularity of the sphere.

Differentiate the coordinate map with respect to its external variables. Its three coordinate derivatives are mutually perpendicular and have squared norms $1, \rho^2, \rho^2 \sin^2 \theta$. Therefore

$$ds^2 = d\rho^2 + \rho^2 d\theta^2 + \rho^2 \sin^2 \theta d\phi^2, \quad J = \rho^2 \sin \theta. \quad (6.2)$$

These are finite identities in restricted-analytic coordinate derivatives. They do not assert an integral transformation theorem for every function on F^3 . On a sphere of radius R , the local area density is $R^2 \sin \theta$ and the tangent metric is obtained by setting $d\rho = 0$.

6.2 Flat triangles retain their usual formulas

Every noncollinear triple of points lies in a unique affine plane. An orthonormal basis of that plane reduces its geometry to F^2 without changing lengths. Let its opposite side lengths be $A, B, C > 0$, its angles $\alpha, \beta, \gamma \in (0, \pi)$, and its area $\Delta > 0$. Then

$$\alpha + \beta + \gamma = \pi, \quad (6.3)$$

$$A^2 = B^2 + C^2 - 2BC \cos \alpha, \quad (6.4)$$

$$\frac{A}{\sin \alpha} = \frac{B}{\sin \beta} = \frac{C}{\sin \gamma} = \frac{ABC}{2\Delta} = 2R_c, \quad (6.5)$$

$$\Delta^2 = s(s - A)(s - B)(s - C), \quad s = (A + B + C)/2. \quad (6.6)$$

These are the usual classical identities [6, §4.42(ii)], now with field-valued lengths.

For completeness, put the first two vertices at $(0, 0)$ and $(C, 0)$, and the third in the upper half-plane. If the ray from the first vertex to the third has phase α and the ray from the second to the third has phase δ , orientation gives $0 < \alpha < \delta < \pi$. The remaining angles are $\pi - \delta$ and $\delta - \alpha$, proving equation (6.3). Expanding the squared difference proves equation (6.4). Computing the same area from each pair of sides proves the first equalities in equation (6.5). The circumcenter is the solution of two perpendicular bisector equations, and the chord relation gives $A = 2R_c \sin \alpha$. Finally squaring $2\Delta = BC \sin \alpha$ and substituting the cosine law gives

$$16\Delta^2 = (A + B + C)(-A + B + C)(A - B + C)(A + B - C),$$

which proves Heron's formula.

The triangle inequalities are strict exactly in the noncollinear case. None of these statements requires the side lengths to be finite. For example, a triangle can have sides of scale ω and an altitude of scale $\omega^{-\omega}$; its nonzero area is the product scale, not automatically zero or infinite.

7 The surreal sphere and its intrinsic angular distance

Let

$$S_R^2(C) = \{C + Ru : \mathbf{u} \cdot \mathbf{u} = 1\}.$$

A great circle is its intersection with a plane through C . Translating and rescaling reduces all angular calculations to the unit sphere $S^2(F)$.

7.1 Minor arcs without a global sine function

For unit vectors $\mathbf{u} \neq -\mathbf{v}$, define the minor arc by the algebraic set

$$\left\{ \frac{(1-t)\mathbf{u} + t\mathbf{v}}{\|(1-t)\mathbf{u} + t\mathbf{v}\|} : 0 \leq t \leq 1 \right\}. \quad (7.1)$$

The denominator is nonzero: vanishing would make the endpoints antipodal unless they coincide. If $a = \text{ang}(\mathbf{u}, \mathbf{v}) \in (0, \pi)$ and

$$\mathbf{e} = \frac{\mathbf{v} - \cos a \mathbf{u}}{\sin a},$$

the same arc has the trigonometric parametrization

$$\mathbf{g}(s) = \cos s \mathbf{u} + \sin s \mathbf{e}, \quad 0 \leq s \leq a. \quad (7.2)$$

To see equality, work in the oriented plane basis $(\mathbf{u}, \mathbf{e})$. Normalized positive combinations of $(1, 0)$ and $(\cos a, \sin a)$ have exactly the phases between 0 and a , as is checked by the signs of their two oriented determinants with the endpoints. This also proves uniqueness of the minor arc. Antipodal points instead have a family of semicircles, indexed by unit directions perpendicular to either endpoint.

Define the angular distance on the unit sphere and the physical distance on $S_R^2(C)$ by

$$d_S(\mathbf{u}, \mathbf{v}) = \text{acos}(\mathbf{u} \cdot \mathbf{v}), \quad d_R(C + R\mathbf{u}, C + R\mathbf{v}) = R d_S(\mathbf{u}, \mathbf{v}). \quad (7.3)$$

The length assigned to a minor arc is Ra . Subarcs add because their finite phase differences add. One complete great circle has length $2\pi R$.

Theorem 7.1 (Angular triangle inequality). d_S is a positive-definite F -valued distance on $S^2(F)$. For $a = d_S(\mathbf{u}, \mathbf{v})$,

$$\|\mathbf{u} - \mathbf{v}\| = 2 \sin(a/2), \quad \frac{2}{\pi}a \leq \|\mathbf{u} - \mathbf{v}\| \leq a. \quad (7.4)$$

A minor great-circle arc minimizes the sum of arc lengths among all finite chains of minor great-circle arcs with the same endpoints.

Proof. Only the triangle inequality needs explanation. Set $a = d_S(\mathbf{u}, \mathbf{v})$ and $b = d_S(\mathbf{v}, \mathbf{w})$. If $a + b \geq \pi$ the claim follows from $d_S \leq \pi$. Otherwise write

$$\mathbf{u} = \cos a \mathbf{v} + \sin a \mathbf{e}, \quad \mathbf{w} = \cos b \mathbf{v} + \sin b \mathbf{f}$$

with unit perpendicular vectors, treating zero angles separately. Cauchy–Schwarz gives $\mathbf{e} \cdot \mathbf{f} \geq -1$, hence $\mathbf{u} \cdot \mathbf{w} \geq \cos(a + b)$. Cosine is decreasing on $[0, \pi]$, so $d_S(\mathbf{u}, \mathbf{w}) \leq a + b$. The chord identity follows by squaring. The inequalities follow from $2t/\pi \leq \sin t \leq t$ for $0 \leq t \leq \pi/2$, valid in the restricted-analytic field. Iterating the triangle inequality proves the finite-chain minimizing assertion. $\square$

This is an intrinsic distance construction with an explicit minimizing class. No supremum over arbitrary partitions, no real-valued length metric, and no unrestricted class of nondefinable curves has been silently introduced.

7.2 Gnomonic and stereographic charts

The gnomonic chart on the northern open hemisphere is

$$\text{Gn}(p, q) = \frac{(p, q, 1)}{\sqrt{1 + p^2 + q^2}}, \quad (p, q) = (u_1/u_3, u_2/u_3). \quad (7.5)$$

Great circles become straight lines because $\mathbf{n} \cdot \mathbf{u} = 0$ becomes $n_1 p + n_2 q + n_3 = 0$. Positive-cone spherical triangles become ordinary affine convex triangles. The chart is valid for arbitrary $p, q \in F$, including infinite values near the horizon. With $\mathbf{p} = (p, q)$ its metric on a sphere of radius R is

$$ds^2 = R^2 \frac{(1 + \|\mathbf{p}\|^2) \|d\mathbf{p}\|^2 - (\mathbf{p} \cdot d\mathbf{p})^2}{(1 + \|\mathbf{p}\|^2)^2}, \quad J_{\text{Gn}} = \frac{R^2}{(1 + \|\mathbf{p}\|^2)^{3/2}}. \quad (7.6)$$

Both follow by direct differentiation of equation (7.5).

Stereographic projection from the north pole has inverse

$$\text{St}(p, q) = \frac{(2p, 2q, p^2 + q^2 - 1)}{1 + p^2 + q^2}. \quad (7.7)$$

It covers the sphere minus that pole, with metric

$$ds^2 = \frac{4R^2(dp^2 + dq^2)}{(1 + p^2 + q^2)^2}.$$

The scalar multiple of $dp^2 + dq^2$ proves conformality, including at infinitesimal and infinite chart coordinates. A spherical circle cut out by $\mathbf{n} \cdot \mathbf{u} = h$ becomes

$$(n_3 - h)(p^2 + q^2) + 2n_1 p + 2n_2 q - (n_3 + h) = 0,$$

a circle or line. Infinite chart coordinates are not the omitted pole; they describe genuine points infinitesimally near it.

8 Spherical triangles and the Gram-matrix calculus

Definition 8.1 (Proper spherical triangle). Three linearly independent unit vectors $\mathbf{u}, \mathbf{v}, \mathbf{w}$ determine the closed spherical triangle consisting of the normalized nonzero vectors in

$$\{\lambda\mathbf{u} + \mu\mathbf{v} + \nu\mathbf{w} : \lambda, \mu, \nu \geq 0\}.$$

Its edges are the minor great-circle arcs. Its interior angles are the angles between the inward-pointing edge tangents at each vertex.

Such a triangle lies in an open hemisphere. Indeed, if $P = [\mathbf{u} \ \mathbf{v} \ \mathbf{w}]$, then $\mathbf{h} = P^{-\top}(1, 1, 1)^\top$ has positive dot product with every nonzero vector in the positive cone. This algebraic observation fixes which region and which angles are intended, even when its real standard part is degenerate. All sides and angles of a proper triangle lie strictly between 0 and π .

Use [equation \(1.4\)](#) and put

$$\mathcal{G} = P^\top P = \begin{pmatrix} 1 & z & y \\ z & 1 & x \\ y & x & 1 \end{pmatrix}, \quad D = \det \mathcal{G} = 1 + 2xyz - x^2 - y^2 - z^2 = \tau^2 > 0. \quad (8.1)$$

Write $s_a = \sqrt{1 - x^2} = \sin a$ and cyclically s_b, s_c . The unit tangents at $\mathbf{u}$ pointing toward $\mathbf{v}$ and $\mathbf{w}$ are

$$T_{uv} = \frac{\mathbf{v} - z\mathbf{u}}{s_c}, \quad T_{uw} = \frac{\mathbf{w} - y\mathbf{u}}{s_b}. \quad (8.2)$$

Consequently

$$\cos \alpha = \frac{x - yz}{s_b s_c}, \quad \sin \alpha = \frac{d}{s_b s_c}, \quad \alpha = \text{atan2}(d, x - yz), \quad (8.3)$$

with cyclic formulas for β, γ . The common positive denominator in the first two formulas is irrelevant to `atan2`.

Theorem 8.2 (Spherical sine and cosine laws). *Every proper spherical triangle satisfies*

$$\cos a = \cos b \cos c + \sin b \sin c \cos \alpha, \quad (8.4)$$

$$\frac{\sin \alpha}{\sin a} = \frac{\sin \beta}{\sin b} = \frac{\sin \gamma}{\sin c} = \frac{d}{\sin a \sin b \sin c}. \quad (8.5)$$

Cyclic permutations give the other cosine laws. A useful additional identity is

$$\sin a \cos \beta = \cos b \sin c - \sin b \cos c \cos \alpha. \quad (8.6)$$

Proof. The dot product of the two tangents in [equation \(8.2\)](#) is $(x - yz)/(s_b s_c)$. Their cross product is parallel to $\mathbf{u}$ and has norm $|\det[\mathbf{u} \ \mathbf{v} \ \mathbf{w}]|/(s_b s_c)$. This proves [equation \(8.3\)](#), and hence both main laws. For [equation \(8.6\)](#), substitute the cosine law into the right-hand side and reduce it to $(y - xz)/s_c$, which is the left-hand side by its cyclic tangent formula. $\square$

The sine and cosine laws are classical [6, §4.42(iii)]. Their proof here uses only finite vector identities plus the already constructed finite angle representatives. The same calculation over a field with square roots still gives the algebraic sine–cosine data even without naming angles.

8.1 Polar triangles and the cosine law for angles

Relabel the vertices so that $\tau > 0$ and define inward unit side normals

$$\mathbf{n}_a = \frac{\mathbf{v} \times \mathbf{w}}{s_a}, \quad \mathbf{n}_b = \frac{\mathbf{w} \times \mathbf{u}}{s_b}, \quad \mathbf{n}_c = \frac{\mathbf{u} \times \mathbf{v}}{s_c}. \quad (8.7)$$

For example $\mathbf{n}_a \cdot \mathbf{u} = d/s_a > 0$, and its dot products with the other two vertices vanish. These normals are independent and positively oriented: their determinant is $d^2/(s_a s_b s_c) > 0$.

Theorem 8.3 (Polar duality). *The triangle with vertices $\mathbf{n}_a, \mathbf{n}_b, \mathbf{n}_c$ has sides and angles*

$$a^* = \pi - \alpha, \quad b^* = \pi - \beta, \quad c^* = \pi - \gamma, \quad \alpha^* = \pi - a, \quad \beta^* = \pi - b, \quad \gamma^* = \pi - c. \quad (8.8)$$

Its polar is the original triangle. In particular,

$$\cos \alpha = -\cos \beta \cos \gamma + \sin \beta \sin \gamma \cos a, \quad (8.9)$$

and cyclically.

Proof. By equation (3.3), $\mathbf{n}_b \cdot \mathbf{n}_c = (yz - x)/(s_b s_c) = -\cos \alpha$. The side is in $(0, \pi)$, so it equals $\pi - \alpha$. Also $(\mathbf{w} \times \mathbf{u}) \times (\mathbf{u} \times \mathbf{v}) = \tau \mathbf{u}$; after normalization the corresponding inward polar normal is $\mathbf{u}$. This proves involutivity and the angle relations. Apply the side cosine law to the polar triangle and substitute equation (8.8). $\square$

8.2 Half-angle, right-triangle, and haversine formulas

Let $s = (a + b + c)/2$. The identity

$$D = 4 \sin s \sin(s - a) \sin(s - b) \sin(s - c) \quad (8.10)$$

follows by factoring

$$D = (\cos(b - c) - \cos a)(\cos a - \cos(b + c)).$$

For a proper triangle $s < \pi$ and $s - a, s - b, s - c > 0$, as proved in the next section. The half-angle formulas therefore have unambiguous positive roots:

$$\sin(\alpha/2) = \sqrt{\frac{\sin(s - b) \sin(s - c)}{\sin b \sin c}}, \quad (8.11)$$

$$\cos(\alpha/2) = \sqrt{\frac{\sin s \sin(s - a)}{\sin b \sin c}}. \quad (8.12)$$

They follow by substituting the cosine law into $\sin^2(\alpha/2) = (1 - \cos \alpha)/2$ and $\cos^2(\alpha/2) = (1 + \cos \alpha)/2$. Their ratio gives the half-angle tangent formula. Applying the same formulas to the polar triangle gives half-side formulas in terms of the angles.

For a right triangle with $\gamma = \pi/2$, the Napier relations can be written in the particularly safe form

$$\cos c = \cos a \cos b, \quad \sin a = \sin c \sin \alpha, \quad \sin b = \sin c \sin \beta, \quad (8.13)$$

$$\cos \alpha = \cos a \sin \beta, \quad \cos \beta = \cos b \sin \alpha, \quad \cos c = \cot \alpha \cot \beta. \quad (8.14)$$

They follow from the side and angle cosine laws and the sine law. The last expression is defined since $\sin \alpha, \sin \beta > 0$. Formulas involving $\tan a$, $\tan b$, or $\tan c$ require additional nonzero-cosine checks. For instance a proper right spherical triangle can itself have a side of length $\pi/2$.

Define $\text{hav } t = \sin^2(t/2)$. Rearranging the cosine law gives

$$\text{hav } a = \text{hav}(b - c) + \sin b \sin c \text{ hav } \alpha. \quad (8.15)$$

This formulation separates small positive quantities rather than obtaining them by subtracting two almost equal cosines.

9 Existence, reconstruction, and ambiguous data

Theorem 9.1 (Complete side-data criterion and reconstruction). *For $a, b, c \in (0, \pi)$ the following are equivalent:*

1. *They are the angular sides of a proper spherical triangle.*
2. *The matrix $\mathcal{G}$ in [equation \(8.1\)](#) is positive definite.*
3. *$D > 0$.*
4. *$a < b + c$, $b < c + a$, $c < a + b$, and $a + b + c < 2\pi$.*

When these conditions hold, a positively oriented realization is

$$\mathbf{u} = (1, 0, 0), \quad \mathbf{v} = (z, s_c, 0), \quad \mathbf{w} = \left(y, \frac{x - yz}{s_c}, \frac{\sqrt{D}}{s_c} \right). \quad (9.1)$$

It is unique up to an orthogonal transformation; the positively oriented realizations form one $\text{SO}(3, F)$ orbit.

Proof. A genuine triple has $\mathcal{G} = P^\top P$ with P invertible, hence positive definiteness. Conversely $1 > 0$ and $1 - z^2 > 0$ are its first two leading principal minors, so $D > 0$ is exactly the remaining condition in Sylvester's criterion. Alternatively [equation \(9.1\)](#) directly constructs a realization: the identity $D = (1 - y^2)(1 - z^2) - (x - yz)^2$ makes the last vector unit and its dot products are y, x . Its determinant is $\sqrt{D} > 0$.

To prove the inequality criterion, use [equation \(8.10\)](#). Here $0 < s < 3\pi/2$ and each $s - a, s - b, s - c$ belongs to $(-\pi/2, \pi)$. At most one of these three numbers is nonpositive. If $s \geq \pi$, all three are positive because $a, b, c < \pi$, whereas $\sin s \leq 0$, giving $D \leq 0$. If $s < \pi$, positivity of D is equivalent to all three differences being positive. This is exactly the stated set of inequalities.

Finally, if P and P' are two realizations, then $Q = P'P^{-1}$ satisfies $Q^\top Q = I$ because their Gram matrices agree. The sign of its determinant is the ratio of their orientation signs. $\square$

The perimeter restriction is essential. The three inequalities alone are not sufficient: sides all equal to $3\pi/4$ satisfy the ordinary triangle inequalities but have perimeter exceeding 2π .

9.1 SAS, AAA, and ASA

Any two sides $b, c \in (0, \pi)$ and included angle $\alpha \in (0, \pi)$ determine exactly one proper triangle up to isometry. An explicit realization is

$$\mathbf{u} = (1, 0, 0), \quad \mathbf{v} = (\cos c, \sin c, 0), \quad \mathbf{w} = (\cos b, \sin b \cos \alpha, \sin b \sin \alpha).$$

Its determinant is $\sin b \sin c \sin \alpha > 0$. Compute a from the dot product and the remaining angles from [equation \(8.3\)](#). This avoids any ambiguous inverse sine.

By polar duality, angles $\alpha, \beta, \gamma \in (0, \pi)$ occur exactly when

$$\alpha + \beta + \gamma > \pi, \quad \beta + \gamma - \alpha < \pi, \quad (9.2)$$

with the last inequality also imposed cyclically. They then determine the triangle up to isometry. In particular, *three angles fix the shape and angular size* on a fixed sphere; there is no nontrivial similarity scaling of angular sides. The condition that the angle sum exceed π alone is not sufficient. Two angles and their included side (ASA) reduce to SAS for the polar triangle and likewise determine a unique proper triangle for inputs in $(0, \pi)$.

9.2 The spherical SSA ambiguity has an exceptional family

Suppose that $a, b, \alpha \in (0, \pi)$ are given, with a opposite α . Set

$$P = \cos b, \quad Q = \sin b \cos \alpha, \quad X = \cos a.$$

The unknown side c must solve

$$PC + QS = X, \quad C^2 + S^2 = 1, \quad S > 0, \quad (C, S) = (\cos c, \sin c). \quad (9.3)$$

Every solution gives a proper triangle via the SAS construction, whose opposite side is the prescribed a .

Proposition 9.2 (Exact SSA alternatives). *If $L = P^2 + Q^2 > 0$, SSA data have zero, one, or two solutions up to isometry. The candidate unit-circle pairs are*

$$(C, S) = \frac{X}{L}(P, Q) \pm \frac{\sqrt{L - X^2}}{L}(-Q, P), \quad (9.4)$$

provided $L - X^2 \geq 0$; only candidates with $S > 0$ are retained, and duplicate candidates are counted once.

If $P = Q = 0$, there are no solutions unless $a = \pi/2$. If $a = \pi/2$, then $b = \alpha = \pi/2$ as well, and every $c \in (0, \pi)$ gives a solution, with $\beta = \pi/2$ and $\gamma = c$.

Proof. For $L > 0$, intersect the line with the unit circle by decomposing a point into components parallel and perpendicular to (P, Q) . This gives [equation \(9.4\)](#); a line has at most two intersections. The upper-half condition is exactly $c \in (0, \pi)$. If $P = Q = 0$, the original equation is $0 = X$. Since $b \in (0, \pi)$ and $\sin b > 0$, the vanishing of P, Q means $b = \alpha = \pi/2$. When also $a = \pi/2$, take $\mathbf{u} = (1, 0, 0)$, $\mathbf{v} = (\cos c, \sin c, 0)$, $\mathbf{w} = (0, 0, 1)$. Their determinant is $\sin c > 0$. The angle formulas give the claimed remaining angles. $\square$

Thus a blanket assertion that spherical SSA always has at most two solutions misses a genuine continuous exceptional family. The polar construction gives the corresponding AAS ambiguity. The effect is already present over $\mathbb{R}$; surreal coordinates add arbitrarily small perturbations of the exceptional data, for which branch-sensitive computation is especially important.

9.3 Latitude, longitude, and bearings

For latitude $\varphi \in [-\pi/2, \pi/2]$ and longitude λ modulo 2π , the unit direction is

$$(\cos \varphi \cos \lambda, \cos \varphi \sin \lambda, \sin \varphi).$$

Two such points have separation a given by

$$\cos a = \sin \varphi_1 \sin \varphi_2 + \cos \varphi_1 \cos \varphi_2 \cos(\lambda_2 - \lambda_1), \quad (9.5)$$

$$\text{hav } a = \text{hav}(\varphi_2 - \varphi_1) + \cos \varphi_1 \cos \varphi_2 \text{hav}(\lambda_2 - \lambda_1). \quad (9.6)$$

At a nonpolar starting point and for a distinct nonantipodal endpoint, the initial bearing measured from north toward east is

$$\text{atan2}(\cos \varphi_2 \sin \Delta\lambda, \cos \varphi_1 \sin \varphi_2 - \sin \varphi_1 \cos \varphi_2 \cos \Delta\lambda). \quad (9.7)$$

Indeed these are the east and north components of the projected tangent to the target. Undefined bearings at poles, coincident points, or antipodes are geometric ambiguities, not numerical failures. The physical great-circle distance remains Ra for any positive surreal radius.

10 Spherical centers, medians, and Ceva

Assume $\tau = d > 0$ and retain the matrix P and inward normals of [equation \(8.7\)](#).

10.1 Circumcenter and incenter

Put $\mathbf{h} = P^{-\top}(1, 1, 1)^\top$. Then

$$\mathbf{o} = \frac{\mathbf{h}}{\|\mathbf{h}\|}, \quad \rho_c = \arccos \frac{1}{\|\mathbf{h}\|} \quad (10.1)$$

is the center and angular radius of the circumcircle with radius less than $\pi/2$. All three dot products with $\mathbf{o}$ equal $1/\|\mathbf{h}\| > 0$. Cauchy–Schwarz makes this number at most one, and distinct vertices make it strictly less than one. Every equidistant unit center solves $P^\top \mathbf{o} = k(1, 1, 1)^\top$, so the two possibilities are exactly $\mathbf{o}$ and $-\mathbf{o}$, with radii ρ_c and $\pi - \rho_c$. The chosen circumcenter need not lie inside the triangle.

The incenter has the particularly simple weighted-vector expression

$$\mathbf{j} = \frac{s_a \mathbf{u} + s_b \mathbf{v} + s_c \mathbf{w}}{\|s_a \mathbf{u} + s_b \mathbf{v} + s_c \mathbf{w}\|}, \quad r_i = \arcsin \frac{d}{\|s_a \mathbf{u} + s_b \mathbf{v} + s_c \mathbf{w}\|}. \quad (10.2)$$

To prove this, take dot products of its numerator with the three inward normals. Each equals d . Hence $\mathbf{j}$ has equal positive signed distances from the three side great circles, and it lies inside the positive cone. Those signed angular distances are the inverse sines in [equation \(10.2\)](#). Uniqueness follows by solving the nonsingular linear system $N^\top \mathbf{j} = k(1, 1, 1)^\top$ for the inward normal matrix N . Equivalently, the inside angle bisectors below concur at this point. The physical circumradius and inradius along the sphere are $R\rho_c$ and Rr_i , not ρ_c and r_i themselves.

10.2 Medians and angle bisectors

The midpoint of a minor edge from $\mathbf{v}$ to $\mathbf{w}$ is $(\mathbf{v} + \mathbf{w})/\|\mathbf{v} + \mathbf{w}\|$. The three vertex-to-opposite-midpoint great circles concur at

$$\frac{\mathbf{u} + \mathbf{v} + \mathbf{w}}{\|\mathbf{u} + \mathbf{v} + \mathbf{w}\|}.$$

This follows by writing the common numerator as a vertex plus the sum of the other two. It is a median center, not an assertion that it is the surface-area centroid.

At $\mathbf{u}$, an internal angle bisector has tangent direction $T_{uv} + T_{uw}$. Its opposite-side intersection is proportional to $s_b\mathbf{v} + s_c\mathbf{w}$. If that point is D_a on the edge vw , then

$$\frac{\sin d_S(\mathbf{v}, D_a)}{\sin d_S(D_a, \mathbf{w})} = \frac{\sin c}{\sin b}. \quad (10.3)$$

This is a ratio of sines of angular subarcs, not in general a ratio of their angular lengths.

Theorem 10.1 (Spherical Ceva). *For interior points D_a, D_b, D_c on the minor edges vw, wu, uv , the three cevians concur inside the triangle if and only if*

$$\frac{\sin d_S(\mathbf{v}, D_a)}{\sin d_S(D_a, \mathbf{w})} \frac{\sin d_S(\mathbf{w}, D_b)}{\sin d_S(D_b, \mathbf{u})} \frac{\sin d_S(\mathbf{u}, D_c)}{\sin d_S(D_c, \mathbf{v})} = 1. \quad (10.4)$$

Proof. Write D_a as the normalization of $\lambda\mathbf{v} + \mu\mathbf{w}$ with $\lambda, \mu > 0$. Cross products with the endpoints show that its first sine ratio is μ/λ . Do the same on the other two edges. A positive linear combination $A\mathbf{u} + B\mathbf{v} + C\mathbf{w}$ belongs to the first cevian exactly when $C/B = \mu/\lambda$. The cyclic conditions have a positive solution precisely when their product is one. Normalizing that solution gives the interior concurrence point. This is ordinary barycentric Ceva expressed in spherical sine ratios. $\square$

11 Spherical excess, solid angles, and polygon area

Define

$$M = 1 + x + y + z, \quad B_0 = \sqrt{M^2 + D}. \quad (11.1)$$

An important positive identity is

$$M^2 + D = 2(1+x)(1+y)(1+z) > 0. \quad (11.2)$$

The sign of M must not be discarded. It determines whether the triangle's area is smaller or larger than a quarter of the full sphere.

Theorem 11.1 (Branch-correct spherical excess). *For a proper spherical triangle,*

$$E := \alpha + \beta + \gamma - \pi = 2 \operatorname{atan} 2(d, M), \quad 0 < E < 2\pi. \quad (11.3)$$

In particular, $E < \pi$, $E = \pi$, or $E > \pi$ according as $M > 0$, $M = 0$, or $M < 0$. The associated half-angle data are

$$\cos(E/2) = \frac{M}{B_0}, \quad \sin(E/2) = \frac{d}{B_0}. \quad (11.4)$$

Proof. The polar triangle has perimeter less than 2π by [theorem 9.1](#), so $3\pi - (\alpha + \beta + \gamma) < 2\pi$. Each interior angle is less than π . Hence $0 < \alpha + \beta + \gamma - \pi < 2\pi$.

For an algebraic evaluation, work in $F[i]$, and put $A_1 = x - yz$, $A_2 = y - zx$, $A_3 = z - xy$ and $T = (1 - x)(1 - y)(1 - z)$. Direct multiplication, using $d^2 = 1 + 2xyz - x^2 - y^2 - z^2$, gives

$$2(A_1 + id)(A_2 + id)(A_3 + id) + T(M + id)^2 = 0. \quad (11.5)$$

This can be checked by separately comparing real and imaginary parts; they reduce respectively to

$$2(A_1A_2A_3 - D(A_1 + A_2 + A_3)) + T(M^2 - D) = 0$$

and $A_1A_2 + A_2A_3 + A_3A_1 - D + TM = 0$. By [equation \(8.3\)](#), the product of the three angle phases is the product of the first three factors in [equation \(11.5\)](#), divided by $(1 - x^2)(1 - y^2)(1 - z^2)$. Since that denominator equals $T(M^2 + D)/2$, we obtain

$$\text{cis}(\alpha + \beta + \gamma) = -\frac{(M + id)^2}{M^2 + D}.$$

It follows that E and $2 \operatorname{atan2}(d, M)$ have the same phase. Both lie in $(0, 2\pi)$, where phase is injective, so they are equal. The sign alternatives and [equation \(11.4\)](#) follow immediately. $\square$

When $M > 0$, [equation \(11.3\)](#) can be written $2 \operatorname{atan}(d/M)$. When $M < 0$, the same uncorrected one-argument arctangent would give a negative number instead of the correct value in $(\pi, 2\pi)$. This is a geometric branch error, not merely a floating-point issue.

11.1 L'Huilier's formula

For $s = (a + b + c)/2$, all four half-arguments below lie in $(0, \pi/2)$. We have the exact formula

$$\tan(E/4) = \sqrt{\tan(s/2) \tan((s - a)/2) \tan((s - b)/2) \tan((s - c)/2)}. \quad (11.6)$$

Here the root is positive, even for infinitesimal excess or for excess infinitesimally close to 2π .

To prove it, use $\tan(E/4) = d/(B_0 + M)$ from [equation \(11.4\)](#). The product-to-sum identity

$$B_0 + M = 8 \cos(s/2) \cos((s - a)/2) \cos((s - b)/2) \cos((s - c)/2)$$

follows by expansion, using $B_0 = 4 \cos(a/2) \cos(b/2) \cos(c/2) > 0$ from [equation \(11.2\)](#). Replace the four full-angle sines in [equation \(8.10\)](#) by twice the corresponding sine–cosine products. After division by $(B_0 + M)^2$, one obtains the square of [equation \(11.6\)](#). Positivity supplies the indicated root.

11.2 Area as a precise finite-polygon functional

Definition 11.2 (Spherical polygon area). For a proper triangle on $S_R^2(C)$ define its area to be R^2E . For a convex geodesic polygon contained in an open hemisphere, triangulate it by minor great-circle diagonals and add these triangle areas.

Theorem 11.3 (Consistency, positivity, and finite additivity). *The definition is independent of triangulation. For a convex polygon with n boundary vertices and interior angles $\alpha_1, \dots, \alpha_n$,*

$$\text{Area}(P) = R^2 \left(\sum_{j=1}^n \alpha_j - (n - 2)\pi \right). \quad (11.7)$$

It is positive for a nonempty two-dimensional polygon, invariant under orthogonal transformations, and additive under finite geodesic polygonal dissections with disjoint interiors, with edges and vertices assigned area zero. It extends by such dissections to finite unions of spherical geodesic polygons.

Proof. Use a gnomonic chart to make the polygon an ordinary convex polygon. In a triangulation with boundary vertices only, summing all the triangle angles gives precisely the sum of boundary angles, and there are $n - 2$ triangles. This proves [equation \(11.7\)](#) and independence of the triangulation.

A refinement can introduce interior vertices or subdivide boundary edges. At an interior vertex the angles sum to 2π ; at an inserted straight boundary vertex the angle is π . If a planar triangulation has I interior vertices, B boundary vertices and T_Δ triangles, the finite Euler count is $T_\Delta = 2I + B - 2$. Thus the extra angles cancel exactly the extra triangle subtractions. A common finite refinement of two dissections proves additivity and consistency. Positivity follows by adding the positive triangle excesses. Orthogonal invariance follows from invariance of dot products and interior angles. For regions not contained in one open hemisphere, intersect with the eight closed coordinate octants, subdividing edges where necessary. Each octant lies in the open hemisphere whose normal has its three coordinate signs. Thus all the resulting pieces admit the preceding construction, and common refinement proves the same consistency. No topological compactness is needed for this explicit finite subdivision. $\square$

An octant triangle has three right angles, hence excess $\pi/2$. Eight octants therefore give

$$\text{Area}(S_R^2) = 4\pi R^2, \quad \text{Area}(\text{hemisphere}) = 2\pi R^2. \quad (11.8)$$

A lune of angle $\lambda \in (0, \pi)$ splits along the equator into two triangles with angles $\lambda, \pi/2, \pi/2$. Its area is $2R^2\lambda$. These values reproduce the usual geometric normalization of area.

For a circular cap of angular radius θ , the usual elementary chartwise primitive of the area density gives the canonical expression

$$A_{\text{cap}}(\theta) = 2\pi R^2(1 - \cos \theta), \quad 0 \leq \theta \leq \pi.$$

This is the extension of a specific classical analytic formula, obtained by evaluating the primitive $-\cos \theta$ of $\sin \theta$. It is not an assertion that arbitrary subsets have a surreal measure or that ordinary Darboux sums define this value. Our finite-additivity theorem concerns the geodesic polygonal class just constructed.

11.3 Solid angle of three arbitrary rays

Let $\mathbf{A}, \mathbf{B}, \mathbf{C} \in F^3$ be linearly independent vectors from an observation point, with lengths $A, B, C > 0$. Their spherical triangle has solid angle

$$\Omega = 2 \operatorname{atan2}(|\det[\mathbf{A} \ \mathbf{B} \ \mathbf{C}]|, ABC + C(\mathbf{A} \cdot \mathbf{B}) + A(\mathbf{B} \cdot \mathbf{C}) + B(\mathbf{C} \cdot \mathbf{A})). \quad (11.9)$$

Normalize each ray to a unit vector in [equation \(11.3\)](#); multiplying both arguments by the positive number ABC proves the formula. This is the branch-complete form of the classical plane-triangle solid-angle formula associated with van Oosterom and Strackee [7]. The formula is independent of positive rescaling of any individual ray. Replacing the absolute determinant by its signed value gives the oriented angle, with sign set by the ray order.

Solid angle is dimensionless and finite, whereas the corresponding area $R^2\Omega$ can have any surreal scale. A determinant that is infinitesimal relative to ray lengths does not by itself imply small solid angle: when the second argument is negative, the angle can instead be close to 2π .

12 Differential geometry and parallel transport

All derivatives in this section are with respect to explicit chart or curve parameters. Scalar parameters such as R are held fixed. These derivatives are not an intrinsic derivation of the scalar field **No**. The calculations are identities between restricted-analytic germs and rational operations; no existence theorem for arbitrary differential equations is being invoked.

12.1 Connection, geodesics, and curvature

For a sphere centered at zero, its tangent plane at P is $T_P S_R^2 = \{X : P \cdot X = 0\}$. Orthogonally project ambient derivatives to obtain the connection

$$\nabla_X Y = D_X Y + \frac{X \cdot Y}{R^2} P. \quad (12.1)$$

Indeed differentiating $P \cdot Y = 0$ gives $P \cdot D_X Y = -X \cdot Y$, so the displayed correction makes the result tangent. The product rule shows metric compatibility; symmetry of the ambient derivative gives zero torsion.

For a unit tangent vector V at P , a unit-speed great-circle geodesic is

$$\gamma(s) = \cos(s/R)P + R \sin(s/R)V, \quad s/R \in \mathcal{O}. \quad (12.2)$$

Direct differentiation gives $\|\gamma'\| = 1$ and $\gamma'' = -\gamma/R^2$, hence $\nabla_{\gamma'} \gamma' = 0$. All minor geodesic segments have $0 \leq s/R \leq \pi$, so they are included. No oscillatory value at an infinitely large s/R is required.

With curvature convention $\mathcal{R}(X, Y)Z = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X, Y]} Z$, substitution of [equation \(12.1\)](#) and cancellation of commuting ambient derivatives gives

$$\mathcal{R}(X, Y)Z = \frac{1}{R^2}((Y \cdot Z)X - (X \cdot Z)Y). \quad (12.3)$$

Thus the Gaussian curvature is exactly $1/R^2$. In particular a sphere of infinite radius has positive infinitesimal curvature, not zero curvature. An infinitesimal sphere has infinite curvature. Both statements are literal field identities.

The exponential map at P , for $X \in T_P S_R^2$ with $r = \|X\|$ and r/R finite, is

$$\text{Exp}_P(X) = \cos(r/R)P + R \sin(r/R) \frac{X}{r},$$

with value P at $r = 0$. For $0 \leq r < \pi R$, it is bijective onto the sphere minus the antipode. Its inverse for $Q \neq -P$ is

$$\text{Log}_P(Q) = \frac{a}{\sin a}(Q - \cos a P), \quad a = \arccos(P \cdot Q/R^2), \quad (12.4)$$

with the removable value zero at $Q = P$. Taking norms and using the unique minor arc verifies both identities.

12.2 Explicit parallel transport

For unit directions $\mathbf{u} \neq -\mathbf{v}$ and $X \cdot \mathbf{u} = 0$, define

$$\mathcal{P}_{uv}(X) = X - \frac{X \cdot \mathbf{v}}{1 + \mathbf{u} \cdot \mathbf{v}}(\mathbf{u} + \mathbf{v}). \quad (12.5)$$

This is the parallel transport along the minor great-circle arc. To verify it, choose the tangent to the arc and its constant binormal as a basis of the initial tangent plane. The tangent rotates with the great circle and has zero covariant derivative; the binormal is ambient-constant and also has zero covariant derivative. The endpoint values of these two parallel fields are exactly [equation \(12.5\)](#). Alternatively, restricting [equation \(5.5\)](#) to $\mathbf{u}^\perp$ reduces algebraically to the same formula. It follows that transport is an isometry between tangent planes. This specifies the canonical transport by explicit analytic continuation along the parametrized arc. It does not assert uniqueness among arbitrary intrinsically differentiable fields: the locally constant counterexample in [section 15](#) explains that distinction. Within the definable restricted-analytic class, uniqueness follows because the two coefficients in the parallel frame have zero derivative and hence are constant. For a radius- R sphere the unit directions are used but tangent vectors may retain their physical lengths.

12.3 A quaternion proof of spherical holonomy

Use the positive orientation $\tau = d > 0$. The unit quaternion of the minimal rotation from $\mathbf{u}$ to $\mathbf{v}$ is

$$q_{uv} = \frac{(1 + z, \mathbf{u} \times \mathbf{v})}{\sqrt{2(1 + z)}}. \quad (12.6)$$

Define q_{vw} and q_{wu} cyclically. Quaternion multiplication gives the especially informative identity

$$q_{wu}q_{vw}q_{uv} = \frac{(M, d\mathbf{u})}{\sqrt{M^2 + D}}. \quad (12.7)$$

For a direct verification, multiply the three numerators using [equation \(5.6\)](#). The scalar component is $2M$ and the vector component is $2d\mathbf{u}$; the denominator is $\sqrt{8(1+x)(1+y)(1+z)} = 2\sqrt{M^2 + D}$. The simplifications use only unit norms and the vector triple-product identities.

Theorem 12.1 (Holonomy equals spherical excess). *Parallel transport around the positively oriented boundary $\mathbf{u} \rightarrow \mathbf{v} \rightarrow \mathbf{w} \rightarrow \mathbf{u}$ is rotation of the initial tangent plane through angle E modulo 2π . Equivalently,*

$$\text{holonomy phase} = \text{cis } E, \quad E = \frac{\text{Area}(\triangle)}{R^2}.$$

Proof. Each edge transport is the restriction of the corresponding quaternion rotation. By [equations \(11.4\)](#) and [\(12.7\)](#), their product is the quaternion $(\cos(E/2), \sin(E/2)\mathbf{u})$, which acts on $\mathbf{u}^\perp$ by rotation through E . The area equality is [definition 11.2](#). $\square$

This is a finite-triangle version of the curvature–area holonomy relation, proved without an unconstructed global integral. The observed tangent-plane rotation records E modulo 2π , while the area branch has already been fixed by $0 < E < 2\pi$. Reversing the boundary orientation changes the holonomy sign but not the unoriented area.

12.4 Curvature correction in normal coordinates

Let p, q be vectors in the tangent plane at P , with $r = \|p\|$, $s = \|q\|$, and $(r + s)/R$ infinitesimal. The squared intrinsic distance between their exponential images satisfies

$$\begin{aligned} d_R(\text{Exp}_P p, \text{Exp}_P q)^2 &= \|p - q\|^2 - \frac{r^2 s^2 - (p \cdot q)^2}{3R^2} \\ &\quad + O\left(\frac{(r + s)^6}{R^4}\right). \end{aligned} \quad (12.8)$$

The error estimate is absolute and uniform over the relative directions; it does not claim a small *relative* error when $p - q$ is exceptionally small.

To derive it, for $r, s \neq 0$ use the spherical cosine law with $\cos \alpha = (p \cdot q)/(rs)$. Expanding to degree four in $r/R, s/R$ gives

$$\cos(d_R/R) = 1 - \frac{\|p - q\|^2}{2R^2} + \frac{r^4 + s^4 + 6r^2s^2 - 4(p \cdot q)(r^2 + s^2)}{24R^4} + O((r + s)^6/R^6).$$

The function $\arccos(1 - h)^2 = 2h + h^2/3 + O(h^3)$ has a real analytic extension through $h = 0$. Substitution yields [equation \(12.8\)](#). Expressions with $r = 0$ or $s = 0$ are obtained by their removable analytic values. Uniform real-germ remainder estimates transfer after rescaling by R . The coefficient $-1/(3R^2)$ explicitly records positive curvature.

13 Infinitesimal scales and geometric conditioning

The extra information carried by surreal coordinates is not a new algebraic identity for an ordinary triangle. It is the ability to retain exact rank, orientation, angle, and size when several different infinitesimal scales occur in the same configuration.

13.1 A valuation dictionary

Write v for the Hahn valuation, with $v(0) = +\infty$. For a nonzero vector $X \in F^3$,

$$v(\|X\|) = \min_j v(X_j). \quad (13.1)$$

Indeed, in $\sum_j X_j^2$ the summands of smallest valuation have positive leading coefficients, so their leading terms cannot cancel. Taking a positive square root halves the resulting valuation.

For unit vectors u, v , put $a = d_S(u, v)$. The chord bounds and their antipodal versions give

$$v(\|u - v\|) = v(a) \quad (a > 0), \quad (13.2)$$

$$v(\|u + v\|) = v(\pi - a) \quad (a < \pi). \quad (13.3)$$

For $0 < a < \pi$ one also has

$$v(\sin a) = v(a) + v(\pi - a). \quad (13.4)$$

To check the last assertion, when both endpoints are separated from a by positive real amounts all three valuations are zero. When a is infinitesimal, $\sin a/a = 1 + O(a^2)$ and $\pi - a$ is a unit; the case $\pi - a$ infinitesimal is identical after reflection. These exhaust the possibilities by standard parts.

Consequently a proper spherical triangle satisfies the exact scale laws

$$\begin{aligned} v(d) &= v(\sin b) + v(\sin c) + v(\sin \alpha) \\ &= v(\sin c) + v(\sin a) + v(\sin \beta) \\ &= v(\sin a) + v(\sin b) + v(\sin \gamma). \end{aligned} \quad (13.5)$$

A small determinant can therefore come from a small side, a nearly antipodal side, a small interior angle, or an angle close to π . Neither its standard part nor its valuation alone identifies which geometric mechanism is responsible.

13.2 A shape-independent local area estimate

The following estimate does not assume that a shrinking triangle has angles bounded away from zero. This matters over the surreals: the area of a triangle can be smaller than every ordinary power of its diameter.

Theorem 13.1 (Uniform gnomonic area comparison). *Let $p_0, p_1, p_2 \in F^2$ be noncollinear, let $\rho = \max_i \|p_i\|$, and suppose $0 < \rho \leq 1/4$. On the unit sphere take the vertices $u_i = (p_i, 1)/\sqrt{1 + \|p_i\|^2}$. Set*

$$H = \det[p_1 - p_0 \quad p_2 - p_0], \quad A_{\text{flat}} = \frac{|H|}{2}.$$

The spherical excess E of their minor-arc triangle satisfies

$$\left| \frac{E}{A_{\text{flat}}} - 1 \right| \leq 4\rho^2. \quad (13.6)$$

In particular, for infinitesimal ρ ,

$$E = \frac{|H|}{2}(1 + O(\rho^2)), \quad v(E) = v(H).$$

On the sphere of radius $R > 0$ the spherical area is $R^2 E$, so $v(\text{Area}) = 2v(R) + v(H)$, without any restriction on the size of R .

Proof. Put $A_i = \sqrt{1 + \|p_i\|^2}$ and

$$B = A_0 A_1 A_2 + (1 + p_0 \cdot p_1)A_2 + (1 + p_1 \cdot p_2)A_0 + (1 + p_2 \cdot p_0)A_1.$$

The determinant and denominator in the solid-angle formula are $d = |H|/(A_0 A_1 A_2)$ and $M = B/(A_0 A_1 A_2)$, respectively. Thus

$$E = 2 \operatorname{atan}(|H|/B). \quad (13.7)$$

Here $B > 0$; the triangle lies near the north pole, not near the hemispherical branch cut.

The inequalities $1 \leq A_i \leq 1 + \rho^2/2$ and $|p_i \cdot p_j| \leq \rho^2$ imply

$$|B - 4| \leq 8\rho^2, \quad B \geq 7/2, \quad |H| \leq 4\rho^2.$$

For example, the deviation of the product $A_0 A_1 A_2$ from 1 is at most $2\rho^2$, and each of the other three terms deviates from 1 by at most $2\rho^2$. For the lower bound one may use the sharper $B \geq 4 - 3\rho^2$.

Set $\zeta = |H|/B$, so $0 < \zeta \leq 2\rho^2$. The ordinary analytic inequality $1 - \zeta^2/3 \leq \operatorname{atan} \zeta / \zeta \leq 1$ holds here by restricted-analytic transfer. Therefore

$$\begin{aligned} \left| \frac{E}{A_{\text{flat}}} - 1 \right| &= \left| \frac{4}{B} \frac{\operatorname{atan} \zeta}{\zeta} - 1 \right| \\ &\leq \frac{16}{7}\rho^2 + \frac{32}{21}\rho^4 \leq 4\rho^2. \end{aligned}$$

Crucially, H was divided out only after the exact factorization of E ; no lower bound on $|H|/\rho^2$ was used. The valuation claims follow because the final multiplicative factor has standard part 1. $\square$

Remark 13.2. For $p_0 = 0$, $p_1 = (\varepsilon, 0)$, and $p_2 = (\varepsilon, \omega^{-\omega})$ with $\varepsilon = \omega^{-1}$, the ratio $|H|/\rho^2$ is smaller than every ordinary positive power of ε . The theorem still gives a relative, rather than merely absolute, estimate for the spherical area. A generic error bound $E = A_{\text{flat}} + O(\rho^4)$ would not give that information.

13.3 Stable vector data and unstable derived data

Proposition 13.3 (Three conditioning estimates). *For unit vectors u, v, u', v' ,*

$$|d_S(u, v) - d_S(u', v')| \leq \frac{\pi}{2} (\|u - u'\| + \|v - v'\|). \quad (13.8)$$

For nonzero X, Y ,

$$\left\| \frac{X}{\|X\|} - \frac{Y}{\|Y\|} \right\| \leq \frac{2\|X - Y\|}{\|X\|}. \quad (13.9)$$

Finally, let $a = d_S(u, v) \in (0, \pi)$ and $\eta = \|u - u'\| + \|v - v'\| < \sin a$. Then $u' \times v' \neq 0$ and

$$\left\| \frac{u \times v}{\|u \times v\|} - \frac{u' \times v'}{\|u' \times v'\|} \right\| \leq \frac{2\eta}{\sin a}. \quad (13.10)$$

Proof. The angular triangle inequality bounds the left side of equation (13.8) by $d_S(u, u') + d_S(v, v')$. Apply the lower chord bound to each summand. For equation (13.9), write the difference as

$$\frac{X - Y}{\|X\|} + Y \left(\frac{1}{\|X\|} - \frac{1}{\|Y\|} \right)$$

and use $|\|X\| - \|Y\|| \leq \|X - Y\|$. For the last estimate,

$$\|u \times v - u' \times v'\| \leq \|(u - u') \times v\| + \|u' \times (v - v')\| \leq \eta.$$

The strict inequality $\eta < \|u \times v\| = \sin a$ prevents the new cross product from vanishing; normalization gives the claimed bound. $\square$

Thus angular distance is uniformly stable under consistent perturbations of the unit vectors. The normal to their spanning plane is not uniformly stable near coincident or antipodal directions: the amplification factor $1/\sin a$ is then infinite. There is no contradiction between these statements; they concern different outputs.

If one perturbs only the scalar cosine as though it were independent of all other data, inverse cosine is also ill-conditioned near an endpoint. Away from the endpoints its derivative is

$$\frac{d}{dc} \arccos c = -\frac{1}{\sqrt{1 - c^2}},$$

whereas for $h > 0$ infinitesimal,

$$\arccos(1 - h) = \sqrt{2h} \left(1 + \frac{h}{12} + O(h^2) \right). \quad (13.11)$$

The square-root law follows by inverting $1 - \cos \theta = \theta^2/2 - \theta^4/24 + \dots$ after extracting $\sqrt{2h}$. It is a genuinely different scale from an ordinary linear perturbation estimate.

14 Worked configurations at several surreal scales

14.1 Infinite lengths with infinitesimal directions

The vectors $X = (\omega, 1, 0)$ and $Y = (\omega, 0, 0)$ have infinite lengths and the finite infinitesimal angle

$$\text{ang}(X, Y) = \text{atan}(\omega^{-1}) = \omega^{-1} - \frac{1}{3}\omega^{-3} + \frac{1}{5}\omega^{-5} - \dots$$

This displayed series is strongly summable because its parameter is infinitesimal. No evaluation of $\sin \omega$ occurs.

For the flat triangle

$$P = 0, \quad Q = (\omega, 0, 0), \quad S = (\omega, \omega^{-\omega}, 0),$$

the area is exactly $\frac{1}{2}\omega^{1-\omega}$, while the angle at P is $\text{atan}(\omega^{-\omega-1})$. The base is infinite, the altitude is smaller than every ordinary negative power of ω , and the area is nevertheless a well-defined positive surreal number. A real standard-part picture alone does not encode this configuration.

14.2 Nearly parallel lines meet their closest points infinitely far away

Let $\varepsilon > 0$ be infinitesimal and let $h = \varepsilon^2$. Consider

$$L_1 = \{(t, 0, 0) : t \in F\}, \quad L_2 = \{(s, 1 + \varepsilon s, h) : s \in F\}.$$

Their squared distance between parameter points is

$$(t - s)^2 + (1 + \varepsilon s)^2 + h^2.$$

Its minimum is attained at $t = s = -1/\varepsilon$ and is h^2 . Hence the lines have distance h , although the standard parts of their *finite-coordinate* points lie on the real lines $y = 0, z = 0$ and $y = 1, z = 0$, whose distance is 1.

The discrepancy is not a failure of the distance formula. The minimizing points have infinite first coordinates and disappear from a finite-coordinate standard-part picture. Taking standard parts and minimizing over an entire affine line are different operations. With $h = 0$, the same lines intersect at a genuine affine point infinitely far from the origin.

14.3 Tangency splits at a square-root scale

Intersect the unit sphere with the line $(t, 1 - \varepsilon, 0)$, where $0 < \varepsilon \in \mathfrak{m}$. The exact solutions are

$$t = \pm \sqrt{2\varepsilon - \varepsilon^2} = \pm \sqrt{2\varepsilon} \left(1 - \frac{\varepsilon}{4} + O(\varepsilon^2)\right).$$

A displacement of valuation $v(\varepsilon)$ splits a double tangency into points separated at valuation $v(\varepsilon)/2$. Replacing $1 - \varepsilon$ by $1 + \varepsilon$ produces no intersection, while $\varepsilon = 0$ gives a single point. All three cases have the same ordinary standard-part tangency.

14.4 An almost planar tetrahedron with an infinite circumradius

Take vertices $0, e_1, e_2, (2, 0, \varepsilon)$ with $0 < \varepsilon \in \mathfrak{m}$. The tetrahedron has volume $\varepsilon/6$. Its circumcenter is

$$C = \left(\frac{1}{2}, \frac{1}{2}, \frac{1}{\varepsilon} + \frac{\varepsilon}{2}\right),$$

since $2C \cdot e_i = 1$ for $i = 1, 2$ and $2C \cdot (2, 0, \varepsilon) = 4 + \varepsilon^2$. The circumradius is

$$\|C\| = \sqrt{\frac{1}{2} + \left(\frac{1}{\varepsilon} + \frac{\varepsilon}{2}\right)^2},$$

which is infinite. Nonzero volume, however small, is enough for uniqueness of the center; it is not enough for a finite conditioning constant.

14.5 A right spherical triangle and the flat limit of an infinite sphere

For positive infinitesimals ε, η , set

$$u = e_3, \quad v = \frac{(\varepsilon, 0, 1)}{A}, \quad w = \frac{(0, \eta, 1)}{B}, \quad A = \sqrt{1 + \varepsilon^2}, \quad B = \sqrt{1 + \eta^2}.$$

The angle at u is exactly $\pi/2$, the adjacent side angles are $c = \text{atan } \varepsilon$ and $b = \text{atan } \eta$, and

$$x = \frac{1}{AB}, \quad y = \frac{1}{B}, \quad z = \frac{1}{A}, \quad d = \frac{\varepsilon\eta}{AB}.$$

The exact excess is therefore

$$E = 2 \text{atan} \frac{\varepsilon\eta}{(A+1)(B+1)} = \frac{\varepsilon\eta}{2} - \frac{\varepsilon^3\eta + \varepsilon\eta^3}{8} + O(\varepsilon\eta(\varepsilon^2 + \eta^2)^2). \quad (14.1)$$

This is a relative expansion with an explicit common area factor. In the symmetric case $\varepsilon = \eta = h$ it becomes

$$E = \frac{h^2}{2} - \frac{h^4}{4} + \frac{7h^6}{48} + O(h^8). \quad (14.2)$$

Now choose $R = \omega$ and $h = \omega^{-1}$, and scale all vertices by R . The physical area is

$$R^2 E = \frac{1}{2} - \frac{1}{4\omega^2} + \frac{7}{48\omega^4} + O(\omega^{-6}),$$

and the physical adjacent side lengths are $\omega \text{atan}(\omega^{-1}) = 1 - 1/(3\omega^2) + \dots$. The sphere has positive infinitesimal curvature ω^{-2} . A bounded patch is asymptotically planar, but its curvature and area corrections remain exact surreal information rather than being discarded.

14.6 A tiny determinant with almost hemispherical area

Let $0 < \varepsilon \in \mathfrak{m}$ and consider

$$u = (1, 0, 0), \quad v = (-1/2, \sqrt{3}/2, 0), \quad w = \frac{(-1/2, -\sqrt{3}/2, \varepsilon)}{\sqrt{1 + \varepsilon^2}}.$$

Writing $A = \sqrt{1 + \varepsilon^2}$, one obtains

$$d = \frac{\sqrt{3}\varepsilon}{2A}, \quad M = \frac{1}{2} - \frac{1}{A} < 0.$$

Thus $D = d^2$ is infinitesimal, but the correct branch gives

$$E = 2\pi - 2 \text{atan} \frac{d}{-M} = 2\pi - 2\sqrt{3}\varepsilon + O(\varepsilon^3). \quad (14.3)$$

The triangle nearly fills a hemisphere rather than collapsing to zero area. Its polar triangle is small. Replacing $\text{atan2}(d, M)$ by the principal $\text{atan}(d/M)$ would give the wrong branch and even the wrong sign for the area. This example and [equation \(14.1\)](#) show why $d \rightarrow 0$ is not a complete description of spherical degeneration.

15 What the development does not silently assume

15.1 The intrinsic order topology is not ordinary real topology

Suppose F is non-Archimedean. Its infinitesimal ideal $\mathfrak{m}$ is both open and closed in the order topology: each infinitesimal has a sufficiently small infinitesimal neighborhood inside $\mathfrak{m}$, while for $x \notin \mathfrak{m}$ a neighborhood of radius $|x|/2$ stays outside $\mathfrak{m}$. Consequently the function

$$f(x) = \begin{cases} 0, & x \in \mathfrak{m}, \\ 1, & x \notin \mathfrak{m} \end{cases}$$

is locally constant and has derivative zero everywhere in the intrinsic ordered-field sense (derivatives with radii from F), but $f(0) \neq f(1)$. Local differentiability alone therefore does not give the ordinary global mean value theorem or uniqueness of an antiderivative.

There is no conflict with the use of restricted-analytic transfer. The externally specified set of all real infinitesimals is not a new definable predicate in the restricted-analytic language used here. Our formulas come from compatible real analytic germs or explicit algebraic definitions, not from arbitrary functions that happen to be locally constant on intrinsic neighborhoods. The same function is locally constant for the full fine topology of **No** as well, since $\mathfrak{m}$ is also open there; [section 26](#) compares the two topologies.

15.2 Ordinary finite partitions do not automatically define an integral

Even the function $f(x) = x$ on $[0, 1] \subset F$ exposes a difficulty. For any partition with an ordinary finite number N of subintervals, put $\Delta_i = x_i - x_{i-1} > 0$. Its upper and lower Darboux sums satisfy

$$U - L = \sum_{i=1}^N \Delta_i^2 \geq \frac{1}{N},$$

by Cauchy–Schwarz and $\sum_i \Delta_i = 1$. Given any positive infinitesimal ε , this difference is always greater than ε , whatever ordinary finite N is chosen. Hence the usual Darboux criterion requiring gaps smaller than *every positive element of F* fails for these partitions. A mesh criterion without a requirement that the relevant partitions exist would instead become vacuous below infinitesimal mesh sizes.

Nevertheless the polynomial antiderivative $x^2/2$ is perfectly well-defined, as are our polygon-area formula and the selected analytic cap primitive. A theory of generalized integration must say which functional, which functions, and which summation or partition notion it uses. This report does not infer a countably additive area measure from finite-polygon additivity. It also does not claim completeness, ordinary topological compactness, arbitrary smooth-ODE existence, or a supremum-based length for every curve in surreal space.

15.3 Finite dimension does not remove size or support conditions

The ambient coordinate count is three, but the class **No**³ remains a proper class. The set-field construction in [section 2](#) allows every finite configuration in the report to be treated in a single honest field and every needed analytic evaluation to be interpreted there. Passing to a larger workspace preserves the answers. Neither this compatibility nor the finite dimension permits unrestricted summation of arbitrary families of surreals.

Likewise, ordinary-size angles do not imply ordinary-size arc lengths. For an infinite R and a positive real a , the minor arc has infinite length Ra . Conversely, a bounded physical arc on an infinite sphere has infinitesimal angular length and is governed by the local expansions above. Choosing a global oscillatory phase at infinite scalar arguments would add another analytic convention; it would not change these coordinate-defined directions or finite geometric angles.

Part II

The rotation group $\mathrm{SO}(3, \mathrm{No})$

Part II is source 09, apart from the material printed once in [sections 2, 3 and 5](#). Its axis theorem, Rodrigues formula, quaternion action, spin map and Cayley chart are [theorem 5.3](#), [equation \(5.7\)](#), [theorem 5.5](#) and [theorem 5.6](#). Vectors are written in plain type, and $G(F) = \mathrm{SO}(3, F)$ for a real closed field F .

Source 09 organizes its material specifically around the three-dimensional rotation group. Centralizers at half-turns, a globally valid matrix Cayley chart with its exceptional locus, infinitesimal conjugacy of finite subgroups, and the differential of the *rotation* exponential are developed together. The ordinary quaternion algebra and spin background is standard [8]. The surreal field and restricted-analytic background is due to Conway, Gonshor, van den Dries–Ehrlich and Ehrlich–Kaplan; the analytic constructions here rely specifically on [1, 3]. The perfectness of the infinitesimal $\mathrm{SO}(3)$ group has a published precedent: Bays–Peterzil explicitly record that every element is a commutator [9, §3.1]. [Theorem 19.3](#) gives an elementary, exact quaternion construction rather than presenting that fact as new.

16 Frames, Euler coordinates, and quaternion matrices

This section collects source 09’s further finite algebra of $G(F)$.

16.1 Products of reflections and transitivity

Reflection in a plane perpendicular to a unit vector u is $S_u = I - 2uu^T$, of determinant -1 . In an oriented perpendicular plane, two such reflections produce the usual doubled planar angle; the assertion is a finite matrix multiplication. Every rotation is a product of two plane reflections: take the two planes through its axis, choosing their normal lines by a half-angle square root on the algebraic circle. The half-turn and identity are included. Alternatively this follows from the explicit quaternion lifts of [section 5.2](#).

Given two unit vectors, complete each to an oriented orthonormal frame. The matrix sending the first frame to the second belongs to $G(F)$. Thus the orbits of $G(F)$ on F^3 are the origin and the spheres of each positive F -radius. The stabilizer of a unit vector n is

$$T_n = \{\mathrm{Rot}(n; c, s) : c^2 + s^2 = 1\} \cong \mathrm{SO}(2, F), \quad G(F)/T_n \cong S^2(F). \quad (16.1)$$

This homogeneous-space statement is algebraic and semialgebraic; it does not assert that these are ordinary real manifolds.

16.2 Euler coordinates and their degeneracy

Write $\text{Rot}_z(c, s) = \text{Rot}(e_3; c, s)$ and $\text{Rot}_y(c, s) = \text{Rot}(e_2; c, s)$. For every $A \in G(F)$ there is a decomposition

$$A = \text{Rot}_z(c_\alpha, s_\alpha) \text{Rot}_y(c_\beta, s_\beta) \text{Rot}_z(c_\gamma, s_\gamma), \quad s_\beta \geq 0. \quad (16.2)$$

To construct it, put $n = Ae_3$, $c_\beta = n_3$, and $s_\beta = \sqrt{n_1^2 + n_2^2}$. When $s_\beta \neq 0$, choose $c_\alpha = n_1/s_\beta$, $s_\alpha = n_2/s_\beta$. Then the first two factors send e_3 to n , and their inverse times A fixes e_3 , giving the last factor. If $s_\beta = 0$, choose any suitable first factor and solve in the remaining stabilizer. The nonuniqueness there is the algebraic form of gimbal lock; infinitesimal s_β also makes these coordinates poorly conditioned. For actual surreals all three circle pairs admit finite angles, as shown in [theorem 5.4](#).

16.3 Quaternion coordinates

For $q = a + bi + cj + dk$ of norm one the rotation matrix is

$$\begin{pmatrix} 1 - 2(c^2 + d^2) & 2(bc - ad) & 2(bd + ac) \\ 2(bc + ad) & 1 - 2(b^2 + d^2) & 2(cd - ab) \\ 2(bd - ac) & 2(cd + ab) & 1 - 2(b^2 + c^2) \end{pmatrix}. \quad (16.3)$$

Composition of rotations is quaternion multiplication in the same order as matrix multiplication, with column-vector convention.

16.4 Complex matrices, projective coordinates, and a caution

The faithful algebra representation

$$a + bi + cj + dk \longmapsto \begin{pmatrix} a + bi & c + di \\ -c + di & a - bi \end{pmatrix} \quad \text{in } M_2(F(i)) \quad (16.4)$$

has determinant $N(q)$ and sends quaternion conjugation to conjugate transpose. Norm-one quaternions are exactly the corresponding $\text{SU}(2, F(i))$ matrices, where conjugation fixes F and sends i to $-i$.

Every nonzero quaternion can be normalized by its positive norm. Therefore

$$G(F) \cong \mathbb{H}_F^\times / F^\times \cong \mathbb{P}^3(F) \quad (16.5)$$

as a group with the quaternion-induced law, and as a semialgebraic space. The second statement concerns F -points: the norm has no zero on $F^4 \setminus \{0\}$. It is *not* an isomorphism of algebraic varieties $G \cong \mathbb{P}^3$. The algebraic projective model is the open subset $N \neq 0$; the norm quadric has points after algebraic closure. This distinction prevents an incorrect inference that the affine algebraic group is projective.

The quaternion quotient also explains why three dimensions are special. It is not necessary to choose an axis continuously to compose rotations. A unit quaternion carries a uniformly bounded four-coordinate representation, with only the unavoidable sign identification.

16.5 A quaternion extraction algorithm without a small divisor

Cayley coordinates become large near a half-turn. A uniformly bounded alternative works directly from A . Set

$$\begin{aligned} d_0 &= 1 + \operatorname{tr} A, & d_1 &= 1 + 2A_{11} - \operatorname{tr} A, \\ d_2 &= 1 + 2A_{22} - \operatorname{tr} A, & d_3 &= 1 + 2A_{33} - \operatorname{tr} A. \end{aligned}$$

By equation (16.3), these are $4a^2, 4b^2, 4c^2, 4d^2$, respectively, for either unit lift. They are nonnegative and sum to four. Choose a largest one, so $d_j \geq 1$, and choose the sign of the lift so that its j -th coordinate is $\sqrt{d_j}/2 \geq 1/2$. The other coordinates are recovered from

$$\begin{aligned} 4ab &= A_{32} - A_{23}, & 4ac &= A_{13} - A_{31}, & 4ad &= A_{21} - A_{12}, \\ 4bc &= A_{12} + A_{21}, & 4bd &= A_{13} + A_{31}, & 4cd &= A_{23} + A_{32}. \end{aligned}$$

All divisions are by a quantity of magnitude at least two. This is an exact ordered-field algorithm with four branches and positive square roots. Ties can be resolved by the first index. It does not depend on a floating-point tolerance, and it does not become singular when the rotation is infinitesimally close to a half-turn.

An axis itself cannot be extracted with comparable conditioning at the identity: there is no preferred axis there. The trace loses first-order information near identity, since $1 - c$ is quadratic in a small angle; the skew coordinates retain the linear part.

17 Conjugacy, centralizers, and maximal axis groups

Theorem 17.1 (Conjugacy and the half-turn exception). *For $A, B \in G(F)$:*

- (i) *They are conjugate in $G(F)$ if and only if $\operatorname{tr} A = \operatorname{tr} B$.*
- (ii) *If $A \neq I$ is not a half-turn and has axis Fn , then $C_{G(F)}(A) = T_n$.*
- (iii) *For a half-turn H_n ,*

$$C_{G(F)}(H_n) = \{B \in G(F) : Bn = \pm n\} \cong O(2, F) \cong T_n \rtimes C_2.$$

- (iv) *The normalizer of T_n is the same line stabilizer. Its quotient by T_n is C_2 , acting by inversion on T_n .*

Proof. Conjugation preserves trace and sends $\operatorname{Rot}(n; c, s)$ to $\operatorname{Rot}(Bn; c, s)$. Equal traces give equal c . The possible s 's differ only in sign; changing n to $-n$ absorbs that sign. Transitivity on unit vectors proves (i), including the half-turn case.

An element commuting with a nonidentity rotation preserves its one-dimensional fixed space, so sends n to n or $-n$. In the first case it is in T_n , which is abelian. In the second case it conjugates the planar rotation to its inverse. A nonidentity circle element equals its inverse exactly at the half-turn. This proves (ii) and (iii). In a frame ending in n , the line stabilizer has matrices $\operatorname{diag}(B, \det B)$ with $B \in O(2, F)$. A half-turn about a perpendicular axis realizes its nontrivial component.

Finally, the common fixed space of all of T_n is Fn , so a normalizer must preserve that line. Every line-preserving rotation either fixes or reverses its orientation, and accordingly centralizes or inverts T_n . This proves (iv). $\square$

In particular, $G(F)$ has trivial center. The centralizers of non-half-turns also show that every nontrivial connected algebraic axis torus is conjugate to T_{e_3} . “Connected” in that sentence refers to the algebraic torus, not to its fine topological point set.

Corollary 17.2 (Commuting rotations). *Two nonidentity rotations commute only in the following cases: they have the same axis, or they are half-turns about perpendicular axes. Both possibilities do occur.*

Proof. If either is not a half-turn, use [theorem 17.1\(ii\)](#). For two half-turns, H_n preserves the line Fm precisely when m is parallel to n or perpendicular to it: its two eigenspaces are Fn and $n^\perp$. $\square$

Thus the three coordinate half-turns and the identity form a Klein four group. The slogan that commuting spatial rotations always share an axis is false, over both the reals and the surreals.

18 The standard-part extension

From now on suppose $F \supseteq \mathbb{R}$. Every matrix entry of a rotation belongs to $\mathcal{O}_F$, so standard part can be applied entrywise to every rotation, not only to a restricted subgroup.

Theorem 18.1 (Real rotation plus an infinitesimal rotation). *There is a split exact sequence*

$$1 \longrightarrow K_F \longrightarrow G(F) \xrightarrow{\text{st}} \text{SO}(3, \mathbb{R}) \longrightarrow 1, \quad K_F = G(F) \cap (I + M_3(\mathfrak{m}_F)). \quad (18.1)$$

Every $A \in G(F)$ has the unique decomposition

$$A = UA_0, \quad A_0 = \text{st}(A), \quad U = AA_0^{-1} \in K_F. \quad (18.2)$$

Consequently $G(F) \cong K_F \rtimes \text{SO}(3, \mathbb{R})$, with the usual conjugation action.

Proof. The equations $A^T A = I$, $\det A = 1$, and matrix multiplication use finitely many sums and products. Applying the ring homomorphism st preserves them. The inclusion of constant real matrices is a section. The kernel is exactly as displayed, and multiplying by $\text{st}(A)^{-1}$ gives the unique decomposition. The transported product on pairs, for $U, V \in K_F$ and $B, B' \in \text{SO}(3, \mathbb{R})$, is

$$(U, B)(V, B') = (U(BVB^{-1}), BB').$$

$\square$

Proposition 18.2 (Exact coordinates on the kernel). *The Cayley map restricts to a bijection $\mathfrak{m}_F^3 \rightarrow K_F$, with multiplication $t \star u = (t + u + t \times u)/(1 - t \cdot u)$. The unit quaternions infinitesimally close to 1 map isomorphically onto K_F .*

Proof. For $t \in \mathfrak{m}_F^3$, [equation \(5.10\)](#) gives $\text{Cay}(t) - I \in M_3(\mathfrak{m}_F)$. Conversely, for $A \in K_F$, $1 + \text{tr } A$ has standard part four, so [equation \(5.12\)](#) is defined and infinitesimal. The denominator in the product law has standard part one. The distinguished spin lift is $(1, t)/\sqrt{1 + \|t\|^2}$, which has standard part 1. Among the two lifts only this one has that standard part, proving the isomorphism. $\square$

The quaternion side of [theorem 18.1](#) and [proposition 18.2](#) is the split sequence $1 \rightarrow U_1 \rightarrow S^3(\mathbf{No}) \rightarrow S^3(\mathbb{R}) \rightarrow 1$ of the surquaternion report (`squat:sub:unitreduction`), whose kernel U_1 is the group of unit quaternions infinitesimally close to 1 (`squat:thm:rotation-log`). The spin map carries it isomorphically onto K .

When $F \neq \mathbb{R}$, the kernel is nontrivial: any nonzero $\varepsilon \in \mathfrak{m}_F$ gives $\text{Cay}(\varepsilon e_1) \neq I$. It is proper because a nonidentity constant real rotation has nontrivial standard part. Thus $\text{SO}(3, \mathbf{No})$ is *not abstractly simple*. Its simple algebraic group and simple Lie algebra are compatible with this fact.

The split extension is not a direct product. For a real rotation B ,

$$B \text{Cay}(t) B^{-1} = \text{Cay}(Bt), \quad (18.3)$$

and this action is visibly nontrivial. Two rotations with the same standard part can therefore interact at finer scales rather than differing by a central error.

The quotient has a useful metric interpretation. For $x \in \mathcal{O}_F^3$, $\text{st}(Ax) = \text{st}(A) \text{st}(x)$. Infinitesimal rotations are invisible on the real shadow of every finite vector. This conclusion need not hold for infinite vectors; [section 27](#) quantifies the failure.

19 A perfect, uniquely divisible infinitesimal group

The word infinitesimal does not mean abelian. Nor does a Baker–Campbell–Hausdorff formula imply that the entire kernel is nilpotent. The following results make both points precise. Throughout this section, $F \supsetneq \mathbb{R}$ is real closed; the same finite proofs apply to the full surreal class.

Lemma 19.1 (No infinitesimal torsion). *If $U \in K_F$ and $U^n = I$ for an ordinary positive integer n , then $U = I$.*

Proof. Write $U = \text{Rot}(n_0; c, s)$ and regard $z = c + is$ as an element of $F(i)$. The trace gives $c \in 1 + \mathfrak{m}_F$. Since $s^2 = 1 - c^2$, also $s \in \mathfrak{m}_F$. The equation $U^n = I$ gives $z^n = 1$. The polynomial $X^n - 1$ already splits into distinct linear factors over the ordinary complex subfield. Hence z is an ordinary complex root of unity. Its standard part is one, so $z = 1$, and $U = I$. $\square$

Theorem 19.2 (Unique divisibility). *For every ordinary $n \geq 1$, the power map $U \mapsto U^n$ is a bijection of K_F . No nontrivial finite group is a homomorphic image of K_F .*

Proof. For $U = \text{Rot}(n_0; c, s)$, choose an n -th root w of $z = c + is$ in the algebraically closed field $F(i)$. Its squared norm satisfies $N(w)^n = N(z) = 1$, and $N(w) > 0$, so $N(w) = 1$. Its coordinates are bounded and $\text{st}(w)$ is an ordinary complex n -th root of unity. Dividing w by this root gives a circle element w_0 infinitesimally close to one with $w_0^n = z$. Rotation around n_0 by this algebraic angle gives a root in K_F .

For uniqueness, the identity case follows from [lemma 19.1](#). Otherwise, if $V^n = U$, then V commutes with U . A nonidentity element of K_F is not a half-turn, so $V \in T_{n_0}$ by [theorem 17.1](#). Two roots in that abelian group have quotient of finite order in K_F , and are equal. Finally, an image of a divisible group is divisible. In a finite image of order m , every m -th power is the identity, so surjectivity of that power map forces the image to be trivial. $\square$

Unique divisibility is a statement about solving one power equation at a time. It does not say that K_F is a vector space over $\mathbb{Q}$, because its group operation is not commutative.

Theorem 19.3 (Every infinitesimal rotation is one commutator). *For every $U \in K_F$ there are $V, W \in K_F$ with*

$$U = (V, W) := VWV^{-1}W^{-1}.$$

Thus $K_F = [K_F, K_F]$. In particular, it is nonabelian and neither solvable nor nilpotent, and it has no nontrivial solvable quotient.

Proof. The identity is immediate. Take the unique lift of $U \neq I$ near one, $q = (c, w) \in \text{Sp}(1, F)$. Then $0 < c < 1$, $c - 1 \in \mathfrak{m}_F$, and $w \neq 0$. Put

$$s = \left(\frac{1-c}{2} \right)^{1/4}, \quad a = \sqrt{1-s^2}, \quad p = (a, se_1), \quad r = (a, se_2).$$

These are unit quaternions near one. Direct quaternion multiplication gives

$$\text{Re}(p, r) = 1 - 2s^4 = c. \tag{19.1}$$

For completeness, a version before the equal-parameter specialization is

$$\text{Re}((a, se_1)(b, te_2)(a, -se_1)(b, -te_2)) = (a^2 + s^2)(b^2 + t^2) - 2s^2t^2.$$

Since (p, r) and q have norm one and the same scalar part, their imaginary vectors have the same nonzero norm. A unit quaternion h conjugates one vector to the other, by the transitivity and spin surjectivity already proved. Therefore

$$q = (hph^{-1}, hrh^{-1}).$$

Conjugation by a unit quaternion preserves infinitesimal imaginary coordinates: its matrix entries are bounded. Thus both factors are still near one. Applying p proves the claim in K_F . A nontrivial perfect group is neither solvable nor nilpotent; every homomorphic image is perfect, and a solvable perfect group is trivial. $\square$

The conclusion has a literature precedent [9, proof of Claim 3.3]; the proof above exposes the exact fourth-root scale used in a commutator construction. The two factors generally live at a *larger infinitesimal magnitude* than their commutator. There is no contradiction with the abelian leading layers established in section 23.

Source 10 (Part III) replaces the balanced fourth-root parameters by two prescribed depths: every $U \neq I$ of depth δ is a commutator of factors of depths α and $\delta - \alpha$ for any $0 < \alpha < \delta$ in the value group (theorem 29.3), using the unequal-parameter identity displayed above.

The whole group $G(F)$ likewise has commutator width one. Choose an algebraic square root B of a rotation A around the same axis, and let H be a half-turn about a perpendicular axis. Then $HBH^{-1} = B^{-1}$, so $(B, H) = B^2 = A$. This latter proof alone does not prove the infinitesimal claim, since the perpendicular half-turn is not infinitesimal.

20 Finite subgroups have no new surreal forms

Lemma 20.1 (Positive matrix square roots over a real closed field). *Every symmetric positive definite matrix over F has a unique symmetric positive definite square root. If $P - I \in M_d(\mathfrak{m}_F)$, its positive square root and inverse square root are also infinitesimally close to I .*

Proof. The symmetric spectral theorem holds over any real closed field, by the argument of [section 4.3](#): an eigenvector $z \neq 0$ over $F(i)$ has eigenvalue $(z^*Pz)/(z^*z) \in F$, and induction on perpendicular complements gives an orthonormal eigenbasis. (Source 09 gave the same argument in full.) For positive definite P , its eigenvalues λ_j are positive, so take $\sqrt{\lambda_j}$ in that basis. Any symmetric positive square root commutes with P , preserves its eigenspaces, and has only the eigenvalue $\sqrt{\lambda_j}$ on each such space; this proves uniqueness.

If every entry of $Y = P - I$ is infinitesimal, a unit eigenvector gives $|\lambda_j - 1| = |x^T Y x| \leq d \max |Y_{ij}|$, by $\sum_i |x_i| \leq \sqrt{d}$. Thus $\lambda_j - 1$, $\sqrt{\lambda_j} - 1$, and $\lambda_j^{-1/2} - 1$ are infinitesimal. Conjugation by the bounded orthonormal eigenbasis preserves this property. $\square$

Theorem 20.2 (Infinitesimal conjugacy of finite subgroups). *Let $H \leq G(F)$ be finite. Standard part is injective on H , and there is $C \in K_F$ such that*

$$C^{-1}hC = \text{st}(h) \quad \text{for every } h \in H. \quad (20.1)$$

In particular, every finite surreal rotation group is conjugate to an ordinary real rotation group by a rotation with standard part I .

Proof. The kernel of $\text{st}|_H$ is finite and lies in the torsion-free group K_F , so it is trivial. Write $\rho(h) = \text{st}(h)$ and average the two representations:

$$B = \frac{1}{|H|} \sum_{h \in H} h\rho(h)^{-1}. \quad (20.2)$$

Each summand is infinitesimally close to I ; hence so is B , and its determinant has standard part one. In particular it is invertible and has positive determinant. Reindexing the finite sum shows $gB = B\rho(g)$ for every $g \in H$.

Set $P = B^T B$. This is symmetric positive definite and infinitesimally close to I . Orthogonality of both representations gives $\rho(g)^T P \rho(g) = P$, so P commutes with $\rho(g)$. By uniqueness of the positive square root, $P^{-1/2}$ commutes with all those matrices as well. Put

$$C = BP^{-1/2}.$$

Then $C^T C = I$, and both determinants in the product are positive, so $\det C = 1$. The preceding lemma gives $C \in K_F$. Finally $gC = C\rho(g)$, which is exactly [equation \(20.1\)](#). $\square$

This proof is finite averaging followed by algebraic polar decomposition. It uses neither an integral over the full group nor a completeness theorem. The same argument works in $O(d, F)$ for any fixed finite dimension, with the appropriate determinant statement.

20.1 Classification and what it does not claim

The familiar finite-subgroup classification of $SO(3, \mathbb{R})$ therefore applies: cyclic groups C_n , rotational dihedral groups D_n of order $2n$, and the tetrahedral, octahedral, and icosahedral groups A_4, S_4, A_5 , together with the trivial group. The word dihedral here denotes a group of *three-dimensional rotations*: the elements outside its principal cyclic subgroup are perpendicular half-turns, not spatial reflections.

One can see the numerical restriction by counting poles on the ordinary unit sphere. If $H \neq 1$, each nonidentity element fixes exactly two points. At a pole the stabilizer is cyclic. If the pole orbits have stabilizer orders $m_1, \dots, m_k \geq 2$, counting incident pairs gives

$$\sum_{j=1}^k (1 - 1/m_j) = 2 - 2/|H|. \quad (20.3)$$

It follows that $k = 2$ or 3 . For $k = 2$, both stabilizers have order $|H|$, giving a cyclic group. For $k = 3$, order the m_j 's; the only possibilities are

$$(2, 2, n), \quad (2, 3, 3), \quad (2, 3, 4), \quad (2, 3, 5),$$

with group orders $2n, 12, 24, 60$. The usual spherical realizations identify them with the dihedral and three polyhedral groups. This last identification is the classical finite rotation-group classification; the new field introduces no additional possibility because of [theorem 20.2](#).

Individual finite-order surreal rotations need not have real axes. A conjugate CHC^{-1} of a real finite group by a nonreal $C \in K_F$ generally has nonreal entries. The theorem says that these are conjugate realizations, not that every finite subgroup is literally contained in the constant matrices. It also says nothing about arbitrary infinite or set-generated subgroups.

21 The Lie algebra and angular velocity

The algebraic tangent space at the identity is computed with the formal dual number κ , satisfying $\kappa^2 = 0$. It is not a nonzero surreal infinitesimal: a field has no such nilpotents. Substituting $I + \kappa X$ in the orthogonality equation gives

$$(I + \kappa X)^T (I + \kappa X) = I + \kappa(X^T + X).$$

Thus

$$\mathfrak{so}(3, F) = \{X \in M_3(F) : X^T = -X\} = \{\widehat{u} : u \in F^3\}. \quad (21.1)$$

Its bracket and adjoint action are exactly

$$[\widehat{u}, \widehat{v}] = \widehat{u \times v}, \quad \text{Ad}_A(\widehat{u}) = \widehat{Au}. \quad (21.2)$$

This identifies the Lie algebra with the cross-product algebra of three-space.

Proposition 21.1 (Simplicity and the spin factor). *The Lie algebra $\mathfrak{so}(3, F)$ is simple. The differential of the spin map is $u \mapsto 2\widehat{u}$ on imaginary quaternions.*

Proof. Let a nonzero ideal contain $u \neq 0$ in cross-product coordinates. Then it contains $u \times v$ for every v . The image of this linear map is exactly $u^\perp$, by [equation \(3.5\)](#). Together with u , these vectors span F^3 , so the ideal is the entire algebra. The algebra is nonabelian. For the spin differential, conjugate x by $1 + \kappa u$: the linear term is $ux - xu = 2u \times x$. The imaginary-quaternion Lie bracket is itself $2u \times v$, consistent with the factor two. $\square$

For a differentiable F -parameter family of rotations $A(t)$, whenever the chosen derivative has the product rule, differentiation of $A^T A = I$ gives

$$\Omega_{\text{space}} = A' A^{-1} \in \mathfrak{so}(3, F), \quad \Omega_{\text{body}} = A^{-1} A' \in \mathfrak{so}(3, F), \quad \Omega_{\text{space}} = A \Omega_{\text{body}} A^{-1}.$$

These are the two angular-velocity conventions. Their vector representatives are related by $\omega_{\text{space}} = A\omega_{\text{body}}$. The algebraic identity remains true for an entrywise field derivation, but the existence or uniqueness of solutions of the resulting differential equation is an additional question. In particular, the Berarducci–Mantova derivation of surreal coefficients is not the variable derivative used in the fine differential of [section 25](#).

A further distinction is essential: $\mathfrak{so}(3, \mathbf{No})$ is a three-dimensional $\mathbf{No}$ -vector space, whereas $\mathfrak{so}(3, \mathbf{m})$ is not closed under multiplication by arbitrary infinite scalars. The latter is the domain of the infinitesimal logarithm, not a replacement for the full algebraic tangent space.

22 Strong exponential, logarithm, and BCH

22.1 A support theorem, not a convergence assumption

Let Γ be a set-sized divisible ordered abelian group, and work in $F_\Gamma = \mathbb{R}((t^\Gamma))$, the Hahn field with well-ordered supports. It is real closed [\[1, 19\]](#). A common surreal workspace uses $t^\gamma = \omega^{-\gamma}$ for an ordered subgroup of surreal exponents. The rational span of the supports of finitely or set-many given normal forms supplies a set-sized divisible exponent group. This localization is larger than the finite-algebra subfield of [proposition 2.1](#); its purpose is to contain the strong sums.

Strong summability is the notion of [section 2](#): a well-ordered union of supports, each exponent in only finitely many members, and coefficientwise sums. It is independent of ordinary sequence convergence. The following all-lengths form of the positive-support lemma is `squat:lem:neumann` (Lemma 7.1 of the surquaternion report), with the same Higman argument; the repository’s `Surreal/HahnSeries/Neumann.lean` proves the support statements for other reports’ labels. Source 09’s statement and proof are kept because they add fixed-size matrix arguments.

Lemma 22.1 (All-length support control). *Let $S \subseteq \Gamma_{>0}$ be well ordered. The additive monoid S^* generated by S is well ordered, and every $\gamma \in \Gamma$ is the sum of only finitely many finite ordered words in S . Consequently every formal power series in finitely many commuting or noncommuting variables can be evaluated strongly at Hahn elements or fixed-size matrices whose entries have positive valuation. The nonconstant support lies in $S^* \setminus \{0\}$.*

Proof. Use Higman’s finite-word lemma for the well-quasi-ordered alphabet S [\[5\]](#). In any infinite sequence of words, an earlier word embeds into a later word with each letter no larger than the corresponding letter. Such an embedding makes the total sum no larger; unless the words are identical, either an extra positive letter or an increased letter makes it strictly larger. Therefore a strictly decreasing sequence of word sums is impossible, and infinitely many distinct words cannot have one fixed sum. This proves the two support assertions.

For the evaluation assertion take the finite union of the positive supports of all input entries. Each coefficient of a fixed total exponent involves only finitely many such words. The finite choices of variables and intermediate matrix indices add only finitely many contributions per word. Thus the formal evaluation is strongly summable, including all word lengths at once. $\square$

The same argument applies to a set-supported surreal calculation after localizing the supports. It justifies rearrangements in what follows; it does not assert that $n\gamma$ is cofinal in an arbitrary ordered value group.

Theorem 22.2 (The infinitesimal exponential and logarithm). *On a divisible Hahn workspace, and on the full surreal class by strong normal-form evaluation, the maps*

$$\text{Exp}_{\mathfrak{m}}(X) = \sum_{n \geq 0} \frac{X^n}{n!}, \quad X \in \mathfrak{so}(3, \mathfrak{m}), \quad (22.1)$$

$$\text{Log}_{\mathfrak{m}}(U) = \sum_{n \geq 1} \frac{(-1)^{n+1}}{n} (U - I)^n, \quad U \in K, \quad (22.2)$$

are inverse bijections $\mathfrak{so}(3, \mathfrak{m}) \leftrightarrow K$. They commute with conjugation by any rotation. Their transported multiplication is the strongly evaluated Baker–Campbell–Hausdorff law, not vector addition.

Proof. The support lemma makes both expressions meaningful. Formal one-variable identities show that exponential and logarithm are inverse on the ambient matrix ideals and groups $M_3(\mathfrak{m})$ and $I + M_3(\mathfrak{m})$. Substitution is legitimate because the nonconstant arguments have positive support.

For skew X , transposition gives $\text{Exp}_{\mathfrak{m}}(X)^T = \text{Exp}_{\mathfrak{m}}(-X)$. The product is I , and the result has standard part I , so its determinant is $+1$, not -1 . Conversely, if $U \in K$, then

$$\text{Log}_{\mathfrak{m}}(U)^T = \text{Log}_{\mathfrak{m}}(U^T) = \text{Log}_{\mathfrak{m}}(U^{-1}) = -\text{Log}_{\mathfrak{m}}(U).$$

The last identity is a formal identity in the one matrix $U - I$. Conjugation commutes with all finite products and with the supported sums. Finally, substituting X, Y in the formal noncommutative identity $\log(e^X e^Y) = \text{BCH}(X, Y)$ is justified by the same word-support argument. $\square$

In vector coordinates, if $\text{Exp}_{\mathfrak{m}}(\hat{u}) \text{Exp}_{\mathfrak{m}}(\hat{v}) = \text{Exp}_{\mathfrak{m}}(\hat{w})$, then

$$\begin{aligned} w &= u + v + \frac{1}{2}u \times v \\ &\quad + \frac{1}{12}(u \times (u \times v) + v \times (v \times u)) + \text{terms of total degree at least four.} \end{aligned} \quad (22.3)$$

The omitted terms are the uniquely specified formal BCH series. If every coordinate of u, v has valuation at least $\gamma > 0$, their sum has valuation at least 4γ . This is a finite-order error certificate, not an assertion of convergence through all positive surreal tolerances.

Proposition 22.3 (Exact bridge to Cayley coordinates). *For $t \in \mathfrak{m}^3$,*

$$\text{Log}_{\mathfrak{m}}(\text{Cay}(t)) = 2 \sum_{j \geq 0} \frac{(-1)^j}{2j+1} (t \cdot t)^j \hat{t}. \quad (22.4)$$

Its leading term is $2\hat{t}$. In particular, exponential coordinates and Cayley coordinates have the same valuation.

Proof. The commuting factors $I + \hat{t}$ and $I - \hat{t}$ give

$$\log((I + \hat{t})(I - \hat{t})^{-1}) = \log(I + \hat{t}) - \log(I - \hat{t}) = 2 \sum_{j \geq 0} \frac{\hat{t}^{2j+1}}{2j+1}.$$

Now use $\hat{t}^{2j+1} = (-t \cdot t)^j \hat{t}$. Every later term has strictly higher valuation than the first. $\square$

23 Valuation layers and noncommutativity at every scale

Use the valuation of [section 2](#): $v(x) = -\max \text{supp}_\omega(x)$, $v(0) = +\infty$, so $v(\omega^{-\gamma}) = \gamma$ and infinitesimals have positive valuation. For a vector or matrix, v is the minimum of the valuations of its entries. On a Hahn workspace this is the ordinary Hahn valuation with value group Γ . For all surreals the value group is a class; statements involving one layer still have finite formulas, and direct-sum notation is localized to set-sized workspaces.

Because all rotation entries have nonnegative valuation, and so do all entries of the inverse, $v(Ax) = v(x)$. The equality follows by applying the two inequalities from A and A^{-1} . Thus a rotation preserves not only the norm, but also the leading scale of every vector.

For $\gamma > 0$, define

$$K_{\geq \gamma} = \{U \in K : v(U - I) \geq \gamma\}, \quad K_{> \gamma} = \{U \in K : v(U - I) > \gamma\}. \quad (23.1)$$

These are normal subgroups of G . If t is the Cayley coordinate of $U \in K$, then

$$v(U - I) = v(t) = v(\text{Log}_m U). \quad (23.2)$$

For the first equality use the rational chart and its inverse, whose denominators have nonzero real standard parts. The second follows from [equation \(22.4\)](#).

Theorem 23.1 (Graded rotation algebra). *For $\alpha, \beta, \gamma > 0$,*

$$(K_{\geq \alpha}, K_{\geq \beta}) \subseteq K_{\geq \alpha + \beta}. \quad (23.3)$$

For surreals, the map

$$\ell_\gamma(U) = 2 \text{st}(\omega^\gamma t(U)) \quad (23.4)$$

induces an isomorphism $K_{\geq \gamma}/K_{> \gamma} \cong (\mathbb{R}^3, +)$. On a Hahn workspace replace $\omega^{-\gamma}$ by t^γ . The commutator between layers is the ordinary cross product:

$$\ell_{\alpha + \beta}((U, V)) = \ell_\alpha(U) \times \ell_\beta(V). \quad (23.5)$$

Conjugation on every layer factors through standard part and is the standard three-dimensional action of $\text{SO}(3, \mathbb{R})$.

Proof. The Cayley law is addition modulo terms of degree at least two. For inputs of valuation at least γ , scaling by ω^γ kills those terms under standard part. Thus ℓ_γ is a homomorphism with kernel $K_{> \gamma}$. It is onto, since $U = \text{Cay}(\frac{1}{2}\omega^{-\gamma}a)$ maps to $a \in \mathbb{R}^3$.

For the commutator it is convenient to use $U = e^X, V = e^Y$ with positive valuation. Formal multiplication gives

$$\text{Log}_m((e^X, e^Y)) = [X, Y] + \text{Lie terms containing at least three letters.}$$

Every nonzero term contains both X and Y . Terms after the bracket have valuation strictly larger than $\alpha + \beta$, whereas the bracket has at least that valuation. Since X has leading vector twice the Cayley vector, [equation \(3.6\)](#) proves [equations \(23.3\) and \(23.5\)](#) with the stated factor two. Lastly [equation \(18.3\)](#) gives $\ell_\gamma(AUA^{-1}) = \text{st}(A)\ell_\gamma(U)$. $\square$

On a set-sized workspace the associated graded Lie algebra is therefore

$$\bigoplus_{\gamma \in \Gamma_{>0}} \mathfrak{so}(3, \mathbb{R}) T^\gamma, \quad [aT^\alpha, bT^\beta] = (a \times b)T^{\alpha+\beta},$$

with vector notation for the coefficients. A leading bracket can vanish when the leading vectors are parallel; the theorem deliberately gives an inclusion of filtration levels, not equality of valuations in every case.

Remark 23.2 (Source 10: the inclusion is an equality of images). The statement of [theorem 23.1](#) is unchanged. For subgroups, source 10 shows that [equation \(23.3\)](#) is an equality, and in a stronger form: over a set-sized real closed field $F \supseteq \mathbb{R}$ with its natural valuation, and over $\mathbf{No}$, the set of single commutators $\{(x, y) : x \in K_{\geq \alpha}, y \in K_{\geq \beta}\}$ already equals $K_{\geq \alpha+\beta}$ ([theorem 29.3](#)). More generally $\{(x, y) : x \in K_{\mathcal{C}}, y \in K_{\mathcal{D}}\} = (K_{\mathcal{C}}, K_{\mathcal{D}}) = K_{\mathcal{C}+\mathcal{D}}$ for arbitrary upper segments of the value group ([theorem 29.4](#)). This concerns the filtration levels only. It does not contradict the vanishing of a leading bracket for parallel leading vectors: the factors realizing a given commutator are chosen for it and have orthogonal leading axes.

The quaternion form of this grading is `squat:thm:graded` (Theorem 8.6 of the surquaternion report), where the graded bracket of leading imaginary coefficients is $2X \times Y$. The two statements agree: the leading imaginary coefficient X of the unit-quaternion lift $(1, t)/\sqrt{1 + \|t\|^2}$ of U is the leading coefficient of its Cayley vector t , so $\ell_\gamma(U) = 2X$. Hence $\ell_{\alpha+\beta}((U, V)) = 2(2X \times Y) = \ell_\alpha(U) \times \ell_\beta(V)$; the factor two is that of the spin differential ([proposition 21.1](#)).

23.1 Why finite jets are nilpotent but the kernel is perfect

For any ordinary integer $N \geq 1$,

$$K_{\geq \gamma} / K_{> N\gamma}$$

is nilpotent of class at most N : an $(N + 1)$ -fold iterated commutator has valuation at least $(N + 1)\gamma > N\gamma$. This is a fixed-scale, finite-depth statement. The full group is the union over *all* positive scales. Its commutator construction can use parameters at scales such as $\gamma/2$, outside a chosen $K_{\geq \gamma}$. [Theorem 19.3](#) therefore remains fully compatible with the nilpotence of these quotients.

It would be incorrect to conclude merely from the BCH description that the whole K is a pronilpotent group in an unspecified topology. In particular, one fixed sequence of multiples $n\gamma$ need not exhaust all valuation scales, and the whole group is perfect rather than residually nilpotent in the sense of its ordinary lower central series.

Remark 23.3 (Source 10: the exact perfect residual). The preceding paragraph is made exact in Part III. For a principal subgroup $K_{\geq \gamma}$ the lower central and derived series are $\Lambda_n(K_{\geq \gamma}) = K_{\geq n\gamma}$ and $K_{\geq \gamma}^{(r)} = K_{\geq 2^r \gamma}$ ([theorem 31.3](#)). Both stabilize at stage ω at the perfect subgroup $K_{\gg \gamma}$ of rotations deeper than every $n\gamma$, which is the largest perfect subgroup of $K_{\geq \gamma}$ ([theorem 31.4](#)). Hence $K_{\geq \gamma}$ is residually nilpotent exactly when γ is an order unit of the value group ([theorem 31.6](#)); over $\mathbf{No}$ this never happens ([corollary 33.5](#)). “One fixed sequence of multiples $n\gamma$ need not exhaust all valuation scales” is thus the failure of the order-unit condition.

23.2 An imported structural perspective

For a set-sized proper real closed extension $F \supsetneq \mathbb{R}$, a theorem of Bays–Peterzil says that the pure group K_F , with parameters allowed, is bi-interpretable with the valued field $(F; +, \cdot, \mathcal{O}_F)$ [9,

Theorem 1.1 and Remark 1.2]. Thus the exact infinitesimal group carries much more information than an abelian tangent space. This is an imported model-theoretic theorem, not proved here, and its statement is about set-sized structures. We do not automatically replace its universe by the full proper class of surreals.

24 Finite angles and canonical infinite phases

24.1 Finite-angle trigonometry from strong sums

For $\theta = r + \varepsilon \in \mathcal{O}$, with $r \in \mathbb{R}$, $\varepsilon \in \mathfrak{m}$, define

$$\text{cis}(\theta) = e^{ir} \sum_{k \geq 0} \frac{(i\varepsilon)^k}{k!} = \cos \theta + i \sin \theta. \quad (24.1)$$

Only the infinitesimal part is strongly expanded; e^{ir} is an ordinary complex constant. This is the finite-angle construction used in the repository [17] and in the restricted-analytic framework [3]. The formal exponential identities prove addition, parity, Pythagoras, and the usual finite trigonometric identities. In particular cis is a homomorphism from $(\mathcal{O}, +)$ to the algebraic surcomplex unit circle. It agrees with the real-centered evaluation of [equation \(2.3\)](#).

Source 09's theorem on finite angles has two halves, both printed in Part I. The map cis is onto the unit circle with kernel exactly $2\pi\mathbb{Z}$: this is [theorem 2.2](#), whose second proof is source 09's. Every $A \in G(\mathbf{No})$ has a representation $A = \text{Rot}(n; \cos \theta, \sin \theta)$ with $0 \leq \theta \leq \pi$, unique for $A \neq I$, with the axis oriented uniquely for $0 < \theta < \pi$: this is [theorem 5.4](#), whose proof is source 09's.

Consequently the stabilizer of any chosen oriented unit axis has the group presentation

$$T_n \cong \mathcal{O}/(2\pi\mathbb{Z}). \quad (24.2)$$

Its finite angles are surreal, not necessarily ordinary real. A rotation can have an infinitesimal angle, or differ infinitesimally from an ordinary angle. There is no need to attach an infinite angle to represent any rotation.

24.2 The Ehrlich–Kaplan global extension does exist

It would nevertheless be wrong to say that no canonical global surreal trigonometry is available. The initial-integer-part construction of Ehrlich–Kaplan [3, §11] supplies one. In terms of normal forms, write

$$\mathbf{No}^\uparrow = \{x : x \text{ has only positive normal-form exponents}\}, \quad \mathbf{Oz} = \mathbf{No}^\uparrow + \mathbb{Z}.$$

The class $\mathbf{Oz}$ is the initial integer part, often called the omnific integers. It is not the class of ordinary ordinal numbers. The zero element is included in $\mathbf{No}^\uparrow$.

The corresponding phase has a particularly transparent expression. For the three-part decomposition $\theta = \theta^\uparrow + r + \varepsilon$, set

$$\text{cis}_{\text{EK}}(\theta) = \text{cis}(r + \varepsilon) = \text{Cos } \theta + i \text{Sin } \theta. \quad (24.3)$$

This is exactly reduction by $2\pi\mathbf{Oz}$: a real multiple of a purely infinite normal form is still purely infinite, so

$$2\pi\mathbf{Oz} = \mathbf{No}^\uparrow + 2\pi\mathbb{Z}, \quad \ker \text{cis}_{\text{EK}} = 2\pi\mathbf{Oz}. \quad (24.4)$$

The decomposition into purely infinite and finite parts is additive. Thus the global phase is a surjective additive-to-multiplicative homomorphism. This construction is canonical for the specified initial-integer-part normalization; it is not an assertion that every conceivable global phase convention agrees with it.

For example,

$$\text{Cos } \omega = 1, \quad \text{Sin } \omega = 0, \quad \text{cis}_{\text{EK}}(\omega + \pi/2) = i.$$

The infinite part is removed before evaluating the finite phase. No divergent Taylor series at ω is being summed. On a fixed axis one therefore has

$$(\mathbf{No}, +)/(2\pi\mathbf{Oz}) \cong T_n, \quad \theta \longmapsto \text{Rot}(n; \text{Cos } \theta, \text{Sin } \theta). \quad (24.5)$$

The finite and global presentations describe the *same* axis subgroup, with different parameter domains and period classes.

The associated spin parameter is

$$q_n(\theta) = \text{Cos}(\theta/2) + n \text{Sin}(\theta/2), \quad \ker q_n = 4\pi\mathbf{Oz}.$$

An ordinary 2π rotation has spin value -1 , while a purely infinite parameter such as ω has spin value $+1$. Indeed, multiplication by any nonzero real preserves $\mathbf{No}^\uparrow$, so the purely infinite parts of $2\pi\mathbf{Oz}$ and $4\pi\mathbf{Oz}$ coincide; their ordinary integer parts differ. This prevents misleading interpretations of an infinite parameter as an ordinary number of physical revolutions.

25 The global rotation exponential and its differential

25.1 A radial map on the whole Lie algebra

For $w \in \mathbf{No}^3$, put $r = \|w\|$. Define

$$\mathcal{E}(\widehat{w}) = \begin{cases} I + \frac{\text{Sin } r}{r} \widehat{w} + \frac{1 - \text{Cos } r}{r^2} \widehat{w}^2, & r \neq 0, \\ I, & r = 0. \end{cases} \quad (25.1)$$

Equivalently, it is the spin image of $\text{Cos}(r/2) + (w/r) \text{Sin}(r/2)$. This is a well-defined global map to $G(\mathbf{No})$, and is equivariant under conjugation. It agrees with the strong matrix exponential for infinitesimal w ; finite w are evaluated using finite trigonometry, not by assuming strong summability of the uncentered matrix power series.

The map is already onto on the finite ball $\|w\| \leq \pi$, by [theorem 5.4](#). On any fixed line of generators it is a homomorphism:

$$\mathcal{E}(\alpha \widehat{n}) \mathcal{E}(\beta \widehat{n}) = \mathcal{E}((\alpha + \beta) \widehat{n}).$$

It is not a homomorphism from the whole additive Lie algebra. The additive domain is abelian while its image is not; the BCH correction is nonzero even for small independent generators.

Theorem 25.1 (Complete fibers of the rotation exponential). *Let $A \neq I$ have the principal representation $A = \text{Rot}(n; \cos \theta, \sin \theta)$, $0 < \theta \leq \pi$. Then*

$$\mathcal{E}^{-1}(A) = \{(\widehat{\theta + 2\pi z})n : z \in \mathbf{Oz}\}. \quad (25.2)$$

At $\theta = \pi$ either choice of axis orientation gives the same set. The identity fiber is

$$\mathcal{E}^{-1}(I) = \{0\} \cup \{\hat{w} : \|w\| \in (2\pi\mathbf{Oz})_{>0}\}. \quad (25.3)$$

Restricting to finite generators replaces $\mathbf{Oz}$ by $\mathbb{Z}$.

Proof. If $\mathcal{E}(\hat{w}) \neq I$, its unique fixed line is Fw , so $w = \lambda n$. Oddness of sine and evenness of cosine show that $\mathcal{E}(\lambda \hat{n}) = \text{Rot}(n; \text{Cos } \lambda, \text{Sin } \lambda)$ for signed λ . Equality of circle values is exactly congruence modulo $2\pi\mathbf{Oz}$, proving equation (25.2). When $\theta = \pi$, negating λ does not change that congruence class. For the identity, the radial circle value must be one, which is precisely the condition in equation (25.3). The finite part of $\mathbf{Oz}$ is $\mathbb{Z}$, giving the last assertion. $\square$

The identity fiber is a union of spheres, not a subgroup of the additive three-dimensional Lie algebra. It must not be called the kernel of a global group homomorphism. Only a fixed-axis restriction has a kernel.

25.2 Fine differentiability

For a function on a surreal finite-dimensional coordinate space, the fine Fréchet derivative at x is a **No**-linear map L such that for every positive surreal η , some positive surreal δ satisfies

$$0 < \|h\| < \delta \implies \|f(x+h) - f(x) - Lh\| < \eta \|h\|.$$

Equivalent finite-dimensional coordinate norms give the same definition. The quantifiers range over all positive surreal scales. This derivative is not defined by ordinary real sequences.

The global phase satisfies the usual local derivative identities $D \text{Sin} = \text{Cos}$, $D \text{Cos} = -\text{Sin}$. Indeed, for infinitesimal h the global addition formulas reduce the increment to $\text{Sin } h$ and $\text{Cos } h$, whose strong Taylor expansions have bounded quadratic remainders. For any fixed positive surreal tolerance, one can choose a still smaller radius. The purely infinite part of the base point causes no obstruction to this local argument. For $w \neq 0$, differentiating the positive square root gives $Dr_w(h) = w \cdot h/r$; the binomial expansion at one justifies its local quadratic remainder even when r is infinite or infinitesimal. Here a quadratic remainder means a bound $C \|h\|^2$ on some neighborhood, with a fixed nonnegative surreal C depending on the base point; C need not be an ordinary real. Choosing a radius smaller than $\eta/(C+1)$ then gives the required fine derivative estimate.

Theorem 25.2 (Differential and exact critical radii). *Right-translate the differential of $\mathcal{E}$ back to the Lie algebra. For $w \neq 0$, $r = \|w\|$, one obtains*

$$(d\mathcal{E}_w(h))\mathcal{E}(\hat{w})^{-1} = \widehat{\mathcal{J}(w)h}, \quad \mathcal{J}(w) = I + \frac{1 - \text{Cos } r}{r^2} \hat{w} + \frac{r - \text{Sin } r}{r^3} \hat{w}^2. \quad (25.4)$$

Here $\mathcal{E}$ is viewed as a function of the vector w , and $\mathcal{J}(0) = I$. For $r \neq 0$,

$$\det \mathcal{J}(w) = \frac{4 \text{Sin}^2(r/2)}{r^2}. \quad (25.5)$$

The nonzero critical radii are exactly $r \in (2\pi\mathbf{Oz})_{>0}$, and the fine differential has rank one there. Otherwise it has rank three.

Proof. Write $w = rn$ and split $h = h_{\parallel} + h_{\perp}$ along n . A radial increment changes only the angle and gives $\mathcal{J}(w)h_{\parallel} = h_{\parallel}$. For a perpendicular increment, $Dr(h_{\perp}) = 0$ and $Dn(h_{\perp}) = h_{\perp}/r$. Differentiate $A = cI + (1 - c)nn^T + s\hat{n}$, with $c = \text{Cos } r$, $s = \text{Sin } r$, and multiply by A^T . The skew matrix that results has vector

$$\mathcal{J}(w)h_{\perp} = \frac{\text{Sin } r}{r}h_{\perp} + \frac{1 - \text{Cos } r}{r}n \times h_{\perp}. \quad (25.6)$$

This calculation can also be made with $n = e_3$, then transported by [equation \(21.2\)](#). Since $\hat{w}^2$ is zero on the axis and $-r^2I$ on the plane, [equation \(25.6\)](#) is precisely [equation \(25.4\)](#). The derivative identities preceding the theorem provide quadratic remainders; finite sums, products, and inverses with nonzero denominators preserve that property locally. At zero the small-generator expansion is $\mathcal{E}(\hat{h}) = I + \hat{h} + O(\|\hat{h}\|^2)$, giving $\mathcal{J}(0) = I$.

On the perpendicular plane, the determinant of [equation \(25.6\)](#) is

$$\frac{\text{Sin}^2 r + (1 - \text{Cos } r)^2}{r^2} = \frac{2(1 - \text{Cos } r)}{r^2} = \frac{4 \text{Sin}^2(r/2)}{r^2}.$$

The axis contributes one. Over an ordered field the planar numerator vanishes exactly when $\text{Sin } r = 0$ and $\text{Cos } r = 1$, equivalently $r \in 2\pi\mathbf{Oz}$. At such a point the whole perpendicular plane is killed and the axis survives, proving the rank statement. $\square$

The singular values of the right-trivialized derivative are

$$1, \quad \frac{2|\text{Sin}(r/2)|}{r}, \quad \frac{2|\text{Sin}(r/2)|}{r}. \quad (25.7)$$

This follows directly from [equation \(25.6\)](#), whose symmetric square is the scalar in [equation \(25.5\)](#) on the perpendicular plane. Thus a noncritical infinite generator can still be very poorly conditioned transversely. For example, at $r = \omega + \pi$ the transverse singular values are $2/(\omega + \pi)$, positive but infinitesimal.

At $w = \omega e_1$, the image is I and the differential has rank one. At the ordinary half-turn radius $r = \pi$, by contrast, the differential is invertible. The ambiguity of a principal logarithm at half-turns is a branch issue, not a critical point of the exponential there. Also note the factor of two relative to the radial unit-quaternion exponential: its imaginary parameter has half the spatial rotation angle. The four-dimensional quaternion radial map in [19] has a different domain and different rank counts.

No inverse function theorem in an unspecified surreal analytic category is inferred solely from nonsingularity of [equation \(25.4\)](#). The formulas and ranks are direct differential statements. Nor are they chain rules for a field derivation acting on the coefficients.

26 Three topologies, and what the spin cover means

This section separates the most important analytical ambiguity. Algebraic identities do not depend on a topology, but connectedness, convergence, compactness, and fundamental groups do.

26.1 The full fine topology

On $\mathbf{No}^d$ use neighborhoods $\|x - a\|_{\infty} < \rho$, with every positive surreal ρ allowed. On matrices and $G(\mathbf{No})$ use the corresponding finite-coordinate topology. For the full class these are neighborhood schemes, or ordinary topologies in a suitably larger-universe formulation with the size restrictions retained.

Theorem 26.1 (Set-sized discreteness in the full surreal topology). *Every set-sized subset of $\mathbf{No}^d$, and hence of $G(\mathbf{No})$, is closed and discrete. Every convergent set-indexed net is eventually equal to its limit. Every set-indexed Cauchy net is eventually constant. Nevertheless $G(\mathbf{No})$ has no isolated points.*

Proof. For a point a and a set-sized subset S , form the set of positive distances $\|s - a\|_\infty$ for $s \in S \setminus \{a\}$. The separated surreal cut with left set $\{0\}$ and that right set has a positive separator ρ smaller than every listed distance. If the list is empty, any positive ρ works. Its ball meets S in at most $\{a\}$. This proves both closedness and discreteness.

A set-indexed net has a set-sized range. Applying the preceding argument at its limit proves eventual equality. For a Cauchy net, apply the cut argument to all positive pairwise distances in its range; these form a set. A Cauchy tail of diameter smaller than the resulting positive separator has only one value. Finally, $\text{Cay}(te_1) \rightarrow I$ as a neighborhood assertion with t arbitrarily small in $\mathbf{No}$, and $\text{Cay}(te_1) \neq I$ for $t \neq 0$. Translations give no isolated points anywhere in the group. $\square$

The general size phenomenon is already emphasized in the repository’s foundations report [20]; the proof here specializes it to rotation coordinates. In a universe-relative version, “set-sized” means small relative to the universe in which the whole surreal carrier is large. It cannot be dropped from the statement.

In particular, an ordinary real interval admits no nonconstant fine-continuous map into $G(\mathbf{No})$: its image is a set, is discrete, and must be connected. An ordinary irrational-angle cyclic subgroup is a countable closed discrete subset of $G(\mathbf{No})$, even though its standard part can be dense in an ordinary real circle. Strongly summed exponential partial sums cannot converge in this fine topology unless they are eventually equal to their limit. This is why the support theorem, rather than a real convergence argument, was required.

26.2 The intrinsic topology of a set-sized workspace

Let $F \supsetneq \mathbb{R}$ be a set-sized real closed field. Its *intrinsic* order topology permits only radii from F . If $F \subseteq \mathbf{No}$, this is not its topology induced from the full surreal class: the latter is discrete by [theorem 26.1](#), while the intrinsic order topology has no isolated points. The distinction applies to $G(F)$ as well.

Even intrinsically, F is totally separated. Its infinitesimal ideal $\mathfrak{m}_F$ is clopen: a positive infinitesimal radius proves openness at zero, and a point with noninfinitesimal magnitude has a ball disjoint from $\mathfrak{m}_F$. Translates and nonzero scalar multiples of $\mathfrak{m}_F$ are clopen. For $a \neq b$, the clopen set $a + |b - a|\mathfrak{m}_F$ contains a but not b . Coordinate projections therefore separate any two points of $G(F)$ by a clopen set. Ordinary real-parameter paths into $G(F)$ are constant, and its ordinary fundamental group at every point is trivial.

This topology is not locally compact. To verify it rather than merely infer it from non-Archimedeaness, note that $\mathbb{R} \subset F$ is closed and discrete: near a real point choose an infinitesimal radius, near a finite nonreal point choose less than its distance to its standard part, and near an infinite point choose a ball avoiding all finite elements. Hence every subset of $\mathbb{R}$ is closed in F . Every neighborhood of the identity in $G(F)$ contains $\{\text{Cay}(ae_1/n) : n \geq 1\}$ for a sufficiently small $a > 0$ in F . This is an infinite closed discrete subset. It is closed even at the boundary of the Cayley chart, since its trace stays a positive F -distance from -1 , by [equation \(5.14\)](#). A compact neighborhood would contain it as a closed subset, which is impossible.

On a Hahn workspace, intrinsic valuation convergence is still meaningful and can be nontrivial. In $\mathbb{R}((t^{\mathbb{Q}}))$, for instance, $t^n \rightarrow 0$, since the ordinary integers are cofinal in $\mathbb{Q}$. In a higher-rank group that conclusion can fail for a given positive exponent. These are statements about the workspace, not convergence in the full surreal fine topology.

26.3 The standard-part quotient topology is discrete

The subgroup K_F is open and closed in the intrinsic topology, since it is the intersection of finitely many conditions $A_{ij} - I_{ij} \in \mathfrak{m}_F$. Thus all fibers of standard part are open. It follows that the quotient topology on

$$G(F)/K_F \cong \mathrm{SO}(3, \mathbb{R}) \quad (26.1)$$

is *discrete*, not the usual real Lie-group topology. Standard part into the usual $\mathrm{SO}(3, \mathbb{R})$ is continuous, but it is not a quotient map onto that usual topology. The constant section from the usual real Lie group is not continuous. The same neighborhood conclusions hold for the full surreal class.

Accordingly, a topological reading of the split extension must use the discrete topology on its real factor. Alternatively, one can deliberately choose the coarser topology pulled back from the usual real group by standard part. That topology has the usual real group as its quotient, but does not separate distinct members of one infinitesimal fiber. These are different choices, not competing proofs about one topology.

26.4 Semialgebraic paths, compactness, and the spin fundamental group

Over a real closed field F , semialgebraic topology restricts maps and homotopies to semialgebraic ones with parameter intervals $[0, 1]_F$. Closed and bounded semialgebraic sets are definably compact, and the matrix equations show that $G(F)$ and $\mathrm{Sp}(1, F)$ have that property. This is not ordinary compactness of all open covers.

There is an explicit semialgebraic path from the identity to every rotation. Choose a unit quaternion lift q with nonnegative scalar part and put

$$q(u) = \frac{(1 - u) + uq}{\sqrt{N((1 - u) + uq)}}, \quad 0 \leq u \leq 1 \text{ in } F. \quad (26.2)$$

The numerator never vanishes: for $0 \leq u < 1$ its scalar part is strictly positive, and at $u = 1$ it is q . Its spin image is the required path. Thus $G(F)$ is semialgebraically path connected even when its ordinary topological connected components are singletons. The parameter interval and the allowed notion of homotopy have changed.

The spin map is a semialgebraic two-sheeted covering. Around the identity use the non-half-turn open set and choose the lift with positive scalar part; the other lift has negative scalar part. Both branches are continuous and semialgebraic by the explicit square-root formulas, and translation supplies such neighborhoods at every point. This is also a local two-sheeted covering for the intrinsic topology of F -points.

Theorem 26.2 (The category in which the familiar fundamental group survives). *For every set-sized real closed field F ,*

$$\pi_1^{\mathrm{sa}}(G(F), I) \cong \mathbb{Z}/2\mathbb{Z}, \quad \pi_1^{\mathrm{sa}}(\mathrm{Sp}(1, F), 1) = 0.$$

In the semialgebraic category the spin map is the simply connected double cover. For a proper real closed extension of $\mathbb{R}$, this is not the ordinary fundamental group of the intrinsic topological space.

Proof. The equations of G , the unit three-sphere, and the spin map are defined over the rationals. The semialgebraic homotopy comparison and real-closed-field extension theorem of Baro–Otero [10, Corollaries 4.3–4.4] identifies their semialgebraic homotopy groups with the ordinary homotopy groups of their real realizations. Over $\mathbb{R}$ the map is the antipodal double cover $S^3 \rightarrow \mathbb{RP}^3$, giving the displayed groups. The local branches just constructed identify the transferred covering. For a proper extension, total separation has already shown that all ordinary loops are constant. $\square$

Likewise, $\pi_2^{\text{sa}}G(F) = 0$ and $\pi_3^{\text{sa}}G(F) \cong \mathbb{Z}$, by the same comparison. A claim about a “surreal universal cover” is meaningful only after specifying this category and, for the full class, the compatible universe or workspace interpretation. The fine topology does not make $G(\mathbf{No})$ into an ordinary real Lie group.

27 Worked examples at surreal scales

27.1 A tiny angle with a macroscopic displacement

Take $\varepsilon = \omega^{-1}$ and

$$A = \text{Cay}(\varepsilon e_3) = \begin{pmatrix} \frac{1-\varepsilon^2}{1+\varepsilon^2} & -\frac{2\varepsilon}{1+\varepsilon^2} & 0 \\ \frac{2\varepsilon}{1+\varepsilon^2} & \frac{1-\varepsilon^2}{1+\varepsilon^2} & 0 \\ 0 & 0 & 1 \end{pmatrix}. \quad (27.1)$$

It belongs to K , is not the identity, and has angle $2 \operatorname{atan} \varepsilon = 2\varepsilon - \frac{2}{3}\varepsilon^3 + \dots$, where the arctangent is the strong local series. Acting on Le_1 gives

$$A(Le_1) - Le_1 = \left(-\frac{2L\varepsilon^2}{1+\varepsilon^2}, \frac{2L\varepsilon}{1+\varepsilon^2}, 0 \right)^T.$$

For $L = \omega$, the transverse displacement has standard part two. For $L = \omega^2$, it is of order ω , although the matrix still has standard part I . An infinitesimal relative error need not be a negligible absolute error at an infinite spatial scale.

Proposition 27.1 (Exact displacement formula). *If $A = \text{Rot}(n; c, s)$ and $x_\perp = x - (n \cdot x)n$, then*

$$\|Ax - x\|^2 = 2(1 - c) \|x_\perp\|^2. \quad (27.2)$$

For a finite angle θ , this is $\|Ax - x\| = 2 |\sin(\theta/2)| \|x_\perp\|$.

Proof. The axis component cancels. On the plane, $(A - I)x_\perp = (c - 1)x_\perp + s(n \times x_\perp)$. The two vectors are perpendicular with equal norms, so the squared norm is $((c - 1)^2 + s^2) \|x_\perp\|^2 = 2(1 - c) \|x_\perp\|^2$. $\square$

27.2 A near-half-turn with an infinite Cayley coordinate

Let $B = \text{Cay}(\omega e_3)$. Then

$$c = -1 + \frac{2}{1 + \omega^2}, \quad s = \frac{2\omega}{1 + \omega^2}, \quad \theta = \pi - 2 \operatorname{atan}(\omega^{-1}).$$

This is not a half-turn, although its standard part is one. The matrix entries are bounded; its Cayley coordinate alone is infinite. Its quaternion lift has scalar part $1/\sqrt{1 + \omega^2}$ and third imaginary part $\omega/\sqrt{1 + \omega^2}$. The largest-coordinate extraction algorithm selects the latter rather than dividing by a tiny scalar coordinate.

27.3 Infinitesimally different axes

The unit vector

$$n = \frac{(1, \varepsilon, \varepsilon^2)^T}{\sqrt{1 + \varepsilon^2 + \varepsilon^4}}$$

defines a half-turn $H_n = 2nn^T - I$ with the same standard part as H_{e_1} . The half-turns are distinct because their fixed lines differ. Their difference is visible already in the off-diagonal entry $(H_n)_{12} = 2\varepsilon/(1 + \varepsilon^2 + \varepsilon^4)$. This is axis information that a real standard-part calculation discards.

27.4 Noncommutativity first appears at a finer scale

For positive infinitesimals ε, δ , let $U = \text{Cay}(\varepsilon e_1)$, $V = \text{Cay}(\delta e_2)$. Then

$$(U, V) = I + 4\varepsilon\delta\widehat{e}_3 + \text{terms divisible by } \varepsilon\delta \text{ of total degree at least three.} \quad (27.3)$$

In valuation language, the remainder has valuation at least $\min(2v(\varepsilon) + v(\delta), v(\varepsilon) + 2v(\delta))$, strictly above the displayed term. This follows either by rational multiplication or by the BCH bracket. The two rotations commute to first order but not exactly.

27.5 Infinite generators and local sensitivity

Put $w = \omega e_1$ and $z = \pi e_2$. Equation [equation \(25.1\)](#) gives $\mathcal{E}(\widehat{w}) = I$ and $\mathcal{E}(\widehat{z}) = H_{e_2}$. Yet

$$\|w + z\| = \sqrt{\omega^2 + \pi^2} = \omega + \frac{\pi^2}{2\omega} + O(\omega^{-3}),$$

so the global phase removes ω and leaves an infinitesimal angle. Consequently

$$\mathcal{E}(\widehat{w + z}) = I + \frac{\pi^2}{2\omega}\widehat{e}_1 + O(\omega^{-2}),$$

not H_{e_2} . Here $O(\omega^{-k})$ means an entrywise bound by a finite real constant times that scale; it is not an unspecified real limit. This illustrates why global radial exponentiation is not additive across noncommuting directions, even when one generator has trivial image.

27.6 A nonreal copy of the octahedral group

Let H_0 be the 24 signed permutation matrices of determinant one, and set $C = \text{Cay}(\varepsilon e_1 + \varepsilon^2 e_2)$. Then

$$H = CH_0C^{-1}$$

is an exact 24-element subgroup with generally nonreal entries. Its standard part is H_0 , and conjugation by $C^{-1} \in K$ restores the ordinary real group. The multiplication table is unchanged; the coordinate realization has been deformed at several infinitesimal scales.

Part III

Valuation cuts and the universal set-sized quotient

Part III is source 10, *Valuation Cuts and the Universal Set-Sized Quotient of Surreal Rotation Groups: exact commutators, normal subgroups, perfect residuals, and surcomplex special unitary groups*. It takes up the direction “normal-subgroup classification compatible with convex valuations” named at the end of source 09 (section 38), and it builds directly on theorem 18.1, on theorem 19.3 with the unequal-parameter identity recorded in its proof, and on theorem 23.1. Material that source 10 recalls from Part II is cited, not repeated.

The infinitesimal rotations of surreal three-space are a meeting point of ordered-field algebra, group theory and size-sensitive foundations. The rotation group has the ordinary real rotations as its standard-part image, but its infinitesimal kernel is nonabelian and perfect (theorem 19.3), while a first-order approximation sees an abelian three-dimensional layer (theorem 23.1). Both statements are true. To reconcile them quantitatively one keeps track of the valuation depth at which a commutator is formed. Part III answers three questions in the precise senses below. How much valuation depth is lost or gained in an *exact*, not leading-order, commutator? Which infinitesimal scales can an abstract normal subgroup contain? Can a set-sized group retain any infinitesimal information from the *full* surreal rotation group? The last answer is genuinely proper-class and has no counterpart for one fixed set-sized field.

In the notation fixed in section 28.4, the principal conclusions are

$$\{(x, y) : x \in K_{\geq \alpha}, y \in K_{\geq \beta}\} = K_{\geq \alpha + \beta}, \quad \langle\langle U \rangle\rangle_G = K_{\geq v(U-I)} \quad (I \neq U \in K),$$

where the left side of the first formula is the image of one word map, not merely the subgroup it generates (theorems 29.3 and 30.3). Consequently the proper normal subgroups of $G(F)$ are classified by valuation cuts (theorem 30.6). For a principal subgroup $K_{\geq \alpha}$,

$$\bigcap_{n \geq 1} \Lambda_n(K_{\geq \alpha}) = \bigcap_{n \geq 0} K_{\geq \alpha}^{(n)} = K_{\gg \alpha} := \{I\} \cup \{U \in K : v(U - I) > n\alpha \text{ for every } n \geq 1\},$$

a perfect subgroup which is the largest perfect subgroup of $K_{\geq \alpha}$; thus $K_{\geq \alpha}$ is residually nilpotent exactly when α is an order unit (theorems 31.4 and 31.6). For the full surreal class,

$$\begin{aligned} \varphi : \mathrm{SO}(3, \mathbf{No}) &\rightarrow \mathcal{T} \text{ a class homomorphism, } \mathcal{T} \text{ a set-sized group} \\ \implies \varphi &= \bar{\varphi} \circ \mathrm{st}, \quad \bar{\varphi} : \mathrm{SO}(3, \mathbb{R}) \rightarrow \mathcal{T} \text{ unique} \end{aligned}$$

(theorem 33.2). Continuity is not among the hypotheses. The same holds for $\mathrm{SO}(n, \mathbf{No})$, $n \geq 3$, and for $\mathrm{SU}(n, \mathbf{No}[i])$, $n \geq 2$ (theorem 34.3), and fails for the circle and for $\mathrm{U}(n, \mathbf{No}[i])$ (proposition 35.1). The main results use finite quaternion and matrix algebra, not convergence, unrestricted trigonometry, or a global surreal exponential.

28 Setting, inheritance, and notation for Part III

28.1 Scalar hypotheses and the natural valuation

Until section 33, F is a *set-sized* real closed field containing $\mathbb{R}$. We normally assume $F \neq \mathbb{R}$ so that positive valuation depths exist; the Archimedean case is the degenerate case $K = \{I\}$. The

finite elements $\mathcal{O} = \mathcal{O}_F$, the infinitesimals $\mathfrak{m} = \mathfrak{m}_F$ and the standard part $\text{st} : \mathcal{O} \rightarrow \mathbb{R}$ are those of [section 2](#). We use the natural additive valuation

$$v : F^\times \longrightarrow \Gamma = F^\times / \mathcal{O}^\times, \quad v(0) = \infty,$$

with $v(x) > 0$ exactly when x is infinitesimal. Every nonzero real constant has value zero. Real closedness makes Γ divisible: positive roots divide arbitrary values by every positive integer. On a Hahn workspace $F = \mathbb{R}((t^\Gamma))$ of [equation \(2.1\)](#) this is the valuation of [section 2](#), with the same exponent group Γ . For all of **No** the value group is the class $(\mathbf{No}, +)$, with $v(\omega^{-\gamma}) = \gamma$; it is written **No** and never Γ , since `docs/NOTATION.md` reserves Γ for a set-sized group.

One elementary property is essential:

$$v(t) \geq v(s), \quad s, t > 0 \iff t/s \in \mathcal{O} \iff t/s \leq M \text{ for some integer } M \geq 1. \quad (28.1)$$

The final bound is by an *ordinary finite integer*. It is why the normal-generation proof of [theorem 30.3](#) works for the natural valuation. Replacing this valuation silently by an arbitrary coarsening would invalidate that argument.

For a vector $u \in F^r$ write $v(u) = \min_j v(u_j)$ and $\|u\| = (\sum_j u_j^2)^{1/2}$. Then, for $u \neq 0$,

$$v(\|u\|) = v(u). \quad (28.2)$$

Indeed, divide all coordinates by a coordinate of minimal value. The residues are real and at least one is nonzero, so the sum of their squared residues is strictly positive: no cancellation of leading squares is possible. This is where the positive-definite form enters.

28.2 Depth and the infinitesimal kernel

Let $G = G(F) = \text{SO}(3, F)$ and let $K = K_F$ be the kernel of standard part, so that $G = K \rtimes \text{SO}(3, \mathbb{R})$ and $U \in K$ exactly when $U - I \in M_3(\mathfrak{m})$ ([theorem 18.1](#); source 10 recalls this splitting and notes that its entrywise argument proves it directly). For a matrix A put $v(A) = \min_{ij} v(A_{ij})$. Matrix multiplication satisfies $v(AB) \geq v(A) + v(B)$. Since a rotation and its inverse have entries in $\mathcal{O}$, conjugation by G preserves matrix valuation exactly, as noted for vectors before [equation \(23.1\)](#). The *depth* of $U \in K$ is $v(U - I)$, with $v(0) = \infty$ for the identity. It follows that

$$\begin{aligned} v(U^{-1} - I) &= v(U - I), \\ v(AUA^{-1} - I) &= v(U - I) \quad (A \in G), \\ v(UV - I) &\geq \min(v(U - I), v(V - I)) \quad (U, V \in K). \end{aligned} \quad (28.3)$$

If $U, V \in K$ have unequal finite depths, the last inequality is an equality: in $UV - I = (U - I) + (V - I) + (U - I)(V - I)$ the unique summand of smaller depth cannot be cancelled by the other two.

The spin map $p : \text{Sp}(1, F) \rightarrow G$ is that of [theorem 5.5](#); for a unit quaternion $q = (a, u)$, $p(q) = (a^2 - u \cdot u)I + 2uu^\top + 2a\hat{u}$ by [equation \(5.7\)](#), so $\text{tr } p(q) = 4a^2 - 1$. The group G acts transitively on unit vectors ([section 16](#)).

Lemma 28.1 (Canonical infinitesimal lift). *Every $U \in K$ has exactly one unit quaternion lift $q_U = (a, u)$ with $\text{st}(q_U) = 1$. It has $a > 0$, $a - 1 \in \mathfrak{m}$ and $u \in \mathfrak{m}^3$. For $U \neq I$,*

$$v(U - I) = v(u) = v(\|u\|), \quad v(1 - a) = 2v(U - I). \quad (28.4)$$

Moreover, two such lifts with the same scalar part are conjugate by a unit quaternion, and their rotations are conjugate in G .

Proof. Existence and uniqueness of the lift are the last clause of [proposition 18.2](#); source 10’s argument is the trace formula: if U is near I , the two lifts have scalar parts with standard parts 1 and -1 , and the positive one has infinitesimal imaginary coordinates by its unit-norm equation.

Set $\delta = v(u) > 0$. The identity $1 - a = (u \cdot u)/(1 + a)$ gives value 2δ . In $p(q_U) - I$ the linear term $2a\hat{u}$ has value δ , while all remaining terms have value at least 2δ . Thus $v(U - I) = \delta$. Alternatively, $p(q) - p(q)^\top = 4a\hat{u}$ prevents cancellation. [Equation \(28.2\)](#) finishes the value assertions.

Equal scalar parts imply equal imaginary norms. For nonzero imaginary parts, a rotation carries one normalized direction to the other, and a unit lift of that rotation ([theorem 5.5](#)) conjugates the quaternions. The zero case is the identity. $\square$

All identities so far are valid over the ordinary reals and over arbitrary real closed fields. Nothing has yet involved a Hahn expansion or a topology.

28.3 What Part III inherits and what it adds

Source 10 inspected this report at commit `465a54b479a1ee842cbf7db1689a7d2f6bfe25e1`; the report directory is unchanged between that commit and the commit at which Part III was added, so its references to Part II (“especially Sections 16–19 and 23”, that is [sections 16 to 19](#) and [section 23](#)) are accurate as printed. Part II already gives the quaternion model, the standard-part splitting, unique divisibility, the single-commutator theorem for K and the graded cross-product bracket. The proof of [theorem 19.3](#) even records the unequal-parameter scalar identity on which the exact valuation refinement below rests. That identity, the spin double cover and the perfectness of the entire kernel are not new here.

D’Andrea and Maffei prove local openness of the commutator map for compact semisimple Lie groups [[11](#), Theorem 2.5]. Bays and Peterzil explicitly record commutator width one for the infinitesimal $\mathrm{SO}(3)$ group in their proof of Claim 3.3 [[9](#)]. Their Theorem 1.1 also recovers a real closed convexly valued field, with parameters, from the infinitesimal group; that is stronger than an abstract recovery of the value group. The normal-lattice reconstruction of [proposition 32.1](#) is a concrete elementary realization, not a claim to have first discovered field recoverability. Source 10 grades its own contributions as follows.

Statement or ingredient	Treatment in Part III
Spin, standard part, perfectness of K	Established background, reproved only as needed; credited to Part II and to published work.
Exact mixed-depth image and arbitrary-cut formula	A refinement of the existing quaternion construction, with full proof and explicit scope.
Normal closures and cut classification	An elementary structural theorem proved here; no general historical priority is asserted.
Perfect residual and stage- ω stabilization	Exact consequences of the cut calculus, including the order-unit criterion.
Universal set-sized quotient and fixed-rank extensions	The main candidate-original surreal-specific contribution of source 10.

The source comparison used the repository catalogue, this report’s README and relevant source sections, and the trigonometry report’s README. The online literature check concentrated on the primary works just cited and on surreal-field background [[12](#)]. It was not an exhaustive review of the literature on anisotropic algebraic groups. Broader searches for normal-subgroup and surreal-quotient terminology were inconclusive, and unrelated hits were not used as evidence.

Failure to locate an identical statement is not a proof of priority; the mathematical content rests on the proofs below.

28.4 Notation, and the renamings made for Part III

The group commutator is $(U, V) = UVU^{-1}V^{-1}$, as in Part II. For subgroups A, B , (A, B) denotes the subgroup generated by such commutators, whereas

$$\{(x, y) : x \in A, y \in B\}$$

is the *set* of single commutators; the two are kept distinct. Square brackets remain Lie brackets (equation (21.2)). An *upper segment* is a subset $\mathcal{C} \subseteq \Gamma_{>0}$ with $\gamma \in \mathcal{C}$ and $\eta \geq \gamma$ implying $\eta \in \mathcal{C}$; the empty segment is allowed. Values in $\Gamma_{>0}$ are denoted $\alpha, \beta, \gamma, \delta, \eta, \varepsilon$; in Part I the letters α, β, γ are spherical angles, while Part II already uses γ for valuations. The normal closure of U in G is $\langle\langle U \rangle\rangle_G$, the subgroup of finite products of conjugates of $U^{\pm 1}$. The following symbols of source 10 were changed so that each has one meaning in this report.

Here	Source 10	Reason
depth $v(U - I)$	$d(g)$	$d = \sqrt{D}$ is the Gram quantity of section 1.4; Part II already writes $v(U - I)$ (equation (23.2))
$(U, V), (A, B), \{(x, y) : x \in A, y \in B\}$	$[g, h], [A, B]$	square brackets are Lie brackets
$K_{\geq \alpha}, K_{> \alpha}, K_{\mathcal{C}}$	$K_{\alpha}, K_{> \alpha}, K_{\mathcal{C}}$	$K_{\geq \gamma}$ is Part II's notation (equation (23.1)); segments are calligraphic because K_F carries a field subscript
$\mathfrak{a}_{\mathcal{C}}$	$I_{\mathcal{C}}$	I is the identity matrix
$K_{> \Delta}, K_{\gg \alpha}$	P_{Δ}, P_{α}	P is the positive matrix of lemma 20.1
Λ_n (lower central series)	γ_n	γ is a valuation
$\mathcal{T}, \mathcal{T}_0$ (target groups)	H, T, H_0	H_n is a half-turn, H and H_0 are finite subgroups in Part II, T_n an axis torus
$p(q)$, spin map	$R(q)$	R is a sphere radius; p as in theorem 5.5
$U_{\gamma} = \text{Cay}(\omega^{-\gamma}e_1)$	$r_{\gamma} = \text{Cay}(\omega^{-\gamma}e_1)$	U denotes elements of K
$\mathbf{No}[i]$, and $F(i)$ for a general F	$\mathbf{No}(i), F(i)$	$\mathbf{SC} = \mathbf{No}[i]$ in docs/NOTATION.md; $F(i) = F[i]$ as in Part II
$\mathbb{T}(\mathbf{No}), U(n, \cdot)$	$U(1, \mathbf{No}(i)), U(n)$	italic U is an element of K ; $\mathbb{T}$ as in section 1.4 and the trigonometry report
$\tilde{K}_F$	M_F	$M_3(F)$ is the matrix algebra
$\ell_{\mathfrak{m}}(u)$	$\ell(u)$	the trigonometry report's name (trigonometry:rot:eq:ellm); ℓ_{γ} is the layer map of theorem 23.1
coef_{η}	c_{η}	c is the cosine coordinate of $\text{Rot}(n; c, s)$
a set $\mathcal{X}$, a set-sized field k	X, E	X is a Lie algebra element, E the spherical excess
W (a Givens block)	B	B is a real rotation in theorem 18.1

The Cayley map is the same in both texts: source 10's $\text{Cay}(x)$ is $\text{Cay}(x)$ of equation (5.10), with canonical lift $(1, x)/\sqrt{1 + x \cdot x}$, and its product law is equation (5.13). No normalization changes.

The factor $\frac{1}{2}$ of the surjectivity argument of [theorem 23.1](#) belongs to the layer map, not to the chart: $\ell_\gamma(\text{Cay}(\frac{1}{2}\omega^{-\gamma}a)) = a$, so $\ell_\gamma(U_\gamma) = 2e_1$. Source 10's Δ_α is kept ([equation \(31.2\)](#)); it is the subgroup that `docs/NOTATION.md` writes H_α in its Hahn–Tate entry, a name not used here because H_n is a half-turn. The depth condition defining $K_{\gg\alpha}$ is the value condition of the error ideal I_κ of the expanding-dynamics report, also recorded in that file. The letter Δ is also a planar area in [equation \(6.6\)](#) and χ_A a characteristic polynomial in [section 5](#); both uses are local to Part I. *Order unit* has the meaning of `docs/NOTATION.md`. The stage index ω of a transfinite series is the first infinite ordinal, which is also the surreal ω .

29 Exact commutators at prescribed valuation depths

Lemma 29.1 (The scalar commutator identity). *For unit quaternions $q_1 = (a, u)$ and $q_2 = (b, w)$,*

$$\text{Re}(q_1, q_2) = 1 - 2 \|u \times w\|^2. \quad (29.1)$$

For arbitrary, not necessarily unit, quaternions,

$$\text{Re}(q_1 q_2 \overline{q_1} \overline{q_2}) = N(q_1)N(q_2) - 2 \|u \times w\|^2. \quad (29.2)$$

Proof. Quaternion multiplication gives $q_1 q_2 - q_2 q_1 = (0, 2u \times w)$. For units,

$$(q_1, q_2) - 1 = (q_1 q_2 - q_2 q_1) q_1^{-1} q_2^{-1}, \quad N((q_1, q_2) - 1) = 4 \|u \times w\|^2.$$

For any unit quaternion r , $N(r - 1) = 2(1 - \text{Re } r)$. This proves [equation \(29.1\)](#). Multiplying by $N(q_1)N(q_2)$ after normalization proves [equation \(29.2\)](#) for nonzero factors, and a zero factor is immediate. $\square$

The identity displayed in the proof of [theorem 19.3](#) is the case $u = se_1$, $w = te_2$ of [equation \(29.2\)](#).

Lemma 29.2 (The depth bound). *If $U, V \in K$, then*

$$v((U, V) - I) \geq v(U - I) + v(V - I). \quad (29.3)$$

Proof. This is the elementwise form of [equation \(23.3\)](#), proved there by a formal BCH computation in a Hahn workspace. Source 10's proof is a finite second route, valid over every real closed $F \supseteq \mathbb{R}$: the identity

$$(U, V) - I = ((U - I)(V - I) - (V - I)(U - I))U^{-1}V^{-1}$$

and the bounded entries of U^{-1}, V^{-1} give the assertion. An identity factor is read with value ∞ . $\square$

Theorem 29.3 (Exact mixed-depth commutator image). *For all $\alpha, \beta \in \Gamma_{>0}$,*

$$\{(x, y) : x \in K_{\geq\alpha}, y \in K_{\geq\beta}\} = K_{\geq\alpha+\beta}. \quad (29.4)$$

More precisely, if $U \neq I$ has depth $\delta \geq \alpha + \beta$, then $U = (V, W)$ with $v(V - I) = \alpha$ and $v(W - I) = \delta - \alpha \geq \beta$.

Proof. The inclusion $\subseteq$ is [lemma 29.2](#). Let (c, z) be the canonical lift of U . Then $0 < c < 1$ and $v(1 - c) = 2\delta$. Put

$$\xi = \sqrt{\frac{1 - c}{2}}, \quad v(\xi) = \delta.$$

Choose any positive $x \in F$ with $v(x) = \alpha$, and set $y = \xi/x$. Both are infinitesimal, since $v(y) = \delta - \alpha \geq \beta > 0$. Therefore

$$q_1 = (\sqrt{1-x^2}, xe_1), \quad q_2 = (\sqrt{1-y^2}, ye_2)$$

are unit quaternions near one. By [equation \(29.1\)](#),

$$\operatorname{Re}(q_1, q_2) = 1 - 2x^2y^2 = 1 - 2\xi^2 = c.$$

The commutator is near one and has the same scalar part as (c, z) . By [lemma 28.1](#), a unit quaternion h conjugates it to (c, z) . Then

$$U = (p(hq_1h^{-1}), p(hq_2h^{-1})).$$

Conjugation preserves depth, so by [equation \(28.4\)](#) the two depths are α and $\delta - \alpha$. The identity target uses two identity factors. $\square$

This is an exact construction: no error term remains and no iterative lifting is needed. The nonlinearity is handled by matching a conjugacy invariant, the scalar part of a unit lift. The balanced fourth-root construction in the proof of [theorem 19.3](#) is the special case in which both depths are $\delta/2$.

29.1 Arbitrary upper segments and valuation ideals

For an upper segment $\mathcal{C} \subseteq \Gamma_{>0}$ define

$$\mathfrak{a}_{\mathcal{C}} = \{0\} \cup \{x \in \mathfrak{m} : v(x) \in \mathcal{C}\}, \quad K_{\mathcal{C}} = \{I\} \cup \{U \in K : v(U - I) \in \mathcal{C}\}. \quad (29.5)$$

The first is an ideal of $\mathcal{O}$ and the second a normal subgroup of G , by [equation \(28.3\)](#). Thus $K_{\geq \alpha} = K_{[\alpha, \infty)}$ and $K_{> \alpha} = K_{(\alpha, \infty)}$ are Part II's subgroups [equation \(23.1\)](#). For upper segments $\mathcal{C}, \mathcal{D}$ the Minkowski sum $\mathcal{C} + \mathcal{D} = \{\alpha + \beta : \alpha \in \mathcal{C}, \beta \in \mathcal{D}\}$ is again upper (if $\gamma \geq \alpha + \beta$, increase the first summand to $\gamma - \beta$), and it is empty if either input is empty.

Theorem 29.4 (Exact cut arithmetic). *For every pair of upper segments,*

$$\{(x, y) : x \in K_{\mathcal{C}}, y \in K_{\mathcal{D}}\} = (K_{\mathcal{C}}, K_{\mathcal{D}}) = K_{\mathcal{C} + \mathcal{D}}. \quad (29.6)$$

Also $\mathfrak{a}_{\mathcal{C}}\mathfrak{a}_{\mathcal{D}} = \mathfrak{a}_{\mathcal{C} + \mathcal{D}}$, where the left side is the usual ideal product.

Proof. For nonidentity factors, [lemma 29.2](#) gives a depth at least a sum of one value in $\mathcal{C}$ and one in $\mathcal{D}$, hence a value in $\mathcal{C} + \mathcal{D}$. Conversely, if a target has depth $\delta \in \mathcal{C} + \mathcal{D}$, write $\delta = \alpha + \beta$ with $\alpha \in \mathcal{C}$, $\beta \in \mathcal{D}$ and apply [theorem 29.3](#). This proves equality of the commutator image with the subgroup $K_{\mathcal{C} + \mathcal{D}}$, and therefore equality with the generated subgroup.

The ideal-product inclusion follows from valuations of products and finite sums. For the reverse inclusion, an element of value $\delta = \alpha + \beta$ can be divided by a positive element of value α ; the quotient has value β . Thus it is already one product, not just a sum of products. $\square$

For example,

$$(K_{\geq \alpha}, K_{> \beta}) = K_{> \alpha + \beta}, \quad (K_{> \alpha}, K_{> \beta}) = K_{> \alpha + \beta}.$$

Divisibility of Γ lets one split any positive excess between the two strict thresholds. The open and closed endpoints matter: replacing all cuts by principal closed ones loses actual subgroup information.

30 Normal generation and the classification of normal subgroups

The arguments of this section concern conjugation by the *full* group G . Conjugating only by K does not allow an arbitrary real change of axis; this distinction is essential throughout.

Lemma 30.1 (Two conjugates fill a definite scalar interval). *Let $A = p(a, se_1)$, where $a^2 + s^2 = 1$, $s > 0$ and $a \geq 0$. If a unit quaternion (b, w) satisfies*

$$1 - 2s^2 \leq b \leq 1, \quad (30.1)$$

then $p(b, w)$ is a product of two G -conjugates of A .

Proof. Set $z = (a^2 - b)/s^2$. The inequalities say exactly $-1 \leq z \leq 1$. Choose unit vectors $u = e_1$ and $u' = ze_1 + \sqrt{1 - z^2}e_2$. The unit quaternions (a, su) and (a, su') are conjugates of (a, se_1) , and their product has scalar part $a^2 - s^2z = b$. Its imaginary norm equals $\|w\|$. Conjugating the product to (b, w) , and conjugating both factors at the same time, proves the result. If that norm is zero, the two quaternions with scalar part $b = \pm 1$ already coincide. $\square$

Lemma 30.2 (Finite powers enlarge the leading coefficient). *Let $q = (a, su)$ be a unit quaternion near one, with $s > 0$ and $\|u\| = 1$. For every ordinary integer $m \geq 1$ write $q^m = (a_m, s_mu)$. Then*

$$\text{st}(s_m/s) = m, \quad s_m > 0, \quad v(s_m) = v(s). \quad (30.2)$$

Proof. The scalar a commutes with su , so the ordinary finite binomial expansion is valid. The coefficient of u is

$$s_m = \sum_{j=0}^{\lfloor (m-1)/2 \rfloor} (-1)^j \binom{m}{2j+1} a^{m-2j-1} s^{2j+1}.$$

After division by s , the term with $j = 0$ has standard part m and every other term is infinitesimal. Positivity and the value equality follow. $\square$

Theorem 30.3 (Normal closure of one infinitesimal rotation). *For $I \neq U \in K$ of depth δ ,*

$$\langle\langle U \rangle\rangle_G = K_{\geq \delta}. \quad (30.3)$$

In fact every $V \in K_{\geq \delta}$ is a product of two G -conjugates of a suitable ordinary positive power U^m .

Proof. Normality of $K_{\geq \delta}$ proves one inclusion. For the other, take canonical unit lifts of U, V and let s, t be their imaginary norms. An identity V is immediate, so assume $t > 0$. Since $v(t) \geq v(s)$, equation (28.1) provides an integer M with $t/s \leq M$. Choose an integer $m > 2M$. By lemma 30.2, the imaginary norm s_m of the canonical lift of U^m satisfies $s_m/s > m/2 > M$. Thus $t < s_m$.

If (b, w) is the canonical lift of V , then $b > 0$ and

$$1 - b = \frac{t^2}{1 + b} \leq t^2 < s_m^2 < 2s_m^2.$$

So b lies in the scalar interval of lemma 30.1, applied to U^m . Hence V is a product of two conjugates of U^m , and belongs to the normal closure of U . $\square$

Proposition 30.4 (No uniform finite normal-generation width). *Fix $I \neq U \in K$. Although its normal closure is $K_{\geq v(U-I)}$, there is no integer L such that every member of this normal closure is a product of at most L conjugates of U or U^{-1} .*

Proof. Let q be the canonical lift of U . The Euclidean quaternion norm satisfies

$$\|q_1 \cdots q_L - 1\| \leq \sum_{j=1}^L \|q_j - 1\|$$

for unit quaternions, by telescoping and multiplicativity of the norm. Conjugates of q and q^{-1} all have the same distance from one. Finite products of such near-one lifts remain near one, so they are the canonical lifts of the corresponding matrix products. For any ordinary $m \geq 1$, [lemma 30.2](#) and $\|q - 1\|^2 = 2s^2/(1 + a)$ show

$$\text{st}\left(\frac{\|q^m - 1\|}{\|q - 1\|}\right) = m.$$

Taking $m > L$ violates the preceding norm bound for a product of at most L conjugates. Therefore U^m cannot be such a product. $\square$

30.1 The full normal-subgroup classification

Lemma 30.5 (Abstract simplicity of the real rotation group). *The abstract group $\text{SO}(3, \mathbb{R})$ is simple.*

Proof. This is classical; source 10 proves it with the lemmas above. Let a normal subgroup contain a nonidentity rotation, with unit lift (a, su) , $a \geq 0$, $s > 0$. By [lemma 30.1](#), it contains all rotations whose positive lift scalar b is sufficiently close to one. An arbitrary real rotation has a real axis and an ordinary real angle θ . For a sufficiently large ordinary integer m , the same-axis rotation by θ/m lies in this small interval, and the original rotation is its m -th power. Only ordinary real trigonometry appears in this real-field lemma. $\square$

Theorem 30.6 (All proper normal subgroups). *The proper normal subgroups of $G = \text{SO}(3, F)$ are exactly the subgroups $K_{\mathcal{C}}$, for upper segments $\mathcal{C} \subseteq \Gamma_{>0}$. Distinct segments give distinct subgroups. In particular, the proper normal subgroups form a chain, and K is the unique largest proper normal subgroup of G .*

Proof. Suppose first that $N \trianglelefteq G$ and $N \subseteq K$. Let $\mathcal{C}$ be the set of depths of nonidentity elements of N . If $\delta \in \mathcal{C}$, [theorem 30.3](#) gives $K_{\geq \delta} \subseteq N$, so every value at least δ also belongs to $\mathcal{C}$. Every positive value is attained by a rotation: take a near-one unit quaternion whose imaginary coordinate has that value. Thus $\mathcal{C}$ is upper and $N = K_{\mathcal{C}}$. The same observation proves that distinct segments give distinct subgroups.

Now suppose N contains $A \notin K$. Choose a unit lift (a, su) with $a \geq 0$. The scalar s is not infinitesimal; otherwise A would have standard part I . For every $V \in K$ with canonical lift (b, w) , $1 - b$ is infinitesimal while $2s^2$ has positive standard part, so $1 - b < 2s^2$. By [lemma 30.1](#), every $V \in K$ belongs to N . Hence $K \subseteq N$, and $\text{st}(N)$ is a nontrivial normal subgroup of $\text{SO}(3, \mathbb{R})$. By [lemma 30.5](#), $\text{st}(N) = \text{SO}(3, \mathbb{R})$. Together with $K \subseteq N$ this gives $N = G$. $\square$

Remark 30.7 (What is not classified). [Theorem 30.6](#) classifies subgroups normal in G . It does not classify all abstract subgroups of G , nor all subgroups normal only in K ; those problems are not settled by this argument. The non-claim of Part II that abstract subgroups are not classified ([S21 in section 37.2](#)) therefore stands. In particular, the results of [section 33](#) about homomorphisms of the full class group must not be asserted for arbitrary homomorphisms defined only on its infinitesimal kernel.

30.2 An exact quotient chart

By [proposition 18.2](#), the Cayley chart is a bijection $\mathfrak{m}^3 \rightarrow K$ with canonical lift $(1, x)/\sqrt{1 + x \cdot x}$, inverse [equation \(5.12\)](#) and transported product $x \star y = (x + y + x \times y)/(1 - x \cdot y)$; all denominators are units of $\mathcal{O}$.

Proposition 30.8 (Coordinates on every cut quotient). *For every upper segment $\mathcal{C}$, the quotient $K/K_{\mathcal{C}}$ has the exact coordinate model $(\mathfrak{m}/\mathfrak{a}_{\mathcal{C}})^3$ with multiplication induced by $\star$. The induced map is an isomorphism of groups, not just a bijection of underlying sets.*

Proof. The condition $UK_{\mathcal{C}} = VK_{\mathcal{C}}$ is equivalent to all entries of $U - V$ belonging to $\mathfrak{a}_{\mathcal{C}}$, because multiplication by the bounded matrices U^{-1} and U preserves that ideal condition. The Cayley formula and its inverse, with unit denominators, show that this is equivalent to coordinatewise congruence of the parameters modulo $\mathfrak{a}_{\mathcal{C}}$. The product law then descends, and its denominator remains invertible in $\mathcal{O}/\mathfrak{a}_{\mathcal{C}}$. $\square$

Thus the classification records actual nonlinear quotient groups, not merely a filtration whose successive vector-space layers happen to be known.

31 Perfect cut subgroups, the lower central series, and the perfect residual

Definition 31.1. For a convex subgroup $\Delta \leq \Gamma$ put

$$\mathcal{C}_{\Delta} = \{\gamma > 0 : \gamma \notin \Delta\}, \quad K_{>\Delta} = K_{\mathcal{C}_{\Delta}}.$$

Convexity means that $0 \leq \gamma \leq \delta \in \Delta$ implies $\gamma \in \Delta$. In particular a positive element outside Δ is larger than every positive element of Δ .

Theorem 31.2 (Classification of perfect ambient-normal subgroups). *For an upper segment $\mathcal{C} \subseteq \Gamma_{>0}$ the following are equivalent:*

- (i) $K_{\mathcal{C}}$ is perfect;
- (ii) $\mathcal{C} + \mathcal{C} = \mathcal{C}$;
- (iii) $\gamma \in \mathcal{C}$ implies $\gamma/2 \in \mathcal{C}$;
- (iv) $\mathcal{C} = \mathcal{C}_{\Delta}$ for a convex subgroup $\Delta \leq \Gamma$.

The subgroup Δ in (iv) is unique. Every such perfect subgroup has commutator width one, including the trivial group by convention.

Proof. The equivalence of (i) and (ii), and the commutator-width assertion, follow from [theorem 29.4](#) and the injectivity of $\mathcal{C} \mapsto K_{\mathcal{C}}$. If $\mathcal{C} + \mathcal{C} = \mathcal{C}$ and $\gamma = a + b$ with $a, b \in \mathcal{C}$, one of a, b is at most $\gamma/2$, so upward closure puts $\gamma/2$ in $\mathcal{C}$. Conversely, (iii) writes every $\gamma \in \mathcal{C}$ as $\gamma/2 + \gamma/2 \in \mathcal{C} + \mathcal{C}$; the reverse inclusion holds for every upper $\mathcal{C}$.

Assume (iii), and define

$$\Delta = \{0\} \cup \{\gamma > 0 : \gamma \notin \mathcal{C}\} \cup \{-\gamma : \gamma > 0, \gamma \notin \mathcal{C}\}.$$

It is symmetric and convex. To prove additive closure, take nonnegative a, b outside $\mathcal{C}$ and let $M = \max(a, b)$. If $a + b \in \mathcal{C}$, upward closure would give $2M \in \mathcal{C}$, and (iii) would give $M \in \mathcal{C}$, a contradiction. Thus the sum stays outside $\mathcal{C}$. The inequality $|x + y| \leq |x| + |y|$ handles arbitrary signs. Hence Δ is a convex subgroup, uniquely determined by its positive part. Finally, if Δ is a convex subgroup and $\gamma > 0$ lies outside it, then $\gamma/2$ cannot lie in it, since a subgroup is closed under doubling. Thus $\mathcal{C}_\Delta$ satisfies (iii). $\square$

The extreme cases are informative. For $\Delta = \{0\}$ the subgroup is all of K , which recovers [theorem 19.3](#). For $\Delta = \Gamma$ it is trivial. Between them, perfect normal subgroups are indexed by proper coarsenings of the natural valuation: in the ordered quotient Γ/Δ a value has positive image exactly when it belongs to $\mathcal{C}_\Delta$. Consequently $K_{>\Delta}$ is the rotation congruence kernel at the maximal ideal of that coarsened valuation, a concrete subgroup interpretation of the convex rank structure.

31.1 Lower central and derived series

Fix $\alpha > 0$. The lower central and derived series of $K_{\geq\alpha}$ are

$$\Lambda_1 = K_{\geq\alpha}, \quad \Lambda_{n+1} = (K_{\geq\alpha}, \Lambda_n); \quad K_{\geq\alpha}^{(0)} = K_{\geq\alpha}, \quad K_{\geq\alpha}^{(r+1)} = (K_{\geq\alpha}^{(r)}, K_{\geq\alpha}^{(r)}),$$

with $\Lambda_n = \Lambda_n(K_{\geq\alpha})$. At a limit ordinal either transfinite series is defined by intersection of its preceding terms.

Theorem 31.3 (Exact finite terms). *For all ordinary $n \geq 1$ and $r \geq 0$,*

$$\Lambda_n(K_{\geq\alpha}) = K_{\geq n\alpha}, \quad K_{\geq\alpha}^{(r)} = K_{\geq 2^r \alpha}. \quad (31.1)$$

For every integer $m \geq 2$, $K_{\geq\alpha}/K_{\geq m\alpha}$ is nilpotent of class exactly $m - 1$. The group $K_{\geq\alpha}$ itself is neither nilpotent nor solvable.

Proof. Induct on n and r using [theorem 29.3](#). Every positive value is attained, so $K_{\geq(n+1)\alpha}$ is a proper subgroup of $K_{\geq n\alpha}$ and $K_{\geq 2^{r+1}\alpha}$ a proper subgroup of $K_{\geq 2^r \alpha}$. All these subgroups contain nonidentity rotations. The quotient by $K_{\geq m\alpha}$ therefore kills the m -th lower central term but not the $(m - 1)$ -st. Neither infinite sequence reaches the identity at a finite step. $\square$

This is the exact form of the nilpotence of finite jets discussed after [theorem 23.1](#). Define

$$\Delta_\alpha = \{\gamma \in \Gamma : |\gamma| \leq n\alpha \text{ for some integer } n \geq 1\}, \quad (31.2)$$

the convex subgroup generated by α , and put $K_{\gg\alpha} = K_{>\Delta_\alpha}$. A value outside Δ_α is larger than every $n\alpha$, so this agrees with the formula for $K_{\gg\alpha}$ at the start of Part III.

Theorem 31.4 (The perfect residual and stage- ω stabilization). *For $\alpha > 0$,*

$$\Lambda_\omega(K_{\geq\alpha}) = K_{\geq\alpha}^{(\omega)} = K_{\gg\alpha}. \quad (31.3)$$

This is a perfect subgroup of commutator width one. Moreover $(K_{\geq\alpha}, K_{\gg\alpha}) = K_{\gg\alpha}$, so both transfinite series stabilize at their ω -th term. The subgroup $K_{\gg\alpha}$ contains every perfect subgroup of $K_{\geq\alpha}$.

Proof. The integer multiples $n\alpha$ and the subsequence $2^r\alpha$ are cofinal in each other. Equation (31.1) gives the two intersections as precisely the elements of depth at least every $n\alpha$. This is equivalent to depth strictly greater than every $n\alpha$, by increasing n by one, and thus gives $K_{\gg\alpha}$.

Perfectness follows from theorem 31.2. For the mixed commutator assertion let $\mathcal{C} = \mathcal{C}_{\Delta_\alpha}$. If $\delta \in \mathcal{C}$, then $\delta - \alpha \in \mathcal{C}$, since $\delta > (n+1)\alpha$ for every n . Therefore $[\alpha, \infty) + \mathcal{C} = \mathcal{C}$, and theorem 29.4 applies; this also covers $\mathcal{C} = \emptyset$.

If $Q \leq K_{\geq\alpha}$ is perfect, induction gives $Q \leq K_{\geq\alpha}^{(r)}$ for all r : from $Q \leq K_{\geq\alpha}^{(r)}$ one has $Q = (Q, Q) \leq K_{\geq\alpha}^{(r+1)}$. Thus $Q \leq K_{\gg\alpha}$. No normality assumption on Q is needed. $\square$

Definition 31.5. A group $\mathcal{T}$ is *residually nilpotent* when $\bigcap_{n \geq 1} \Lambda_n(\mathcal{T}) = \{1\}$. A positive value α is an *order unit* of Γ when $\Delta_\alpha = \Gamma$.

Theorem 31.6 (The universal residually nilpotent quotient). *The quotient $K_{\geq\alpha}/K_{\gg\alpha}$ is residually nilpotent. Every homomorphism from $K_{\geq\alpha}$ to a residually nilpotent group annihilates $K_{\gg\alpha}$ and therefore factors uniquely through that quotient. In particular,*

$$K_{\geq\alpha} \text{ is residually nilpotent} \iff \alpha \text{ is an order unit of } \Gamma. \quad (31.4)$$

Proof. Since $K_{\gg\alpha}$ lies in every lower central term, $\Lambda_n(K_{\geq\alpha}/K_{\gg\alpha}) = K_{\geq n\alpha}/K_{\gg\alpha}$, and the intersection of these is trivial by theorem 31.4. For a homomorphism $f : K_{\geq\alpha} \rightarrow \mathcal{T}$, the image $f(K_{\gg\alpha})$ is perfect. Any perfect subgroup Q of $\mathcal{T}$ lies in every lower central term of $\mathcal{T}$: if $Q \leq \Lambda_n(\mathcal{T})$, then $Q = (Q, Q) \leq (\mathcal{T}, \Lambda_n(\mathcal{T})) = \Lambda_{n+1}(\mathcal{T})$. Residual nilpotence therefore forces $f(K_{\gg\alpha}) = 1$. Finally $K_{\gg\alpha}$ is trivial exactly when there are no positive values outside Δ_α , because each such value is realized by a rotation. $\square$

There is also an exact separation statement. The intersection of the kernels of all homomorphisms from $K_{\geq\alpha}$ into nilpotent groups is $K_{\gg\alpha}$, and the same holds with “solvable” in place of “nilpotent”. The explicit quotients $K_{\geq\alpha}/K_{\geq n\alpha}$, which are nilpotent and hence solvable, separate every element outside $K_{\gg\alpha}$; conversely any image of the perfect subgroup $K_{\gg\alpha}$ in a solvable group is trivial.

This resolves the apparent conflict between finite nilpotent jets and a perfect infinitesimal group. The full kernel has no least positive depth, so its commutator factors may move to half the target depth. A fixed principal subgroup forbids that move below its threshold. In higher rank, some deeper scales nevertheless remain beyond every finite multiple of that threshold, and those scales form the surviving perfect residual.

32 Recovering the value group from the normal-subgroup lattice

This section uses that F is set-sized, so that the collection of subgroups of G is itself a set. Let $\mathcal{L}$ be the complete lattice of normal subgroups of G contained in K , ordered by inclusion. By theorem 30.6 it is the lattice of upper segments of $\Gamma_{>0}$. Arbitrary joins are unions, since in a chain of subgroups every union is a subgroup. An element A of a complete lattice is *compact* if whenever $A \leq \bigvee_{j \in J} B_j$, some finite subfamily already has a join above A ; this is a lattice-theoretic notion, not a topological one.

Proposition 32.1 (Intrinsic principal depths). *The nontrivial compact elements of $\mathcal{L}$ are exactly the principal subgroups $K_{\geq\alpha}$, $\alpha > 0$. On these elements*

$$(K_{\geq\alpha}, K_{\geq\beta}) = K_{\geq\alpha+\beta}, \quad K_{\geq\alpha} \subseteq K_{\geq\beta} \iff \alpha \geq \beta. \quad (32.1)$$

Consequently the abstract group G determines its ordered value group Γ up to ordered-group isomorphism.

Proof. If $K_{\geq\alpha}$ lies in a union of cut subgroups, a rotation of depth exactly α belongs to one member, and that member contains $K_{\geq\alpha}$ by [theorem 30.3](#). Thus $K_{\geq\alpha}$ is compact. If a nonempty upper segment $\mathcal{C}$ has no least element, then $K_{\mathcal{C}} = \bigcup_{\alpha \in \mathcal{C}} K_{\geq\alpha}$, but no finite subfamily has union $K_{\mathcal{C}}$, since a finite subfamily has a least selected value and $\mathcal{C}$ contains a smaller one. Hence $K_{\mathcal{C}}$ is not compact. This proves the characterization, and [equation \(32.1\)](#) follows from the cut calculus; it reverses the order on the positive cone.

The subgroup K is intrinsic to G : by [theorem 30.6](#) it is the unique largest proper normal subgroup. An abstract isomorphism of two such groups therefore preserves K , its normal lattice, the compact elements and their commutator operation. It recovers an order-preserving additive bijection of the positive value cones. The group completion of this cancellative cone is Γ . The case $\Gamma = 0$ is recognized by the absence of nontrivial compact elements. $\square$

This is a second-order invariant built from the whole normal-subgroup lattice. It should not be confused with a first-order interpretation without parameters, nor with a classification of fields by value group; different real closed fields can have the same value group. The stronger bi-interpretability theorem of Bays–Peterzil [9, Theorem 1.1], recalled in [section 23](#), has a different logical content; the proposition supplies an explicit normal-closure and commutator description of the value arithmetic.

33 The full surreal class and the universal set-sized quotient

33.1 Foundational conventions

We now work in a standard set/class foundation such as NBG, as allowed in [section 2](#). Every surreal number is a set-coded sign sequence, and the collection $\mathbf{No}$ of all of them is a proper class. A fixed finite matrix over $\mathbf{No}$ is a set, so $\mathrm{SO}(n, \mathbf{No})$ is a legitimate class group. A *class homomorphism* is a class function satisfying the group equation. Its target in the theorems below is a set in the *same* foundation. The real-closed-field structure of $\mathbf{No}$, its real subfield and its Conway normal forms are established background ([section 2](#); see also [12, Sections 2.2–2.3]). The natural value group is identified with $(\mathbf{No}, +)$ by $v(\omega^{-\gamma}) = \gamma$. These are Conway monomials, not values of an unrestricted analytic power function; a nonzero normal form has a leading monomial, and its value is the negative of the leading exponent.

Every result of [sections 28 to 31](#) stated for individual elements or individual class subgroups has the same finite proof over $\mathbf{No}$. In particular the normal closure of U means the class of finite products of conjugates of $U^{\pm 1}$, not an intersection quantified over a hypothetical set of all class subgroups. No “class of all classes” of subgroups is formed; the complete-lattice construction of [section 32](#) was deliberately restricted to set-sized fields. For class groups, the finite-depth formulas followed by the constant value $K_{\gg\alpha}$ at every stage from ω onward define the stabilized series directly; no recursion theorem for arbitrary sequences of classes is assumed.

Let $G_{\mathbf{No}} = \mathrm{SO}(3, \mathbf{No})$ and $K_{\mathbf{No}} = \ker \mathrm{st}$, with standard part defined entrywise as before. Writing $G_{\mathbf{No}}/K_{\mathbf{No}} \cong \mathrm{SO}(3, \mathbb{R})$ is convenient shorthand. It should not be read as forming a set whose elements are proper-class cosets: the quotient equivalence relation is *represented* by the set-valued map st .

33.2 Proper classes inside every positive value interval

Lemma 33.1 (Large bounded value intervals). *For every positive surreal δ , the interval $\{\gamma \in \mathbf{No} : 0 < \gamma < \delta\}$ is a proper class. More explicitly,*

$$\lambda \mapsto \frac{\delta}{\lambda + 2} \quad (\lambda \in \text{Ord}) \quad (33.1)$$

is an injective class map into that interval, where the ordinals are embedded in $\mathbf{No}$ and the displayed addition and division are field operations.

Proof. The embedded ordinals are strictly increasing. Adding the field element 2, taking reciprocals and multiplying by $\delta > 0$ gives a strictly decreasing family of distinct values, all positive and at most $\delta/2 < \delta$. If the interval were a set, its Hartogs ordinal would contradict the restriction of this injection to that ordinal. $\square$

For $\gamma > 0$ define the explicit small rotation

$$U_\gamma = \text{Cay}(\omega^{-\gamma} e_1). \quad (33.2)$$

Then $v(U_\gamma - I) = \gamma$, and, for $\gamma \neq \eta$,

$$v(U_\gamma U_\eta^{-1} - I) = \min(\gamma, \eta) \quad (33.3)$$

by the unequal-depth equality after [equation \(28.3\)](#). These rotations all have the *same axis*; no complicated configuration is needed to obtain a proper class of distinguishable scales.

Theorem 33.2 (Universal set-sized quotient of surreal rotations). *Let $\mathcal{T}$ be a set-sized group and $\varphi : G_{\mathbf{No}} \rightarrow \mathcal{T}$ a class homomorphism. Then $K_{\mathbf{No}} \subseteq \ker \varphi$, and there is a unique ordinary group homomorphism $\bar{\varphi} : \text{SO}(3, \mathbb{R}) \rightarrow \mathcal{T}$ such that $\varphi = \bar{\varphi} \circ \text{st}$.*

Proof. Suppose, for a contradiction, that $U \in K_{\mathbf{No}}$ has $\varphi(U) \neq 1$, and put $\delta = v(U - I) > 0$. Let $N = \ker \varphi \cap K_{\mathbf{No}}$, a class subgroup normal in $G_{\mathbf{No}}$. It contains no nonidentity element of depth at most δ . Indeed, if $V \in N$ had such a depth ε , [theorem 30.3](#), with its identical class proof, would give $K_{\geq \varepsilon} \subseteq \ker \varphi$ and hence $U \in \ker \varphi$.

For distinct $\gamma, \eta \in (0, \delta)$, [equation \(33.3\)](#) shows that $U_\gamma U_\eta^{-1} \notin N$. Thus $\varphi(U_\gamma) \neq \varphi(U_\eta)$. Combining [equations \(33.1\)](#) and [\(33.2\)](#) gives an injection from every ordinal into the set $\mathcal{T}$. Restricting to the Hartogs ordinal of $\mathcal{T}$ gives a contradiction. Consequently $K_{\mathbf{No}}$ is annihilated.

Define $\bar{\varphi}$ by restricting φ to the constant real rotations. For every $A \in G_{\mathbf{No}}$, $A \text{st}(A)^{-1} \in K_{\mathbf{No}}$, so $\varphi(A) = \bar{\varphi}(\text{st}(A))$. Surjectivity of standard part gives uniqueness. $\square$

The argument uses set/class size only at its last contradiction. Before that point it is finite quaternion algebra and normal generation. In particular the theorem is not an automatic-continuity result disguised by a topological assumption. No global choice is used: the injection [equation \(33.1\)](#) is explicit, and the Hartogs contradiction is choice-free.

Corollary 33.3 (Only one nontrivial set-sized image). *Every nontrivial set-sized homomorphic image of $G_{\mathbf{No}}$ is isomorphic to $\text{SO}(3, \mathbb{R})$. More precisely, the kernel of a nontrivial class homomorphism from $G_{\mathbf{No}}$ to a set-sized group is exactly $K_{\mathbf{No}}$.*

Proof. The factor $\bar{\varphi}$ of [theorem 33.2](#) is trivial or injective, by [lemma 30.5](#). In the nontrivial case its kernel is trivial, so the original kernel is $K_{\mathbf{No}}$ and the image is isomorphic to $\text{SO}(3, \mathbb{R})$. $\square$

Corollary 33.4 (Set actions and set-field representations). *Every class action of $G_{\mathbf{No}}$ on a set factors through st . Every finite-dimensional representation of $G_{\mathbf{No}}$ over a set-sized field factors through st . In particular no such action or representation is faithful.*

Proof. An action on a set $\mathcal{X}$ gives a class homomorphism to the set-sized group $\text{Sym}(\mathcal{X})$. A representation of finite dimension d over a set-sized field k gives one to $\text{GL}(d, k)$, which is also a set. Apply [theorem 33.2](#). The kernel $K_{\mathbf{No}}$ is nontrivial. $\square$

The faithful geometric action on $\mathbf{No}^3$ does not contradict this corollary: its acted-on space is a proper class. Likewise the defining three-dimensional representation over $\mathbf{No}$ does not have a set-sized coefficient field. The result draws a sharp distinction between a finite dimension and a set-sized carrier. The layer maps ℓ_γ of [theorem 23.1](#) are homomorphisms to the set $(\mathbb{R}^3, +)$, but they are defined only on $K_{\geq \gamma}$; the theorem makes no assertion about such maps ([remark 30.7](#)).

33.3 The cardinal obstruction in a fixed field

There is a set-sized precursor. Suppose F is a set-sized real closed extension of $\mathbb{R}$ and $\varphi : \text{SO}(3, F) \rightarrow \mathcal{T}$ does not annihilate a member of K of depth δ . Choose, for each $\gamma \in (0, \delta)_\Gamma$, a positive field element of value γ and its one-axis Cayley rotation. The same proof shows that their images are distinct. Therefore

$$|\mathcal{T}| \geq |(0, \delta)_\Gamma|. \quad (33.4)$$

Choice here is only ordinary set choice. This inequality does not in general force the kernel to contain K : the identity map of $\text{SO}(3, F)$ is a set-sized counterexample. Passing to the full $\mathbf{No}$ changes every nonempty bounded open value interval from a set to a proper class. That, not merely non-Archimedeaness, produces the universal factorization.

33.4 A perfect residual beyond every principal surreal threshold

Corollary 33.5 (No principal surreal infinitesimal group is residually nilpotent). *For every $\alpha \in \mathbf{No}_{>0}$, the group $K_{\geq \alpha} \leq G_{\mathbf{No}}$ has a nontrivial perfect residual $K_{\gg \alpha}$. Its lower central and derived series stabilize at that residual at stage ω .*

Proof. The value $\gamma = \omega\alpha$, a surreal product, satisfies $\gamma > n\alpha$ for every ordinary positive integer n . Thus $U_\gamma \neq I$ belongs to $K_{\gg \alpha}$. Apply [theorems 31.4](#) and [31.6](#), whose finite and countable-intersection arguments apply to these class subgroups as well. $\square$

This result concerns principal congruence subgroups. It does not conflict with the fact that the full $K_{\mathbf{No}}$ is perfect ([theorem 19.3](#)). Nor does [theorem 33.2](#) assert that every homomorphism defined only on $K_{\geq \alpha}$ or on $K_{\mathbf{No}}$ factors through a real group.

34 Surcomplex special unitary groups and higher-dimensional rotations

For any real closed F write $F(i) = F \oplus iF$ with $i^2 = -1$ and the conjugation $\overline{x + iy} = x - iy$; for $F = \mathbf{No}$ this is the surcomplex class $\mathbf{No}[i]$. Put

$$\text{SU}(n, F(i)) = \{V \in M_n(F(i)) : V^*V = I, \det V = 1\}.$$

All real and imaginary coordinates of a unitary matrix are bounded, so standard part gives $\text{st} : \text{SU}(n, \mathbf{No}[i]) \rightarrow \text{SU}(n, \mathbb{C})$. These groups are studied with this designated real form and conjugation, not with an arbitrary automorphic image of the surreal real axis.

34.1 The spin bridge for $\text{SU}(2)$

The representation [equation \(16.4\)](#) identifies unit quaternions over F with $\text{SU}(2, F(i))$; conversely a determinant-one unitary 2×2 matrix has that form, by equating its adjugate inverse with its conjugate transpose. With [theorem 5.5](#) this gives an exact central double cover

$$1 \longrightarrow \{\pm I\} \longrightarrow \text{SU}(2, F(i)) \longrightarrow \text{SO}(3, F) \longrightarrow 1.$$

Let $\widetilde{K}_F = \ker(\text{st} : \text{SU}(2, F(i)) \rightarrow \text{SU}(2, \mathbb{C}))$. The spin map restricts to an isomorphism $\widetilde{K}_F \cong K_F$, because every small rotation has its unique near-one lift ([lemma 28.1](#)) and $-I$ is not near I . In particular $\widetilde{K}_F$ is perfect. Over the full surreal class the same holds for $\widetilde{K}_{\mathbf{No}}$.

Theorem 34.1 (The universal quotient for surcomplex $\text{SU}(2)$). *Every class homomorphism from $\text{SU}(2, \mathbf{No}[i])$ to a set-sized group annihilates $\widetilde{K}_{\mathbf{No}}$ and factors uniquely through standard part onto $\text{SU}(2, \mathbb{C})$.*

Proof. Let ψ be such a homomorphism and $\mathcal{T}_0$ its image, a subset of the set-sized target. The image $Z_0 = \psi(\{\pm I\})$ is central in $\mathcal{T}_0$. Composing with $\mathcal{T}_0 \rightarrow \mathcal{T}_0/Z_0$ gives a homomorphism that factors through the spin quotient $\text{SO}(3, \mathbf{No})$. By [theorem 33.2](#) this induced homomorphism annihilates $K_{\mathbf{No}}$. Hence $\psi(\widetilde{K}_{\mathbf{No}}) \subseteq Z_0$. But $\widetilde{K}_{\mathbf{No}}$ is perfect and Z_0 is abelian, so the image is trivial. Restriction to the constant complex matrices and surjectivity of standard part give the unique factorization. $\square$

Both ingredients are necessary: the finite central quotient passes to rotations, and perfectness removes the possible central image left over after that passage.

34.2 Finite near-identity elimination

Lemma 34.2 (Near-identity Givens factorizations). *Fix an ordinary finite integer $n \geq 2$ and a real closed $F \supseteq \mathbb{R}$.*

- (i) *Every $A \in \text{SO}(n, F)$ with $\text{st}(A) = I$ is a product of at most $n(n-1)/2$ near-identity coordinate-plane rotations.*
- (ii) *Every $V \in \text{SU}(n, F(i))$ with $\text{st}(V) = I$ is a product of at most $n(n-1)/2$ near-identity embedded coordinate $\text{SU}(2)$ matrices.*

The same finite statements hold over the full surreal class.

Proof. For the real case, work down the columns, using rows j, k with $k > j$ to eliminate the entry below the pivot. The current pivot a is near one and the entry b to be eliminated is infinitesimal. Set

$$r = \sqrt{a^2 + b^2}, \quad W = \frac{1}{r} \begin{pmatrix} a & b \\ -b & a \end{pmatrix}.$$

This determinant-one orthogonal block is near I and sends $(a, b)^T$ to $(r, 0)^T$. Embedding it in rows j, k does not change already cleared columns. Since the current matrix remains near I , all subsequent pivots have the same property. After $n(n-1)/2$ such blocks the matrix is upper triangular and

orthogonal, hence diagonal with entries ± 1 . Its standard part is I , so it is exactly I . Inverting the elementary factors gives the asserted product.

In the complex case use

$$r = \sqrt{|a|^2 + |b|^2}, \quad W = \frac{1}{r} \begin{pmatrix} \bar{a} & \bar{b} \\ -b & a \end{pmatrix}. \quad (34.1)$$

Then $W^*W = I$, $\det W = 1$ and $W(a, b)^T = (r, 0)^T$, and the block is near I . Use a block for each pair $j < k$, even if $b = 0$: in that case it normalizes the phase of the pivot. As before the result is upper triangular and unitary, hence diagonal. Its first $n - 1$ diagonal entries are positive real numbers of modulus one, hence equal to one, and the last is one by the determinant condition. Reversing all blocks proves (ii). $\square$

There is no infinite elimination, no limit of products and no analytic inverse-function theorem in this lemma. Positivity ensures that every pivot normalization has a nonzero denominator.

Theorem 34.3 (Universal set-sized quotients in fixed finite rank). *For every fixed ordinary $n \geq 3$, every class homomorphism $\mathrm{SO}(n, \mathbf{No}) \rightarrow \mathcal{T}$ into a set-sized group factors uniquely through $\mathrm{st} : \mathrm{SO}(n, \mathbf{No}) \rightarrow \mathrm{SO}(n, \mathbb{R})$. For every fixed ordinary $n \geq 2$, every class homomorphism $\mathrm{SU}(n, \mathbf{No}[i]) \rightarrow \mathcal{T}$ into a set-sized group factors uniquely through $\mathrm{st} : \mathrm{SU}(n, \mathbf{No}[i]) \rightarrow \mathrm{SU}(n, \mathbb{C})$.*

Proof. In the orthogonal case, restrict the homomorphism to any coordinate copy of $\mathrm{SO}(3, \mathbf{No})$. By [theorem 33.2](#), every near-identity element of that copy is killed. Each coordinate-plane rotation in [lemma 34.2\(i\)](#) belongs to such a copy, by choosing any third coordinate distinct from its two coordinates; this is where $n \geq 3$ is used. Therefore every matrix in the kernel of standard part is a finite product of killed elements. In the unitary case, restrict to each coordinate $\mathrm{SU}(2, \mathbf{No}[i])$ and apply [theorem 34.1](#); part (ii) of the elimination lemma again shows that the entire kernel of standard part is killed. In both cases the real or complex constant inclusion is a section of standard part, and restriction to that section gives the unique factor. $\square$

Corollary 34.4. *For these dimensions, every class action of $\mathrm{SO}(n, \mathbf{No})$ or $\mathrm{SU}(n, \mathbf{No}[i])$ on a set, and every finite-dimensional representation over a set-sized field, factors through the corresponding ordinary compact group. No continuity or definability hypothesis is required.*

Proof. Use the set-sized targets $\mathrm{Sym}(\mathcal{X})$ and $\mathrm{GL}(d, k)$, as in [corollary 33.4](#). $\square$

Part III does not classify the normal subgroups of these higher-rank groups; finite centers and products of simple factors introduce additional structure. The theorem concerns the universal standard-part factorization, and the finite generator argument suffices for that. Part V (source 11) supplies that classification over $\mathbf{No}$: all proper normal subgroups when the Lie algebra is simple, which covers $\mathrm{SO}(n, \mathbf{No})$ with $n \geq 3$, $n \neq 4$, and $\mathrm{SU}(n, \mathbf{No}[i])$ with $n \geq 2$ ([theorem 42.3](#)), and, for semisimple groups such as $\mathrm{SO}(4)$, the subgroups of the infinitesimal kernel that are normal in the whole group ([theorem 42.4](#)).

35 Examples, sharp boundaries, and the circle

35.1 A rank-one workspace

Let $F = \mathbb{R}((t^{\mathbb{Q}}))$, with $v(t^q) = q$; real closedness of these Hahn workspaces is established background ([section 2](#); [12, Corollary 2.1]). Every $\alpha > 0$ is an order unit of $\mathbb{Q}$, so $K_{\gg \alpha} = \{I\}$ and $K_{\geq \alpha}$ is

residually nilpotent. Nevertheless $K_{\geq \alpha}$ is neither nilpotent nor solvable, its quotients $K_{\geq \alpha}/K_{\geq m\alpha}$ have every finite nilpotency class as m varies, and the entire kernel K is perfect. The principal groups become residually nilpotent by fixing a minimum allowed depth, not by making their commutators vanish. For instance, with $\alpha = 1$ a rotation of depth $7/2$ is visible in $K_{\geq 1}/K_{\geq 4}$ but not in a quotient whose cutoff is at most $7/2$. Every nonidentity member is visible in some finite nilpotent quotient because its rational depth is bounded above by an integer: the order-unit condition in concrete form.

35.2 A rank-two workspace with two principal behaviours

Take $\Gamma = \mathbb{Q} \oplus_{\text{lex}} \mathbb{Q}$, with the first coordinate dominant, and $F = \mathbb{R}((t^\Gamma))$. For $\alpha = (0, 1)$,

$$\Delta_\alpha = \{0\} \oplus \mathbb{Q}, \quad K_{\gg \alpha} = \{I\} \cup \{U \in K : v(U - I) = (a, b), a > 0\}.$$

This residual is nontrivial and perfect. A rotation of depth $(1, -100)$, for example, lies beyond every $(0, n)$ and survives every term of the lower central series of $K_{\geq (0,1)}$; its second coordinate does not change that. On the other hand $\beta = (1, 0)$ is an order unit: for any (a, b) , an ordinary integer $n > a$ gives $(a, b) < (n, 0)$. Therefore $K_{\geq (1,0)}$ is residually nilpotent. The same field supports both behaviours: residual nilpotence is a property of the chosen principal scale, not merely of the rank of the field.

35.3 No largest Archimedean scale

Let $\Gamma = \bigoplus_{j \geq 0} \mathbb{Q}e_j$ with finite supports and the largest nonzero index dominant. Given any positive α , all its multiples have support below some index r , and e_{r+1} exceeds every $n\alpha$. Thus no positive value is an order unit, and every principal $K_{\geq \alpha}$ has a nontrivial perfect residual. The full surreal value group has a stronger version of this behaviour: $\omega\alpha$ exceeds all ordinary multiples of any fixed $\alpha > 0$, and every interval $(0, \delta)$ contains a proper class of values. These are separate facts. The first causes nontrivial perfect residuals ([corollary 33.5](#)); the second prevents any set-sized image of the full rotation group from retaining a nonzero infinitesimal ([theorem 33.2](#)).

35.4 The abelian boundary: the theorem fails for the circle

The dimensional restrictions of [theorem 34.3](#) cannot simply be removed. The circle $\text{SO}(2, \mathbf{No}) \cong \mathbb{T}(\mathbf{No}) = \text{U}(1, \mathbf{No}[i])$ has a different answer. This uses one additional established construction: the canonical splitting `trigonometry:eq:circlesplit` of `trigonometry:thm:polar` in the trigonometry report [17],

$$\mathbb{T}(\mathbf{No}) \cong \mathbb{T}(\mathbb{R}) \times (\mathfrak{m}_{\mathbb{R}}, +), \quad u \mapsto (u_0, \ell_{\mathfrak{m}}(u)), \quad u_0 = \text{st}(u), \quad \ell_{\mathfrak{m}}(u) = -i \log(u/u_0). \quad (35.1)$$

Here $\mathfrak{m}_{\mathbb{R}}$ is the class of real infinitesimal surreals. The second coordinate is the infinitesimal angle; it is defined without choosing an ordinary argument. The logarithm is the infinitesimal Hahn logarithm $\log(1 + z) = \sum_{m \geq 1} (-1)^{m+1} z^m / m$, not a logarithm at an arbitrary surcomplex input. Source 10 recalls the proof: the supports of the powers of an infinitesimal are strongly summable by the positive-support lemma ([section 2](#)), so formal exp/log identities evaluate in a set-sized Hahn workspace containing the finitely many inputs. Conjugation commutes with these sums. Since $u/u_0 = (u/u_0)^{-1}$, the logarithm is purely imaginary. Formal logarithmic addition in this commutative group makes $\ell_{\mathfrak{m}}$ a homomorphism to $(\mathfrak{m}_{\mathbb{R}}, +)$, inverse on that factor to the infinitesimal

circle exponential. This is a support-based statement, not convergence in the full surreal fine topology.

Fix a positive surreal η . For $x \in \mathfrak{m}_{\mathbb{R}}$ let $\text{coef}_{\eta}(x) \in \mathbb{R}$ be the coefficient of the Conway monomial $\omega^{-\eta}$ in its normal form. Normal-form addition is coefficientwise [12, Theorem 2.6], so coef_{η} is additive.

Proposition 35.1 (A set-valued circle character detecting infinitesimals). *The map*

$$\chi_{\eta} : \mathbb{T}(\mathbf{No}) \longrightarrow (\mathbb{R}, +), \quad \chi_{\eta}(u) = \text{coef}_{\eta}(\ell_{\mathfrak{m}}(u)) \quad (35.2)$$

is a class homomorphism, vanishes on the ordinary unit circle, and is nonzero on the infinitesimal kernel. Consequently standard part is not the universal set-sized quotient of $\text{SO}(2, \mathbf{No})$ or $\text{U}(1, \mathbf{No}[i])$. The analogous factorization also fails for $\text{U}(n, \mathbf{No}[i])$ for every $n \geq 1$.

Proof. The composition in equation (35.2) is additive by the two preceding facts. For a constant ordinary unit, $u/u_0 = 1$, so it vanishes. For $u = \exp(i\omega^{-\eta})$, evaluated by the infinitesimal series, $\ell_{\mathfrak{m}}(u) = \omega^{-\eta}$ and $\chi_{\eta}(u) = 1$. For $\text{U}(n)$, compose χ_{η} with the determinant and test it on $\text{diag}(u, 1, \dots, 1)$. $\square$

This counterexample is a consequence of the already available torus splitting and normal forms, not an independent priority claim. Restricted to $\{u \in \mathbb{T}(\mathbf{No}) : v(u - 1) \geq \eta\}$, the character χ_{η} induces the canonical layer isomorphism of `trigonometry:rot:thm:graded` (Section 18 of the trigonometry report), the planar counterpart of the layer maps of theorem 23.1. The proposition observes that for the abelian circle this coefficient extends to a homomorphism on the whole group. It shows why semisimple directions behave differently: infinitesimal torus angles have additive coefficient functionals, whereas ambient normal generation in three-dimensional rotations forces a whole proper class of depth layers to survive whenever one survives. This is also the only part of Part III that needs infinitesimal formal exp/log; the universal quotient theorem and all exact commutator and normal-generation results are proved without it. Part V shows that this boundary is exact among compact connected groups: toral factors are the only obstruction to a universal set-sized quotient (theorem 46.5), and for $\text{U}(n, \mathbf{No}[i])$ the separating class quotient is $\text{U}(n, \mathbb{C}) \times \mathfrak{m}_{\mathbb{R}}$ (proposition 47.1).

Part IV

Computation, limits, and provenance

36 Exact computation and formalization routes

Sources 04 and 09 give complementary implementation plans, one for the geometry and one for the rotation group. Source 10 adds a formalization plan and constructive recipes for Part III (sections 36.5 and 36.6). All are printed; none describes existing software. Source 11 gives no formalization plan.

36.1 A coordinate-to-invariants algorithm (source 04)

For three proposed vertices P_1, P_2, P_3 on a sphere of center O and radius $R > 0$, an exact workflow is as follows.

1. Verify $\|P_i - O\|^2 = R^2$ and set $u_i = (P_i - O)/R$. Compute $x = u_2 \cdot u_3$, $y = u_3 \cdot u_1$, $z = u_1 \cdot u_2$, and $\tau = u_1 \cdot (u_2 \times u_3)$.
2. Check $\tau \neq 0$ for a proper triangle. Retain its sign for orientation, and put $d = |\tau|$. Verify the redundant identity $d^2 = 1 + 2xyz - x^2 - y^2 - z^2$ as a useful certificate.
3. Compute the side angles with $a = \text{atan2}(\|u_2 \times u_3\|, x)$ and cyclic variants; compute $\alpha = \text{atan2}(d, x - yz)$ and cyclic variants. All these calls have strictly positive first argument.
4. Compute $M = 1 + x + y + z$ and $E = 2 \text{atan2}(d, M)$. Return physical side lengths Ra, Rb, Rc and area $R^2 E$. Centers, polar vertices, and transport matrices are obtained from the explicit formulas above, with each denominator checked before division.

This is a mathematical algorithm over a field with exact operations, order decisions, positive square roots, and finite-angle evaluation. It is not a claim that every surreal input has a finite machine representation. A general Conway normal form has set-sized support, and a generic element need not come with effective coefficient or support oracles. An actual computer algebra implementation must name its representable subclass and its supported comparison procedure.

For exact finite descriptions, it is useful to separate an algebraic certificate from a displayed numerical approximation. For example, $\tau \neq 0$ can be certified by a leading coefficient even when $\text{st}(\tau) = 0$. Near tangency, an exact discriminant sign should not be replaced by an ordinary floating-point tolerance. Truncated Hahn expansions must retain enough support to decide the leading nonzero term of the predicate being tested; an unresolved cancellation is an unknown sign, not a zero.

36.2 An implementation hierarchy for rotations (source 09)

There is no finite data structure representing every surreal number with a decidable exact equality test. A practical rotation package must name its scalar backend. The finite algebra in this report works over any exact ordered field with the required square roots; analytic operations require more.

For a backend of rational functions in one formal positive infinitesimal, Cayley coordinates are especially convenient: composition, inverse, and the action on vectors are rational formulas. Unit quaternions can be kept projectively, avoiding square roots until normalization is needed. A bounded Puiseux or Hahn-series backend can additionally track leading exponents, standard parts, and finite-order BCH truncations. General supports need an explicit effective representation; a mathematical support theorem is not an algorithm for deciding equality of arbitrary input names.

A robust division of responsibilities is the following.

1. The scalar layer supplies exact arithmetic, order, positive square roots, and, when supported, valuation and standard part.
2. The finite rotation layer supplies matrix, quaternion, algebraic axis-angle, and Cayley representations, with all exceptional branches.
3. The supported-series layer supplies infinitesimal exp/log and BCH with declared error valuations and support certificates.
4. The phase layer distinguishes finite angles from the chosen global Ehrlich–Kaplan phase; it never treats an infinite argument as an uncentered convergent Taylor input.

The exact rational law on $\mathfrak{m}^3$ can often replace BCH entirely. BCH is useful for logarithmic generators and graded errors, not necessary merely to multiply two rotations.

36.3 A layered Lean development for the geometry (source 04)

At source 04’s pin, the repository’s `Surreal/Algebra/Geometry.lean` works with two-coordinate complexifications and field-valued moduli [18]. Its approach suggests a clean separation for the three-dimensional extension. The following are proposed layers, not assertions that these modules already exist or have been compiled. At the merge pin 50cb709 no such module exists.

Layer	Assumptions	Principal obligations
Vector algebra	Commutative ring	Dot/cross identities; determinants; Gram polynomials
Ordered geometry	Ordered field; positive square roots	Norm inequalities; projections; affine incidence; intersections
Orthogonal algebra	Real closed field when needed	Principal axes; axis–angle cosine/sine data; quaternion action
Spherical algebra	Unit vectors; nonzero determinant	Tangent and normal formulas; cosine/sine laws; reconstruction
Finite angles	Compatible restricted-analytic structure	Phase bijection; inverse branches; angle inequalities
Area and transport	Proper minor-arc polygons	Excess branches; finite additivity; edge transport; holonomy
Scale analysis	Valuation and standard part	Noncancellation; asymptotic bounds; degeneration certificates

An appropriate vector type is $\text{Fin}(3) \rightarrow F$ or a three-coordinate structure. Length should initially be an F -valued function, not a real-valued `NormedSpace` norm. Theorems about matrices and finite sums can be proved before introducing any surreal construction. Angles should not enter the algebraic spherical-law layer until the circle-phase bridge is available. Proper-class statements are then obtained externally from compatible set-sized workspaces, rather than by pretending that all of **No** is a small type in one universe.

The exceptional SSA family in section 9.2 is a useful specification test: a solver returning only a list of at most two triangles is mathematically incomplete unless it explicitly excludes the case $\cos b = \sin b \cos \alpha = 0$. Its output type should distinguish no solution, finitely many solutions, and a parametrized family.

36.4 A Lean development plan for the rotation group (source 09)

The repository’s formalization ledger is the authority for which existing statements have actually been matched to compiled Lean declarations [23]. This report does not add a compiled Lean development, and its proofs must not be marked proved in that ledger solely because they appear here or pass symbolic tests.

The natural finite-algebra modules would begin with F^3 , the hat map, and equations (3.5) and (3.6), then prove the distance-isometry theorem and axis theorem. Next come the Hamilton action, spin kernel and explicit surjectivity, the matrix Cayley equivalence, and the centralizer cases. A standard-part interface then gives the split extension and the kernel chart. Finite subgroup rigidity can be formalized using a finite sum and the positive matrix-root interface, without invoking Hahn series.

The torsion-free and unique-divisibility proofs are also finite algebra over a real closed extension of $\mathbb{R}$. The single-commutator construction reduces to one checked quaternion identity, positive

fourth roots, and the already proved transitivity theorem. These are particularly attractive next targets: they deliver significant group structure without depending on a general surreal analytic inverse function theorem.

For the analytic modules, formal noncommutative series must be evaluated only after establishing joint strong summability. The derivative theorem requires a fine Fréchet interface in three coordinates and local quadratic remainders; it cannot be obtained merely by substituting surreal symbols into a theorem whose hypotheses require a real Banach space. Semialgebraic fundamental groups are a further, independent formalization layer.

36.5 A formalization plan for Part III (source 10)

A formalization of Part III can separate four layers. The first is finite quaternion and matrix algebra over a real closed field: norm, spin, conjugacy, the near-identity lift, and the two-conjugate interval. The second is abstract natural-valuation algebra: finite ratios, depth invariance, ideals, upper segments, and the exact word-map image. The third is group structure: normal closures, cut classification, and the lower central and derived series. The last is explicitly size-sensitive: either NBG-style classes or a relative universe theorem with a stated large-interval hypothesis.

The proper-class theorem must not be translated into the false assertion that a large Lean type has no identity homomorphism into a set-sized group in a higher universe. A correct universe-relative version would say: if every relevant bounded interval contains more elements than the permitted target size, then any homomorphism to such a target kills the infinitesimal kernel. The cardinal inequality [equation \(33.4\)](#) is the clean bridge. The ambient universe and the permitted target universe must be named.

At source 10's pin this report recorded no Lean mapping for its Part II rotation statements, and none exists at [7b5f934](#) ([section 1.2](#)). The finite algebra of Part III can therefore be an additional formalization track; source 10 does not claim that existing repository code certifies its results. An implementation should also retain the distinction between the image of the commutator word map and the subgroup generated by that image; silently weakening the former would lose the constructive theorem.

36.6 Constructive recipes from Part III (source 10)

Factoring a target commutator. Given a target U of depth $\delta \geq \alpha + \beta$, extract its positive unit lift (c, z) , compute $\xi = \sqrt{(1-c)/2}$, choose any positive x of value α , and set $y = \xi/x$. Form the two orthogonal-axis unit quaternions of [theorem 29.3](#), calculate their commutator, and rotate its imaginary axis to $z/\|z\|$. Conjugating both factors by that same rotation finishes the exact factorization. The choices are algebraic; the recipe is not a claim that arbitrary surreal inputs have a computable finite encoding or a decidable valuation.

A finite normal-generation certificate. For $U, V \in K$ with $v(V - I) \geq v(U - I)$, extract the imaginary norms s, t of their canonical lifts. Find an ordinary integer M with $t/s \leq M$, and choose $m > 2M$. The positive imaginary norm s_m of U^m exceeds t . If b is the positive scalar lift of V , choose the relative axis dot product $z = (a_m^2 - b)/s_m^2 \in [-1, 1]$. The two factors of [lemma 30.1](#), followed by one simultaneous conjugation, express V as two conjugates of U^m , hence as at most $2m$ conjugates of U . The integer m must be allowed to depend on the target, by [proposition 30.4](#).

Separating an element by a nilpotent quotient. For $V \in K_{\geq \alpha} \setminus K_{\gg \alpha}$, find an ordinary m with $v(V - I) < m\alpha$. Then V survives in $K_{\geq \alpha}/K_{\geq m\alpha}$, a nilpotent group of class $m - 1$. No such integer exists when $V \in K_{\gg \alpha}$; the obstruction is exact, not an implementation failure.

36.7 The companion checks

Each source came with an exact SymPy script. All four are kept byte-identical, under the prefixed names given here.

`code/04-three-space-verify_identities.py` (source 04) checks selected identities and expansions: the vector and Gram identities, the spherical-excess polynomial, the quaternion holonomy numerator, the chart metrics, the local curvature coefficient, and the worked examples. Its recorded run, `data/04-three-space-verification.txt`, passed all 19 test groups. These checks can detect a sign, coefficient, or transcription error in the displayed formulas. They do not prove the analytic transfer theorem, establish Hahn summability, discharge every branch condition, or formally verify the geometry. Those distinctions are part of the mathematical specification, not merely software caveats.

`code/09-so3-verify.py` (source 09) checks the hat identities, quaternion multiplication and rotation formulas, orthogonality and determinant of the Cayley chart, its composition law, the scalar commutator identity used in [theorem 19.3](#), selected leading commutators, and the Jacobian calculation in an axis-adapted frame. It also checks all 576 products of the 24-element octahedral group and an exact rational averaging intertwiner. Its recorded run, `data/09-so3-verification.txt`, passed all 24 checks. Rational specialization tests supplement, but do not replace, symbolic identities. These checks can catch sign errors, factor-of-two errors, and incorrect matrix branches. They do not prove real closedness of **No**, the Neumann support lemma, a global topology theorem, or a Lean kernel judgment. Those claims rest on the mathematical proofs and explicitly imported results stated in the text.

Both recorded runs used Python 3.13.5 and SymPy 1.14.0. For this merge both scripts were rerun with Python 3.14.4 and SymPy 1.14.0. The rerun reproduced 19/19 groups and 24/24 checks; only the version line and the timing line differ. Neither script writes a file.

`code/10-rotation-quotients-verify.py` (source 10) runs 39 named exact checks: 12 groups of general polynomial or rational identities, eight ordinary-power specializations, 16 one-parameter commutator-depth examples, and three unitary Givens identities. The general identities include quaternion norm multiplicativity, [equation \(29.2\)](#), the spin matrix's scaled orthogonality, determinant and multiplication law, the hat-matrix bracket, and the Cayley inverse and product numerator. The one-parameter tests use rational unit quaternions with Cayley parameters t^a and t^b for $1 \leq a, b \leq 4$; exact orders at $t = 0$ give imaginary commutator depth $a + b$ and scalar-defect depth $2(a + b)$, as [lemmas 28.1](#) and [29.2](#) predict. No floating-point threshold is involved. Its recorded run, `data/10-rotation-quotients-verification.json` (Python 3.13.5, SymPy 1.14.0; requirements in `data/10-rotation-quotients-requirements.txt`), passed all 39 checks. For this addition the script was rerun on a copy with Python 3.14.4 and SymPy 1.14.0; it passed 39/39, and its report differs from the recorded one only in the Python version field. Unlike the other two scripts, it writes a JSON report, by default `verification.json` in the current directory, so it should be run with `-output` pointing elsewhere or on a copy. The delivered helpers `code/10-rotation-quotients-build.sh` and `.ps1` call the unprefixed names `verify.py` and `article.tex` in their own directory and overwrite their outputs in place; they do not run from the placed layout and are kept only as delivered. These tests can detect sign errors and algebraic slips. They do not establish a theorem for all cuts or all ordered fields, and they do not verify the finite-power bound for every integer, the subgroup classification, the transfinite intersections, the Hartogs argument or the universal factorization; those are proved in the text.

`code/11-compact-groups-verify.py` (source 11) runs 33 named exact checks with SymPy and the Python standard library. Five concern the SL_2 Weyl and diagonal factorizations, root

scaling by a square, and the root subgroup, as used in [theorem 48.1](#). One is the matrix commutator identity of [lemma 43.1](#). Eleven concern $\mathfrak{so}(3)$ and $\mathfrak{su}(3)$: bases, brackets and bracket-spanning certificates; for $\mathfrak{su}(3)$ they include both a spanning set of brackets and a spanning set of brackets with conjugates of one fixed nonzero Lie vector under explicit rational special unitary matrices, with certificate determinants 16 and 4608/625. Four concern a rational orthogonal Cayley commutator, whose normalized mixed derivative is the Lie bracket; four check the formal circle logarithm through a finite truncation; and eight enumerate the signed-permutation and Gaussian-unit counts of [theorem 49.3](#) for $n \leq 4$ (1, 4, 24, 192 and 1, 8, 96, 1536). Its recorded run, `data/11-compact-groups-verification_results.json` (Python 3.13.5, SymPy 1.14.0; the requirement `sympy>=1.12,<2` is in `data/11-compact-groups-requirements.txt`), passed all 33 checks. For this addition the script was rerun on a copy with Python 3.14.4 and SymPy 1.14.0; it passed 33/33, and its report differs from the recorded one only in the Python version field. By default the script writes `verification_results.json` next to itself, so it should be run with `-output` pointing elsewhere or on a copy. The delivered helper `code/11-compact-groups-build.sh` calls the unprefixed names `verify.py` and `article.tex` in its own directory; it does not run from the placed layout and is kept only as delivered. These are representative finite checks of the local mechanisms, not substitutes for the all-type Lie-algebra lemma. They do not verify class replacement, Nash implicit-function transfer, the normal-generation proofs, or novelty.

37 Non-claims and limits, by source

A merged report must not silently upgrade any source. This section records every limitation, disclaimer and priority caveat stated by any of the four manuscripts. Most also appear at the point of use.

37.1 Source 04 (Part I)

- G1. No priority claim is made, and no named open problem is claimed to be resolved. This covers in particular the uniform gnomonic estimate ([theorem 13.1](#)), the exceptional-family classification ([proposition 9.2](#)), and their combination into a surreal development. Repeated classical identities are not presented as new results merely because their arguments are surreal.
- G2. Established surreal analytic machinery and classical identities are cited: [1, 2, 3, 4, 6, 7]. The proofs supply the precise extensions used.
- G3. The symbolic checks test selected polynomial identities and formal coefficients only. They do not verify the whole development or its foundations ([section 36.7](#)).
- G4. No Lean formalization is claimed. The layers of [section 36.3](#) are proposals, not assertions that the modules exist or have been compiled.
- G5. Real-valued metric completeness, ordinary Riemann integration, and a global oscillatory differential equation are not imported without additional hypotheses.
- G6. The finite theory does not assert that no distinguished global sine or cosine exists ([remark 2.3](#)). It needs no phase for infinite inputs. The zero-centred sine series at ω is not strongly summable, whatever global definition is chosen.
- G7. The distance is field-valued, not an ordinary real-valued metric, and F^3 is not made a Banach space over $\mathbb{R}$.

- G8. The “double cover” of [section 5](#) is an algebraic two-to-one homomorphism, not a topological covering assertion in the fine topology.
- G9. The spherical-coordinate metric and Jacobian are finite identities in restricted-analytic coordinate derivatives. They are not an integral transformation theorem for every function on F^3 .
- G10. The minimizing property of minor arcs concerns finite chains of minor great-circle arcs. No supremum over arbitrary partitions, no real-valued length metric, and no unrestricted class of nondefinable curves is introduced.
- G11. Area is a finitely additive functional on geodesic polygons. The cap formula extends one specific analytic formula. It asserts neither that arbitrary subsets have a surreal measure nor that Darboux sums define it, and no countably additive area measure is inferred.
- G12. Derivatives in [section 12](#) are taken with respect to external chart or curve parameters. They are not an intrinsic derivation of **No**, and no existence theorem for arbitrary differential equations is invoked.
- G13. Parallel transport is canonical by explicit analytic continuation. It is unique within the definable restricted-analytic class, not among arbitrary intrinsically differentiable fields.
- G14. The normal-coordinate estimate [equation \(12.8\)](#) is absolute. It claims no small relative error when $p - q$ is exceptionally small.
- G15. The coordinate-to-invariants algorithm does not claim that every surreal input has a finite machine representation. An implementation must name its representable subclass and comparison procedure.
- G16. Completeness, ordinary topological compactness, existence for arbitrary smooth ODEs, and a supremum-based length for every curve are not claimed. Finite dimension does not permit unrestricted summation of arbitrary families of surreals. Ordinary-size angles do not imply ordinary-size arc lengths, and a global phase at infinite arguments would be an additional convention.
- G17. The repository comparison is not an exhaustive audit of every file, or of the published work on non-Archimedean geometry. The repository is a research source, not a substitute for published foundations.
- G18. No completed repository integration, general-purpose surreal computer algebra, or formal verification is asserted.

37.2 Source 09 (Part II)

- S1. The manuscript is an AI-assisted research exposition, subject to mathematical review. It makes no claim of new priority and no claim of complete machine-checked formalization.
- S2. No new priority is claimed for classical quaternion theory or for the perfectness of the infinitesimal rotation group. Bays–Peterzil record that every element is a commutator [[9](#), §3.1], and [theorem 19.3](#) gives an elementary exact construction of that fact.
- S3. The quaternion spin description, standard-part reduction of unit quaternions, infinitesimal exponential and logarithm, and valuation filtration in the surquaternion report, finite angles in the trigonometry report, and the foundations report’s distinction of strong summation from fine convergence are starting points, not discoveries of the source.

- S4. The localization result, [proposition 2.1](#), does not assert that the subfield is closed under strong sums, or that its intrinsic topology is the one induced from **No**.
- S5. The homogeneous-space statement [equation \(16.1\)](#) is algebraic and semialgebraic, not a statement about ordinary real manifolds.
- S6. $G(F) \cong \mathbb{P}^3(F)$ is a statement about F -points, not an isomorphism of algebraic varieties.
- S7. Connectedness and absolute simplicity of the algebraic group imply neither abstract simplicity of $G(F)$ nor connectedness of its fine topology.
- S8. Unique divisibility of K_F does not make it a $\mathbb{Q}$ -vector space.
- S9. [Theorem 20.2](#) gives conjugate realizations. It does not say that finite subgroups are literally constant matrices, and it says nothing about infinite or set-generated subgroups.
- S10. The dual number κ is formal, not a surreal infinitesimal. For an angular-velocity equation, existence and uniqueness of solutions is a separate question. The Berarducci–Mantova derivation is not the variable derivative of [section 25](#).
- S11. The BCH truncation valuations are finite-order error certificates, not convergence through all positive surreal tolerances. The support lemma does not assert that $n\gamma$ is cofinal.
- S12. [Theorem 23.1](#) gives an inclusion of filtration levels, not equality of valuations in every case. The BCH description does not make K pronilpotent in an unspecified topology.
- S13. The Bays–Peterzil bi-interpretability theorem is imported, not proved, and it concerns set-sized structures. It is not automatically transferred to the full class.
- S14. The Ehrlich–Kaplan phase is canonical for its specified normalization. This is not an assertion that every conceivable global phase convention agrees with it.
- S15. The identity fiber of the global rotation exponential is not a kernel, and that exponential is not a homomorphism on the additive Lie algebra.
- S16. No inverse function theorem is inferred from nonsingularity of [equation \(25.4\)](#), and the derivative formulas are not chain rules for a field derivation.
- S17. The size restriction in [theorem 26.1](#) cannot be dropped. The spin fundamental group $\mathbb{Z}/2\mathbb{Z}$ is semialgebraic, not the ordinary fundamental group of the intrinsic or fine space. Definable compactness is not ordinary compactness. The fine topology does not make $G(\mathbf{No})$ an ordinary real Lie group, and a “surreal universal cover” is meaningful only after a category is specified.
- S18. No finite data structure represents every surreal number with decidable equality, and a support theorem is not an equality algorithm.
- S19. No compiled Lean development is added. The proofs must not be marked proved in the formalization ledger solely because they appear here or pass symbolic tests.
- S20. The checks do not prove real closedness of **No**, the Neumann support lemma, a topology theorem, or a Lean kernel judgment ([section 36.7](#)).
- S21. There is no claim to classify all abstract subgroups or automorphisms, to construct a Haar measure on the full class, or to obtain a physical model merely from a change of scalars.
- S22. The repository was not rebuilt for the source, and the source changed no repository file.

37.3 Source 10 (Part III)

- C1. The proofs are mathematical proofs in an AI-assisted, unrefereed manuscript, not machine-checked proofs. No Lean proof and no independent referee check has been performed.
- C2. Commutator width one of the whole kernel K is already in the literature (Bays–Peterzil, proof of Claim 3.3 [9]) and in this report ([theorem 19.3](#)). The spin double cover, the standard-part splitting, the perfectness of K and the unequal-parameter scalar identity are not presented as new. D’Andrea–Maffei’s local openness of the commutator map [11, Theorem 2.5] is cited for historical comparison.
- C3. No named historical open problem is claimed solved, and priority for the new formulations is not certified. No identical formulation of the universal quotient theorem was located in the sources inspected, but this is not an exhaustive priority certification; failure to locate an identical statement is not a proof of priority.
- C4. The literature check concentrated on the primary works cited and on surreal-field background; it was not an exhaustive review of anisotropic algebraic groups. Broader searches were inconclusive, unrelated hits were not used as evidence, and no claim is made about the scope or non-applicability of general papers on orthogonal groups over rings that were surfaced without a content check.
- C5. [Theorem 30.6](#) classifies subgroups normal in the full G , not all abstract subgroups and not subgroups normal only in K . The universal factorization has the same domain restriction: nothing is claimed for homomorphisms defined only on $K_{\geq \alpha}$ or on $K_{\mathbf{No}}$ ([remark 30.7](#)).
- C6. The normal closure theorem and the cut classification are elementary structural theorems proved here; no general historical priority is asserted.
- C7. The value-group recovery [proposition 32.1](#) is a concrete elementary realization, not a first discovery of valued-field recoverability; Bays–Peterzil’s Theorem 1.1 is stronger. It uses the whole subgroup lattice, not a first-order formula without parameters, is not a classification of fields by value group, and is restricted to set-sized fields.
- C8. The circle and $U(n)$ counterexample ([proposition 35.1](#)) is a consequence of the established torus splitting and normal-form coefficient extraction, not an independent priority claim. It is a support-based statement, not convergence in the fine topology, and it is the only part that uses infinitesimal formal $\exp/\log$.
- C9. The 39 checks are finite polynomial and rational identities and one-parameter examples. They do not verify arbitrary real closed fields, all cuts, the finite-power bound for every integer, the classification, the transfinite series, the Hartogs argument, the universal factorization, or any Lean formalization ([section 36.7](#)).
- C10. The class theorems use NBG-style foundations. “Set-sized” means a set in the same foundation, not a higher-universe type. A Lean version must be universe-relative with a stated large-interval hypothesis and named universes; it must never assert that a large type has no homomorphism into a higher universe ([section 36.5](#)).
- C11. No class of all classes is formed. $G_{\mathbf{No}}/K_{\mathbf{No}} \cong \mathrm{SO}(3, \mathbb{R})$ is shorthand for representation by st , and the stabilized series over $\mathbf{No}$ are defined directly; no recursion theorem for sequences of classes is assumed.

- C12. The universal quotient theorem is not an automatic-continuity result, and it does not say that infinitesimals have no geometric effect: the faithful action on the proper class $\mathbf{No}^3$ and the defining representation over $\mathbf{No}$ remain.
- C13. The set-sized cardinal bound [equation \(33.4\)](#) does not force the kernel to contain K ; the identity map is a counterexample. Only ordinary set choice is used there, and no global choice is used in the class theorem.
- C14. The natural valuation is essential. Replacing it by an arbitrary coarsening would invalidate the normal-generation argument.
- C15. Only $\mathrm{SO}(n, \mathbf{No})$, $n \geq 3$, and $\mathrm{SU}(n, \mathbf{No}[i])$, $n \geq 2$, for a fixed ordinary n and the designated conjugation, are treated. The normal subgroups of these higher-rank groups are not classified, no uniform reductive classification is claimed, and the higher-rank factorization gives no higher-rank commutator width one at prescribed depths.
- C16. The constructive recipes of [section 36.6](#) are algebraic. They do not claim that arbitrary surreal inputs have a computable finite encoding or a decidable valuation.
- C17. The repository was read at [465a54b](#) through a connector; no clone or local build was made. The comparison was targeted, not line-by-line: this 71-page report was not read in full, and the trigonometry report's theorem inventory was read through its README. Source 10 does not claim that repository code certifies its results.
- C18. The questions of [section 38.1](#) are left open: K -normal subgroups and the set-sized images of K itself, exact word images in other anisotropic groups, a uniform reductive classification, and first-order accessibility of the lattice reconstruction.

37.4 The merge

- M1. The merge adds no mathematical result. Each statement comes from one of the sources, and each statement proved by both is marked in [section 1.1](#). Merge-written remarks only relate statements to each other or to other reports (for example the factor-two comparison after [theorem 23.1](#)). [Theorem 5.4](#) and [theorem 5.1](#) combine clauses that the two sources stated separately; the proofs printed are the sources' proofs.
- M2. The renamings of [section 1.4](#) change symbols, not statements.
- M3. [Theorem 5.5](#) is cited from the surquaternion report and not reproved. Its proof there is outside the scope of this report's review.
- M4. The report has not been refereed. None of its labels is mapped to a Lean declaration.
- M5. The addition of source 10 adds no mathematical result. Part III is source 10's; the recalls it shared with Part II are cited instead of printed ([section 1.1](#)). The remarks [remarks 23.2](#) and [23.3](#), the pointer after [theorem 19.3](#), and the comparisons with the trigonometry report in [section 35.4](#) relate statements; they change no statement or label of Parts I and II.
- M6. The renamings of [section 28.4](#) change symbols, not statements. The status paragraph in [section 38](#) re-scopes a direction of further work; it does not claim that any other direction has been addressed.
- M7. The addition of source 11 adds no mathematical result. Part V is source 11's; the results it shares with Part III and with the omnific reports are cited or printed once with every proving

source named (section 1.1). Sections 50.1 and 50.2, the one-line comparisons after Part V statements, the two paragraphs added to Part III and the status notes in sections 38 and 38.1 relate statements. They change no statement or label of Parts I–IV.

- M8. Part V is printed after Part IV so that no existing number changes. The renamings of section 39.3 change symbols, not statements. The wording of theorem 42.3 restores its intended meaning, a proper subgroup that may be a set or a class, and the change is disclosed after it.
- M9. C15 and C18 are source 10's statements about source 10, and they stand as such. Part V now supplies, for compact connected groups over **No**, the higher-rank normal subgroups and the uniform classification that they mention; the status notes in section 38.1 say what remains open. Every other non-claim is unchanged.

37.5 Source 11 (Part V)

- P1. The manuscript is AI-assisted, not independently refereed and not Lean-verified. No Lean source is supplied and no Lean build was performed. Its proofs are mathematical proofs, not machine-checked ones.
- P2. The 33 finite checks do not prove the Nash-transfer, normal-generation or class arguments. They do not verify class replacement, Nash implicit-function transfer, the normal-generation proofs, or novelty, and they are not substitutes for the all-type Lie-algebra lemma (section 36.7).
- P3. Historical priority is not certified, and no named longstanding open problem is claimed solved. The candidate-original package is narrowly the exact all-set invisible subgroup, the separating class quotient and the semisimplicity criterion; general priority for the supporting valuation normal-subgroup calculations is left open. Bibliographic searches and a targeted repository comparison cannot certify absence from the literature.
- P4. The repository comparison was targeted. The repository was read at 9693b28 through a connector; a local clone failed and was not used as evidence. The 306-kilobyte article, the 96-page predecessor and the other reports were not read line by line. No repository build and no complete review of its source code is claimed. Source 11's proofs do not rely on Part III's calculations, which it treats as unreviewed.
- P5. The $SO(3)$, $SO(n)$, $SU(n)$ and circle and $U(n)$ results of Part III are prior results, not claimed as new. Several special cases and much of the surrounding structure have published or repository precedents.
- P6. The central and semisimple decomposition of infinitesimals and valued-field interpretability are Bays and Peterzil's [9]; no new interpretability claim is made. The infinitesimal logarithm, local commutator openness [11], quantitative normal generation (Kramer, Nikolov and Segal [13]) and the signed-permutation classification are credited, not claimed. Kramer's ordinary and countable small-target questions are not identified with the proper-class question of Part V. The textbooks [15, 16, 14] are background references, checked through publisher records and not claimed as fully read.
- P7. The SL_n argument is the classical simplicity mechanism; its proper-class consequence is a contrast, not a new simplicity theorem. Determinant is not claimed to be a set-sized universal quotient, and its abstract set-valued characters are not classified.

- P8. Compactness is an assumption about the real group. $G = \mathbf{G}(\mathbf{No})$ is not asserted compact for the fine topology.
- P9. The number of conjugates in normal generation is not uniformly bounded, and no infinite or surreal-length word or infinite product is used; only ordinary finite powers are.
- P10. The segment theorems concern subgroups normal in the full ambient group. They do not classify subgroups normal only in μ_G , or arbitrary subgroups, and must not be used as a substitute for that problem.
- P11. The mixed commutator theorem concerns generated subgroups, not single-commutator images; neither local openness nor Part III's $\mathrm{SO}(3)$ result is substituted for it. The graded proposition describes the graded bracket only, not the whole group law.
- P12. A nontrivial perfect residual is not asserted in rank-one Hahn fields.
- P13. The actions corollary does not classify the abstract representations of G_0 .
- P14. The coefficient maps separate central infinitesimals but are not claimed to exhaust the abstract additive dual.
- P15. Universal does not mean faithful: the identity homomorphism detects every infinitesimal, but its target is a proper class.
- P16. “Points over $\mathbf{Oz}$ ” needs care for groups not defined over $\mathbb{Z}$; the compact collapse concerns entrywise integral matrices in a chosen realization. No independent novelty is claimed for the signed-permutation classification.
- P17. Exact normal closure of one element is false for an element supported in one simple factor of a semisimple group, and false for a central torus.
- P18. G/μ_S is shorthand for the class of pairs $G_0 \times \mu_Z$. No class of proper-class cosets and no product over all class homomorphisms is formed.
- P19. No completeness, saturation or topological convergence in $\mathbf{No}$ is imported; only finite first-order statements transfer. No global exponential on $\mathbf{No}[i]$, no extension of the logarithm to infinite angles, no infinite-dimensional Hilbert theory and no classification of arithmetic homomorphisms from $\mathbf{Oz}$ is used or established.
- P20. The finite-quotient triviality of compact connected groups is a familiar observation and is not the source of the all-set theorem.
- P21. Left open: an exact single-commutator theorem at two prescribed depths in all compact semisimple types; a small-observation classification for noncompact real reductive groups; a classification of μ_G -normal subgroups; and a hierarchy of small-target theorems for a fixed set-sized real closed Hahn field ([section 50.3](#)).

38 Conclusion and directions

The first two paragraphs are source 04's conclusion, the next five are source 09's, one status paragraph is written for the addition of source 10, and the rest is source 10's. Status notes written for the addition of source 11 (Part V) follow the first status paragraph and the questions of [section 38.1](#).

Three-dimensional surreal geometry has a large, rigorous core that is both familiar and genuinely more informative than its real standard-part picture. Affine incidence, orthogonality, quadrics,

rotations, and finite volumes are ordered-field algebra. Canonical finite angles supply the usual trigonometric language for all directions, even when coordinates and physical lengths are infinite. Spherical geometry then admits exact sine and cosine laws, reconstruction criteria, centers, a finite-polygon area functional, and a curvature–holonomy identity.

The indispensable extra care concerns branch choices, degeneracy, and the meaning of analysis. A zero standard part is not a zero determinant; a small determinant need not mean small spherical area; a locally constant, intrinsically differentiable function need not be globally constant; and a formal integral symbol is not a measure. With those distinctions made explicit, infinitesimal geometry becomes a calculus of exact scales rather than a collection of heuristic limiting pictures.

The correct analogue of the ordinary spatial rotation group is not a new multiplication invented for surreal coordinates: it is the positive-definite special orthogonal class group $\mathrm{SO}(3, \mathbf{No})$. Its axes, algebraic angles, quaternion double cover, and rational charts have exact finite formulas. Every matrix entry is bounded by one, but the entries can retain arbitrarily fine surreal information.

The strongest structural contrast with the ordinary group is the extension $K \rtimes \mathrm{SO}(3, \mathbb{R})$. The real quotient does not exhaust the geometry. The kernel has a full hierarchy of cross-product layers, yet the entire kernel is perfect and uniquely divisible. It contains no finite torsion and no nontrivial finite quotient. Finite subgroups have no genuinely new forms: finite averaging and polar normalization conjugate them to real subgroups.

Finite angles suffice to represent every rotation. The canonical global phase also exists, but its period class is $2\pi \mathbf{Oz}$, not just ordinary multiples of 2π . The resulting radial rotation exponential has explicitly described fibers and an exact Jacobian. Infinite generators can be critical or severely ill-conditioned without producing unbounded rotation matrices.

Finally, the phrase “like $\mathrm{SO}(3)$ ” has a category-dependent answer. The algebraic group is three-dimensional and connected. Over a set-sized real closed field it is semialgebraically connected and definably compact, with spin fundamental group $\mathbb{Z}/2\mathbb{Z}$. In the non-Archimedean intrinsic topology it is totally separated and not locally compact. In the full surreal fine topology, every set-sized subset is closed and discrete. None of these assertions contradicts the others.

Further work should therefore specify a category before posing a problem. Substantial directions include normal-subgroup classification compatible with convex valuations (now answered for subgroups normal in the full group; see the next paragraph); effective multi-scale rotation algorithms with support and conditioning certificates; integration of the finite proofs into the repository’s Lean development; and comparison of semialgebraic, valued-field, and fine analytic representation theories. We do not claim to classify all abstract subgroups or automorphisms, construct a Haar measure on the full class, or obtain a physical model merely from a change of scalars.

Status of the first direction after source 10. For subgroups normal in the full group $G(F)$, over a set-sized real closed field $F \supseteq \mathbb{R}$ with its natural valuation, the classification is [theorem 30.6](#): the proper normal subgroups are the cut subgroups $K_{\mathcal{C}}$, and the perfect ones correspond exactly to convex subgroups of the value group ([theorem 31.2](#)). What remains open is re-scoped as follows: subgroups normal only in K , abstract subgroups of $G(F)$ and automorphisms (the non-claim above stands), normal subgroups of $\mathrm{SO}(n)$ and $\mathrm{SU}(n)$ in higher rank, and the other directions listed, which source 10 does not address.

Status after source 11. Over $\mathbf{No}$, Part V classifies the proper normal subgroups of every compact connected group with simple Lie algebra, including $\mathrm{SO}(n, \mathbf{No})$ with $n \geq 3$, $n \neq 4$, and $\mathrm{SU}(n, \mathbf{No}[i])$ with $n \geq 2$, and the subgroups of the infinitesimal kernel normal in the whole group for every compact semisimple group ([theorems 42.3 and 42.4](#)). The higher-rank item above is therefore answered over $\mathbf{No}$ for subgroups normal in the full group. Part V states its results over $\mathbf{No}$ only;

source 11 notes that its local normal-generation proof also applies over a fixed set-sized real closed Hahn field but does not state the classification there. Subgroups normal only in the infinitesimal kernel, abstract subgroups and automorphisms remain open in every type.

The exact identity $\{(x, y) : x \in K_{\mathcal{C}}, y \in K_{\mathcal{D}}\} = K_{\mathcal{C}+\mathcal{D}}$ turns infinitesimal rotation theory into a calculus of valuation cuts. Normal generation makes those cuts intrinsic to the ambient abstract group, and convex subgroups identify exactly which cuts are perfect. Principal groups retain a perfect residual precisely beyond all finite multiples of their initial scale. These statements make precise the difference between nilpotent finite jets, residual nilpotence in one scale, and perfectness across arbitrarily larger infinitesimal magnitudes.

For the full surreal class an additional size mechanism appears. A normal kernel that misses even one infinitesimal rotation must distinguish a proper class of smaller valuation depths, which a set-sized target cannot do. Standard part is therefore universal among all set-sized group images of the stated surreal and surcomplex semisimple matrix groups. This is an algebraic obstruction, not a choice of topology and not a statement that infinitesimals have no geometric effect; their full nontrivial action remains available on proper-class spaces and over proper-class scalar systems.

38.1 Questions left open by source 10

The four questions below are source 10's. Source 11 (Part V) bears on the first three, and the status note after each of them is written for its addition.

Normal subgroups of the infinitesimal group itself. Part III classifies G -normal subgroups inside K , not all K -normal subgroups. The leading-axis action of an infinitesimal conjugator is much more restricted than that of a macroscopic rotation. A full intrinsic normal lattice of K , and a corresponding analysis of its own arbitrary set-sized images, require additional ideas; the universal quotient theorem for G does not answer them. *Status after source 11:* still open, in every compact type. Part V also classifies only subgroups normal in the full group, it lists the μ_G -normal subgroups among its open problems (section 50.3), and its theorems concern homomorphisms of the full group, not homomorphisms defined only on μ_G .

Exact word images in other anisotropic groups. The quaternion scalar identity reduces an exact commutator equation in $\mathrm{SO}(3)$ to one real scalar constraint. It is natural to ask for exact mixed-cut word images in other compact real forms, not merely inclusion bounds or local openness. The higher-dimensional universal factorization needed only generation by rank-one blocks, and does not itself provide higher-dimensional commutator width one at prescribed depths. *Status after source 11:* partly addressed. For every compact connected semisimple group the *generated* mixed commutator subgroups are exact, $(\mu_{\geq\alpha}, \mu_{\geq\beta}) = \mu_{\geq\alpha+\beta}$ and $(\mu_{\mathcal{C}}, \mu_{\mathcal{D}}) = \mu_{\mathcal{C}+\mathcal{D}}$ (theorem 43.2 and corollary 43.3). Exact single-commutator images at prescribed depths, and commutator width, remain open outside $\mathrm{SO}(3)$; source 11 lists the single-commutator theorem as open.

A uniform reductive classification. Theorems 33.2 and 34.3, together with proposition 35.1, suggest a structural distinction between semisimple and toral factors of compact real algebraic groups. A general statement must account for finite central overlaps, specify the real form, and state its class foundation. Only the orthogonal and special unitary cases above are proved, not a theorem about every reductive class group. *Status after source 11:* answered for connected real linear algebraic groups with compact connected real points (theorem 39.1). The invisible subgroup is the derived infinitesimal subgroup μ_S , the central torus contributes the separating factor μ_Z , and a universal set-sized quotient exists exactly for semisimple groups. The statement accounts for finite central overlaps through the isogeny $\mathbf{S} \times \mathbf{Z} \rightarrow \mathbf{G}$, uses the compact real form with its designated conjugation, and is stated in NBG with choice. Still open: noncompact real reductive groups, which

source 11 also lists as open, and groups whose real points are not connected.

Which quotient information is first-order accessible? The compact elements of the whole normal-subgroup lattice recover value addition, by a construction that quantifies over subgroups. Relating this reconstruction to definable data, without replacing it by the already stronger published field-interpretation theorem, is a separate model-theoretic task.

Part V

Surreal compact groups: small representations and normal structure

Part V is source 11, *Small Representations and Normal Structure of Surreal Compact Groups: a semisimplicity criterion, valuation-cut classification, and contrasts with split groups and omnific lattices* (23 September 2026). It is printed after Part IV rather than before it, so that no section, theorem or equation number of Parts I–IV changes (section 1.1). Part V extends the class-group theorems of Part III from $\mathrm{SO}(3)$, $\mathrm{SO}(n)$ and $\mathrm{SU}(n)$ to every compact connected real algebraic group. For that class it answers the question “A uniform reductive classification” of section 38.1. Results that Parts II and III already prove are cited. Where source 11 proves a special case by a different method, both proofs are kept and the relation is stated.

A surreal matrix group is a proper-class object, but a permutation group on a set, a matrix group over an ordinary field, and an ordinary abstract group all have set-sized underlying collections. This raises a representation-theoretic question that does not arise for ordinary Lie groups:

Which distinctions between surreal group elements can be detected by an abstract homomorphism into a set-sized group? Is there one set-sized quotient through which every such observation factors?

No continuity, definability, measurability, or support-preservation hypothesis is imposed on these homomorphisms. The restriction concerns only the size of the target, so the answer is not a disguised continuity theorem.

39 Setting, the main theorem, and notation for Part V

39.1 Hypotheses and the main theorem

Throughout Part V, G is a finite-dimensional connected real linear algebraic group with compact connected $G(\mathbb{R})$. We evaluate its defining equations in the real closed surreal field and write

$$G = G(\mathbf{No}), \quad G_0 = G(\mathbb{R}), \quad \mu_G = \ker(\mathrm{st} : G \longrightarrow G_0).$$

Let S denote the connected semisimple derived algebraic group and let $Z = Z(G)^0$ denote the connected central torus. Set $\mu_S = S(\mathbf{No}) \cap \mu_G$ and $\mu_Z = Z(\mathbf{No}) \cap \mu_G$. Compactness here is an assumption about the *real* group. We do not assert that G is compact for the surreal fine topology. For $G = \mathrm{SO}(3)$ these are the groups $G_{\mathbf{No}}$, $\mathrm{SO}(3, \mathbb{R})$ and $K_{\mathbf{No}}$ of section 33.

Theorem 39.1 (Small-observation theorem). *Under these hypotheses, the following assertions hold.*

- (i) Every class homomorphism $\varphi : G \rightarrow \mathcal{T}$ to a set-sized group kills μ_S .
- (ii) There is a surjective class homomorphism

$$\varpi : G \longrightarrow G_0 \times \mu_Z \quad \text{with} \quad \ker \varpi = \mu_S.$$

Every nonidentity point of $G_0 \times \mu_Z$ is separated from the identity by a homomorphism to a set-sized group.

- (iii) A universal set-sized quotient of G exists exactly when $\dim Z = 0$, or equivalently when G is semisimple. In this case $\text{st} : G \rightarrow G_0$ is universal.
- (iv) The subgroup in (i) is intrinsically $\mu_S = (\mu_G, \mu_G)$.

As in Part III (section 28.4), the commutator (A, B) of two subgroups is the subgroup generated by single commutators, and $\{(x, y) : x \in A, y \in B\}$ is the set of single commutators. The distinction matters in the valuation results below. A universal set-sized quotient means a homomorphism to a set group through which every homomorphism to every set group factors uniquely. Assertions (i) and (ii) are theorem 46.4 with proposition 46.3, (iii) is theorem 46.5, and (iv) is proposition 46.1.

When G is semisimple, theorem 39.1 identifies all actions on all sets with ordinary actions of G_0 . When a central torus is present, arbitrarily fine central infinitesimals can be observed one coefficient at a time, and no set is large enough to collect all these observations into one universal group. Thus the torus is an exact obstruction, not merely one counterexample. For semisimple G the invisible subgroup is all of μ_G , in agreement with theorems 33.2, 34.1 and 34.3; for tori the separating maps are the characters of proposition 35.1. Section 50.1 states the agreement with Parts II and III point by point.

Logical dependency of the proofs. The proofs have three independent ingredients. A local Nash chart transfers finite real implicit-function assertions to the surreal field. Simplicity of the real Lie algebra makes a suitable word map submersive after division by its infinitesimal scale. Finally, every positive surreal interval contains sets of arbitrarily large cardinality, which forces collisions under any map to a fixed set. Only the last ingredient is specifically proper-class in nature. The central torus is handled by the strongly summable logarithm near one.

39.2 What Part V inherits and what it adds

Source 11 compared its results with the repository at commit 9693b28 (full hash in section B), read through a connector after a local clone failed. It read the repository catalogue, which then listed 51 reports, this report's README, and 10-rotation-quotients-SOURCE_AUDIT.md. It did not read the 306-kilobyte `article.tex` or the 96-page predecessor line by line. This report did not change between that commit and the placement of source 11, so source 11's descriptions of Part III ("Sections 28–35", a 96-page predecessor, the single-commutator theorem for $\text{SO}(3)$) are accurate.

The following are inherited and are not claimed as new by source 11. From this report: the valuation-cut calculus, normal closures, mixed single-commutator images, perfect residuals and the universal set-sized quotient for $\text{SO}(3, \mathbf{No})$, the extensions to $\text{SO}(n)$ and $\text{SU}(n)$, and the circle and $\text{U}(n)$ boundary (Part III). From the literature: local openness of the commutator map for compact semisimple groups [11]; the central and semisimple decomposition of infinitesimals and the model-theoretic recovery of valued fields [9]; small abstract quotients and quantitative normal generation in ordinary Lie and locally compact groups, including a theorem of Nikolov and Segal, surveyed by Kramer [13]; the infinitesimal surreal logarithm [12, 16]; the signed-permutation classification; and

the classical simplicity mechanism for special linear groups. The specific proper-class and set-sized classification is the subject of Part V; it is not a claim to have introduced infinitesimal Lie theory.

Source 11 did not see three items of the collection. They are credited where used. The omnific-groups report [21] was already in the tree at 9693b28 but was not yet listed in the catalogue that source 11 read. Its Part I contains the omnific points of compact groups used in [section 49](#), and its `ogl:el:prop:finiteside` is compared with Part V in [section 50.2](#). The omnific Diophantine report [22] proves the orthogonal count of [theorem 49.3\(i\)](#) as `odg:cor:orthogonal`. The trigonometry report's circle splitting [17] is [lemma 46.2](#).

Statement or ingredient	Treatment in Part V
Standard-part splitting, central decomposition, infinitesimal logarithm, signed-permutation counts	Established background, reproved only as needed; credited.
Nash normal generation in every compact simple type	A uniform route; the $\mathrm{SO}(3)$ case is theorem 30.3 , proved there by quaternions with a width bound.
Ambient-normal classification in simple type and in finite products	Extends Part III beyond $\mathrm{SO}(3)$, whose case is theorem 30.6 ; general priority left open.
Generated mixed commutator subgroups, series, perfect residuals	All semisimple types; weaker for $\mathrm{SO}(3)$ than the single-commutator results of Part III; general priority left open.
Exact invisible subgroup, separating class quotient, semisimplicity criterion	The main candidate-original contribution of source 11.
Triviality of set-sized images of $\mathrm{SL}_n(\mathbf{No})$	The classical simplicity mechanism; presented as a contrast.
Compact omnific collapse	Elementary; the orthogonal case is in the collection already.

The provenance comparison of source 11 was targeted rather than exhaustive. No identical all-compact-group formulation was found in the material it inspected. That is a limited novelty assessment, not a proof of priority.

39.3 Notation, and the renamings made for Part V

Part V keeps the conventions of [sections 1.4](#) and [28.4](#). The symbols of source 11 listed in the two tables below were changed so that each has one meaning in this report. No normalization changes. The new symbols are μ_G , μ_S , μ_Z , the depth $\nu(g) = v(g - I)$, and the subgroups $\mu_{\geq \alpha}$, $\mu_{> \alpha}$, $\mu_{\mathcal{C}}$ of [sections 41](#) and [42](#). For $\mathbf{G} = \mathrm{SO}(3)$ they are Part III's $K_{\mathbf{No}}$, the depth $v(U - I)$, and $K_{\geq \alpha}$, $K_{> \alpha}$, $K_{\mathcal{C}}$. The letters μ, ν also appear as local scalars in Part I, for example in its Ceva computation. Upper segments here are upward closed subclasses $\mathcal{C} \subseteq \mathbf{No}_{>0}$, possibly empty and possibly proper classes. The map gr_{α} of [proposition 43.5](#) has the normalization of Part II's layer map: for $U \in K_{\geq \alpha} \leq \mathrm{SO}(3, \mathbf{No})$, $\mathrm{gr}_{\alpha}(U)$ is the hat matrix of $\ell_{\alpha}(U)$, because $\mathrm{Cay}(t) - I = 2\hat{t} + \text{higher terms}$.

Here	Source 11	Reason
Nash chart σ , chart coordinates σ^{-1}	$q, c = q^{-1}$	q is a unit quaternion and $p(q)$ the spin map; c is the cosine coordinate of $\text{Rot}(n; c, s)$; source 11 also uses q for its quotient
class quotient ϖ , unitary quotient ϖ_n	q, Q_n	as above; Q_n is a kernel in ogl:el:prop:finiteside
$\omega^{-\gamma}$	t^γ	Part III writes Conway monomials over No this way; t is a Cayley coordinate
$\text{coef}_\gamma, \chi_{j,\gamma}$	$[t^\gamma], \lambda_{j,\gamma}$	the coefficient and the circle character χ_η of proposition 35.1
$g_0 = \text{st}(g)$	r_g, r	r also named $\dim \mathbf{Z}$, the number of simple factors and a matrix
$\dim \mathbf{Z}$; an index set $\mathcal{I}$ of simple factors	r	as above
$h^{1/m}$	$r_m(h)$	as for r_g
$\dim \mathfrak{g}$	d	$d = \sqrt{D}$ is the Gram quantity
$\text{diag}(c, c^{-1}, 1, \dots, 1)$	$d(c)$	as above
$\mathcal{T}$ (target group)	H	H_n is a half-turn
$(g, h), (A, B)$	$[g, h], [A, B]$	square brackets are Lie brackets
$\Lambda_n(\mathcal{Y}), \mathcal{Y}^{(n)}$	$\gamma_n K, D^n K$	γ is a value, D the Gram determinant
$\mathcal{Y}$ (a generic group)	K	K is the kernel of standard part
Q (a perfect subgroup)	P	P is a positive matrix; Q as in theorem 31.4

Here	Source 11	Reason (continued)
upper segment	upper cut	Part III's term (section 28.4)
$\mathbf{No}[i], \mathbf{U}(n, \cdot), \mathbb{T}(F)$	$\mathbf{No}(i), \mathbf{U}(n), T(F)$	as in section 28.4
$\ell_{\mathfrak{m}}, \exp(i\theta)$ on $\mathfrak{m}_{\mathbb{R}}$	L, E	the trigonometry report's names; E is the spherical excess
$\mathcal{X}$ (a set)	X	X is a Lie algebra element
$\mathcal{W}$ (a word map)	W	W is a Givens block
$B_j \in G_0$	k_j	k is a set-sized field
$\mathcal{A} \leq Z(G_0)$	$A \leq Z_0$	A is a group element; Z_0 is local to the proof of theorem 34.1
Ψ (a Nash map)	F	F is a real closed field
$\mathbb{K} \in \{\mathbf{No}, \mathbf{No}[i]\}$	F	as above; $\mathbf{No}[i]$ is not real closed
ϱ, ϱ_0 (real radii)	η, ϵ	η, ϵ are values
λ_-, λ_+ (real constants)	a, b	a, b are scales below
gr_α	ℓ_α	ℓ_γ is the layer map of theorem 23.1
$\mathcal{U} \subseteq G_0, \mathbb{B}_{\varrho_0} \subseteq \mathfrak{g}$	$U, B_\epsilon(0)$	U is an element of K , B a real rotation
$k, \mathcal{V}$ (field, vector space)	K, V	K as above; V is an element of K

39.4 Scalars, size and standard part

The scalars are those of [section 33.1](#). Every surreal number has a Conway normal form

$$x = \sum_{\gamma \in A} x_\gamma \omega^{-\gamma}, \quad x_\gamma \in \mathbb{R}, \quad A \subseteq \mathbf{No} \text{ a well-ordered set}, \quad (39.1)$$

with the Hahn rules for addition and multiplication [[12](#), Section 2.3], [[15](#), [16](#)]. The monomials $\omega^{-\gamma}$ are Conway monomials, with $\omega^{-\alpha-\beta} = \omega^{-\alpha}\omega^{-\beta}$; they are not obtained by substituting surreal

exponents into a chosen global exponential. In particular $\omega^{-\gamma}$ is infinitesimal for *every* $\gamma > 0$, even when γ is itself infinitesimal. The support of an individual number is a set; no proper-class family is summed. The valuation is $v(x) = \min \text{supp}(x)$, with $v(0) = +\infty$, and $\mathcal{O} = \{x : v(x) \geq 0\}$, $\mathfrak{m} = \{x : v(x) > 0\}$ are the finite and the infinitesimal numbers; next to surcomplex numbers $\mathfrak{m}$ is written $\mathfrak{m}_{\mathbb{R}}$, as in [equation \(35.1\)](#). Every $x \in \mathcal{O}$ is uniquely $\text{st}(x) + \varepsilon$ with $\text{st}(x) \in \mathbb{R}$ and $\varepsilon \in \mathfrak{m}$, and standard part is a ring homomorphism acting coordinatewise on vectors and matrices. For a vector or matrix X put $v(X) = \min_j v(X_j)$. Multiplication by an invertible real matrix preserves this value, and for $\alpha \in \mathbf{No}$ the vector $X/\omega^{-\alpha}$ is finite exactly when $v(X) \geq \alpha$. For surcomplex numbers the normal form has coefficients in $\mathbb{C}$, and $v(x + iy) = \min\{v(x), v(y)\}$. Conjugation and infinitesimal logarithms are used, but no global surcomplex exponential or phase.

The foundation is NBG with choice; [section 33.1](#) uses the same NBG-style conventions. Elements of $\mathbf{No}$, matrices and individual finite words are sets; subgroups may be classes, and a class homomorphism has a class graph whose restriction to a set has a set-sized image. “Killed by every set-sized homomorphism” is a pointwise property: for every set group $\mathcal{T}$ and every class homomorphism to $\mathcal{T}$ a stated equality holds. No product indexed by all class homomorphisms is formed, and no class whose elements are proper classes. Where a quotient would have proper-class cosets, an explicit class of set-coded representatives is used, or only equality of cosets is spoken of.

Source 11 proves that for every $\delta > 0$ and every set cardinal, the interval $(0, \delta)$ contains a set of more than that many elements, using the values $\delta/(2 + \xi)$ for the ordinals ξ below a larger cardinal. This is [lemma 33.1](#), with the same map, and is not repeated. It is stronger than the statement that $\mathbf{No}$ is a proper class: normal-generation arguments need many scales *below a prescribed positive depth*.

Choose a faithful real algebraic matrix representation of G . One may choose an orthogonal representation, but most arguments need only bounded matrix entries and bounded inverse entries. Ordinary compactness of G_0 supplies real bounds for both. Those bounds are first-order statements in real closed fields and therefore remain true over $\mathbf{No}$.

Proposition 39.2 (Standard-part splitting). *All entries of every $g \in G$ and g^{-1} are finite. Coordinatewise standard part gives a surjective group homomorphism*

$$\text{st} : G \longrightarrow G_0.$$

The inclusion of real matrices splits it, and multiplication gives $G = \mu_G \rtimes G_0$.

Proof. Transfer the real bounds just described. Polynomial equations defining the group are preserved by standard part. If the chosen affine description also uses the inverse of the determinant, boundedness of g^{-1} ensures that $\text{st}(g)$ is invertible: taking standard parts of $gg^{-1} = I$ gives an explicit inverse. Thus $\text{st}(g) \in G_0$. Matrix multiplication commutes with standard part, and real points are fixed. Finally $g = (g \text{st}(g)^{-1}) \text{st}(g)$ is the unique required decomposition. $\square$

For $G = \text{SO}(3)$ this is [theorem 18.1](#), and the same entrywise argument proves it.

Remark 39.3 (Transfer is not completeness). An ordered-field sentence with finitely many real parameters transfers to $\mathbf{No}$ by real-closed-field quantifier elimination. Alternatively, every finite tuple used in such a sentence may be placed in a set-sized real closed subfield containing its parameters. No sequential completeness, order completeness, saturation assumption, or topological convergence in $\mathbf{No}$ is imported.

40 Finite Nash charts and transfer at infinitesimal scales

A Nash map is a real analytic semialgebraic map on an open semialgebraic set. This category is used because its graphs can be interpreted in every real closed field, while real differential calculus supplies the local existence statements. Standard Nash-manifold and semialgebraic transfer background is in [14].

Write $\mathfrak{g} = \text{Lie}(G_0)$ and identify it with $\mathbb{R}^{\dim \mathfrak{g}}$ by a real basis. There is a Nash chart

$$\sigma : \mathbb{B}_{\varrho_0} \longrightarrow \mathcal{U} \subseteq G_0, \quad \sigma(0) = I, \quad D\sigma_0(X) = X, \quad (40.1)$$

on a real ball $\mathbb{B}_{\varrho_0} \subseteq \mathfrak{g}$, with inverse σ^{-1} . Here the derivative is viewed in the ambient matrix algebra. A real algebraic matrix group is a smooth real algebraic manifold; local algebraic-coordinate projection and the real implicit function theorem give such a chart. Only a fixed sufficiently small chart is used. Over **No** the symbols σ, σ^{-1} mean the interpretations of their semialgebraic graphs, not arbitrary extensions of real analytic functions.

Lemma 40.1 (Chart and matrix depths agree). *Over **No** the chart contains all of μ_G , and σ^{-1} identifies μ_G with $\mathfrak{m}^{\dim \mathfrak{g}}$. For $g \in \mu_G$,*

$$v(\sigma^{-1}(g)) = v(g - I).$$

*Moreover, for infinitesimal X , $\sigma(X) = I + X + O(\|X\|^2)$. The remainder statement means an actual inequality with a fixed real constant on a sufficiently small real neighbourhood, transferred to **No**.*

Proof. A smaller real neighbourhood of the identity lies in $\mathcal{U}$, and every infinitesimal point lies there. Both σ and σ^{-1} preserve the identity under standard part, so they give the stated bijection. The invertible derivative and bounded first derivatives of the chart and its inverse on smaller neighbourhoods give positive real constants λ_-, λ_+ with

$$\lambda_- \|\sigma^{-1}(g)\| \leq \|g - I\| \leq \lambda_+ \|\sigma^{-1}(g)\|.$$

All finite-dimensional coordinate norms are equivalent by real constants. Taking natural valuations yields the equality. The second-order estimate is ordinary Taylor's theorem on a smaller real neighbourhood; its finite inequality transfers. $\square$

Lemma 40.2 (Removal of a scale factor). *Suppose $\Psi(a, u)$ is Nash near $(0, u_0)$ and $\Psi(0, u) = 0$. Then $\Psi(a, u)/a$, for $a \neq 0$, extends to a Nash map near $(0, u_0)$. If $\Psi(a, b, u)$ vanishes whenever $a = 0$ and whenever $b = 0$, then $\Psi(a, b, u)/(ab)$ has a Nash extension. The statement holds coordinatewise for vector-valued maps.*

Proof. Analytic division by the coordinate a gives an analytic extension, with value $\partial_a \Psi(0, u)$ on the hyperplane. This extension is semialgebraic: off the hyperplane its graph is a quotient of semialgebraic graphs, and on the hyperplane the derivative is semialgebraic, since its defining limit is a first-order condition over the real field. Thus it is Nash. In the second assertion, divide by a first. The result still vanishes on $b = 0$, including $a = 0$ by continuity, so division by b applies. $\square$

Lemma 40.3 (Parametric local solvability). *Let $\Psi(p, y)$ be a Nash map near $(p_0, 0)$, where p ranges over a finite-dimensional real parameter space and y over $\mathfrak{g}$, with $\Psi(p_0, 0) = 0$ and $D_y \Psi(p_0, 0)$ invertible. There are positive real radii such that, for p close to p_0 and z sufficiently close to zero, the equation $\Psi(p, y) = z$ has a solution y close to zero, unique in a smaller prescribed neighbourhood. This assertion holds with the same real radii over **No**.*

Proof. Apply the ordinary real inverse function theorem to $(p, y) \mapsto (p, \Psi(p, y))$. Shrink to semi-algebraic neighbourhoods and real balls on which existence and uniqueness hold. The statement that each (p, z) in the chosen balls has exactly one corresponding y in the chosen ball is first-order, because the graph of Ψ is semialgebraic. Transfer that statement. No completeness-dependent analytic theorem is applied directly inside **No**. $\square$

40.1 Roots as a finite rescaling device

Lemma 40.4 (Unique infinitesimal roots). *For every ordinary integer $m \geq 1$, the map $h \mapsto h^m$ is a bijection of μ_G . If $h^{1/m}$ denotes the unique infinitesimal m th root, then*

$$\sigma^{-1}(h^{1/m}) = \frac{1}{m}\sigma^{-1}(h) + O(\|\sigma^{-1}(h)\|^2), \quad v(h^{1/m} - I) = v(h - I). \quad (40.2)$$

The constants in the remainder may depend on the fixed integer m .

Proof. In chart coordinates the power map is $P_m(X) = \sigma^{-1}(\sigma(X)^m)$, with $P_m(0) = 0$ and derivative mI . Apply lemma 40.3 to its local inverse. The inverse contains every infinitesimal input and has infinitesimal output, by standard part and local uniqueness. Any two infinitesimal roots lie in the same real uniqueness neighbourhood. Taylor's theorem for the real inverse has linear part I/m ; transfer gives equation (40.2). The quadratic remainder has strictly larger valuation than the nonzero linear term, which proves the depth assertion. $\square$

For $G = \text{SO}(3)$, unique divisibility of K is theorem 19.2.

Lemma 40.5 (From a small real output ball to a whole valuation ball). *Let $a > 0$ be infinitesimal and let $N \leq G$ be any subgroup. Suppose that for some positive real ϱ ,*

$$h \in \mu_G, \quad \|\sigma^{-1}(h)/a\| < \varrho \quad \implies \quad h \in N.$$

Then every $h \in \mu_G$ with $v(\sigma^{-1}(h)) \geq v(a)$ belongs to N .

Proof. The vector $z = \sigma^{-1}(h)/a$ is finite. Choose an ordinary integer m so large that $\|\text{st}(z)\|/m < \varrho/2$. By lemma 40.4,

$$\frac{\sigma^{-1}(h^{1/m})}{a} = \frac{z}{m} + O\left(\frac{\|\sigma^{-1}(h)\|^2}{a}\right).$$

The last term is infinitesimal, since $\sigma^{-1}(h)/a$ is finite and $\sigma^{-1}(h)$ is infinitesimal. Consequently $h^{1/m}$ satisfies the displayed hypothesis, and $h = (h^{1/m})^m \in N$. The exponent m is an ordinary finite integer, although it may depend on h/a . No infinite product is used. $\square$

41 Exact normal generation at one valuation scale

For $g \in \mu_G$ define the depth $\nu(g) = v(g - I)$, with $\nu(I) = +\infty$. For $\alpha > 0$ put

$$\mu_{\geq \alpha} = \{g \in \mu_G : \nu(g) \geq \alpha\}, \quad \mu_{> \alpha} = \{g \in \mu_G : \nu(g) > \alpha\}.$$

The group is suppressed from the subscript when unambiguous.

Lemma 41.1 (Elementary filtration bounds). *For $g, h \in \mu_G$ and $A \in G$,*

$$\nu(gh) \geq \min\{\nu(g), \nu(h)\}, \quad \nu(g^{-1}) = \nu(g), \quad \nu(AgA^{-1}) = \nu(g).$$

Consequently $\mu_{\geq \alpha}$ and $\mu_{> \alpha}$ are normal in G .

Proof. Use $gh - I = (g - I) + (h - I) + (g - I)(h - I)$ and $g^{-1} - I = -g^{-1}(g - I)$; all matrices involved have finite entries. For conjugation, $AgA^{-1} - I = A(g - I)A^{-1}$ gives one inequality, and conjugating back by A^{-1} gives the reverse one. The inverse argument is similar. $\square$

For $G = \text{SO}(3)$ these are [equation \(28.3\)](#).

Lemma 41.2 (The Lie-algebra spanning mechanism). *Assume $\mathfrak{g}$ is simple and nonabelian. For every nonzero $X \in \mathfrak{g}$,*

$$\text{span}_{\mathbb{R}}\{[\text{Ad}(B)X, Y] : B \in G_0, Y \in \mathfrak{g}\} = \mathfrak{g}. \quad (41.1)$$

If $s \in G_0$ is not central, then also

$$\text{span}_{\mathbb{R}}\{(\text{Ad}(BsB^{-1}) - I)Y : B \in G_0, Y \in \mathfrak{g}\} = \mathfrak{g}. \quad (41.2)$$

For any semisimple $\mathfrak{g}$, the span of all brackets $[X, Y]$ is $\mathfrak{g}$.

Proof. The span of the adjoint orbit of a nonzero X is a nonzero $\text{Ad}(G_0)$ -invariant subspace. Differentiating the adjoint action makes it an ideal, hence all of $\mathfrak{g}$. Bracketing that span with $\mathfrak{g}$ gives $[\mathfrak{g}, \mathfrak{g}] = \mathfrak{g}$, proving [equation \(41.1\)](#).

The span in [equation \(41.2\)](#) is likewise invariant. It is nonzero, since the kernel of the adjoint representation of a connected Lie group is its center. Simplicity again applies. Finally a semisimple Lie algebra is a direct sum of nonabelian simple ideals, each equal to its derived algebra. $\square$

Theorem 41.3 (Exact normal closure of an infinitesimal). *Suppose $\text{Lie}(G_0)$ is simple and nonabelian. If $I \neq g \in \mu_G$, then*

$$\langle\langle g \rangle\rangle_G = \mu_{\geq \nu(g)}. \quad (41.3)$$

Thus every infinitesimal at least as deep as g is a finite product of conjugates of g and g^{-1} in the full group.

Proof. The right-hand side is normal and contains g , so it contains the normal closure. We prove the reverse inclusion.

Let $\alpha = \nu(g)$, $a = \omega^{-\alpha}$ and $u = \sigma^{-1}(g)/a$. By [lemma 40.1](#), u is finite with nonzero standard part $X \in \mathfrak{g}$. Using [lemma 41.2](#), choose $\dim \mathfrak{g}$ pairs $(B_j, Y_j) \in G_0 \times \mathfrak{g}$ such that $[\text{Ad}(B_1)X, Y_1], \dots, [\text{Ad}(B_{\dim \mathfrak{g}})X, Y_{\dim \mathfrak{g}}]$ form a real basis of $\mathfrak{g}$.

For real variables a, u, y near 0, $X, 0$, form the Nash word

$$\mathcal{W}(a, u, y) = \prod_{j=1}^{\dim \mathfrak{g}} \left(B_j \sigma(au) B_j^{-1}, \sigma(y_j Y_j) \right), \quad (41.4)$$

with the factors in the displayed order and $(x, y) = xyx^{-1}y^{-1}$. The chart coordinate $\sigma^{-1}(\mathcal{W})$ vanishes at $a = 0$, so $\Psi(a, u, y) = a^{-1}\sigma^{-1}(\mathcal{W}(a, u, y))$ has a Nash extension by [lemma 40.2](#). The derivative of Ψ in y_j at $(0, X, 0)$ is $[\text{Ad}(B_j)X, Y_j]$. This follows either by differentiating the matrix word twice or by retaining its bilinear term:

$$(I + aX' + O(a^2), I + yY' + O(y^2)) = I + ay[X', Y'] + \text{terms of higher total degree}.$$

Thus $D_y\Psi(0, X, 0)$ is invertible.

Apply [lemma 40.3](#). For some real $\varrho > 0$, our infinitesimal a and $u \approx X$ have the following property: whenever $h \in \mu_G$ and $\|\sigma^{-1}(h)/a\| < \varrho$, a finite vector y solves $\Psi(a, u, y) = \sigma^{-1}(h)/a$. Since $\sigma(au) = g$, the word [equation \(41.4\)](#) then expresses h as a product of commutators, each involving a conjugate of g . Each such commutator lies in $\langle\langle g \rangle\rangle_G$.

Finally apply [lemma 40.5](#) with $N = \langle\langle g \rangle\rangle_G$. It includes every h with $v(\sigma^{-1}(h)) \geq \alpha$, which is exactly the desired valuation ball. $\square$

For $G = \mathrm{SO}(3)$ this is [theorem 30.3](#). Part III proves that case by the quaternion two-conjugate interval and obtains more: every element of $K_{\geq \delta}$ is a product of two conjugates of one power U^m . The proof above uses no quaternions and applies to every compact simple type, but gives no such width bound.

Remark 41.4 (What the theorem does not assert). The number of conjugates need not be uniformly bounded for all elements of $\mu_{\geq \alpha}$: the normalized target may have an arbitrarily large real standard part. The proof gives a fixed local word and then an ordinary finite power. It does not turn an infinite or surreal-valued word length into a legitimate group word. This is one reason to keep the rescaling lemma explicit. For $\mathrm{SO}(3)$, [proposition 30.4](#) proves that no uniform width exists.

Theorem 41.5 (Normal generation from a noncentral standard part). *Under the same simple-Lie-algebra hypothesis, if $g \in G$ has noncentral standard part, then $\langle\langle g \rangle\rangle_G = G$.*

Proof. Put $s = \mathrm{st}(g)$ and choose pairs (B_j, Y_j) using [equation \(41.2\)](#). Consider

$$(x, y) \mapsto \sigma^{-1} \left(\prod_{j=1}^{\dim g} (B_j x B_j^{-1}, \sigma(y_j Y_j)) \right)$$

near $(s, 0)$. Its derivative in y has columns $(\mathrm{Ad}(B_j s B_j^{-1}) - I)Y_j$, so it is invertible. The parametric local solvability lemma shows that the normal closure of $g \approx s$ contains all points in a fixed positive real neighbourhood of the identity over $\mathbf{No}$. In particular it contains μ_G and the real points of that neighbourhood.

In the connected real group G_0 , the subgroup generated by any identity neighbourhood is open. Its other cosets are open, so it is also closed and must be all of G_0 . Therefore the normal closure contains G_0 . By [proposition 39.2](#), it contains $G = \mu_G G_0$. $\square$

42 Normal subgroups and valuation segments

Definition 42.1. An *upper segment* is an upward closed subclass $\mathcal{C} \subseteq \mathbf{No}_{>0}$; the empty subclass is allowed. Define

$$\mu_{\mathcal{C}} = \{I\} \cup \{g \in \mu_G : v(g) \in \mathcal{C}\}.$$

An upper segment need not have a realized endpoint and need not be a set.

Theorem 42.2 (Ambient-normal infinitesimal subgroups). *If $\mathrm{Lie}(G_0)$ is simple and nonabelian, the subgroups of μ_G normal in G are exactly the $\mu_{\mathcal{C}}$ for upper segments $\mathcal{C}$. The segment is unique, and inclusions of subgroups correspond to inclusions of segments.*

Proof. The filtration bounds show that $\mu_{\mathcal{C}}$ is a subgroup normal in G . Conversely, let $N \leq \mu_G$ be normal in G and let $\mathcal{C}$ be the class of finite values $v(g)$ for $I \neq g \in N$. If $\alpha \in \mathcal{C}$, the exact

normal-closure theorem gives $\mu_{\geq \alpha} \subseteq N$. Every positive value β is realized by $\sigma(\omega^{-\beta}X)$ for a fixed nonzero real Lie vector X , so $\mathcal{C}$ is upward closed and $N = \mu_{\mathcal{C}}$. Realization of every value also proves uniqueness and the inclusion assertion. $\square$

Normality in the *full ambient group* is essential. This theorem does not classify subgroups normal only in μ_G , much less arbitrary subgroups of μ_G .

Theorem 42.3 (All proper normal subgroups in simple type). *Suppose $\text{Lie}(G_0)$ is simple and nonabelian, and let $Z(G_0)$ be the center of G_0 , a finite group. Every proper normal subgroup $N \triangleleft G$ (a set or a proper class) has a unique expression*

$$N = \mathcal{A}\mu_{\mathcal{C}}, \quad \mathcal{A} \leq Z(G_0), \quad \mathcal{C} \subseteq \mathbf{No}_{>0} \text{ an upper segment.} \quad (42.1)$$

Conversely, every subgroup displayed in equation (42.1) is proper and normal in G .

Proof. By theorem 41.5, every member of a proper normal subgroup has central standard part. Write such a member as $g = zh$, with $z \in Z(G_0)$ and $h \in \mu_G$. Real central elements remain central over $\mathbf{No}$ by transfer. If m is the finite order of z , then $g^m = h^m \in N$.

If $h \neq I$, lemma 40.4 gives $\nu(h^m) = \nu(h)$, and theorem 41.3 applied to h^m puts h in N . If $h = I$ this already holds. Hence $z = gh^{-1} \in N$ as well. It follows that

$$N = (N \cap Z(G_0))(N \cap \mu_G).$$

Apply theorem 42.2 to the second factor. The intersection of the two factors is trivial, by standard part, which proves uniqueness. Conversely these products are normal, and their standard parts lie in the proper subgroup $Z(G_0) \neq G_0$, so they are proper. $\square$

Source 11 states this theorem for “every proper class subgroup N ”. The intended reading is a proper subgroup, which may be a class; read literally, “proper class” would exclude the set-sized subgroups $N = \mathcal{A}$ with $\mathcal{C} = \emptyset$, which the theorem includes. The statement above is worded accordingly. For $G = \text{SO}(3)$ the center is trivial and the theorem is theorem 30.6. For $\text{SO}(n)$ with $n \geq 3$ and $n \neq 4$, and for $\text{SU}(n)$ with $n \geq 2$, it classifies all proper normal subgroups of $\text{SO}(n, \mathbf{No})$ and $\text{SU}(n, \mathbf{No}[i])$; the centers are $\{\pm I\}$ for even n , trivial for odd n , and the n th roots of unity, respectively.

42.1 Several simple factors and the absence of diagonal normals

Standard compact Lie structure gives a finite central isogeny from a product of compact connected almost simple groups onto any connected compact semisimple group. Such a map has invertible derivative and is a local Nash isomorphism. It therefore identifies the infinitesimal subgroup with the product of the factor infinitesimal groups, and the finite central kernel has no nonidentity infinitesimal point. These are familiar structural facts; compare [9, Section 5]. The cut calculation adds the following.

Theorem 42.4 (Product classification). *Let G be compact connected semisimple, with simple Lie factors indexed by a finite set $\mathcal{I}$. In the infinitesimal product identification $\mu_G \simeq \prod_{i \in \mathcal{I}} \mu_i$, every subgroup $N \leq \mu_G$ normal in G is uniquely*

$$N = \prod_{i \in \mathcal{I}} \mu_{i, \mathcal{C}_i}$$

for upper segments $\mathcal{C}_i$. Thus no diagonal coupling of two simple factors survives ambient normality, even when the factors are isomorphic.

Proof. It is enough to work in the finite central covering product. Take $n = (n_i)_{i \in \mathcal{I}} \in N$ and fix i with $n_i \neq I$. Let $\alpha_i = \nu(n_i)$ and let $X_i \neq 0$ be its normalized leading Lie vector. There is $B_i \in G_{i,0}$ with $(\text{Ad}(B_i) - I)X_i \neq 0$, since the Lie center is zero. Commuting n with the element that is B_i in factor i and the identity elsewhere gives an element of N supported only in factor i , with depth exactly α_i there. Its ambient normal closure contains the pure i th-factor element n_i , by [theorem 41.3](#).

Thus N contains each component of each of its elements. There are only finitely many factors, so N is the product of its intersections with the factors. Each intersection is classified by [theorem 42.2](#). Pulling through the infinitesimal isomorphism gives the result for G . $\square$

For $\text{SO}(4)$, which has two simple Lie factors, the ambient-normal infinitesimal subgroups are indexed by two segments, not one. Part III treats $\text{SO}(4, \mathbf{No})$ only through the universal quotient ([theorem 34.3](#)), which agrees: both segments must be killed by a set-sized image.

43 Exact mixed commutator subgroups

Lemma 43.1 (The matrix commutator bound). *For $g, h \in \mu_G$,*

$$\nu((g, h)) \geq \nu(g) + \nu(h).$$

Proof. The exact identity

$$(g, h) - I = (gh - hg)g^{-1}h^{-1} = ((g - I)(h - I) - (h - I)(g - I))g^{-1}h^{-1}$$

and finiteness of the inverse matrices give the inequality. $\square$

This is the identity of source 10's proof of [lemma 29.2](#), which Part III states for $\text{SO}(3)$; it holds verbatim for every G .

Theorem 43.2 (Two prescribed scales). *Let G be compact connected semisimple. For any positive surreal depths α, β ,*

$$(\mu_{\geq \alpha}, \mu_{\geq \beta}) = \mu_{\geq \alpha + \beta}. \quad (43.1)$$

The left-hand side is the generated commutator subgroup; the theorem does not assert equality with the set $\{(x, y) : x \in \mu_{\geq \alpha}, y \in \mu_{\geq \beta}\}$ of single commutators.

Proof. The inclusion from left to right follows from [lemma 43.1](#) and the filtration bounds. For the converse, choose $\dim \mathfrak{g}$ pairs (X_j, Y_j) whose brackets form a real basis of the semisimple Lie algebra, using [lemma 41.2](#). Set $a = \omega^{-\alpha}$, $b = \omega^{-\beta}$ and consider

$$\mathcal{W}(a, b, y) = \prod_{j=1}^{\dim \mathfrak{g}} (\sigma(aX_j), \sigma(by_jY_j)).$$

For real parameters near zero the chart coordinate $\sigma^{-1}(\mathcal{W})$ vanishes on both axes $a = 0$ and $b = 0$. Thus $\Psi(a, b, y) = \sigma^{-1}(\mathcal{W}(a, b, y))/(ab)$ has a Nash extension. Its y -derivative at $(0, 0, 0)$ has columns $[X_j, Y_j]$ and is invertible. By [lemma 40.3](#), all targets h for which $\sigma^{-1}(h)/(ab)$ belongs to a fixed small real ball are represented by this word.

The solved parameters y_j remain finite. Therefore $\sigma(aX_j) \in \mu_{\geq \alpha}$ and $\sigma(by_jY_j) \in \mu_{\geq \beta}$, and all those targets lie in the left-hand subgroup of [equation \(43.1\)](#). Apply [lemma 40.5](#) with scale ab to obtain every target of depth at least $\alpha + \beta$. $\square$

For upper segments put $\mathcal{C} + \mathcal{D} = \{\alpha + \beta : \alpha \in \mathcal{C}, \beta \in \mathcal{D}\}$. If either segment is empty, the sum is empty. Otherwise it is upward closed: an excess in a sum can be added to its first summand.

Corollary 43.3 (Segment calculus). *For compact connected semisimple G and upper segments $\mathcal{C}, \mathcal{D}$,*

$$(\mu_{\mathcal{C}}, \mu_{\mathcal{D}}) = \mu_{\mathcal{C} + \mathcal{D}}.$$

In particular, μ_G is perfect.

Proof. The commutator bound gives the forward inclusion for each generator and then for finite products. Conversely, if $\nu(h) = \gamma \in \mathcal{C} + \mathcal{D}$, choose $\alpha \in \mathcal{C}, \beta \in \mathcal{D}$ with $\alpha + \beta = \gamma$ and apply [theorem 43.2](#). For the final assertion use $\mathcal{C} = \mathcal{D} = \mathbf{No}_{>0}$ and split every positive depth into two equal halves. $\square$

Remark 43.4 (Position relative to local openness and to Part III). D’Andrea and Maffei prove local openness of the single-commutator map [[11](#), Theorem 2.5]. The argument above is deliberately weaker concerning single commutators, but it makes the two valuation constraints explicit and arbitrary. For $\mathrm{SO}(3)$, Part III proves the stronger single-commutator statements $\{(x, y) : x \in K_{\geq \alpha}, y \in K_{\geq \beta}\} = K_{\geq \alpha + \beta}$ and $\{(x, y) : x \in K_{\mathcal{C}}, y \in K_{\mathcal{D}}\} = K_{\mathcal{C} + \mathcal{D}}$ by quaternion algebra ([theorems 29.3](#) and [29.4](#)), and the perfectness of K is [theorem 19.3](#). Neither that special result nor ordinary local openness alone is silently substituted for [theorem 43.2](#).

43.1 The associated graded Lie algebra

Proposition 43.5 (Leading Lie coefficients). *For every $\alpha > 0$, the map*

$$\mathrm{gr}_{\alpha} : \mu_{\geq \alpha} \longrightarrow \mathfrak{g}, \quad \mathrm{gr}_{\alpha}(g) = \mathrm{st}(\omega^{\alpha}(g - I))$$

is a surjective homomorphism to the additive real Lie algebra, with kernel $\mu_{>\alpha}$. If $g \in \mu_{\geq \alpha}$ and $h \in \mu_{\geq \beta}$, then

$$\mathrm{gr}_{\alpha + \beta}((g, h)) = [\mathrm{gr}_{\alpha}(g), \mathrm{gr}_{\beta}(h)].$$

Proof. The chart estimate implies that the leading coefficient belongs to $\mathfrak{g}$, and $\sigma(\omega^{-\alpha}X)$ realizes every $X \in \mathfrak{g}$. The product term $(g - I)(h - I)$ has value at least $2\alpha > \alpha$ when both inputs have depth at least α , which proves additivity at that level. The kernel is exactly the strict inequality in depth. In the exact commutator identity the inverse factors have standard part I , so the coefficient at $\alpha + \beta$ is the ordinary matrix Lie bracket of the two leading coefficients. $\square$

For $G = \mathrm{SO}(3)$ this is [theorem 23.1](#) with its bracket, in the normalization recorded in [section 39.3](#). The proposition describes the graded bracket, not the entire noncommutative group law. The exact subgroup statements need the submersion argument, since a leading term alone does not control the higher-order remainder.

44 Derived series and perfect residuals

For a group $\mathcal{Y}$ write $\mathcal{Y}^{(0)} = \mathcal{Y}$, $\mathcal{Y}^{(n+1)} = (\mathcal{Y}^{(n)}, \mathcal{Y}^{(n)})$, and $\Lambda_1(\mathcal{Y}) = \mathcal{Y}$, $\Lambda_{n+1}(\mathcal{Y}) = (\Lambda_n(\mathcal{Y}), \mathcal{Y})$, as in [section 31](#).

Theorem 44.1 (Exact series). *For compact connected semisimple G and any upper segment $\mathcal{C}$,*

$$\Lambda_n(\mu_{\mathcal{C}}) = \mu_{n\mathcal{C}}, \quad \mu_{\mathcal{C}}^{(n)} = \mu_{2^n \mathcal{C}},$$

where $n\mathcal{C}$ is the n -fold sum. Furthermore

$$\bigcap_{n < \omega} \mu_{\mathcal{C}}^{(n)} = \mu_{\mathcal{C}_{\infty}}, \quad \mathcal{C}_{\infty} = \{\gamma > 0 : \gamma/m \in \mathcal{C} \text{ for every ordinary } m \geq 1\}. \quad (44.1)$$

This subgroup is perfect and contains every perfect subgroup of $\mu_{\mathcal{C}}$. Thus the transfinite derived series stabilizes by stage ω .

Proof. The finite identities follow inductively from [corollary 43.3](#). For an upper segment,

$$n\mathcal{C} = \{\gamma > 0 : \gamma/n \in \mathcal{C}\}. \quad (44.2)$$

Indeed, if γ is a sum of n members of $\mathcal{C}$, their minimum is at most their average, so the average is in $\mathcal{C}$. Conversely use n equal summands. The powers of two are cofinal among positive ordinary integers, so intersecting the derived segments gives exactly [equation \(44.1\)](#).

The segment $\mathcal{C}_{\infty}$ is closed under halving. Hence $\mathcal{C}_{\infty} + \mathcal{C}_{\infty} = \mathcal{C}_{\infty}$, and its subgroup is perfect by [corollary 43.3](#). If $Q \leq \mu_{\mathcal{C}}$ is perfect, induction gives $Q \leq \mu_{\mathcal{C}}^{(n)}$ for every n , without any normality assumption on Q . Thus $Q \leq \mu_{\mathcal{C}_{\infty}}$, which proves maximality and stabilization. $\square$

For $G = \mathrm{SO}(3)$ and $\mathcal{C} = [\alpha, \infty)$ this is [theorems 31.3](#) and [31.4](#): then $\mathcal{C}_{\infty} = \{\gamma : \gamma > n\alpha \text{ for every } n\}$ and $\mu_{\mathcal{C}_{\infty}} = K_{\gg \alpha}$. Part III also has commutator width one for these subgroups and the lower-central residual; the theorem above states the derived-series residual for every segment and every semisimple type.

Corollary 44.2 (No nontrivial solvable ambient segment). *Every nonempty upper segment $\mathcal{C} \subseteq \mathbf{No}_{>0}$ has nonempty $\mathcal{C}_{\infty}$. Hence every nontrivial ambient-normal infinitesimal subgroup in simple type is neither solvable nor nilpotent.*

Proof. Choose $\alpha \in \mathcal{C}$. Then $\gamma = \omega\alpha$ satisfies $\gamma/m > \alpha$ for every ordinary $m \geq 1$, so $\gamma \in \mathcal{C}_{\infty}$. The corresponding perfect subgroup is nontrivial because every positive depth is realized. A solvable group has no nontrivial perfect subgroup, and a nilpotent group is solvable. $\square$

This is the all-type and all-segment form of [corollary 33.5](#).

Example 44.3 (A ball whose finite derived stages never stabilize). For $\mathcal{C} = [\alpha, +\infty)$ with $\alpha > 0$,

$$\mu_{\geq \alpha}^{(n)} = \mu_{\geq 2^n \alpha}, \quad \mu_{\geq \alpha}^{(\omega)} = \{I\} \cup \{g : \nu(g) > m\alpha \text{ for all } m \geq 1\}.$$

Every finite inclusion is strict. The residual is nevertheless nontrivial and perfect, containing elements of depth $\omega\alpha$. These statements use multiplication and inequalities of surreal *values*; they involve no infinite group product.

Remark 44.4 (Why a rank-one model can behave differently). The nonemptiness argument uses a value larger than every ordinary multiple of a prescribed value, which is unavailable in an Archimedean value group. The local mixed-segment calculation remains valid over suitable real closed extensions, but the existence of this particular nontrivial residual must not be inferred in every rank-one Hahn field. The rank-one workspace of [section 35](#) is an example: there $K_{\gg \alpha} = \{I\}$.

45 Why every small representation kills semisimple infinitesimals

We now use the size of the full surreal value group. The proof gives a specific finite obstruction to any proposed nontrivial image of an infinitesimal; it does not argue merely from the size of G .

Lemma 45.1 (Separated infinitesimal points). *Fix $0 \neq X \in \mathfrak{g}$. For distinct positive surreal values β, γ ,*

$$\nu\left(\sigma(\omega^{-\beta}X)\sigma(\omega^{-\gamma}X)^{-1}\right) = \min\{\beta, \gamma\}.$$

Proof. Suppose $\beta < \gamma$. The chart expansion gives $\sigma(\omega^{-\beta}X) = I + \omega^{-\beta}X +$ a matrix of strictly higher value, and similarly for the other factor and its inverse. In the product, the term $\omega^{-\beta}X$ is the unique lowest-value contribution; the second scale is strictly deeper and cannot cancel it. The case $\gamma < \beta$ is symmetric. $\square$

This is the all-type form of [equation \(33.3\)](#).

Theorem 45.2 (Semisimple infinitesimal invisibility). *Let G be compact connected semisimple, $\mathcal{T}$ a set-sized group, and $\varphi : G \rightarrow \mathcal{T}$ an arbitrary class homomorphism. Then $\varphi(\mu_G) = \{1\}$.*

Proof. First suppose the real Lie algebra is simple. Fix $h \in \mu_G$, $h \neq I$, and let $\alpha = \nu(h)$. By [lemma 33.1](#), choose a set of more than $|\mathcal{T}|$ distinct values $\beta_j \in (0, \alpha)$. The points $g_j = \sigma(\omega^{-\beta_j}X)$ form a set for a fixed nonzero real X . Two have the same image under φ . Therefore $g_j g_k^{-1} \in \ker \varphi$, and by [lemma 45.1](#) its depth is $\min\{\beta_j, \beta_k\} < \alpha$.

The kernel is normal in the full group. Its containment of this element and [theorem 41.3](#) imply that it contains every element of depth at least that minimum, including h . Since h was arbitrary, all of μ_G is killed.

For a semisimple group, use a finite central covering product of compact almost simple groups and restrict φ along each factor homomorphism. The first part kills the factor's infinitesimal subgroup. These factor infinitesimals generate the entire infinitesimal subgroup downstairs, by the local isomorphism of the central isogeny. Thus φ kills μ_G . $\square$

Corollary 45.3 (Universal set-sized quotient in the semisimple case). *For compact connected semisimple G , every homomorphism $\varphi : G \rightarrow \mathcal{T}$ with $\mathcal{T}$ a set group has the unique form*

$$\varphi = \varphi_0 \circ \text{st}, \quad \varphi_0 : G_0 \rightarrow \mathcal{T}.$$

Proof. Take φ_0 to be the restriction to real points. Writing $g = h \text{st}(g)$ with $h \in \mu_G$ and applying [theorem 45.2](#) gives the factorization. Surjectivity of standard part makes it unique. $\square$

For $\text{SO}(3, \mathbf{No})$, $\text{SU}(2, \mathbf{No}[i])$, $\text{SO}(n, \mathbf{No})$ with $n \geq 3$ and $\text{SU}(n, \mathbf{No}[i])$ with $n \geq 2$ this is [theorems 33.2, 34.1 and 34.3](#). Those proofs are different: Part III kills the kernel of $\text{SO}(3, \mathbf{No})$ by quaternion normal generation and a Hartogs argument, passes to $\text{SU}(2)$ through the spin quotient and perfectness, and reaches higher rank by near-identity Givens factorization. The proof above treats every compact simple type directly. Both routes are kept.

Corollary 45.4 (Actions and linear representations). *Every action of G on a set is obtained uniquely by standard-part pullback of an action of G_0 . In particular, for any set-sized field k and any set-sized k -vector space $\mathcal{V}$, every abstract linear representation $G \rightarrow \text{GL}_k(\mathcal{V})$ factors through standard part.*

Proof. For an action on a set $\mathcal{X}$, the target $\text{Sym}(\mathcal{X})$ is a set group. For a linear representation, $\text{GL}_k(\mathcal{V})$ is also a set. Apply [corollary 45.3](#). $\square$

This statement is stronger than a classification of continuous finite-dimensional complex representations: it applies to arbitrary set dimensions and arbitrary abstract homomorphisms. It is weaker in another direction, since it does not classify the resulting abstract representations of the ordinary real group. For the groups of Part III, the finite-dimensional case is [corollaries 33.4](#) and [34.4](#).

46 Central tori and the exact obstruction

46.1 Splitting the infinitesimal kernel

For a compact connected group, standard structure theory gives a finite central isogeny

$$S \times Z \longrightarrow G, \quad (s, z) \longmapsto sz,$$

where S is semisimple, Z is a compact torus, and $S \cap Z$ is finite. The infinitesimal decomposition is a classical input; its local proof is included to avoid hidden saturation assumptions.

Proposition 46.1 (Central-semisimple infinitesimal product). *Multiplication gives an isomorphism*

$$\mu_S \times \mu_Z \xrightarrow{\sim} \mu_G. \quad (46.1)$$

The central factor commutes with all of G , and $(\mu_G, \mu_G) = \mu_S$.

Proof. The derivative of the multiplication map at the identity is the direct-sum isomorphism $\text{Lie}(S) \oplus \text{Lie}(Z) \rightarrow \mathfrak{g}$. Local Nash inversion gives a unique infinitesimal lift of every infinitesimal point. The finite kernel has no nonidentity infinitesimal point, which also shows injectivity. Centrality is preserved under real-closed-field transfer. Finally the central factor has no commutators, and μ_S is perfect by [corollary 43.3](#). $\square$

The decomposition itself is in Bays and Peterzil [9, Section 5]; it is not offered here as a new fact.

46.2 Logarithmic coordinates on the central infinitesimals

Let $\mathbb{T}(F) = \{z \in F(i) : z\bar{z} = 1\}$ be the algebraic circle over a real closed field. Over $\mathbf{No}$ its infinitesimal subgroup is

$$\mu_{\mathbb{T}} = \{z \in \mathbb{T}(\mathbf{No}) : z - 1 \in \mathfrak{m} + i\mathfrak{m}\},$$

the group $\mathbb{T}_{\mathfrak{m}}$ of `trigonometry:rot:eq:ellm`.

Lemma 46.2 (The infinitesimal circle logarithm). *For $z \in \mu_{\mathbb{T}}$ and $\theta \in \mathfrak{m}_{\mathbb{R}}$ the maps*

$$\ell_{\mathfrak{m}}(z) = -i \sum_{n \geq 1} \frac{(-1)^{n+1}}{n} (z - 1)^n, \quad \exp(i\theta) = \sum_{n \geq 0} \frac{(i\theta)^n}{n!}$$

are well-defined strong Hahn sums. They are inverse group isomorphisms $\ell_{\mathfrak{m}} : \mu_{\mathbb{T}} \rightarrow (\mathfrak{m}_{\mathbb{R}}, +)$ and $\exp(i \cdot) : (\mathfrak{m}_{\mathbb{R}}, +) \rightarrow \mu_{\mathbb{T}}$.

Proof. This is the infinitesimal factor of the trigonometry report's canonical circle splitting `trigonometry:eq:circlesplit` in `trigonometry:thm:polar` [17], restricted to $\text{st}(z) = 1$, where it is the map ℓ_m of `trigonometry:rot:eq:ellm`; Part III recalls it as [equation \(35.1\)](#). Source 11 gives the same proof as the one recalled there. The union of the finitely many input supports is a well-ordered subset of the positive value group; the positive-support lemma ([section 2](#)) makes the power-series families strongly summable, with set supports, so formal identities may be evaluated and regrouped coefficientwise. They show that the two series are inverse, that $\log(zw) = \log z + \log w$ for commuting near-one arguments, and that conjugation commutes with both series. Since $z\bar{z} = 1$, $\overline{\log z} = \log \bar{z} = \log(z^{-1}) = -\log z$, so $-i \log z$ is real and infinitesimal. Conversely $\exp(i\theta)\overline{\exp(i\theta)} = 1$. $\square$

The ordinary infinitesimal surreal logarithm and its Hahn interpretation are standard [12, 16]. No extension to infinite angles enters the quotient theorem.

Proposition 46.3 (Coefficient separation of a central torus). *After a choice of real torus coordinates,*

$$\mu_Z \simeq (\mathfrak{m}_{\mathbb{R}}, +)^{\dim Z}.$$

For each coordinate j and each positive surreal γ , coefficient extraction

$$\chi_{j,\gamma}(\theta_1, \dots, \theta_{\dim Z}) = \text{coef}_{\gamma}(\theta_j)$$

is a homomorphism to the set group $(\mathbb{R}, +)$. This family separates points. If $\dim Z > 0$, μ_Z is a proper class.

Proof. A compact real algebraic torus is a product of algebraic circles; equivalently one may use a real torus isogeny, which is an isomorphism on infinitesimals. Apply [lemma 46.2](#) in each coordinate. Coefficient extraction is additive by the normal-form addition rule. Any nonzero coordinate has a nonempty support, so one of its coefficients is nonzero. Finally $\mathfrak{m}_{\mathbb{R}}$ contains the distinct monomials $\omega^{-\gamma}$ for all $\gamma > 0$, and [lemma 33.1](#) precludes a set-sized underlying collection. $\square$

For the circle, $\chi_{1,\eta}$ is the character χ_{η} of [proposition 35.1](#) restricted to $\mu_{\mathbb{T}}$, and on the layer $\{u \in \mu_{\mathbb{T}} : v(u - 1) \geq \gamma\}$ the map $\chi_{1,\gamma}$ induces the layer isomorphism `trigonometry:rot:thm:graded`.

46.3 The class quotient and its universal property for observations

For $g \in G$ put $g_0 = \text{st}(g)$ and use [proposition 46.1](#) to write uniquely

$$gg_0^{-1} = s_g z_g, \quad s_g \in \mu_S, \quad z_g \in \mu_Z.$$

Define

$$\varpi(g) = (g_0, z_g) \in G_0 \times \mu_Z. \tag{46.2}$$

This defines the quotient by concrete pairs of sets, rather than by treating proper-class cosets as elements of another class.

Theorem 46.4 (Exact classification of set-sized observations). *The map ϖ of [equation \(46.2\)](#) is a surjective class homomorphism with kernel μ_S . Every homomorphism from G to a set group factors uniquely through ϖ . For every $g \notin \mu_S$ there is a homomorphism from G to a set group that does not kill g .*

Proof. The decomposition $G = \mu_G \rtimes G_0$ is compatible with $\mu_G = \mu_S \times \mu_Z$. Conjugation by G_0 preserves μ_S and is trivial on μ_Z , since the latter is central. Therefore the central infinitesimal component of a product is the product of the two central components, and ϖ is a homomorphism. The point (g_0, z) is represented by zg_0 , which proves surjectivity, and the kernel is exactly μ_S .

Let $\varphi : G \rightarrow \mathcal{T}$ have set-sized target. Restricting to $S(\mathbf{No})$ and using [theorem 45.2](#) shows that φ kills μ_S . Its unique factor through the explicit quotient is $\bar{\varphi}(g_0, z) = \varphi(zg_0)$. To separate $g \notin \mu_S$, first check its real component. If $g_0 \neq I$, the homomorphism $\text{st} : G \rightarrow G_0$ separates it. If $g_0 = I$, then $z_g \neq I$, and one of the homomorphisms of [proposition 46.3](#), composed with the second component of ϖ , separates it with target $(\mathbb{R}, +)$. $\square$

Equivalently, a homomorphism into a set group $\mathcal{T}$ amounts to homomorphisms $\varphi_0 : G_0 \rightarrow \mathcal{T}$ and $\varphi_Z : \mu_Z \rightarrow \mathcal{T}$ whose images commute, with $\varphi(g) = \varphi_0(g_0)\varphi_Z(z_g)$. This is a classification in terms of the ordinary real factor and a divisible abelian class factor. It does not claim that every additive map on $\mathfrak{m}_{\mathbb{R}}$ is a finite combination of coefficient extractions. The coefficient maps are used for separation, not as a description of the entire algebraic dual.

Theorem 46.5 (The semisimplicity criterion). *The class group G has a universal set-sized quotient if and only if G is semisimple. When it exists, standard part is such a quotient.*

Proof. If G is semisimple, μ_Z is trivial, and [corollary 45.3](#) applies.

Conversely, suppose $\dim Z > 0$ and let $v : G \rightarrow \mathcal{T}_0$ be a proposed universal homomorphism with $\mathcal{T}_0$ a set group. The first part of [theorem 46.4](#) gives $\mu_S \subseteq \ker v$. If some $g \notin \mu_S$ lay in $\ker v$, the separating homomorphism supplied by that theorem could not factor through v . Hence $\ker v = \mu_S$. Its restriction to μ_Z would then be injective, since $\mu_Z \cap \mu_S = \{I\}$. This injects a proper class into a set, which is impossible by replacement and [proposition 46.3](#).

For a compact connected algebraic group, the connected center has dimension zero exactly when the group is semisimple. This completes the equivalence. $\square$

Remark 46.6 (A limitation specific to small targets). The identity homomorphism of G detects all of its infinitesimal elements. Its target is a proper class, so it does not contradict the criterion. Conversely, standard part has a large kernel even though it is universal among maps to sets in the semisimple case. Universal does not mean faithful.

47 Examples over surreal and surcomplex scalars

47.1 Classical and exceptional semisimple groups

For $n \geq 2$, every homomorphism $SU(n, \mathbf{No}[i]) \rightarrow \mathcal{T}$ to a set group factors uniquely through $SU(n, \mathbb{C})$. For $n \geq 3$ the analogous statement holds for $SO(n, \mathbf{No})$ and $SO(n, \mathbb{R})$. These are the cases of Part III ([theorem 34.3](#)). The group $SO(4)$ is semisimple with two simple Lie factors, so its infinitesimal normal subgroups use two segments, not one. In contrast, $SO(2)$ is a torus and fails the set-universality criterion.

The same theorem applies to the compact symplectic groups and to compact real forms of the exceptional simple algebraic groups. No case-by-case root-system calculation is required. Finite central coverings and finite central quotients change the real group that appears as the universal recipient but do not alter the infinitesimal argument. The orthogonal and special unitary instances overlap Part III; the uniform all-type statement follows from the general proof here.

47.2 An explicit formula for the unitary obstruction

Let $g \in \mathbf{U}(n, \mathbf{No}[i])$ and put $g_0 = \text{st}(g) \in \mathbf{U}(n, \mathbb{C})$. Then $\det(g)/\det(g_0)$ is a near-one circle element. Define

$$\varpi_n(g) = \left(g_0, \ell_{\mathfrak{m}} \left(\frac{\det g}{\det g_0} \right) \right) \in \mathbf{U}(n, \mathbb{C}) \times \mathfrak{m}_{\mathbb{R}}. \quad (47.1)$$

Proposition 47.1 (The exact unitary small quotient). *The map ϖ_n is a surjective class homomorphism with kernel $\mu_{\mathbf{SU}(n)}$. All homomorphisms from $\mathbf{U}(n, \mathbf{No}[i])$ to set groups factor through it, and its points are separated by such homomorphisms. In particular, $\mathbf{U}(n, \mathbf{No}[i])$ has no universal set-sized quotient.*

Proof. Both determinant and standard part are homomorphisms, and $\ell_{\mathfrak{m}}$ is additive on the near-one circle, so ϖ_n is a homomorphism. Given (g_0, θ) , its preimage is $\exp(i\theta/n)g_0$, with the scalar $\exp(i\theta/n)$ multiplying the identity matrix. The determinant of that scalar matrix is $\exp(i\theta)$, so the formula gives exactly the desired point. The kernel condition is $\text{st}(g) = I$ and $\det(g) = 1$. The remaining assertions are [theorems 46.4](#) and [46.5](#). $\square$

For example, for every fixed $\gamma > 0$ the map

$$g \mapsto \text{coef}_{\gamma} \ell_{\mathfrak{m}} \left(\frac{\det g}{\det \text{st}(g)} \right) \in (\mathbb{R}, +)$$

is a group homomorphism with a set target. It detects the central matrix $\exp(i\omega^{-\gamma}/n)I_n$, whose standard part is the identity. Different coefficients need not be assembled into a single set-valued faithful map. This refines the last clause of [proposition 35.1](#): the failure of the standard-part factorization for $\mathbf{U}(n, \mathbf{No}[i])$ is now an exact class quotient.

47.3 Products and finite observations

For $G = \mathbf{SU}(2, \mathbf{No}[i]) \times \mathbb{T}(\mathbf{No})$, the invisible subgroup is $\mu_{\mathbf{SU}(2)} \times \{1\}$, while a separating class quotient is $\mathbf{SU}(2, \mathbb{C}) \times \mathbb{T}(\mathbb{R}) \times \mathfrak{m}_{\mathbb{R}}$. This example separates three issues: real nonabelian information survives, nonabelian infinitesimal information is invisible to sets, and central infinitesimal information remains visible.

All compact connected groups nevertheless have trivial finite quotients. For each ordinary $m \geq 1$, the m th-power map on a compact connected real Lie group is surjective, by placing a point in a maximal torus. This is a first-order assertion for fixed m and transfers to G . If $\mathcal{T}$ is finite, choose m divisible by the orders of all its elements. For every $g = x^m$, any homomorphism to $\mathcal{T}$ gives $\varphi(g) = \varphi(x)^m = 1$. Thus central infinitesimals need genuinely infinite set targets to be detected. This familiar finite-quotient observation is not the source of the much stronger all-set theorem.

48 The split contrast: special linear groups have no small images

Compactness constrained conjugation to preserve infinitesimal depth. A split torus can change scale, and the outcome is opposite.

Theorem 48.1 (Triviality of all set-sized special-linear images). *Let $\mathbb{K} = \mathbf{No}$ or $\mathbb{K} = \mathbf{No}[i]$, and let $n \geq 2$. Every class homomorphism $\varphi : \mathbf{SL}_n(\mathbb{K}) \rightarrow \mathcal{T}$ to a set-sized group is trivial.*

Proof. For $a \in \mathbb{K}$ write $u(a) = I + aE_{12}$. The map $a \mapsto \varphi(u(a))$ cannot be injective, because $\mathbb{K}$ contains sets larger than $\mathcal{T}$. Hence the kernel contains $u(a)$ for some $a \neq 0$. Direct multiplication with the determinant-one diagonal matrix $\text{diag}(c, c^{-1}, 1, \dots, 1)$ gives

$$\text{diag}(c, c^{-1}, 1, \dots, 1) u(a) \text{diag}(c, c^{-1}, 1, \dots, 1)^{-1} = u(c^2 a). \quad (48.1)$$

If $\mathbb{K} = \mathbf{No}[i]$, every nonzero b/a is a square, so all $u(b)$ lie in the kernel. If $\mathbb{K} = \mathbf{No}$, every positive b/a is a square, and the other sign is obtained from $u(-a) = u(a)^{-1}$; again every $u(b)$ is killed.

Conjugation by determinant-one signed permutation matrices moves this root subgroup to every off-diagonal elementary subgroup, with a possible sign absorbed into the parameter. Thus the kernel contains all elementary matrices $I + bE_{ij}$, $i \neq j$.

For completeness, those elementary matrices generate $\text{SL}_n(\mathbb{K})$. Gaussian elimination uses elementary row additions and determinant-one row interchanges to reduce to a diagonal matrix. The necessary two-by-two diagonal blocks are also elementary. With

$$u(a) = \begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix}, \quad l(b) = \begin{pmatrix} 1 & 0 \\ b & 1 \end{pmatrix}, \quad w(a) = u(a)l(-a^{-1})u(a) = \begin{pmatrix} 0 & a \\ -a^{-1} & 0 \end{pmatrix},$$

one has $w(a)w(-1) = \text{diag}(a, a^{-1})$. This finishes elimination inside the elementary subgroup and proves that the kernel is all of $\text{SL}_n(\mathbb{K})$. $\square$

Corollary 48.2 (Determinant is the only source of small general-linear images). *For $\mathbb{K} = \mathbf{No}$ or $\mathbf{No}[i]$ and $n \geq 2$, every homomorphism from $\text{GL}_n(\mathbb{K})$ to a set group factors uniquely through $\det : \text{GL}_n(\mathbb{K}) \rightarrow \mathbb{K}^\times$. Every action of $\text{SL}_n(\mathbb{K})$ on a set is trivial.*

Proof. The kernel of determinant is $\text{SL}_n(\mathbb{K})$, which is killed by [theorem 48.1](#). The determinant is onto, which gives unique factorization. For actions use the set group $\text{Sym}(\mathcal{X})$. $\square$

The determinant target is still a proper class. The corollary does not classify its abstract set-valued characters or assert that determinant itself is a set-sized universal quotient. The elementary argument is closely related to the classical simplicity mechanism for projective special linear groups; its proper-class consequence is presented as a contrast, not as a new simplicity theorem.

49 Omnific lattices inside compact matrix groups

An omnific integer has Conway normal form $z = z_0 + \sum_{\alpha > 0} z_\alpha \omega^\alpha$ with $z_0 \in \mathbb{Z}$, $z_\alpha \in \mathbb{R}$ and reverse-well-ordered set support on the positive exponents. The ring is $\mathbf{Oz}$; this is the ordinary omnific-integer definition, not the valuation ring $\mathcal{O}$ and not a nonstandard model of the ordinary integers. The results of this section are already in the collection, except the unitary count; they are printed with source 11's short proofs and with credit.

Lemma 49.1 (Finite omnific integers). *One has $\mathbf{Oz} \cap \mathcal{O} = \mathbb{Z}$ and $\mathbf{Oz}[i] \cap (\mathcal{O} + i\mathcal{O}) = \mathbb{Z}[i]$.*

Proof. A nonzero positive-exponent part of an omnific integer has an infinite leading monomial, so the number is not finite. Thus a finite omnific integer has only its ordinary integral constant term. Apply this argument to both coordinates in the complexification. $\square$

The omnific-groups report contains this as `ogl:lem:bounded` [21]: bounded coordinates are constant, and $\mathbf{Oz} \cap \mathbb{R} = \mathbb{Z}$.

For a real matrix group not defined over $\mathbb{Z}$, the notation “points over $\mathbf{Oz}$ ” can mislead: $\mathbf{Oz}$ does not contain all real constants. The following statement is explicitly about entrywise integral matrices in a chosen real representation. For the classical orthogonal and unitary equations below, the equations themselves have integral coefficients, so the usual ring-of-points notation is legitimate.

Proposition 49.2 (Compact integral collapse). *In any fixed bounded real matrix realization of a compact group,*

$$G \cap \mathrm{GL}_N(\mathbf{Oz}) = G_0 \cap \mathrm{GL}_N(\mathbb{Z}).$$

This group is finite. In an orthogonal realization the same assertion can be stated entrywise, since transposition supplies the inverse.

Proof. An element on the left has finite entries by compactness, and its omnific entries are ordinary integers by lemma 49.1. Its inverse has the same properties, so it belongs to $\mathrm{GL}_N(\mathbb{Z})$. Being a real matrix, it is in G_0 . The reverse inclusion is immediate. There are only finitely many integral matrices within any fixed real coordinate bounds, which proves finiteness. In an orthogonal realization the inverse has exactly the transposed entries, so the entrywise condition already gives invertibility over the indicated ring. $\square$

For a closed subgroup scheme of $\mathrm{GL}_{N,\mathbb{Z}}$ with compact real points, the equality is the rigid case of `ogl:alg:thm:real` [21], which characterizes all real algebraic groups with no new omnific points.

Theorem 49.3 (Explicit orthogonal and unitary omnific groups). *For every ordinary $n \geq 1$:*

- (i) $\mathrm{SO}_n(\mathbf{Oz})$ consists exactly of the signed permutation matrices of determinant one, and has order $2^{n-1}n!$.
- (ii) $\mathrm{SU}_n(\mathbf{Oz}[i])$ consists exactly of the monomial matrices with nonzero entries in $\{1, -1, i, -i\}$ and determinant one, and has order $4^{n-1}n!$.

Proof. For an orthogonal matrix each entry has absolute value at most one, so an omnific entry is in $\{-1, 0, 1\}$. The sum of the squared entries in each column is one, so exactly one entry is 1 or -1 , and orthogonality puts these entries in distinct rows. For each permutation, choosing the first $n - 1$ signs arbitrarily determines the last sign from the determinant condition. This gives $2^{n-1}n!$.

For a unitary matrix, the real and imaginary coordinates of every entry are finite omnific integers, hence ordinary integers. Each column has a sum of squares of these integer coordinates equal to one, so exactly one entry is a Gaussian unit and all the others vanish. Distinct columns use distinct rows. After fixing the permutation and the first $n - 1$ Gaussian units, the determinant condition fixes the last unit, giving $4^{n-1}n!$. $\square$

Part (i) is `odg:cor:orthogonal` of the omnific Diophantine report [22], proved there by sources 01 and 02 of that report from the row sums, and reproved in the omnific-groups report through the column bound (`ogl:alg:sec:examples`, with $|\mathrm{SO}_3(\mathbf{Oz})| = 24$). Part (ii) is an explicit instance of `ogl:alg:thm:real`, since the unitary group is a real algebraic group with compact real points; the omnific-groups report notes that the complex orthogonal group $\mathrm{SO}_3(\mathbf{Oz}[i])$ is not compact in this sense and does acquire new points, so the two statements do not conflict. No independent novelty is claimed for the signed-permutation classification. These assertions are elementary consequences of normal form and positive definiteness. They supply a boundary for the main theory: the surreal

compact group has a proper-class infinitesimal kernel, but its omnific lattice contains no nonidentity infinitesimal matrix at all. Large surreal analytic symmetry does not by itself produce large compact omnific arithmetic.

50 Consistency, contrasts, and what remains

50.1 Consistency with Parts II and III

This subsection is written for the addition; it relates statements and adds none. Source 11 and Part III were written independently. Their statements agree wherever both apply, and Part III is stronger in some places.

- *Semisimple groups.* For semisimple G , μ_Z is trivial and the invisible subgroup of [theorem 39.1](#) is all of μ_G . For $SO(3, \mathbf{No})$, $SU(2, \mathbf{No}[i])$, $SO(n, \mathbf{No})$ with $n \geq 3$ and $SU(n, \mathbf{No}[i])$ with $n \geq 2$ this is [theorems 33.2](#), [34.1](#) and [34.3](#). The proofs are different ([section 45](#)). For $SO(4)$, Part III uses Givens factorization and Part V uses two segments ([theorem 42.4](#)); the conclusions agree.
- *Tori.* The points separated by [proposition 46.3](#) are those separated by [proposition 35.1](#): source 11's coefficient of t^γ , with $t^\gamma = \omega^{-\gamma}$, is Part III's coef_γ . For $U(n)$, Part III's statement that standard part is not universal becomes the exact class quotient $U(n, \mathbb{C}) \times \mathfrak{m}_{\mathbb{R}}$ of [proposition 47.1](#).
- *Normal subgroups and series.* For $SO(3)$, the segment subgroups μ_C , the ambient-normal classification, the lower central and derived series μ_{nC} , $\mu_{2^n C}$ and the perfect residual μ_{C_∞} agree with [theorems 30.6](#), [31.3](#) and [31.4](#); for $C = [\alpha, \infty)$ the residual is $K_{\gg \alpha}$.
- *Where Part III is stronger.* For $SO(3)$, Part III proves single-commutator exactness ([theorems 29.3](#) and [29.4](#)), commutator width one for perfect segment subgroups ([theorem 31.2](#)), width two for normal generation ([theorem 30.3](#)), and the universal residually nilpotent quotient ([theorem 31.6](#)). Source 11 claims none of these for other types.
- *Where Part V is more general.* It treats every compact connected real algebraic group, including the compact symplectic and exceptional types, finite central quotients and products; all proper normal subgroups in simple type ([theorem 42.3](#)); every segment, not only principal and convex ones, in [theorem 44.1](#); and representations on set-sized vector spaces of any dimension ([corollary 45.4](#)).

50.2 Relation to the omnific-groups report

This subsection is written for the addition. For $n \geq 3$, the omnific-groups report [21] proves as `ogl:el:prop:finiteside` that every class homomorphism from $SL_n(\mathcal{O}_{\text{fin}}(k))$ to a set-sized group factors through standard part onto $SL_n(k)$, and it compares that statement with Part III in `ogl:el:rem:e3`. Neither that proposition nor [theorem 39.1](#) implies the other. A homomorphism defined on a compact subgroup of $SL_n(\mathcal{O}_{\text{fin}})$ need not extend to the whole group, and conversely the proposition says nothing about homomorphisms defined only on a compact subgroup. The two are consistent. A compact torus inside $SL_3(\mathcal{O}_{\text{fin}}(\mathbb{R}))$ carries the set characters of [proposition 46.3](#), which do not extend. The group $SL_n(\mathcal{O}_{\text{fin}}(\mathbb{R}))$ is a non-normal subgroup of $SL_n(\mathbf{No})$ with set-sized image $SL_n(\mathbb{R})$, while $SL_n(\mathbf{No})$ has only trivial set-sized images ([theorem 48.1](#)); restricting the trivial maps causes no conflict. The compact omnific collapse of [section 49](#) agrees with Part I of the omnific-groups report.

50.3 What the results settle, and what remains

The representation-theoretic question posed at the beginning of Part V has an exact answer in the compact connected real algebraic setting. The invisible subgroup is not all infinitesimals in general: it is precisely their derived subgroup. Removing it leaves a real compact group and a central abelian infinitesimal group. The former is a set; the latter is pointwise observable by set targets but cannot itself fit faithfully into a set. This proves the semisimplicity criterion and explains both sides of it.

The valuation results determine the ambient-normal infinitesimal subgroups in each simple type and in their finite semisimple products. They also determine mixed *generated* commutator subgroups and transfinite perfect residuals. Their proofs neither use nor establish a global exponential on $\mathbf{No}[i]$, an infinite-dimensional Hilbert theory, or a classification of all arithmetic homomorphisms from $\mathbf{Oz}$.

Several substantive extensions remain outside the proved scope. The first is a general exact *single*-commutator theorem with two prescribed valuation bounds in all compact semisimple types. The second is a comparably explicit small-observation classification for arbitrary noncompact real reductive groups, where compact derived factors and split root subgroups interact with a more complicated real torus. The third is a classification of normal subgroups of μ_G without ambient normality; the segment theorem must not be used as a substitute for that problem. The status notes in [section 38.1](#) record these problems against the questions left open by source 10.

A further size-sensitive direction is to replace the full surreal class by a fixed set-sized real closed Hahn field. The local normal-generation proof still applies, but the collision step only eliminates infinitesimals whose lower-scale interval is larger than the chosen target. A formulation with explicit interval cardinalities should produce a hierarchy of small-target theorems rather than the absolute all-set conclusion proved here, compare [equation \(33.4\)](#). This direction isolates exactly which part of the argument is lost on passing from $\mathbf{No}$ to a single small workspace.

50.4 Proof audit and foundations checklist

Source 11 lists the delicate points at which a superficially similar argument could fail.

No analytic convergence is transferred. The normal-generation and mixed-commutator proofs use real Nash maps and first-order local existence statements with fixed real radii. The divided maps extend as Nash maps by [lemma 40.2](#). Real Taylor estimates enter only as finite inequalities. Strong Hahn summation is used separately, for the infinitesimal central logarithm, where positive supports justify it.

The target normalization can be large but only finitely large. Depth $\nu(h) \geq \alpha$ implies that $\sigma^{-1}(h)/\omega^{-\alpha}$ is finite, not that it lies in a fixed small real ball. [Lemma 40.5](#) closes exactly this gap: an ordinary root and a finite power enlarge local surjectivity to the whole valuation ball.

Simplicity is used only where needed. Exact normal closure of one infinitesimal requires a simple nonabelian Lie algebra. It is false in that form for a semisimple element supported in only one simple factor, and false for a central torus. The mixed commutator theorem needs only semisimplicity. Product classification is proved by extracting individual components, not by assuming that a subdirect product is a direct product.

The collision argument is set-sized at every execution. For a fixed target $\mathcal{T}$, one set of scales of cardinality greater than $|\mathcal{T}|$ is chosen. Its image is a set and ordinary pigeonhole applies. No set theorem is applied directly to an unbounded proper-class family.

Quotients use explicit representatives. The class quotient [equation \(46.2\)](#) has underlying class

$G_0 \times \mu_Z$ of pairs of sets. The notation G/μ_S is shorthand for that model, not a collection of proper-class cosets.

Central separation is not a dual-classification theorem. Normal-form coefficients separate $\mathfrak{m}_{\mathbb{R}}$. They need not exhaust all abstract additive functionals, and the nonexistence of a set-sized universal recipient needs separation only.

Scalar and arithmetic conventions are distinct. The finite ring $\mathcal{O}$, its infinitesimal ideal $\mathfrak{m}$, the omnific integers $\mathbf{Oz}$ and the ordinary integers $\mathbb{Z}$ are not identified. In particular $\mathbf{Oz} \cap \mathcal{O} = \mathbb{Z}$ is a conclusion from normal form, not a convention.

Verification boundary. The companion program checks finite matrix identities, selected real Lie spanning calculations, formal logarithm coefficients and finite counts (section 36.7). It does not verify class replacement, Nash implicit-function transfer, the normal-generation proofs, or novelty. No Lean source is supplied and no Lean build was performed. Source 11 asks independent review to focus on theorems 41.3, 43.2, 46.4 and 46.5.

A Formula and convention reference for Part II

Object	Convention or formula
Rotation group	$G(F) = \{A : A^T A = I, \det A = 1\}$, acting on column vectors.
Hat map	$\hat{u}v = u \times v$, with the standard right-handed cross product.
Rodrigues	$\text{Rot}(n; c, s) = cI + (1 - c)nn^T + s\hat{n}$.
Half-turn	$H_n = 2nn^T - I$; its centralizer is a line stabilizer, not just an axis circle.
Spin map	$p(a, u) = (a^2 - \ u\ ^2)I + 2uu^T + 2a\hat{u}$, for $a^2 + \ u\ ^2 = 1$.
Cayley chart	$\text{Cay}(t) = ((1 - t \cdot t)I + 2tt^T + 2\hat{t})/(1 + t \cdot t)$.
Cayley product	$t \star u = (t + u + t \times u)/(1 - t \cdot u)$, where the denominator is nonzero.
Infinitesimal kernel	$K = \ker \text{st}$, identified with all $\mathfrak{m}^3$ by the Cayley chart.
Logarithmic leading term	$\text{Log}_{\mathfrak{m}} \text{Cay}(t) = 2\hat{t} + \text{higher terms}$.
Valuation	$v(\omega^{-\gamma}) = \gamma$; infinitesimals have $v > 0$.
Graded coordinate	$\ell_{\gamma}(U) = 2 \text{st}(\omega^{\gamma}t(U))$.
Group commutator	$(U, V) = UVU^{-1}V^{-1}$.
Lie bracket	$[\hat{u}, \hat{v}] = \widehat{u \times v}$.
Finite phase period	$\ker \text{cis} = 2\pi\mathbb{Z}$, on $\mathcal{O}$.
Global phase period	$\ker \text{cis}_{\text{EK}} = 2\pi\mathbf{Oz}$, on $\mathbf{No}$.
Global exponential	Radial map $\mathcal{E}$ of equation (25.1); its identity fiber is not an additive kernel.
Fine Jacobian	Right-trivialized as $(d\mathcal{E}_w h)\mathcal{E}_w^{-1} = \widehat{\mathcal{J}(w)h}$.
Critical radii	$r \in (2\pi\mathbf{Oz})_{>0}$, rank one; $r = 0$ is regular.

B Source, dependency, and verification ledger

B.1 Pins and files

Item	Value
Source 04 pin	d22a5b35d5b3040e870c3cfd6d1c8f7259e094b0
Source 09 pin	dcf86662b574988e75d08d515d3a2a5ba7bbc0d7
Placement	5fe7f8d (both manuscripts placed, source 04's text as the starting file)
Merge written at	50cb709
Source 04 files	code/04-three-space-verify_identities.py, data/04-three-space-verification.txt, data/04-three-space-requirements.txt
Source 09 files	code/09-so3-verify.py, data/09-so3-verification.txt, data/09-so3-requirements.txt
Source 10 pin	465a54b479a1ee842cbf7db1689a7d2f6bfe25e1
Placement of source 10	7b5f934 (files only; this article unchanged)
Part III written at	7b5f934
Source 10 files	code/10-rotation-quotients-verify.py, code/10-rotation-quotients-build.sh, code/10-rotation-quotients-build.ps1, data/10-rotation-quotients-verification.json, data/10-rotation-quotients-requirements.txt, 10-rotation-quotients-SOURCE_AUDIT.md
Source 11 pin	9693b28c24e6fcb317ce47969a40185c5dfef402
Placement of source 11	66d7e55 (files only; this article unchanged)
Part V written at	66d7e55
Source 11 files	code/11-compact-groups-verify.py, code/11-compact-groups-build.sh, data/11-compact-groups-verification_results.json, data/11-compact-groups-requirements.txt, 11-compact-groups-SOURCE_AUDIT.md

The manuscripts themselves, their PDFs, READMEs and source 09's checksum list are not shipped. The code and data files are byte-identical to the delivered ones. Source 10 delivered no checksum list; its source and novelty audit is shipped verbatim as `10-rotation-quotients-SOURCE_AUDIT.md`. That file names the delivered article and files by their unprefixed names. Source 11 also delivered no checksum list. Its manuscript, PDF and README are not shipped; its source and proof audit is shipped verbatim as `11-compact-groups-SOURCE_AUDIT.md`, which likewise uses the unprefixed names.

B.2 Source 04: inspected sources and dependencies

At its pin, source 04 specifically inspected the trigonometry report and the Lean module `Surreal/Algebra/Geometry.lean`. The repository is a research source, not a substitute for published foundations. The source comparison is not an exhaustive audit of every file or of all published work on non-Archimedean geometry.

The main dependencies of Part I are these. Real closedness and the restricted-analytic Hahn-field model theorem are imported from [1], with support-combinatorial cautions informed by its erratum [2]. The canonical global extension available from an initial integer part is attributed to [3]; Part I

needs only its finite restriction. The classical spherical identities are recorded in [6]; their algebraic proofs and sign conditions are provided here. The solid-angle expression is classical and attributed to [7]. No priority claim is made for the remaining formulations, including the uniform gnomonic estimate, the exceptional-family classification, or their combination into a surreal development. The proofs are mathematical proofs, not Lean proof terms.

B.3 Source 09: inspected sources and dependency ledger

At its pin, source 09 reviewed the documentation index, the introductory status of the formalization ledger, the surquaternion README and the relevant surquaternion source sections. Its bibliography separates the report directories identified through that index from the source sections actually inspected. It did not rebuild the repository.

Result family	Essential assumptions	Evidence in this report
Isometries, axes, spin, Cayley, centralizers	Real closed ordered field; finite matrices	Direct finite proofs and selected symbolic identities; spin cited from <code>squat.thm.rotations</code> .
Standard-part splitting	Real closed $F \supseteq \mathbb{R}$	Direct ring-homomorphism proof.
Torsion-free and unique divisibility	Real closed $F \supseteq \mathbb{R}$	Algebraic circle and root proofs.
Perfect infinitesimal group	Proper real closed extension; fourth roots	Explicit commutator construction; known precedent [9].
Finite-subgroup rigidity	Finite group, real closed $F \supseteq \mathbb{R}$	Averaging and positive polar factor, fully proved.
Strong exp/log and BCH	Set-supported normal forms or a divisible Hahn workspace	Support proof using Higman's lemma; formal identities.
Global phase	Ehrlich–Kaplan initial-integer-part normalization	Imported normalization, explicitly expressed and analyzed.
Rotation exponential differential	Fine variable derivatives and local Taylor remainder bounds	Direct radial/perpendicular calculation.
Full fine discreteness	Full surreal cut property; small index sets	Direct separating-radius proof.
Semialgebraic homotopy	Set-sized real closed field	Imported comparison theorem [10] and explicit spin branches.
Machine formalization	A compiled proof assistant development	Not supplied; no such verification is claimed.

B.4 Source 10: inspected sources and dependency ledger

At its pin 465a54b, source 10 read, through a connector and without a clone or build, the repository and documentation catalogues, this report’s README with its theorem inventory, provenance and limitations, the source sections of Part II on quaternions and conjugacy, the infinitesimal kernel, unique divisibility and the commutator-width-one proof (article lines 2650–2780 and 2830–3030 at that pin), and the trigonometry report’s README, including the statement of its Theorem 4.2. The `docs/new` directory then held only its procedural README. Published sources inspected were [11, 9, 12].

Result	Main input	Essential boundary
Mixed-depth image	Quaternion scalar identity; equal-scalar conjugacy	Positive depths; real closedness; natural valuation fixed throughout.
Normal closure	Two conjugates; ordinary powers	Natural finite-ratio property; conjugation by the full G .
All proper normals	Normal closure; real simplicity	Subgroups normal in G , not merely in K .
Perfect-cut classification	Exact Minkowski-sum law	Upper segments; divisible ordered value group.
Perfect residual	Finite series formulas; convex subgroup Δ_α	Principal threshold $\alpha > 0$; stage ω is an intersection.
Value reconstruction	Compact normal-lattice elements	Set-sized field; whole subgroup lattice, not a claimed first-order formula.
Universal set quotient	Normal closure; proper-class value interval	Full No ; class homomorphism; target a set in the same foundation.
Surcomplex $SU(2)$	Spin quotient; perfect small kernel	Fixed conjugation; finite central image removed explicitly.
Higher fixed ranks	Near-identity finite Givens products	$SO(n)$, $n \geq 3$; $SU(n)$, $n \geq 2$; ordinary finite n .
Circle counterexample	Canonical infinitesimal logarithm; normal-form coefficient	Additional established strong-summation construction; not used by the main theorem.

B.5 Source 11: inspected sources and dependency ledger

At its pin 9693b28, source 11 read through a connected GitHub reader, without a successful clone or build, the recursive tree and root README, the documentation catalogue (then 51 reports), the `docs/new` directory (then only its procedural README), this report’s directory listing and README, and `10-rotation-quotients-SOURCE_AUDIT.md` in full. Published sources inspected were [11], including a visual inspection of the PDF, [9], with Section 5, Fact 5.3 and Claims 5.4–5.5 read explicitly, [13] and [12]. Publisher records were checked for [16, 15, 14].

Result	Main input	Essential boundary
Standard-part splitting	Real bounds from compactness; transfer	Compactness of the real group only.
Nash toolkit	Real inverse function theorem; semialgebraic graphs	Finite first-order statements with real radii; no completeness.
Normal closure	Spanning lemma; division by the scale; roots	Simple nonabelian Lie algebra; conjugation by the full G ; no width bound.
Proper normal subgroups	Normal closure; noncentral normal generation	Simple type; finite center; ambient normality.
Product classification	Finite central isogeny; component extraction	Semisimple; subgroups of μ_G normal in G .
Mixed commutator subgroups	Bracket spanning; division by ab	Generated subgroups, not single commutators.
Series and residuals	Segment calculus	Derived residual at stage ω ; values above all ordinary multiples.
Semisimple invisibility	Normal closure; proper-class value intervals	Full No ; class homomorphisms; set targets.
Central tori	Infinitesimal circle logarithm; coefficients	Separation only, not the additive dual.
Semisimplicity criterion	Class quotient; separation; replacement	Connected real linear algebraic group with compact connected real points.
SL_n triviality	Root scaling by squares; elementary generation	Classical mechanism; presented as a contrast.
Compact omnific collapse	$\mathbf{Oz} \cap \mathcal{O} = \mathbb{Z}$	Entrywise integral matrices in a chosen realization.

References

- [1] L. van den Dries and P. Ehrlich, *Fields of surreal numbers and exponentiation*, *Fundamenta Mathematicae* **167** (2001), no. 2, 173–188. doi:10.4064/fm167-2-3.
- [2] L. van den Dries and P. Ehrlich, *Erratum to “Fields of surreal numbers and exponentiation”*, *Fundamenta Mathematicae* **168** (2001), no. 3, 295–297. doi:10.4064/fm168-3-5.
- [3] P. Ehrlich and E. Kaplan, *Surreal ordered exponential fields*, *Journal of Symbolic Logic* **86** (2021), no. 3, 1066–1115, especially §11. doi:10.1017/jsl.2021.59. See also the citation-correction erratum, **87** (2022), no. 2, 871, doi:10.1017/jsl.2022.5. Source 09 cites the preprint version, <https://arxiv.org/abs/2002.07739v3>.
- [4] B. H. Neumann, *On ordered division rings*, *Transactions of the American Mathematical Society* **66** (1949), 202–252. [Primary journal archive](#).
- [5] G. Higman, *Ordering by divisibility in abstract algebras*, *Proceedings of the London Mathematical Society* (3) **2** (1952), 326–336. doi:10.1112/plms/s3-2.1.326.

- [6] F. W. J. Olver et al., eds., *NIST Digital Library of Mathematical Functions*, §4.42, “Solution of Triangles,” especially §4.42(iii), “Spherical Triangles.” [Official online section](#), consulted September 22, 2026.
- [7] A. van Oosterom and J. Strackee, *The solid angle of a plane triangle*, IEEE Transactions on Biomedical Engineering **BME-30** (1983), no. 2, 125–126. doi:10.1109/TBME.1983.325207.
- [8] J. Voight, *Quaternion Algebras*, Graduate Texts in Mathematics **288**, Springer, 2021. Author’s official and corrected editions: <https://jvoight.github.io/quat.html>.
- [9] M. Bays and Y. Peterzil, *Definability in the group of infinitesimals of a compact Lie group*, Confluentes Mathematici **11**, no. 2 (2019), 3–23. doi:10.5802/cml.58. <https://arxiv.org/abs/1901.10831>.
- [10] E. Baro and M. Otero, *On o-minimal homotopy groups*, Quarterly Journal of Mathematics **61**, no. 3 (2010), 275–289. doi:10.1093/qmath/hap011. <https://arxiv.org/abs/0810.0365>.
- [11] A. D’Andrea and A. Maffei, *Commutators of small elements in compact semisimple groups and Lie algebras*, Journal of Lie Theory **26** (2016), no. 3, 683–690. <https://www.heldermann-verlag.de/jlt/jlt26/andla2e.pdf>. Source 10 uses Theorem 2.5 for historical comparison; source 11 cites the same local openness.
- [12] O. Bournez and Q. Guilmant, *Surreal fields stable under exponential and logarithmic functions*, arXiv:2201.08199v1 (2022). <https://arxiv.org/abs/2201.08199>. Source 10 uses Sections 2.2–2.3, Corollary 2.1 and Theorem 2.6 for the real-closed-field, Hahn and normal-form background; source 11 uses Section 2.3 and the near-one logarithm.
- [13] L. Kramer, *On small abstract quotients of Lie groups and locally compact groups*, arXiv:1405.2711v3 (2016). <https://arxiv.org/abs/1405.2711>. Source 11 cites it for quantitative normal generation, including a theorem of Nikolov and Segal.
- [14] J. Bochnak, M. Coste and M.-F. Roy, *Real Algebraic Geometry*, Ergebnisse der Mathematik und ihrer Grenzgebiete (3) **36**, Springer, 1998.
- [15] J. H. Conway, *On Numbers and Games*, second edition, A K Peters, 2001; first edition, Academic Press, 1976.
- [16] H. Gonshor, *An Introduction to the Theory of Surreal Numbers*, London Mathematical Society Lecture Note Series **110**, Cambridge University Press, 1986.
- [17] VladimirReshetnikov/Surreal, *Trigonometry on the Surcomplex Plane: Canonical Phases, Arbitrary-Scale Geometry, and Infinitesimal Degeneration*, docs/surcomplex/trigonometry/article.tex. Source 04 inspected it at d22a5b35d5b3; source 09 identified it through the documentation index at dcf86662b574. Labels cited in Parts I and II are those present at 50cb709. Source 10 read its README at 465a54b; Part III cites trigonometry:thm:polar, trigonometry:eq:circlesplit and trigonometry:rot:thm:graded, present at 465a54b and at 7b5f934. Part V cites trigonometry:thm:polar, trigonometry:eq:circlesplit, trigonometry:rot:eq:ellm and trigonometry:rot:thm:graded, present at 66d7e55; source 11 itself did not cite this report.
- [18] VladimirReshetnikov/Surreal, *Finite geometry over an ordered base field*, Surreal/Algebra/Geometry.lean, inspected by source 04 at d22a5b35d5b3.

- [19] VladimirReshetnikov/Surreal, *Surquaternions: Algebra, Infinitesimal Geometry, and Support-Controlled Analysis*, docs/surquaternions/surquaternions/article.tex. Source 09 inspected the README and the relevant algebraic and phase sections at [dcf86662b574](#). Theorem numbers cited here are those at 50cb709.
- [20] VladimirReshetnikov/Surreal, *Foundations*, docs/foundations-and-computation/foundations/; topology and size conventions, identified by source 09 through the documentation index at [dcf86662b574](#).
- [21] VladimirReshetnikov/Surreal, *Omnific Groups and Lattices: algebraic groups, elementary quotients, and missing lattice minima over the omnific integers*, docs/surreal/omnific-groups-and-lattices/article.tex. Part V cites ogl:lem:bounded, ogl:alg:thm:real, ogl:alg:sec:examples, ogl:el:prop:finiteside and ogl:el:rem:e3, present at 66d7e55. The report was in the tree at source 11's pin 9693b28 but not in the catalogue source 11 read.
- [22] VladimirReshetnikov/Surreal, *Omnific Integers and Omnific-Diophantine Geometry*, docs/surreal/omnific-diophantine-geometry/article.tex. Part V cites odg:cor:orthogonal, present at 66d7e55.
- [23] VladimirReshetnikov/Surreal, *Formalization coverage and dependency ledger*, docs/FORMALIZATION.md; see also docs/README.md. Source 09 inspected the introductory status definitions at [dcf86662b574](#).